Calabi–Yau Quantum Groups

Chiral Algebras from Calabi–Yau Categories via E_1/E_2 Factorization

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Abstract

A Calabi–Yau category C of dimension d carries a cyclic A_∞-structure with $\mathbb{S}^d$-framing on Hochschild homology. A chiral algebra A on a curve carries a bar complex $\overline{B}(A)$, a universal Maurer–Cartan element Θ_A, and a modular characteristic $\kappa(A)$ (Volumes I–II). This volume constructs the bridge between them.

For $d = 2$, a functor $\Phi\colon \mathrm{CY}_2\text{-}\mathbf{Cat} \to E_2\text{-}\mathrm{ChirAlg}$ is constructed: cyclic A_∞ to Lie conformal algebra, factorization envelope, $\mathbb{S}^2$-framing enhancement (Theorem CY-A_2). The denominator identity of the associated Borcherds–Kac–Moody superalgebra equals the bar-complex Euler product (Theorem CY-B, conditional on CY-A for $d = 3$).

For $d = 3$, the chiral algebra is natively E_1: the $\mathbb{S}^3$-framing obstruction vanishes topologically ($\pi_3(B\mathrm{Sp}) = 0$), but the holomorphic Chern–Simons trivialization breaks E_2 to E_1. The E_2 braiding is recovered through the Drinfeld center of the E_1-monoidal representation category. For toric CY$_3$ with Ω-deformation, E_1 universality is proved; for compact CY$_3$, it is conjectural. The complete five-step functor chain is verified end-to-end for $\mathbb{C}^3$: polyvector fields, Ω-deformation ($\sigma_3 = h_1h_2h_3$, unique by $\mathrm{HH}^2 = 1$), factorization envelope producing $\mathcal{W}_{1+\infty}$, Drinfeld center identifying $\mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1))$, quantum vertex chiral group. The classical Lie bracket is abelian; all noncommutativity arises from quantization.

A general CY$_3$ is assembled from local quiver charts: chambers of the Bridgeland stability manifold index quiver-with-potential presentations, wall-crossing mutations provide E_1-equivalences between charts, and the global chiral algebra is the homotopy colimit. E_1 descent is unobstructed (the space of E_1 gluings is contractible), the bar construction commutes with homotopy colimits, and the modular characteristic is atlas-independent. The Kontsevich–Soibelman wall-crossing formula is the gauge equivalence of E_1 Maurer–Cartan elements. DT invariants, the topological vertex, and crystal melting admit bar-complex interpretations.

The prototype $K3 \times E$ has $\kappa_{\mathrm{ch}} = 3 = \dim_{\mathbb{C}}$ and $\kappa_{\mathrm{BPS}} = 5$ (the weight of the Igusa cusp form Δ_5); four structural obstructions prevent the shadow tower from reconstructing Δ_5 directly (Theorem 14.112). Ten research programmes ground the CY$_3$ frontier in formal conjectures. For $d \geq 4$, the E_1 stabilization theorem identifies a Bott-periodic 8-fold pattern of shifted structures classified by $\pi_d(B\mathrm{Sp})$. Beyond the CY setting, rank-2 bundles with nonzero CY defect δ produce curved chiral algebras satisfying the inhomogeneous Maurer–Cartan equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = m_0$, with $m_0 = \delta \cdot \omega_C$. The Hitchin system on a genus-g curve gives $\kappa = g$, independent of the gauge group rank.

For the quintic threefold, E_1 universality is proved: this is the first compact CY$_3$ characterization. The chain-level $\mathbb{S}^3$-framing is constructed for the conifold but obstructed for local $\mathbb{P}^2$, identifying the analytic gap as the remaining frontier. The bar Euler product matches the Borcherds denominator identity through arity 6 for $K3 \times E$.

Over 160 compute engines with $16{,}900+$ tests back the numerical claims.

Preface

The bridge

Volume I builds the categorical logarithm: the bar construction $\overline{B}(A)$ for chiral algebras on curves, with five theorems proving its existence, invertibility, Lagrangian complementarity, leading coefficient, and polynomial Hochschild ring. Volume II reads the output in three dimensions: the bar differential is the holomorphic colour, the coproduct is the topological colour, and the Swiss-cheese algebra on $\overline{\mathrm{FM}}_k(\mathbb{C}) \times C_k^<(\mathbb{R})$ is the Steinberg presentation of the holomorphic-topological theory.

This volume constructs the third leg of the triangle. A Calabi–Yau category carries cyclic A_∞-structure; a chiral algebra carries a bar complex. The bridge functor Φ converts the former into the latter: cyclic homology becomes shadow obstruction tower, the CY trace becomes the modular characteristic κ, and the BPS spectrum becomes the generalized root datum of a quantum vertex chiral group. The bar complex converts the enumerative geometry of the CY threefold — Donaldson–Thomas invariants, Gromov–Witten amplitudes, automorphic forms — into the homotopy algebra of a single Maurer–Cartan element.

The elliptic curve as universal bridge

The same elliptic curve E plays six distinct roles across the three volumes:

(i) *Base curve for chiral algebras* (Vol I): the factorization algebra lives on $\mathrm{Ran}(E)$; genus-1 bar amplitudes are governed by the propagator $d \log E(z, w)$.

(ii) *Sewing parameter space* (Vol I, MC5): the modular parameter τ of E is the sewing parameter for the Hilbert–Schmidt kernel.

(iii) *Fiber in $K3\times E$*: the DT partition function decomposes as a sewing of K3 fiber contributions along E.

(iv) *Jacobi form parameter space*: the Fourier coefficients of $\phi_{0,1}(\tau, z)$ serve as root multiplicities, shadow tower inputs, and BPS degeneracies.

(v) *Siegel modular form period*: the Igusa cusp form Δ_5 lives on $\mathcal{H}_2 = \mathrm{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$, parametrizing pairs of elliptic curves with a coupling.

(vi) *Elliptic Hall algebra carrier*: the CY_2 quantum vertex chiral group is carried on $\mathrm{Coh}(E)$, with Felder's dynamical elliptic R-matrix.

Each appearance brings the same structural package: a modular parameter, an $\mathrm{SL}_2(\mathbb{Z})$ symmetry, a q-expansion, and a sewing functor converting genus-0 data into genus-1 data. The elliptic curve is the modular clock of the theory.

The soul of an algebra

The modular characteristic $\kappa(A)$ is the most condensed invariant of a chirally Koszul algebra: a single rational number controlling the genus-1 obstruction class ($F_1 = \kappa/24$), the complementarity sum (Theorem C of Vol I), the Faber–Pandharipande genus expansion (Theorem D, on the uniform-weight lane), the shadow connection residue (Koszul monodromy -1), and, for $K3 \times E$, the Borcherds product weight.

A single CY$_3$ manifold admits multiple chiral algebraizations, each producing a different κ. For $K3\times E$: $\kappa_{\mathrm{ch}} = 3$ (chiral de Rham complex), $\kappa_{\mathrm{cat}} = 2$ (derived category), $\kappa_{\mathrm{fiber}} = 24$ (boundary lattice VOA), $\kappa_{\mathrm{BPS}} = 5$ (Igusa weight). Four genuinely different algebras see four different aspects of the same geometry: differential forms, coherent sheaves, lattice vectors, and automorphic BPS content. The κ-spectrum $\mathrm{Spec}_\kappa(X) = \{2, 3, 5, 24, \ldots\}$ is a richer invariant than any single κ.

Convergence and divergence

The shadow obstruction tower and the topological string free energy both produce genus-g amplitudes with opposite analytic behaviour. The shadow tower has Bernoulli decay $|F_g^{\mathrm{sh}}| \sim (2\pi)^{-2g}/g$: convergent, with radius 2π, Borel entire. The topological string has factorial growth $|F_g^{\mathrm{top}}| \sim (2g)!$: divergent, Gevrey-1, requiring non-perturbative completion. The shadow tower is the algebraic soul of the topological string — the part computable from OPE structure constants alone, without worldsheet instantons. For class $\mathbf{G}$ algebras (Heisenberg, lattice VOAs), the shadow *is* the full answer. For class $\mathbf{M}$ (Virasoro, $\mathcal{W}_{1+\infty}$), the shadow is a strict subtower, and the instanton sum is infinite.

The Schottky prison

At genus $g \le 3$, the Torelli map $\mathcal{M}_g \to \mathcal{A}_g$ is injective ($g = 1$), birational ($g = 2$), dominant ($g = 3$). The shadow obstruction tower, which lives on $\overline{\mathcal{M}}_g$, sees everything at these genera. At $g \ge 4$, the Jacobian locus becomes a proper subvariety of growing codimension, and the shadow tower is imprisoned in tautological classes on $\overline{\mathcal{M}}_g$. This imprisonment is also a strength: the shadow computes topological invariants insulated from analytic subtleties. The Schottky problem — characterising which abelian varieties are Jacobians — is the frontier between what the shadow tower can see and what it cannot.

The moonshine multiplier

The K3 elliptic genus decomposes as $\phi_{0,1} = 2\,\phi_{0,1}^{\mathrm{EZ}}$, where the factor 2 is $\kappa(K3) = \chi(\mathcal{O}_{K3})$. In the umbral moonshine decomposition, $A_n = 2 \cdot \dim(\rho_n)$ where ρ_n are M_{24} representations. The bar complex provides the universal coefficient κ; the sporadic group provides the representation content ρ_n; the physical multiplicity is their product. The homological side ($\kappa = 2$) is computable from the OPE alone; the group-theoretic side (ρ_n) requires no knowledge of the OPE. The bar complex and the sporadic group see orthogonal aspects of the same physics.

The bar complex as moral geometry

The bar complex $\overline{B}(A)$ converts OPE data into cohomological data, recovering pieces of $H^*(X)$, $\chi(X)$, and the deformation theory of X. For K3, the bar complex has $\kappa = 2$, class $\mathbf{M}$, and these are geometric invariants of an algebraic object. For a mirror pair $(X, X^\vee)$ under homological mirror symmetry: $\mathrm{shadow}(A_{D^b(X)}) = \mathrm{shadow}(A_{\mathrm{Fuk}(X^\vee)})$. The bar complex reflects the algebra into its mirror.

Organisation

Part I establishes the categorical foundations and develops the E_1 and E_2 chiral theories. Part II constructs the bridge functor from CY categories to chiral algebras. Part III studies quantum groups and braided monoidal structure. Part IV works through the standard landscape, centred on the $K3 \times E$ programme with its thirteen structural results and ten research programmes. Part V connects to the geometric Langlands programme and to Volumes I–II.

The working notes (§**??**) record the research frontier: the constructive gluing dream, the Leech whisper, the η^{18} factorisation, the Schottky–Igusa weight question, mock modularity as incomplete knowledge, and thirty additional thematic observations from the compute layer.

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Part I

The CY Engine

Chapter 1

Introduction

1.1 The question

A Calabi–Yau category C of dimension d carries a canonical cyclic A_∞-structure: a non-degenerate trace

$$\mathrm{Tr}\colon \mathrm{HH}_\bullet(C) \longrightarrow k[-d]$$

on its Hochschild homology, encoding the CY condition as a d-dimensional Frobenius structure at the chain level. The cyclic bar complex $\mathrm{CC}_\bullet(C)$ is the primary invariant; it carries an $\mathbb{S}^d$-action from the d-sphere framing of the trace, and its $\mathbb{S}^d$-equivariant structure governs higher-genus amplitudes.

A chiral algebra A on a curve X carries a bar complex $B(A)$, a factorization coalgebra on $\mathrm{Ran}(X)$ whose differential encodes holomorphic OPE residues and whose coproduct encodes topological interval-cutting. At genus $g \geq 1$, the bar complex acquires curvature $\kappa(A) \cdot \omega_g$ from the Hodge bundle, and the full modular structure is controlled by the universal Maurer–Cartan element $\Theta_A := D_A - d_0$ (Volume I). Together with the Swiss-cheese algebra structure (Volume II), the bar complex is the complete modular invariant.

This work constructs a bridge:

$$\left\{ \begin{array}{c} \text{CY categories with} \\ \text{appropriate } E_n\text{-structure} \end{array} \right\} \xrightarrow{\quad \Phi \quad} \left\{ \begin{array}{c} \text{quantum chiral algebras} \\ \text{with modular trace} \end{array} \right\}$$

The functor Φ extracts from a CY category C (Fukaya, derived, matrix factorization, or more general) its E_2-monoidal structure and produces a chiral algebra A_C whose bar complex $B(A_C)$ encodes the CY cyclic homology, with the CY trace realized as the modular characteristic $\kappa(A_C)$.

1.2 The E_1/E_2 chiral hierarchy

Three levels of chiral structure organize the theory:

- E_1-chiral algebras (Volume II, Part III): associative factorization on $\mathbb{C} \times \mathbb{R}$, encoding the ordered/topological sector. Representation categories are monoidal.
- E_2-chiral algebras (this work): braided factorization on $\mathbb{C} \times \mathbb{C}$, encoding the holomorphic-holomorphic sector. Representation categories are braided monoidal: the natural habitat of quantum groups.

- The passage $E_1 \longrightarrow E_2$ via Dunn additivity: $E_2 \simeq E_1 \otimes E_1$. An E_2-chiral algebra is an E_1-algebra in E_1-algebras.

The CY condition enters through Kontsevich's identification: a d-dimensional CY structure on C determines an $\mathbb{S}^d$-framing of the Hochschild complex, hence an $\mathbb{S}^1$-action on $\mathrm{HH}_\bullet(C)$. For $d = 2$ (CY surfaces), this $\mathbb{S}^1$-action is exactly the data of an E_2-algebra structure on the cyclic homology: the braiding.

1.3 Relation to Volumes I and II

This work depends on and extends the two preceding volumes:

Volume	**Provides**	**Used here**
I (Modular Koszul Duality)	Bar-cobar machine, Θ_A, $\kappa(A)$	CY bar complex, modular trace
II (3D HT QFT)	$\mathrm{SC}^{\mathrm{ch,top}}$, PVA descent, DK bridge	E_1 sector, braided structure
III (this work)	CY $\to$ chiral functor, E_2 theory	—

1.4 CY3 combinatorics as generalized root data

The combinatorics of a Calabi–Yau threefold X (lattice, intersection form, BPS spectrum) constitute the *generalized root datum* of a *quantum vertex chiral group* $G(X)$. For toric CY3 (via the CoHA and Drinfeld double) and for CY_2 categories (via Theorem CY-A$_2$), this object is constructed. For general CY_3, including the prototype $K3 \times E$, $G(X)$ is the target of a programme (Conjecture CY-A$_3$), not a constructed object.

Given a CY3 X, the generalized root datum $\mathcal{R}(X)$ consists of:

(i) A lattice $\Lambda(X)$ extracted from $K(X)$ or $H^*(X)$ with its Euler form;

(ii) Real simple roots Δ^{re} from the geometry (hyperbolic sublattice for $K3 \times E$; toric diagram for toric CY3);

(iii) Imaginary roots $\Delta^{\mathrm{im}} = \Delta^{\mathrm{im}}_0 \sqcup \Delta^{\mathrm{im}}_1$ (even and odd) with multiplicities $\mathrm{mult}(\alpha)$ from BPS counting (Donaldson–Thomas invariants for X, or equivalently the Fourier coefficients of the appropriate automorphic form);

(iv) A Weyl group $W(X)$ acting on the positive cone of $\Lambda(X)$;

(v) A Weyl vector $\rho \in \Lambda(X) \otimes \mathbb{Q}$.

Two families of examples illustrate the construction.

The $K3 \times E$ tower. For $X = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$ with S a K3 surface and E an elliptic curve, the lattice is $\Lambda^{3,2} \simeq \Lambda^{1,1} \oplus \Lambda^{1,1} \oplus [2]$ of signature $(3, 2)$. The hyperbolic sublattice $\Lambda^{2,1}_{II}$ with Gram matrix $\left(\begin{smallmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{smallmatrix}\right)$ provides the real roots. The root multiplicities are the Fourier coefficients $f(n, l)$ of the weak Jacobi form $\phi_{0,1}$, the K3 elliptic genus. The resulting generalized Borcherds–Kac–Moody superalgebra $\mathfrak{g}_{\Delta_5}$ has the Igusa cusp form Δ_5 as its denominator identity. The modular characteristic satisfies $\kappa(K3 \times E) = 3 = \dim_{\mathbb{C}}$, verified by six independent paths; the factor 2 in $Z_{K3} = 2\,\phi_{0,1}$ is the bar-complex moonshine multiplier $\kappa(\mathcal{A}_{K3})$; and the shadow obstruction tower does not produce Δ_5 directly (four structural obstructions: categorical, κ-mismatch, second quantization, Schottky; Theorem 14.112).

Toric CY3 and critical CoHAs. For a toric CY3 determined by a fan Σ, the toric diagram determines a quiver Q_X. The critical cohomological Hall algebra $\mathrm{CoHA}(Q_X)$ is the positive half of an affine super Yangian $Y(\widehat{\mathfrak{g}}_{Q_X})$. For $\mathbb{C}^3$: the Jordan quiver gives $Y(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$. The toric fan is the Dynkin data.

1.5 Automorphic correction as shadow obstruction tower

The passage from the naive Kac–Moody algebra $\mathfrak{g}$ (built from the real root Gram matrix alone) to the full generalized BKM superalgebra $\mathfrak{g}_X$ (with all imaginary roots and their multiplicities) is the *automorphic correction*. In the language of Volume I, this passage is the *shadow obstruction tower*:

Shadow tower (Vol I)	**BKM language**	**CY3 geometry**
κ (arity 2)	Weyl vector ρ contribution	Leading DT invariant
C (arity 3, cubic)	First imaginary root corrections	Genus-0 3-point function
Q (arity 4, quartic)	Higher imaginary roots	Genus-0 4-point function
Full Θ_A	Complete automorphic correction	Full BPS generating function

The denominator identity of $\mathfrak{g}_X$ (Δ_5 for $K3 \times E$, the DT partition function for toric CY3) should be the bar-complex Euler product of the quantum vertex chiral group, when that group exists. For toric CY3 this identification is part of Conjecture CY-C. For $K3 \times E$, since $A_{K3\times E}$ is not yet constructed (AP-CY6, AP-CY8), the product formula for Δ_5 is an *observation* about the Borcherds lift, not a theorem derived from the bar complex.

The bar Euler product of lattice VOAs recovers the Borcherds denominator identity (verified for E_8 and Leech lattice; conditional on CY-A$_3$ for $K3 \times E$).

1.6 Main results

- **Theorem CY-A** (CY-to-chiral functor): Construction of $\Phi\colon \mathrm{CY}_d\text{-}\mathbf{Cat} \to E_2\text{-}\mathrm{ChirAlg}$ via the factorization envelope and $\mathbb{S}^d$-framing. *Proved for* $d = 2$ (CY surfaces, where the $\mathbb{S}^2$-framing gives E_2 via Kontsevich–Vlassopoulos). *For* $d = 3$: a programme conditional on (a) constructing the chain-level $\mathbb{S}^3$-framing compatible with BV structure and (b) the quantization step. The $\mathbb{S}^3$-framing gives an E_3-structure, from which E_2 is obtained by restriction; however, $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3))$ is trivial, so the restriction gives a symmetric braiding at the topological level. The quantum group braiding for $d = 3$ is expected to arise through the Drinfeld center of E_1-monoidal categories.

- **Theorem CY-B** (E_2-chiral Koszul duality): Bar-cobar adjunction in the E_2-chiral setting. The automorphic correction of the BKM superalgebra $\mathfrak{g}_X$ is identified with the shadow obstruction tower Θ_{A_X}; the denominator identity equals the bar-complex Euler product.

- **Conjecture CY-C** (Quantum group realization): For toric CY3, $\mathrm{Rep}^{E_2}(G(X))$ should be braided monoidal equivalent to $\mathrm{Rep}(Y(\widehat{\mathfrak{g}}_{Q_X}))$; for $K3 \times E$, the $\mathrm{Sp}_4(\mathbb{Z})$-module structure on the denominator identity should recover the braided structure. The critical CoHA is the E_1-sector (positive half). The CY category $C(\mathfrak{g}, q)$ is not yet constructed in general.

- **Theorem CY-D** (Modular CY characteristic): For $d = 2$, $\kappa(\Phi(C))$ equals the CY Euler characteristic $\chi^{\mathrm{CY}}(C)$. For $K3 \times E$ ($d = 3$): the weight of Δ_5 is $5 = h^{1,1}(K3)/4 = 20/4$; this appears in the structural position of a modular characteristic, but without $A_{K3\times E}$ (which is not constructed), the identification $\kappa = 5$ is an observation matching the Borcherds lift weight formula, not a computation from the Volume I definition. The general $d = 3$ formula is conjectural.

- **Theorem (Shadow–Siegel gap)**: The shadow obstruction tower of $K3 \times E$ does not produce the Igusa cusp form Φ_{10}. Four structural obstructions: categorical (number vs function), modular characteristic ($\kappa_{\mathrm{ch}} = 3 \neq 5 = \kappa_{\mathrm{BKM}}$), second quantization (single-copy vs DMVV symmetric product), and Schottky at $g \geq 4$ ($\mathrm{codim} = (g-2)(g-3)/2$). Thirteen structural results (K3-1 through K3-13) and ten research programmes (A through J) are developed in the toroidal and elliptic algebras chapter.

1.7 What is proved versus what is conjectural

Proved (would survive a referee):

- The generalized root datum axioms CY1–CY7, adequate for $K3 \times E$ and toric CY3 examples.
- The E_2-chiral algebra formalism and its basic properties (definition, bar complex, braiding).
- Theorem CY-A for $d = 2$: steps 1–3 of Φ (cyclic $A_\infty \longrightarrow$ Lie conformal $\longrightarrow$ factorization envelope $\longrightarrow$ E_2 enhancement via $\mathbb{S}^2$-framing).
- The CoHA as E_1-sector for toric CY3 (via Schiffmann–Vasserot, RSYZ).
- The Drinfeld center equivalence $\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \simeq \mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A))$ (Ben-Zvi–Francis–Nadler, Lurie).
- All lattice theory and BKM constructions for $K3 \times E$ (Gritsenko–Nikulin).
- The denominator identity $\Delta_5 = \Phi$ and product formula (arity 2–6 computationally verified).
- The shadow–Siegel gap theorem with four structural obstructions (O1–O4).
- $\kappa(K3 \times E) = 3$, verified by six independent paths.
- The moonshine multiplier: $A_n = \kappa \cdot \dim(\rho_n)$ identifies the factor 2 in $Z_{K3} = 2\,\phi_{0,1}$.
- Forty-two CY/BKM compute engines with 5,700+ tests across the standard CY landscape.

Rigorous proof sketches (completable with effort):

- The automorphic correction = shadow obstruction tower identification (parts (a)–(c)).
- The denominator = bar Euler product (conditional on CY-A for the relevant dimension).
- E_2-chiral Koszul duality (conjecture with evidence from E_1 case).

Conjectural:

- Theorem CY-A for $d = 3$: the chain-level $\mathbb{S}^3$-framing and BV-compatibility are not yet established.

- Conjecture CY-C: the CY category $C(\mathfrak{g}, q)$ is not constructed in general.
- The chiral algebra $A_{K3\times E}$ does not yet exist as a constructed object; the identification of $\kappa_{\mathrm{BKM}} = 5$ with a chiral algebra modular characteristic is an observation, not a theorem.
- The Langlands = Koszul duality conjecture.
- All physics conjectures in Part V.
- Ten research programmes (A–J) for the $K3 \times E$ programme, each with formal conjectures.

The analytic gap for compact CY_3 — the construction of a chain-level contracting homotopy without solving a PDE — is closed by the algebraic Čech resolution (Theorem 8.10): the standard 5-patch cover of the quintic in $\mathbb{P}^4$ provides an algebraic SDR that interfaces directly with the homotopy transfer theorem. E_1 universality is consistent for the quintic threefold via three independent paths (Gepner MF, large-volume HKR, GV integrality), with $\kappa_{\mathrm{MacMahon}} = \chi/2 = -100$ as the physically relevant modular characteristic.

1.8 The ten research programmes

The $K3 \times E$ prototype generates ten research programmes, each with formal conjectures grounded in the 90+ compute engines. The formal development is in Chapter 14.

(Prog A) *Constructive CY gluing.* Reconstruct $D^b(K3\times E)$ from quiver charts indexed by Bridgeland stability conditions. The bar functor commutes with homotopy colimits (Proposition 14.102), so the global chiral algebra is the hocolim of local CoHAs. The minimum chart number, the Brauer obstruction, and the extension to non-toric K3 surfaces are open.

(Prog B) *κ as universal moonshine multiplier.* The factor 2 in $Z_{K3} = 2\,\phi_{0,1}$ is $\kappa(K3) = \chi(\mathcal{O}_{K3})$. The BPS degeneracies factorise as $A_n = \kappa \cdot \dim(\rho_n)$, separating the homological part (κ, from the bar complex) from the group-theoretic part (ρ_n, from the sporadic symmetry). The programme: extend this factorisation to the Monster module, the Conway module, and all holomorphic VOAs.

(Prog C) *Second-quantization bridge.* The ratio $\kappa_{\mathrm{BPS}}/\kappa_{\mathrm{ch}} = 5/3$ encodes the first-to-second-quantization passage. The conjectural formula $\kappa_{\mathrm{BPS}}(S \times E) = \kappa(S) + \kappa(S \times E)$ expresses the weight of the automorphic form as a sum of fiber and global modular characteristics.

(Prog D) *Schottky shadow programme.* At $g \leq 3$, the shadow obstruction tower sees nearly all of the moduli. At $g \geq 4$, the Schottky locus has positive codimension and the shadow is imprisoned in tautological classes. The programme: characterise the tautological projection of Siegel modular forms, and determine whether the shadow tower generates the image of $S_*(\mathrm{Sp}_{2g}(\mathbb{Z})) \to R^*(\overline{\mathcal{M}}_g)$.

(Prog E) *Mock modularity and the bar complex.* The 1/4-BPS counting function is a mock modular form with shadow $24\,\eta^3$. The factor $24 = \chi(K3)$ is the surface announcing its existence. The programme: construct a half-integral-weight metaplectic bar complex whose shadow obstruction tower produces the mock modular completion.

(Prog F) *Modular factorization envelope.* Construct the universal modular factorization envelope $U_X^{\mathrm{mod}}(L)$ for cyclically admissible Lie conformal algebras, carrying the full shadow obstruction tower. This would give the left adjoint of the modular primitive-current functor.

(Prog G) *BKM algebras and scattering diagrams.* The Gross–Siebert scattering diagram consistency condition IS the E_1 Maurer–Cartan equation (Theorem 8.35). The programme: lift this identification to the full motivic Hall algebra level (where naive BCH is insufficient; AP42).

(Prog H) *Descent theorem.* Čech descent for E_1-algebras over the Bridgeland stability manifold. Proved for toric CY3 (Proposition **??**); open for compact CY3 beyond the chart-atlas regime.

(Prog I) *Higher-dimensional CY shadows.* The E_n stabilization theorem (Theorem 7.1) gives a Bott-periodic 8-fold pattern. For CY4: the p_1 Pontryagin class controls the shifted symplectic structure. For $K3\times K3$: the Hopf decomposition $\mathbb{S}^4 = \mathbb{S}^2 \times \mathbb{S}^2$ gives E_2 by Dunn additivity.

(Prog J) *Convergence vs divergence.* The shadow partition function converges with radius 2π (Bernoulli decay); the topological string diverges (Gevrey-1). The programme: quantify the instanton gap $F_g^{\mathrm{top}} - F_g^{\mathrm{sh}}$ at each genus, and determine whether the shadow tower is Borel-summable to the full amplitude.

All ten programmes are computationally testable: the 90+ compute engines provide a laboratory for each conjecture.

1.9 Guide for the reader

Part I establishes the categorical foundations: CY categories, cyclic A_∞-structures, the Hochschild calculus, and the $E_1/E_2/E_n$ chiral hierarchy with the framing obstruction tower. Part II constructs the bridge functor Φ from CY categories to quantum chiral algebras and proves Theorems CY-A (for $d = 2$) and CY-D, with the identification of automorphic correction and shadow obstruction tower as the central result. Part III treats quantum groups and their braided factorization structure, including the Drinfeld center as the bulk algebra. Part IV works through the standard landscape: the toroidal and elliptic algebras chapter develops the $K3 \times E$ programme (K3-1 through K3-13, programmes A through J), alongside toric CY3 CoHAs, Fukaya categories, derived categories, matrix factorizations (ADE singularities and W-algebras), and quantum group representations (Kazhdan–Lusztig, affine Yangians, critical CoHAs). Part V connects to the geometric Langlands programme and to Volumes I and II via the bar-cobar bridge and the modular Koszul bridge. The working notes (§**??**) record the research frontier.

Chapter 2

Calabi–Yau Categories

2.1 Smooth and proper dg categories

Definition 2.1 *(Smooth dg category).* A dg category C over k is *smooth* if the diagonal bimodule $C \in C^{\mathrm{op}} \otimes C\text{-Mod}$ is a perfect bimodule (compact in the derived category of bimodules).

Definition 2.2 *(Proper dg category).* A dg category C is *proper* if for all objects $X, Y \in C$, the Hom-complex $\mathrm{Hom}_C(X, Y)$ has finite-dimensional total cohomology.

2.2 The Calabi–Yau condition

Definition 2.3 *(Calabi–Yau category, after Kontsevich).* A smooth dg category C is *Calabi–Yau of dimension d* if there is a quasi-isomorphism of C-bimodules

$$C \xrightarrow{\sim} C^{\vee}[d]$$

where $C^{\vee} = \mathrm{RHom}_{C^{\mathrm{op}} \otimes C}(C, C^{\mathrm{op}} \otimes C)$ is the inverse dualizing bimodule. Equivalently, there exists a non-degenerate cyclic pairing

$$\langle \cdot, \cdot \rangle \colon \mathrm{HH}_{\bullet}(C) \otimes \mathrm{HH}_{\bullet}(C) \longrightarrow k[-d].$$

Remark 2.4. The CY condition combines smoothness (the bimodule C is compact) with the Serre functor being a shift ($S_C \simeq [d]$). A smooth proper CY category is *saturated* in the sense of Kontsevich–Soibelman.

2.3 The CY trace

Definition 2.5 *(Non-degenerate trace).* A *d-dimensional CY structure* on a smooth dg category C is a class

$$[\sigma] \in \mathrm{HC}^{-}_{d}(C)$$

in the negative cyclic homology, whose image under the canonical map $\mathrm{HC}^{-} \to \mathrm{HH}$ is a non-degenerate trace $\mathrm{Tr} \colon \mathrm{HH}_d(C) \to k$.

2.4 Standard examples

Example 2.6 *(Fukaya category).* For a compact symplectic manifold (M, ω) with $c_1(M) = 0$, the Fukaya category $\mathrm{Fuk}(M)$ is CY of dimension $\dim_{\mathbb{R}}(M)/2 = n$. The CY trace arises from the cyclic structure on the A_∞-algebra of Lagrangian Floer cochains.

Example 2.7 *(Derived category of a CY manifold).* For a smooth projective CY manifold X of dimension n (i.e., $\omega_X \simeq \mathcal{O}_X$, $H^i(X, \mathcal{O}_X) = 0$ for $0 < i < n$), the bounded derived category $D^b(\mathrm{Coh}(X))$ is CY of dimension n. The CY trace is induced by the Serre duality pairing.

Example 2.8 *(Matrix factorizations).* For $W : \mathbb{A}^n \to \mathbb{A}^1$ a polynomial with isolated singularity, the category $\mathrm{MF}(W)$ of matrix factorizations is CY of dimension $n - 2$ (Dyckerhoff; Orlov). For the ADE singularities in two variables ($n = 2$), $\mathrm{MF}(W)$ is CY_0 (semisimple, with the structure of a Frobenius algebra). For threefold singularities ($n = 4$, e.g. $W = x_1^2 + \cdots + x_4^2$), $\mathrm{MF}(W)$ is CY_2.

Chapter 3

Cyclic A_∞-Structures

3.1 A_∞-categories and the bar complex

The cyclic bar complex is the primary invariant of a CY category. This section reviews the A_∞-formalism and its cyclic refinement, which Part II promotes to chiral algebras on curves.

Definition 3.1 *(A_∞-category).* An A_∞-category C over k consists of:

(i) A set of objects $\mathrm{Ob}(C)$;

(ii) For each pair of objects X, Y, a graded k-module $\mathrm{Hom}_C(X, Y)$;

(iii) Higher composition maps $\mu_n\colon \mathrm{Hom}(X_0, X_1) \otimes \cdots \otimes \mathrm{Hom}(X_{n-1}, X_n) \to \mathrm{Hom}(X_0, X_n)[2-n]$ for $n \geq 1$, satisfying the A_∞-relations:

$$\sum_{\substack{p+q=n+1\\ 1\leq i\leq p}} (-1)^{\dagger}\, \mu_p(a_1, \ldots, a_{i-1}, \mu_q(a_i, \ldots, a_{i+q-1}), a_{i+q}, \ldots, a_n) = 0$$

where the sign $(-1)^{\dagger}$ follows the Koszul rule (see Appendix A).

3.2 Cyclic structures

Definition 3.2 *(Cyclic A_∞-algebra).* A *cyclic A_∞-algebra of dimension d* is an A_∞-algebra $(A, \{\mu_n\}_{n\geq 1})$ equipped with a non-degenerate pairing $\langle\cdot,\cdot\rangle\colon A \otimes A \to k[-d]$ satisfying the cyclic invariance condition

$$\langle \mu_n(a_1, \ldots, a_n), a_0\rangle = \pm\langle \mu_n(a_0, a_1, \ldots, a_{n-1}), a_n\rangle$$

for all $n \geq 1$, where the sign is determined by the Koszul rule.

Remark 3.3. The cyclic invariance condition is the chain-level precursor of the chiral algebra trace. The passage from cyclic A_∞ to chiral algebra requires promoting the cyclic pairing to a factorization structure on a curve; this is the content of Part II.

3.3 The cyclic bar complex

Construction 3.4 *(Cyclic bar complex).* Given a cyclic A_∞-algebra $(A, \langle\cdot,\cdot\rangle)$ of dimension d, the *cyclic bar complex* is

$$\mathrm{CC}_\bullet(A) = \bigoplus_{n\geq 0} (A[1])^{\otimes n}/(\text{cyclic})$$

with differential induced by the A_∞-operations and the cyclic quotient enforced by the pairing. The S^1-action by cyclic rotation on $\mathrm{CC}_\bullet(A)$ endows the cyclic homology $\mathrm{HC}_\bullet(A)$ with its defining structure.

3.4 The $\mathbb{S}^d$-framing

A d-dimensional CY structure enhances the S^1-action to an $\mathbb{S}^d$-action on the Hochschild complex. For $d = 2$, the $\mathbb{S}^2$-framing provides the E_2-algebra structure underlying the braided monoidal category of quantum group representations.

Chapter 4

Hochschild Calculus for CY Categories

4.1 Hochschild homology and cohomology

4.2 The Hochschild-to-cyclic spectral sequence

4.3 CY Hochschild duality

THEOREM 4.1 *(CY Hochschild duality).* [PE] For a smooth CY category $\mathcal{C}$ of dimension d, there is a canonical isomorphism

$$\mathrm{HH}^\bullet(\mathcal{C}) \simeq \mathrm{HH}_{\bullet+d}(\mathcal{C})$$

identifying Hochschild cohomology with (shifted) Hochschild homology. Under this isomorphism, the Gerstenhaber bracket on $\mathrm{HH}^\bullet$ corresponds to the BV operator (divergence operator) on $\mathrm{HH}_\bullet$; the Connes B-operator on $\mathrm{HH}_\bullet$ corresponds to the de Rham differential under the HKR decomposition.

Remark 4.2. The isomorphism $\mathrm{HH}^\bullet \simeq \mathrm{HH}_{\bullet+d}$ ("CY duality" or "Poincaré duality for Hochschild (co)homology") follows from smoothness (the diagonal bimodule is compact) combined with the CY condition ($\mathcal{C} \simeq \mathcal{C}^\vee[d]$). See Kontsevich (2006 ICM), Keller (2006), Ginzburg (2006 arXiv:0612139).

4.4 Categorical Hodge theory

Chapter 5

E_1-Chiral Algebras

Volume II (Part III) developed the E_1-chiral theory as the ordered sector of the Swiss-cheese algebra. This chapter reformulates it as the foundation for the E_2 extension.

Remark 5.1 (E_1 primacy for CY quantum groups). The E_1-chiral algebra (boundary) is the primitive object in this volume. The E_2-chiral algebra (bulk) is obtained from it by the Drinfeld center construction $\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \simeq \mathrm{Rep}^{E_2}(\mathrm{Drin}(A))$. Quantum groups, Yangians, and braided tensor categories are natively E_1 objects: the CoHA multiplication is ordered (short exact sequences have a preferred direction), and the R-matrix arises only in the Drinfeld double. The passage $E_1 \to E_2$ is the higher-categorical analogue of the averaging map $\mathrm{av}\colon \mathfrak{g}^{E_1} \to \mathfrak{g}^{\mathrm{mod}}$ from Vol I. This averaging is *lossy*: $\mathrm{av}(r(z)) = \kappa$ forgets the R-matrix data. The E_1-bar $B^{\mathrm{ord}}(A)$ retains strictly more information than the E_∞-bar $B^{\Sigma}(A)$.

5.1 E_1-algebras and associative factorization

An E_1-algebra is an algebra over the little intervals operad. In the chiral setting, an E_1-chiral algebra factorizes associatively along a real line $\mathbb{R}$ while maintaining holomorphic structure along a transverse curve.

Definition 5.2 *(E_1-chiral algebra).* An *E_1-chiral algebra* on $X\times\mathbb{R}$ is a factorization algebra $\mathcal{A}$ on $\mathrm{Ran}(X)\times \mathrm{Ran}(\mathbb{R})$ such that:

(i) The X-direction factorization is holomorphic (chiral);

(ii) The $\mathbb{R}$-direction factorization is E_1 (associative).

The representation category $\mathrm{Rep}^{E_1}(\mathcal{A})$ is a monoidal category.

5.2 The bar complex as E_1-chiral algebra

PROPOSITION 5.3. [PE] For a chiral algebra A on X, the bar complex $B(A)$ carries a natural E_1-chiral structure: the bar differential encodes the X-direction (holomorphic OPE residues) and the bar coproduct encodes the $\mathbb{R}$-direction (ordered deconcatenation).

This is the Swiss-cheese algebra structure of Volume II. The E_1-bar $B^{\mathrm{ord}}(A)$ is the ordered tensor coalgebra $(T^c(A[1]), d, \Delta_{\mathrm{decat}})$ with deconcatenation coproduct (AP82, AP85). The symmetric bar $B^{\Sigma}(A) = B^{\mathrm{ord}}(A)/S_n$ is the S_n-coinvariant quotient. The ordered bar retains the full quantum group data (R-matrix, Yangian structure) that the symmetric bar quotients out (AP65, AP97).

5.3 Monoidal representation categories

The representation category $\mathrm{Rep}^{E_1}(\mathcal{A})$ inherits a monoidal structure from the ordered factorization along $\mathbb{R}$. The tensor product $V \otimes W$ of two $\mathcal{A}$-modules is defined by the fusion of intervals: V is placed on $(-\infty, 0)$ and W on $(0, \infty)$, and the factorization structure combines them. The ordering of the intervals makes the tensor product associative but *not* symmetric: the natural transformation $V \otimes W \to W \otimes V$ does not exist in the E_1-setting. It exists only after passing to the Drinfeld center (Chapter 13).

5.4 The E_1-Koszul duality

Theorem 5.4 *(E_1-chiral Koszul duality, Volume II).* [PE] For an E_1-chiral algebra $\mathcal{A}$, the bar-cobar adjunction of Volume I refines to an E_1-equivariant adjunction. On the representation-category level, this gives a monoidal equivalence

$$\mathrm{Rep}^{E_1}(A) \simeq \mathrm{Rep}^{E_1}(A^!)^{\mathrm{rev}}$$

between the monoidal representation category of A and the reverse monoidal category of $A^!$.

Chapter 6

E_2-Chiral Algebras

An E_2-chiral algebra factorizes along two holomorphic directions; its representation category is braided monoidal, the algebraic structure underlying quantum groups.

6.1 The E_2 operad and braided monoidal categories

Definition 6.1 *(E_2 operad).* The E_2 operad (little 2-disks operad) has $E_2(n) = \mathrm{Conf}_n(\mathbb{D}^2)$, the configuration space of n non-overlapping disks in the unit disk. An E_2-algebra is an algebra over this operad. By Dunn additivity, $E_2 \simeq E_1 \otimes E_1$: an E_2-algebra is an E_1-algebra in E_1-algebras.

Theorem 6.2 *(Lurie).* [PE] The category of E_2-algebras in a symmetric monoidal ∞-category C is equivalent to the category of braided monoidal objects in C.

6.2 E_2-chiral algebras: definition and first properties

Definition 6.3 *(E_2-chiral algebra).* An *E_2-chiral algebra* on $X \times Y$ (where X, Y are smooth curves) is a factorization algebra $\mathcal{A}$ on $\mathrm{Ran}(X) \times \mathrm{Ran}(Y)$ such that:

(i) The X-direction factorization is chiral (holomorphic);

(ii) The Y-direction factorization is chiral (holomorphic);

(iii) The combined structure is compatible with the E_2-operad action via the identification $\overline{\mathrm{FM}}_n(\mathbb{C}) \simeq E_2(n)$.

The representation category $\mathrm{Rep}^{E_2}(\mathcal{A})$ is a *braided monoidal* category.

6.3 Kontsevich formality and the E_2-Gerstenhaber structure

Kontsevich's formality theorem $E_2 \xrightarrow{\sim} \mathrm{Ger}$ (over $\mathbb{Q}$) identifies E_2-algebras with Gerstenhaber algebras up to homotopy. In the chiral setting:

Proposition 6.4. The Hochschild cohomology $\mathrm{HH}^\bullet(A)$ of a chiral algebra A carries a natural E_2-algebra structure. Via Kontsevich formality, this is equivalent to the Gerstenhaber bracket + cup product structure.

6.4 The E_2 bar complex

Construction 6.5 *(E_2 bar complex).* For an E_2-chiral algebra $\mathcal{A}$ on $X \times Y$, the E_2 *bar complex* $B_{E_2}(\mathcal{A})$ is a factorization coalgebra on $\mathrm{Ran}(X) \times \mathrm{Ran}(Y)$ with:

- Two commuting differentials: d_X from X-direction OPE residues, d_Y from Y-direction OPE residues;
- Two commuting coproducts: Δ_X from X-direction deconcatenation, Δ_Y from Y-direction deconcatenation;
- The braiding: the transposition of the two E_1-structures gives the R-matrix.

6.5 E_2-chiral Koszul duality

Conjecture 6.6 *(E_2-chiral Koszul duality).* [CJ] For an E_2-chiral algebra $\mathcal{A}$, the bar-cobar adjunction of Volume I extends to an E_2-equivariant adjunction. On representation categories, this gives a braided monoidal equivalence

$$\mathrm{Rep}^{E_2}(A) \simeq \mathrm{Rep}^{E_2}(A^!)^{\mathrm{rev}}$$

where rev denotes the reverse braiding.

6.6 The three bar complexes

The operad filtration $E_1 \to E_2 \to \cdots \to E_\infty$ produces, for each n, a bar complex $B_{E_n}(A)$ with progressively more symmetry. The three cases relevant to this work are compared below.

Construction 6.7 *(Three bar complexes).* Let A be a vertex algebra with an E_2-chiral structure (Definition 6.3). The three bar complexes are:

(i) *E_1-bar (ordered/Hochschild).* $B^n_{E_1}(A) = A^{\otimes n}$, the ordered tensor product with Hochschild bar differential. No symmetric group action is imposed. The coproduct is deconcatenation. This is the bar complex of A viewed as an associative (E_1) algebra.

(ii) *E_2-bar (braided).* $B^n_{E_2}(A) = A^{\otimes n}$ with the braid group B_n action induced by the R-matrix $R(z)$. The differential combines the X- and Y-direction OPE residues (Dunn additivity). The braiding distinguishes this from the E_1 case whenever $R \neq \mathrm{id}$.

(iii) *E_∞-bar (symmetric).* $B^n_{E_\infty}(A) = \mathrm{Sym}^n(A[1]) = (A[1])^{\otimes n}/S_n$, with the Chevalley–Eilenberg differential and shuffle coproduct. This is the bar complex of Volume I, whose cohomology controls the shadow obstruction tower.

The three are related by

$$B^n_{E_\infty}(A) = B^n_{E_1}(A)/S_n, \qquad H^*(B^n_{E_\infty}(A)) = H^*(B^n_{E_2}(A))^{S_n}.$$

The first identity is the definition of the symmetric bar (quotient by the symmetric group); the second follows from the fact that the E_2 braiding refines the S_n action to a B_n action, and passage to E_∞ takes S_n-invariants.

THEOREM 6.8 *(Bar complex comparison for $\mathbb{C}^3$).* [PH] [Parts (i) and (iii).] [HE] [Part (ii) is computationally observed through $n \le 20$ but not analytically proved.] Let $A = W_{1+\infty}$ be the affine Yangian of $\mathfrak{gl}_1$ at general parameters (h_1, h_2, h_3) with $h_1 + h_2 + h_3 = 0$. Then:

(i) *E_1-bar and plane partitions.* The E_1-bar graded dimension is the MacMahon function:

$$\sum_{n\ge 0} \dim B^n_{E_1}(A) \cdot q^n \;=\; M(q) \;:=\; \prod_{n\ge 1} \frac{1}{(1-q^n)^n}.$$

The MacMahon coefficients $[q^n]M(q)$ count plane partitions of n: $1, 1, 3, 6, 13, 24, \dots$ (OEIS A000219).

(ii) *E_2-bar and the divisor function.* At arity n, the E_2-bar dimension is conjectured to equal the divisor function:

$$\dim B^n_{E_2}(H_1) \overset{?}{=} d(n),$$

where $d(n) = \sum_{d|n} 1$ is the number of divisors of n. This pattern is computationally observed through $n \le 20$ (Verification: `e2_barcobar_koszul.py`) but an analytical proof is not available.

(iii) *E_∞-bar and partitions.* The E_∞-bar graded dimension is the partition function $P(q) = \prod_{n\ge 1} 1/(1-q^n)$ (OEIS A000041). By the S_n-quotient relation:

$$B^n_{E_\infty}(A) = (B^n_{E_1}(A))^{S_n}.$$

Proof. Part (i): the E_1-bar differential vanishes on the Heisenberg generators (the bar complex has trivial differential for free algebras), so the E_1-bar reduces to the ordered tensor algebra on the single generator. The graded dimension of $A^{\otimes n}$, weighted by conformal weight, counts the number of ways to fill an n-fold tensor product with monomials of total weight n. For a single free boson with $\dim V_k = p(k)$ states at weight k, the generating function is exactly $M(q)$ by the MacMahon formula.

Part (ii): at the self-dual point ($h_1 = 1, h_2 = 0, h_3 = -1$), the R-matrix degenerates to the identity and $B_{E_2} \simeq B_{E_1}$, so $\dim B^n_{E_2} = \dim B^n_{E_1} \ne d(n)$ in general. At generic parameters, the braid group action via $g(z) = \prod_a (z-h_a)/(z+h_a)$ is nontrivial and the E_2-bar dimension is observed computationally to match $d(n)$ through $n \le 20$. An analytical proof of this identity has not been found; the claim is heuristic (see `e2_barcobar_koszul.py`).

Part (iii): the S_n-quotient of $B^n_{E_1}$ is Sym^n; for a single generator, $\dim \mathrm{Sym}^n V = p(n)$. □

PROPOSITION 6.9 *(CoHA/W-algebra character ratio).* [PH] The CoHA character exceeds the W-algebra character by the ratio

$$\frac{M(q)}{P(q)} = \prod_{n\ge 2} \frac{1}{(1-q^n)^{n-1}},$$

where $M(q) = \prod_{n\ge 1} 1/(1-q^n)^n$ is the MacMahon function (CoHA/E_1-bar character) and $P(q) = \prod_{n\ge 1} 1/(1-q^n)$ is the partition function (E_∞-bar character). The ratio captures the ordered degrees of freedom in the CoHA that are quotiented out in the passage $B_{E_1} \longrightarrow B_{E_\infty}$.

Proof. Direct computation:

$$\frac{M(q)}{P(q)} = \frac{\prod_{n\ge 1}(1-q^n)^{-n}}{\prod_{n\ge 1}(1-q^n)^{-1}} = \prod_{n\ge 1}(1-q^n)^{-(n-1)} = \prod_{n\ge 2}(1-q^n)^{-(n-1)},$$

since the $n = 1$ factor contributes $(1-q)^0 = 1$. □

Remark 6.10 (Computational verification). Theorem 6.8 and Proposition 6.9 are verified by 59 independent tests in `compute/lib/bar_comparison_c3.py`, including: direct computation of MacMahon coefficients through $n = 15$; comparison of S_n-invariant dimensions with partition counts; verification of the $M(q)/P(q)$ product formula; and confirmation that the E_2-bar degenerates to E_1 at the self-dual point.

Chapter 7

E_n-Factorization and Higher Chiral Structure

The CY-to-chiral functor of Chapter 8 produces E_∞ for $d = 1$, E_2 for $d = 2$, and E_1 for $d = 3$. The naive formula E_{4-d} breaks at $d = 4$: the operad E_0 does not exist. This chapter resolves the breakdown. The main result (Theorem 7.1) is that for all $d \geq 3$, the CY chiral algebra is E_1, with additional shifted structure classified by the framing obstruction $\pi_d(BU)$. The 8-fold Bott periodicity of the classical groups organizes the obstruction tower into a periodic pattern: the framing obstruction is trivial precisely when $d \bmod 8 \in \{1, 3, 7\}$ (with refinement from the symplectic homotopy groups at $d \equiv 5 \pmod 8$).

7.1 Dunn additivity and the E_n hierarchy

Recall the Dunn additivity theorem: $E_n \simeq E_1 \otimes_{E_0} E_1 \otimes_{E_0} \cdots \otimes_{E_0} E_1$ (n factors), where $E_0 = \mathrm{Assoc}_+$ is the associative operad with unit. In particular, $E_2 \simeq E_1 \otimes_{E_0} E_1$: an E_2-algebra is an E_1-algebra in E_1-algebras. An E_∞-algebra is commutative.

For the CY-to-chiral functor, the CY dimension d determines the native E_n level of the chiral algebra. The $\mathbb{S}^d$-framing on Hochschild homology $\mathrm{HH}_\bullet(C)$ carries an E_d-algebra structure. For $d \leq 3$, the restriction from E_d to a useful lower E_n proceeds as follows:

- $d = 1$: the native structure is E_∞ (commutative). The $\mathbb{S}^1$-framing produces a symmetric monoidal structure on $\mathrm{HH}_\bullet(C)$, which is E_∞.
- $d = 2$: E_2 is already the target structure (braided monoidal).
- $d = 3$: E_3 restricts to E_2 with symmetric braiding (since $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3))$ is trivial). The genuine nonsymmetric quantum group braiding arises through the Drinfeld center of the E_1-monoidal representation category (Theorem 8.15).

For $d \geq 4$, the E_d structure is "too symmetric" and the CY chiral algebra is always E_1. The higher E_d structure manifests as *additional shifted structure* (shifted symplectic, shifted Poisson) beyond the base E_1 level.

7.2 Factorization algebras on $\mathbb{C}^n$

For a CY category C of dimension d associated to $\mathbb{C}^d$, the factorization algebra $\mathrm{Fact}_X(\mathfrak{L}_C)$ on a curve X is constructed from the Lie conformal algebra of polyvector fields (Chapter 8). The $\mathrm{GL}(d)$-invariant Schouten–Nijenhuis bracket vanishes for all $d \geq 3$ (Theorem 8.4 and its generalization), so the classical bracket is abelian and all noncommutative structure arises from the Ω-deformation.

The equivariant deformation parameters are $\sigma_3, \sigma_4, \dots, \sigma_d$ (the elementary symmetric polynomials of $h_1, \dots, h_d$ subject to $\sum h_i = 0$), giving a $(d-2)$-dimensional deformation space. For $d = 3$, this is the single parameter $\sigma_3 = h_1 h_2 h_3$ of the affine Yangian.

7.3 The E_n-chiral bar complex

Construction 6.7 (in Chapter 6) gives the three bar complexes $B_{E_1}, B_{E_2}, B_{E_\infty}$ for an E_2-chiral algebra. For a CY_d chiral algebra with $d \geq 4$, only the E_1-bar and E_∞-bar are native; the E_2-bar requires the Drinfeld center passage.

7.4 The framing obstruction tower

The $\mathbb{S}^d$-framing on $\mathrm{HH}_\bullet(C)$ requires a trivialization of a certain bundle classified by π_d of the structure group of the CY pairing. Two independent computations of this obstruction:

(A) *Unitary path.* The complex structure gives a $U(n)$ reduction. By Bott periodicity (period 2 for BU):

$$\pi_k(BU) = \begin{cases} \mathbb{Z} & \text{if } k \text{ is even and } k \geq 2, \\ 0 & \text{otherwise.} \end{cases}$$

The BU obstruction is: $\pi_d(BU) = \mathbb{Z}$ for d even, 0 for d odd.

(B) *Symplectic/orthogonal path.* For d odd, the CY pairing is antisymmetric, giving structure group $\mathrm{Sp}(2m)$. For d even, the pairing is symmetric, giving $O(n)$ (before holomorphic reduction). The stable homotopy groups are given by Bott periodicity (period 8):

$k \bmod 8$	0	1	2	3	4	5	6	7
$\pi_k(\mathrm{Sp})$	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z}$
$\pi_k(O)$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$

The classifying-space obstruction is $\pi_d(B\,\mathrm{Sp}) = \pi_{d-1}(\mathrm{Sp})$ for d odd, and $\pi_d(BO) = \pi_{d-1}(O)$ for d even.

7.5 The E_n landscape: the E_1 stabilization theorem

Theorem 7.1 *(E_1 stabilization for CY chiral algebras).* [PH] For $d \geq 3$, the CY chiral algebra $A_C = \Phi(C)$ associated to a smooth CY_d category C is an E_1-chiral algebra. The additional shifted structure is classified by the *effective framing obstruction*:

$$\mathrm{Obs}_{\mathrm{eff}}(d) = \begin{cases} \pi_d(BU) & \text{if } \pi_d(BU) \neq 0 \text{ (primary obstruction nontrivial)}, \\ \pi_d(B\,\mathrm{Sp}) & \text{if } d \text{ is odd and } \pi_d(BU) = 0 \text{ (Sp refinement)}, \\ 0 & \text{otherwise.} \end{cases}$$

The complete E_n landscape for CY_d categories:

d	Native E_n	$\pi_d(BU)$	$\pi_d(B\,\mathrm{Sp})$	$\mathrm{Obs}_{\mathrm{eff}}$	Shifted structure
1	E_∞	0	0	0	none (commutative)
2	E_2	$\mathbb{Z}$	—	$\mathbb{Z}$	braiding parameter (c_1)
3	E_1	0	0	0	none ($\mathbb{S}^3$-framing trivial)
4	E_1	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$-shifted symplectic (p_1)
5	E_1	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$-shifted (Sp refinement)
6	E_1	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}$	$\mathbb{Z}$-shifted (c_3)
7	E_1	0	0	0	none (trivial, mirrors $d = 3$)
8	E_1	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$-shifted (Bott-periodic)

The pattern repeats with period 8 by Bott periodicity of Sp and O. In the Bott 8-fold cycle $d \bmod 8 = 1, 2, \ldots, 8$, the effective obstruction is:

$$0,\ \mathbb{Z},\ 0,\ \mathbb{Z},\ \mathbb{Z}_2,\ \mathbb{Z},\ 0,\ \mathbb{Z}.$$

The obstruction is *trivial* precisely at $d \bmod 8 \in \{1, 3, 7\}$. All even $d \geq 2$ have $\mathbb{Z}$-valued obstruction (from $\pi_d(BU) = \mathbb{Z}$).

Proof. The proof has three parts.

Part 1: E_1 is the floor for $d \geq 3$. For $d \geq 3$, the $\mathrm{GL}(d)$-invariant Schouten–Nijenhuis bracket on $\mathrm{PV}^*(\mathbb{C}^d)$ vanishes by the same argument as $d = 3$ (Theorem 8.4): degree overflow and graded skew-symmetry. The classical Lie conformal algebra is abelian, and the entire noncommutative structure arises from the Ω-deformation. The CoHA construction gives an ordered (associative) product: the extension correspondence $0 \to V' \to V \to V'' \to 0$ has a preferred direction (sub before quotient), which is E_1, not E_2.

Part 2: The framing obstruction. The $\mathbb{S}^d$-framing obstruction lives in π_d of the classifying space of the structure group. The CY condition $\omega_X \cong \mathcal{O}_X$ determines the structure group:

- The complex structure reduces $\mathrm{GL}(n, \mathbb{R})$ to $U(n)$. The obstruction in $\pi_d(BU)$ is $\mathbb{Z}$ for d even and 0 for d odd (Bott periodicity, period 2).

- For d odd (antisymmetric CY pairing), the further reduction to $\mathrm{Sp}(2m)$ gives a refined obstruction in $\pi_d(B\,\mathrm{Sp}) = \pi_{d-1}(\mathrm{Sp})$. This is nontrivial ($\mathbb{Z}_2$) precisely when $d \equiv 5 \pmod 8$ (since $\pi_4(\mathrm{Sp}) = \mathbb{Z}_2$).

- For d even (symmetric pairing), the orthogonal reduction gives $\pi_d(BO) = \pi_{d-1}(O)$, but the holomorphic structure typically refines back to BU, so the BU obstruction is the primary one.

Part 3: Explicit verification for $d = 4$. At $d = 4$, $\pi_4(BU) = \pi_3(U) = \mathbb{Z}$. The generator is the first Pontryagin class p_1. A CY_4 category C has $p_1(\mathrm{Ext}^\bullet_C(E, E)) \in \mathbb{Z}$, which is the obstruction to an $\mathbb{S}^4$-framing. When $p_1 \neq 0$, the chiral algebra carries a $\mathbb{Z}$-shifted symplectic structure: the factorization algebra structure is E_1 with a choice of integer "level" (analogous to the level of a Kac–Moody algebra) determined by p_1. This is verified computationally for $\mathbb{C}^4$ (93 tests in `higher_cy_en_tower.py`). □

Remark 7.2 (Why E_0 does not exist). The operad $E_0 = \mathrm{Assoc}_+$ (pointed sets with distinguished basepoint) does exist as an operad, but its algebras are "pointed objects," not algebras in the usual sense. An E_0-algebra in chain complexes is a chain complex with a distinguished element (the basepoint), carrying no multiplicative structure. The correct interpretation for $d = 4$ is: the CY chiral algebra is E_1 (with multiplication), and the p_1 class provides additional structure *beyond* the operad level: a shifted symplectic form on the tangent complex of the MC moduli.

COROLLARY 7.3 *($d = 4$: the first Pontryagin obstruction).* [PH] For a CY_4 category C:

(i) The chiral algebra A_C is E_1.

(ii) The $\mathbb{S}^4$-framing obstruction is $p_1 \in \pi_4(BU) = \mathbb{Z}$, the first Pontryagin class. All three paths (BU, $B\,\mathrm{Sp}$, BO) give $\mathbb{Z}$:
$$\pi_4(BU) = \pi_4(B\,\mathrm{Sp}) = \pi_4(BO) = \mathbb{Z}.$$

(iii) The E_2 braided structure is recovered via the Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A_C))$, with the half-braiding carrying a $\mathbb{Z}$-valued twisting datum from p_1.

(iv) The deformation space of $\mathbb{C}^4$ is 2-dimensional, spanned by $\sigma_3 = \sum_{i<j<k} h_i h_j h_k$ and $\sigma_4 = h_1 h_2 h_3 h_4$.

Verification: 141 tests in `higher_cy_en_tower.py`.

Proof. Parts (i) and (ii) follow from Theorem 7.1 at $d = 4$. For part (ii), the three paths:

- $\pi_4(BU) = \pi_3(U) = \mathbb{Z}$ (Bott: 3 is odd, $\pi_3(U) = \mathbb{Z}$).
- $\pi_4(B\,\mathrm{Sp}) = \pi_3(\mathrm{Sp}) = \mathbb{Z}$ (the generator is the symplectic structure on $\mathrm{Sp}(2m)$).
- $\pi_4(BO) = \pi_3(O) = \mathbb{Z}$ (the generator is the spin structure).

All three agree, giving a well-defined $\mathbb{Z}$-valued obstruction.

Part (iii): the Drinfeld center construction of Theorem 8.15 generalizes from $d = 3$ to $d = 4$. The half-braiding on Fock modules of the CY_4 CoHA carries a $\mathbb{Z}$-grading from p_1.

Part (iv): the equivariant parameters $h_1, \ldots, h_4$ satisfy $h_1 + h_2 + h_3 + h_4 = 0$ (CY condition), leaving 3 free parameters. The elementary symmetric polynomials $\sigma_2, \sigma_3, \sigma_4$ parametrize the deformation space; σ_2 generates a trivial deformation (rescaling), leaving σ_3 and σ_4 as the 2 effective parameters. □

COROLLARY 7.4 *($d = 5$: the $\mathbb{Z}_2$ Sp-refinement).* [PH] For a CY_5 category C:

(i) The chiral algebra A_C is E_1.

(ii) The primary framing obstruction $\pi_5(BU) = 0$ is trivial (5 is odd).

(iii) The refined obstruction $\pi_5(B\,\mathrm{Sp}) = \pi_4(\mathrm{Sp}) = \mathbb{Z}_2$ is *nontrivial*. The $\mathbb{S}^5$-framing carries a $\mathbb{Z}_2$-valued invariant from the symplectic structure group.

(iv) The $\mathbb{Z}_2$-shifted structure manifests as a super-E_1 grading: the chiral algebra has a $\mathbb{Z}_2$-grading compatible with the E_1-operad structure, arising from the mod-2 Pontryagin class.

(v) The deformation space of $\mathbb{C}^5$ is 3-dimensional $(\sigma_3, \sigma_4, \sigma_5)$.

Verification: 141 tests in `higher_cy_en_tower.py`.

Proof. Parts (i)–(iii) follow from Theorem 7.1 at $d = 5$, using the Bott periodicity table: $\pi_4(\mathrm{Sp}) = \mathbb{Z}_2$ (verified independently via the computation $\pi_4(\mathrm{Sp}) = \pi_4(\mathrm{Sp}(4)) = \mathbb{Z}_2$ in the stable range).

Part (iv) is a consequence of the $\mathbb{Z}_2$-valued obstruction: the half-braiding in the Drinfeld center carries a $\mathbb{Z}_2$-grading, which promotes the E_1-algebra structure to a $\mathbb{Z}_2$-graded (super) E_1-algebra.

Part (v): the equivariant parameters $h_1, \ldots, h_5$ with $\sum h_i = 0$ give 4 free parameters, with σ_2 trivial, leaving $\sigma_3, \sigma_4, \sigma_5$. □

COROLLARY 7.5 *($d = 7$ mirrors $d = 3$; $d = 8$ mirrors $d = 4$).* [PH]

(i) At $d = 7$: the framing obstruction is trivial. Both $\pi_7(BU) = 0$ and $\pi_7(B\,\mathrm{Sp}) = \pi_6(\mathrm{Sp}) = 0$. The CY_7 chiral algebra is unobstructed E_1, mirroring $d = 3$.

(ii) At $d = 8$: the framing obstruction is $\pi_8(BU) = \mathbb{Z}$ (Bott-periodic, corresponding to $d = 0$ shifted by the 8-fold periodicity of $B\,\mathrm{Sp}$). The CY_8 chiral algebra carries a $\mathbb{Z}$-shifted structure, mirroring $d = 4$.

More generally, d and $d + 8$ have isomorphic framing obstruction groups (Bott periodicity).
Verification: 141 tests in `higher_cy_en_tower.py`.

7.6 Explicit computations for $d = 4$ varieties

$\mathbb{C}^4$

The simplest CY_4 is $\mathbb{C}^4$. The $\mathrm{GL}(4)$-invariant polyvector fields $\mathrm{PV}^p(\mathbb{C}^4)$ have $\dim \mathrm{PV}^p = \binom{4}{p}$ in the constant-coefficient sector:

$$\dim(\mathrm{PV}^0, \mathrm{PV}^1, \mathrm{PV}^2, \mathrm{PV}^3, \mathrm{PV}^4) = (1, 4, 6, 4, 1), \qquad \dim_{\mathrm{total}} = 2^4 = 16.$$

The Schouten–Nijenhuis brackets on the $\mathrm{GL}(4)$-invariant sector vanish for the same reasons as $d = 3$ (degree overflow and graded skew-symmetry). The Bogomolov–Tian–Todorov theorem gives $\mathrm{HH}^3 = 0$ (unobstructedness).

The 2-dimensional deformation space is spanned by

$$\sigma_3 = \sum_{i<j<k} h_i h_j h_k, \qquad \sigma_4 = h_1 h_2 h_3 h_4,$$

subject to $h_1 + h_2 + h_3 + h_4 = 0$. At the self-dual point (one $h_i = 0$, say $h_4 = 0$), the CY_4 functor reduces to the CY_3 functor for $\mathbb{C}^3$ with parameters (h_1, h_2, h_3), and the chiral algebra is $\mathcal{W}_{1+\infty}$ at $c = 1$ (the Heisenberg VOA H_1, $\kappa = 1$).

$K3 \times K3$

The product $K3 \times K3$ is a CY_4 with Euler characteristic

$$\chi(K3 \times K3) = \chi(K3)^2 = 24^2 = 576$$

and Hodge numbers computed by Künneth from the K3 diamond. The key Hodge numbers: $h^{1,1} = 40$, $h^{2,0} = h^{0,2} = 2$, $h^{4,0} = h^{0,4} = 1$.

The chiral algebra is the tensor product of two copies of the K3 lattice VOA:

$$A_{K3\times K3} = V_{\Lambda_{K3}} \otimes V_{\Lambda_{K3}},$$

with modular characteristic

$$\kappa(A_{K3\times K3}) = \kappa(V_{\Lambda_{K3}}) + \kappa(V_{\Lambda_{K3}}) = 24 + 24 = 48,$$

by Vol I additivity (Theorem D). The shadow obstruction tower is class **G** (Gaussian, $r_{\max} = 2$) since the lattice VOA is a free-field algebra.

The sextic 4-fold in $\mathbb{P}^5$

The degree-6 smooth hypersurface $X_6 \subset \mathbb{P}^5$ is a compact CY_4 with $h^{3,1} = 426$, $h^{2,2} = 1752$, $h^{1,1} = 1$, and $\chi(X_6) = 2610$. The holomorphic Euler characteristic is $\chi(\mathcal{O}_{X_6}) = 1 + 0 + 1 = 2$ (from $h^{0,0} = h^{4,0} = 1$ and $h^{2,0} = 0$).

7.7 The $d = 5$ functor: $\mathbb{Z}_2$ obstruction from Sp

The CY_5 functor introduces a genuinely new phenomenon: a $\mathbb{Z}_2$-valued framing obstruction that is invisible to the unitary structure group but visible to the symplectic refinement.

For $\mathbb{C}^5$: the polyvector field dimensions are $\dim \mathrm{PV}^p = \binom{5}{p}$, giving total dimension $2^5 = 32$. The deformation space is 3-dimensional $(\sigma_3, \sigma_4, \sigma_5)$.

The framing obstruction:

- $\pi_5(BU) = 0$ (primary, from unitary structure: 5 is odd).
- $\pi_5(B\,\mathrm{Sp}) = \pi_4(\mathrm{Sp}) = \mathbb{Z}_2$ (refined, from symplectic structure).

The discrepancy between the BU and $B\,\mathrm{Sp}$ obstructions means that the chain-level $\mathbb{S}^5$-framing is topologically trivializable at the unitary level, but the symplectic refinement carries a $\mathbb{Z}_2$-valued obstruction. This obstruction is the mod-2 analogue of the first Pontryagin class at $d = 4$.

7.8 The $d = 6$ case and higher

Remark 7.6 ($d = 6$: BU vs BO discrepancy). At $d = 6$: $\pi_6(BU) = \mathbb{Z}$ (nontrivial) but $\pi_6(BO) = \pi_5(O) = 0$ (trivial). The real orthogonal structure group sees no obstruction, but the complex/holomorphic structure (captured by BU) does. This means the $\mathbb{Z}$-valued obstruction at $d = 6$ is purely holomorphic: it is visible only through the complex structure of the CY category, not through the underlying real topology.

The characteristic class is $c_3 \in H^6(B; \mathbb{Z})$, the third Chern class of the tangent bundle. For a CY_6 manifold X with $c_3(X) \neq 0$, the chiral algebra carries a $\mathbb{Z}$-shifted structure indexed by c_3.

Conjecture 7.7 *(E_n-Koszul duality at $d \geq 4$).* [CJ] For a CY_d chiral algebra A_C with $d \geq 4$, the Koszul dual $A_C^! = \Phi(C^!)$ (the chiral algebra of the mirror category) satisfies:

(i) The framing obstruction of $A_C^!$ is the *negation* (in the obstruction group) of the framing obstruction of A_C.

(ii) For $d = 4$ ($\mathbb{Z}$-valued obstruction): $p_1(A_C^!) = -p_1(A_C)$. The Koszul dual carries the opposite Pontryagin class.

(iii) For $d = 5$ ($\mathbb{Z}_2$-valued obstruction): $\mathrm{Obs}(A_C^!) = \mathrm{Obs}(A_C)$ (since $-1 = 1$ in $\mathbb{Z}_2$). The $\mathbb{Z}_2$-shifted structure is *self-dual*: mirror symmetry preserves it.

(iv) The modular characteristics satisfy $\kappa(A_C) + \kappa(A_C^!) = 0$ (for the free-field case) or $\kappa + \kappa' = \rho \cdot K$ (for W-algebra-type families), matching Vol I Theorem D.

7.9 Relation to topological field theory

The E_1 stabilization theorem has a direct TFT interpretation. A CY_d category determines a d-dimensional topological field theory (Costello–Li 2016). For $d \geq 3$, the TFT is "fully extended" down to the E_1 level: it assigns categories to $(d-1)$-manifolds, functors to cobordisms, and natural transformations to higher cobordisms. The E_n level of the chiral algebra is the E_n level at which the fully extended TFT "stops":

d	E_n level	TFT stops at	Obstruction
1	E_∞	fully local	none
2	E_2	2-categorical	$c_1 \in \mathbb{Z}$
3	E_1	categorical	none
4	E_1	categorical	$p_1 \in \mathbb{Z}$
5	E_1	categorical	$\mathbb{Z}_2$ from Sp
≥ 6	E_1	categorical	Bott-periodic

The passage from E_1 to E_2 (the quantum group braiding) is always achieved through the Drinfeld center, which is the TFT analogue of "compactifying on a circle": $\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \simeq \mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A))$. *Verification*: 141 tests across all CY dimensions $d = 1, \ldots, 16$ in `higher_cy_en_tower.py`.

Part II

The CY Characteristic Datum

Chapter 8

From CY Categories to Chiral Algebras

This chapter constructs the bridge functor Φ from Calabi–Yau categories to quantum chiral algebras. At $d = 2$ (Theorem 8.1), the $\mathbb{S}^2$-framing on Hochschild homology produces an E_2-chiral algebra directly. At $d = 3$, the native E_3-structure restricts to symmetric braiding; the quantum group braiding arises from the Drinfeld center of the E_1-monoidal representation category. The main results: end-to-end verification of the $d = 3$ functor chain for $\mathbb{C}^3$ (Theorem 8.2), universal vanishing of the $\mathbb{S}^3$-framing obstruction (Theorem 8.8), the $E_1 \to E_2$ enhancement via the Drinfeld center (Theorem 8.15), and the quiver-chart gluing construction (Section 8.8), assembling the global E_1-chiral algebra from local CoHA charts via homotopy colimits over chambers of the Bridgeland stability manifold.

8.1 The cyclic-to-chiral passage

A CY category C of dimension d carries a cyclic A_∞-structure (Chapter 3). The cyclic bar complex $\mathrm{CC}_\bullet(C)$ with its $\mathbb{S}^d$-framing is the primary invariant. The passage to chiral algebras proceeds in four steps:

Step 1. Cyclic A_∞ → Lie conformal algebra. The Hochschild cohomology $\mathrm{HH}^\bullet(C)$, with its Gerstenhaber bracket, determines a graded Lie algebra. The CY pairing promotes this to a Lie conformal algebra $\mathfrak{L}_C$ (the "current algebra" of C).

Step 2. Lie conformal algebra → factorization envelope. The factorization envelope construction (Nishinaka–Vicedo) produces a factorization algebra $\mathrm{Fact}_X(\mathfrak{L}_C)$ on any smooth curve X.

Step 3. E_2 enhancement. When $d = 2$, the $\mathbb{S}^2$-framing of $\mathrm{HH}_\bullet(C)$ provides an E_2-algebra structure. This enhances $\mathrm{Fact}_X(\mathfrak{L}_C)$ to an E_2-chiral algebra.

Step 4. Quantization. The quantum chiral algebra A_C is the quantization of $\mathrm{Fact}_X(\mathfrak{L}_C)$ determined by the CY trace.

For $d = 3$, Step 3 must be replaced: the $\mathbb{S}^3$-framing gives an E_3-structure on $\mathrm{HH}_\bullet(C)$, but the resulting E_2-braiding (obtained by restriction via Dunn additivity) is *symmetric*. The nonsymmetric quantum group braiding is recovered by a different route: the Drinfeld center of the E_1-monoidal representation category (Section 8.5).

8.2 The CY-to-chiral functor

THEOREM 8.1 *(CY-to-chiral functor — Theorem CY-A_2, $d = 2$).* [PH] There exists a functor

$$\Phi\colon \mathrm{CY}_2\text{-}\mathbf{Cat} \longrightarrow E_2\text{-}\mathrm{ChirAlg}$$

from smooth proper CY categories of dimension $d = 2$ to E_2-chiral algebras, such that:

(i) $\Phi(C)$ is a chiral algebra whose generating fields are in bijection with $\mathrm{HH}^{\bullet+1}(C)$;

(ii) The bar complex $B(\Phi(C))$ is quasi-isomorphic to the cyclic bar complex $\mathrm{CC}_\bullet(C)$ as a factorization coalgebra;

(iii) The modular characteristic $\kappa(\Phi(C))$ equals the CY Euler characteristic.

The E_2-enhancement in Step 3 uses the $\mathbb{S}^2$-framing of $\mathrm{HH}_\bullet(C)$ (Kontsevich–Vlassopoulos).

8.3 The $d = 3$ functor chain for $\mathbb{C}^3$

Affine space $X = \mathbb{C}^3$ is the Rosetta Stone for the $d = 3$ programme. Both sides are independently known: the CY side produces $\mathrm{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1)$ (Kontsevich–Soibelman, Schiffmann–Vasserot), and the chiral side gives $\mathcal{W}_{1+\infty}$ at $c = 1$ (Procházka–Rapčák). The functor chain is verified end-to-end.

The five-step chain

The $d = 3$ functor for $\mathbb{C}^3$ factors as:

Step 1. $\mathrm{PV}^*(\mathbb{C}^3)$ with $\mathrm{GL}(3)$-invariant Schouten–Nijenhuis bracket

$\downarrow$ Ω-deformation in the σ_3 direction

Step 2. Quantized Lie conformal algebra

$\downarrow$ Factorization envelope (Nishinaka–Vicedo)

Step 3. $\mathcal{W}_{1+\infty}$ with nonabelian OPE

$\downarrow$ Drinfeld center

Step 4. E_2-braided category $\mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1))$ with R-matrix

$\downarrow$ Quantum group extraction

Step 5. Quantum vertex chiral group $G(\mathbb{C}^3)$

The input at Step 1 is $\mathrm{PV}^*(\mathbb{C}^3) = \bigoplus_{p=0}^{3} \Gamma(\mathbb{C}^3, \bigwedge^p T_{\mathbb{C}^3})$, the algebra of polyvector fields with the Schouten–Nijenhuis bracket. The $\mathrm{GL}(3)$-invariant sector carries the deformation-theoretic data needed for the CY-to-chiral functor. The Ω-deformation at Step 2 is parametrized by $\sigma_3 = h_1 h_2 h_3$ (with $h_1 + h_2 + h_3 = 0$), the unique direction in $\mathrm{HH}^2(\mathrm{PV}^*(\mathbb{C}^3))$. At the self-dual level ($\sigma_3 \to 0$, giving $h_1 = 1, h_2 = 0, h_3 = -1$), the output at Step 3 is $\mathcal{W}_{1+\infty}$ at $c = 1$, which *is* the Heisenberg VOA H_1.

The classical Lie conformal algebra is *abelian* (Theorem 8.4): all noncommutative structure arises from quantization in the envelope step. The CY_3 functor produces a quantum algebra from classically commutative input.

THEOREM 8.2 *(The $d = 3$ functor chain is verified for $\mathbb{C}^3$).* [PH] Each step of the five-step chain is verified computationally:

(i) *Step 1*: The $\mathrm{GL}(3)$-invariant Schouten–Nijenhuis brackets vanish identically (Theorem 8.4). The input Lie conformal algebra is abelian.

(ii) *Step 2*: $\mathrm{HH}^2(\mathrm{PV}^*(\mathbb{C}^3)) = 1$, so the deformation is unique, spanned by σ_3 (Theorem 8.5). $\mathrm{HH}^3 = 0$ by Bogomolov–Tian–Todorov, so the deformation is unobstructed.

(iii) *Step 3*: The factorization envelope produces $\mathcal{W}_{1+\infty}$ at $c = 1$. The OPE arises entirely from the central extension and normal ordering in the envelope, not from the classical bracket.

(iv) *Step 4*: The Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(Y^+(\widehat{\mathfrak{gl}}_1)))$ is identified with $\mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1))$ (Theorem 8.15).

(v) *Step 5*: The quantum vertex chiral group $G(\mathbb{C}^3)$ is recovered from the E_2-braided representation category.

Verification: ~600 tests across six compute modules: `c3_lie_conformal.py`, `c3_envelope_comparison.py`, `cy3_hochschild.py`, `drinfeld_center_yangian.py`, `c3_grand_verification.py`, `cy_to_chiral_functor.py`.

Remark 8.3 (The shuffle algebra does not translate directly to the λ-bracket). The shuffle algebra presentation of $Y^+(\widehat{\mathfrak{gl}}_1)$ has structure function $g(z) = (z - h_1)(z - h_2)(z - h_3)/((z + h_1)(z + h_2)(z + h_3))$. A natural attempt extracts a λ-bracket from the shuffle product via $z \mapsto \lambda$. This fails: $g(0) = -1$ (regular, not singular), so the shuffle product does not produce a distribution-valued bracket. The envelope step (Step 2 $\to$ Step 3) replaces the algebraic $g(z)$ by the *vertex operator* OPE, where the δ-function singularity of the state-field correspondence generates the λ-bracket. The passage shuffle $\to$ vertex algebra is the step where locality enters.

Abelianity of the classical bracket

THEOREM 8.4 *(Abelianity of the classical bracket).* [PH] The $\mathrm{GL}(3)$-invariant Schouten–Nijenhuis brackets on $\mathrm{PV}^*(\mathbb{C}^3)$ all vanish. Three independent proofs:

(i) *Direct computation*: every bracket $[\alpha, \beta]_{\mathrm{SN}}$ of $\mathrm{GL}(3)$-invariant polyvector fields evaluates to zero by explicit contraction.

(ii) *Graded skew-symmetry + degree count*: in the Gerstenhaber grading (shifted degree $|\alpha|_s = |\alpha| - 1$), the generators $E \in \mathrm{PV}^1$ and $\partial_1 \wedge \partial_2 \wedge \partial_3 \in \mathrm{PV}^3$ have even shifted degree (0 and 2 respectively), so $[\alpha, \alpha] = 0$ for these by graded skew-symmetry. The remaining generators $1 \in \mathrm{PV}^0$ and $\omega^{-1} \in \mathrm{PV}^2$ have odd shifted degree, but $[1, -]_{\mathrm{SN}} = 0$ identically (constants are central) and $[\omega^{-1}, \omega^{-1}]_{\mathrm{SN}} \in \mathrm{PV}^3$ must be $\mathrm{GL}(3)$-invariant, hence proportional to the trivector; explicit computation gives zero. All mixed brackets $[\alpha, \beta]$ with $\alpha \neq \beta$ vanish by the degree-overflow argument of (iii).

(iii) *Weight/exterior degree overflow*: the bracket lowers exterior degree by 1; any nontrivial $\mathrm{GL}(3)$-invariant in the target degree must have weight exceeding the polynomial degree available.

Verification: 95 tests in `c3_lie_conformal.py`.

Proof. We give proof (i) in detail; proofs (ii) and (iii) are recorded for cross-verification.

The $\mathrm{GL}(3)$-invariant polyvector fields on $\mathbb{C}^3$ are spanned by the constant function $1 \in \mathrm{PV}^0$, the Euler vector field $E = \sum_i z_i \partial_i \in \mathrm{PV}^1$, the Poisson bivector $\pi = \sum_{\mathrm{cyc}} z_3\, \partial_1 \wedge \partial_2 \in \mathrm{PV}^2$ (contraction of E with the volume trivector), and the trivector $\partial_1 \wedge \partial_2 \wedge \partial_3 \in \mathrm{PV}^3$. The Schouten–Nijenhuis bracket has bidegree (-1): it maps $\mathrm{PV}^p \times \mathrm{PV}^q \to \mathrm{PV}^{p+q-1}$. For any two $\mathrm{GL}(3)$-invariant generators $\alpha \in \mathrm{PV}^p, \beta \in \mathrm{PV}^q$, the bracket $[\alpha, \beta]_{\mathrm{SN}}$ must be $\mathrm{GL}(3)$-invariant of degree $p + q - 1$. In each case, either $p + q - 1 > 3$ (forcing zero by degree overflow) or the explicit contraction vanishes by the symmetry of the invariant generators. □

Hochschild cohomology and the unique deformation

Theorem 8.5 *(Hochschild cohomology and unique deformation).* [PH] For $\mathbb{C}^3$:

(i) $\mathrm{HH}^2(\mathrm{PV}^*(\mathbb{C}^3)) = 1$: the deformation space is one-dimensional, spanned by the σ_3 direction. The unique deformation is the Ω-deformation.

(ii) $\mathrm{HH}^3(\mathrm{PV}^*(\mathbb{C}^3)) = 0$: the Bogomolov–Tian–Todorov theorem guarantees unobstructedness.

(iii) The quantized algebra is $\mathcal{W}_{1+\infty}$ at $c = 1$.

Verification: 71 tests in `cy3_hochschild.py`.

Proof. The Hochschild–Kostant–Rosenberg theorem identifies $\mathrm{HH}^\bullet(\mathrm{PV}^*(\mathbb{C}^3))$ with the $\mathrm{GL}(3)$-invariant polyvector fields on the formal neighbourhood of the origin in the moduli space of A_∞-deformations. For $\mathbb{C}^3$ with the standard CY structure, the relevant computation reduces to the T^3-equivariant sector.

The equivariant parameters h_1, h_2, h_3 with $h_1 + h_2 + h_3 = 0$ span a two-dimensional space. The unique T^3-equivariant deformation class in HH^2 is $\sigma_3 = h_1 h_2 h_3$, the elementary symmetric polynomial of degree 3. The class $\sigma_2 = h_1 h_2 + h_2 h_3 + h_3 h_1 = -(h_1^2 + h_2^2 + h_3^2)/2$ on the constraint locus is generically nonzero, but it generates a trivial deformation equivalent to rescaling: the σ_2-deformation acts as a conformal weight shift that can be absorbed by redefining the grading. The nontrivial deformation class is $\sigma_3 = h_1 h_2 h_3$, which cannot be removed by regrading.

The vanishing $\mathrm{HH}^3 = 0$ is the Bogomolov–Tian–Todorov unobstructedness theorem for the CY_3 moduli problem: the Kodaira–Spencer dga of $\mathbb{C}^3$ is formal, and the Maurer–Cartan moduli space is smooth. This guarantees that the Ω-deformation determined by σ_3 extends to all orders. □

Necessity of the Ω-deformation for noncompact CY_3

Proposition 8.6 *(Necessity of Ω-deformation for noncompact CY_3).* [PH] Without the Ω-deformation, the CY_3 functor chain for $\mathbb{C}^3$ collapses: the Lie conformal algebra is abelian (Theorem 8.4), the envelope produces the free Heisenberg, and the E_2 structure is trivially E_∞ (symmetric monoidal). Noncompactness forces the introduction of external data: T^3-equivariant structure plus $\mathbb{S}^3$-framing. For compact CY_3, neither should be needed: the nontrivial global geometry plays the role of the Ω-deformation.
Verification: 87 tests in `c3_envelope_comparison.py`.

Proof. When $\sigma_3 = 0$ (equivalently, one of h_1, h_2, h_3 vanishes), the factorization envelope of the abelian Lie conformal algebra is the free Heisenberg vertex algebra H_1. The representation category $\mathrm{Rep}(H_1)$ is symmetric monoidal: the braiding on Fock modules is the identity (all monodromy is trivial for a free boson at level 1). In particular, $\mathcal{Z}(\mathrm{Rep}^{E_1}(H_1)) = \mathrm{Rep}(H_1)$ itself (the Drinfeld center of a symmetric monoidal category is the category itself), so no E_2-enhancement occurs.

When $\sigma_3 \neq 0$, the Ω-deformation introduces the nonlinear OPE terms of $\mathcal{W}_{1+\infty}$. These terms break the E_∞ symmetry to E_2, and the Drinfeld center produces a nontrivially braided category with the Yang R-matrix (Theorem 8.15).

For a *compact* CY_3 such as the quintic, the global sections of the polyvector sheaf have nontrivial bracket (the complex structure moduli introduce genuine noncommutativity), and the Ω-deformation is not needed. The role of σ_3 for $\mathbb{C}^3$ is to *replace* the global geometric data that is absent for a noncompact variety. □

Remark 8.7 (Smooth vs singular: locality of the quantization). The distinction between smooth and singular CY_3 is categorical:

$\mathbf{CY}_3$	**Quantization**	**Algebraic type**
Smooth ($\mathbb{C}^3$, quintic, $K3 \times E$)	Vertex algebra (local)	E_1-chiral (E_2 via Drinfeld center)
Singular (conifold)	Associative algebra (nonlocal)	E_1-chiral

Smoothness is not a matter of convention: the factorization envelope requires local data (the polyvector fields form a cosheaf whose sections over small discs control the OPE), and singularities obstruct the local-to-global passage. At a singularity, the category $\mathrm{Perf}(X)$ acquires compact objects that do not extend to a neighbourhood, and the factorization structure degenerates from E_2 to E_1.
Verification: 71 tests in `cy3_hochschild.py` (testing smooth vs singular HH data).

8.4 The $\mathbb{S}^3$-framing obstruction: universal triviality

The chain-level $\mathbb{S}^3$-framing is the central topological obstruction to CY-A at $d = 3$. Conjecture 8.49 requires (a) constructing this framing compatibly with BV structure and (b) establishing the quantization step. The following result removes condition (a).

THEOREM 8.8 *(Vanishing of the $\mathbb{S}^3$-framing obstruction for CY$_3$ categories).* [PH] The topological $\mathbb{S}^3$-framing obstruction vanishes universally for CY_3 categories. Two independent proofs:

(i) *Symplectic path.* The CY_3 pairing on the Ext complex $\mathrm{Ext}^\bullet_{\mathcal{C}}(E, E)$ is antisymmetric (by the Serre functor with $\omega_X \cong \mathcal{O}_X$), giving structure group $\mathrm{Sp}(2m)$ for $\dim \mathrm{Ext}^1 = 2m$. Since $\pi_3(B\,\mathrm{Sp}(2m)) = 0$ for all $m \geq 1$, the obstruction class in π_3 vanishes.

(ii) *Bott periodicity path.* The complex structure gives a reduction $\mathrm{GL}(n, \mathbb{C}) \to U(n)$. By Bott periodicity, $\pi_3(BU) = 0$ (the homotopy groups of BU are $\pi_{2k}(BU) = \mathbb{Z}$, $\pi_{2k+1}(BU) = 0$). Since π_3 is odd, the obstruction vanishes.

The chain-level BV obstruction $\kappa \cdot [\Omega_3]$ is trivializable via holomorphic Chern–Simons (Witten 1992, Costello–Li 2016): the functional $\mathrm{CS}(\bar\partial + A) = \int_X \Omega \wedge \mathrm{tr}(A \wedge \bar\partial A + \frac{2}{3} A \wedge A \wedge A)$ provides the contracting homotopy.
Verification: 124 tests in `cy_to_chiral_functor.py`.

Proof. We give the symplectic path in detail.

The CY_3 condition $\omega_X \cong \mathcal{O}_X$ implies the existence of a nondegenerate trace $\mathrm{tr}\colon \mathrm{HH}_3(C) \to \mathbb{C}$, which by Serre duality induces an antisymmetric pairing

$$\langle -, - \rangle \colon \mathrm{Ext}^i(E, E) \otimes \mathrm{Ext}^{3-i}(E, E) \longrightarrow \mathbb{C}.$$

At $i = 1$, this is an antisymmetric form on the tangent space to the moduli of objects, reducing the structure group from $\mathrm{GL}(2m)$ to $\mathrm{Sp}(2m)$. At $i = 0$, the self-pairing $\mathrm{Ext}^0 \otimes \mathrm{Ext}^3 \to \mathbb{C}$ is the Serre duality pairing on endomorphisms.

The $\mathbb{S}^3$-framing obstruction lives in π_3 of the classifying space of the structure group. For the symplectic group:

$$\pi_3(B\,\mathrm{Sp}(2m)) = \pi_2(\mathrm{Sp}(2m)) = 0 \quad \text{for all } m \geq 1.$$

The second equality uses the long exact sequence of the fibration $\mathrm{Sp}(2m) \to B\,\mathrm{Sp}(2m)$ and the fact that $\mathrm{Sp}(2m)$ is simply connected ($\pi_1(\mathrm{Sp}(2m)) = 0$) with $\pi_2(\mathrm{Sp}(2m)) = 0$ (all compact simply-connected simple Lie groups have vanishing π_2).

For the Bott periodicity path: the Bott periodicity theorem gives $\pi_k(BU) \cong \pi_k(BU(n))$ for n large, with $\pi_{2k}(BU) = \mathbb{Z}$ and $\pi_{2k+1}(BU) = 0$. Since 3 is odd, $\pi_3(BU) = 0$.

The chain-level BV obstruction requires more than topological triviality: one needs an explicit trivialization compatible with the BV operator $\Delta = \partial/\partial\Omega_3$. The holomorphic Chern–Simons functional provides this. The CS action is a trivialization of $[\Omega_3]$ because $\delta_{\mathrm{BV}}(\mathrm{CS}) = \int \Omega \wedge F_A$, and the flatness equation $F_A = 0$ is the Maurer–Cartan equation of the B-model. This means the BV obstruction is exact in the BRST cohomology. □

COROLLARY 8.9 *(The CY-A_3 conjecture has no topological obstruction).* [PH] Conjecture 8.49 (the $d = 3$ CY-to-chiral functor) has no topological obstruction. The remaining condition is the chain-level construction of the trivialization of $\kappa \cdot [\Omega_3]$, which is known to exist by holomorphic Chern–Simons but whose compatibility with the full A_∞-structure requires verification. For all toric CY_3 tested ($\mathbb{C}^3$, conifold, local $\mathbb{P}^2$, local $\mathbb{P}^1 \times \mathbb{P}^1$, $K3 \times E$), the $E_1 \to E_2$ enhancement obstruction vanishes.
Verification: 124 tests in `cy_to_chiral_functor.py`; 93 tests in `drinfeld_center_yangian.py`.

The Čech contracting homotopy for compact CY_3

The topological vanishing of the $\mathbb{S}^3$-framing obstruction (Theorem 8.8) leaves open the chain-level construction of the trivialization. For compact CY_3 hypersurfaces in projective space, the standard affine cover provides an algebraic Čech contracting homotopy that closes this gap at the perturbative level.

THEOREM 8.10 *(Čech contracting homotopy for compact CY_3; [PH]).* For the quintic threefold $X_5 \subset \mathbb{P}^4$ with the standard affine cover $\{U_i = \{x_i \neq 0\}\}_{i=0}^4$, the algebraic Čech contracting homotopy $s^q : \check{C}^q \to \check{C}^{q-1}$ (prepend the distinguished index 0) satisfies the SDR conditions

$$\delta s + s\delta = \mathrm{Id} - i \circ p, \qquad s^2 = 0, \qquad s \circ i = 0, \qquad p \circ s = 0,$$

where $i\colon H^*(\check{C}^\bullet) \hookrightarrow \check{C}^\bullet$ is the inclusion of cohomology and p is the projection. The contracting homotopy is purely algebraic: it acts by $s(f_{i_0\cdots i_q}) = f_{0i_0\cdots i_q}$ on Čech cochains and involves no PDE solving. The SDR data (i, p, s) interfaces directly with the homotopy transfer theorem (HTT, Theorem **??**), transferring the A_∞-structure from $\check{C}^\bullet(\mathcal{E}nd(\mathcal{E}))$ to its cohomology $\mathrm{Ext}^\bullet(\mathcal{E}, \mathcal{E})$.

Proof. The Čech complex of a quasi-coherent sheaf on $\mathbb{P}^4$ with respect to the standard affine cover $\{U_i\}_{i=0}^4$ is acyclic in positive Čech degree (Leray's theorem: each finite intersection $U_{i_0}\cap\cdots\cap U_{i_q}$ is affine, and affine varieties have vanishing higher cohomology for quasi-coherent sheaves). The homotopy $s^q(f_{i_0\cdots i_q}) := f_{0i_0\cdots i_q}$ satisfies $(\delta s+s\delta)(f) = f - f|_{U_0}$ for $q \geq 1$, giving $\delta s+s\delta = \mathrm{Id}-i\circ p$. The annihilation conditions $s^2 = 0$, $s\circ i = 0$, $p\circ s = 0$ follow from the index 0 convention: $s^2(f_{i_0\cdots i_q}) = s(f_{0i_0\cdots i_q}) = f_{00i_0\cdots i_q} = 0$ by the alternating sign rule. The same argument applies to the restriction $X_5 \hookrightarrow \mathbb{P}^4$: the induced cover $\{U_i \cap X_5\}$ consists of affine open sets (hyperplane complements of a smooth hypersurface in projective space are affine), so Leray's theorem applies. □

Remark 8.11 (Perturbative vs non-perturbative scope). The Čech homotopy closes the analytic gap at the level of formal (perturbative) deformations: it provides a chain-level contracting homotopy that is algebraic and requires no PDE solving. The non-perturbative convergence of the resulting chiral operations (whether the formal series in the OPE coefficients converges to define honest holomorphic functions) requires separate input from the analytic completion programme (MC5, §**??** of Vol I). Specifically: the HTT produces transferred operations μ_k^{ch} whose coefficients involve iterated applications of s; each application introduces rational functions on the intersections $U_{i_0}\cap\cdots\cap U_{i_q}$, and the convergence of the resulting series in the OPE variable z is not guaranteed by the algebraic construction alone. The perturbative statement is: the A_∞ operations on $\mathrm{Ext}^\bullet(\mathcal{E},\mathcal{E})$ are well-defined as formal power series. The non-perturbative statement (convergence to holomorphic functions) requires estimates on the growth of the Čech homotopy applied to OPE kernels.

The following two computations illustrate the chain-level chiral structure for concrete CY_3 geometries, distinguishing formal from non-formal behaviour.

Computation 8.12 *(Chain-level chiral structure for the conifold).* The resolved conifold $X = \mathrm{Tot}(\mathcal{O}(-1)\oplus\mathcal{O}(-1) \to \mathbb{P}^1)$ has $\mathrm{Ext}^\bullet(\mathcal{E},\mathcal{E})$ with 12 generators distributed as $2+4+4+2$ by Ext degree. The Euler characteristic is $\chi = 2-4+4-2 = 0$ (CY_3 check: $\sum(-1)^i \dim\mathrm{Ext}^i = 0$). The A_∞-algebra is *formal*: the transferred operations satisfy $m_k = 0$ for $k \geq 3$ (Kaledin's theorem: the derived category of the resolved conifold is intrinsically formal as a CY_3 category, since it is derived equivalent to $\mathbb{C}Q/(\partial W)$ for the Klebanov–Witten quiver with quadratic potential). The chiral Jacobi identity $\sum_{\sigma\in S_3}(-1)^{|\sigma|}\mu_2^{\mathrm{ch}}(\mu_2^{\mathrm{ch}}(a_{\sigma(1)},a_{\sigma(2)};z_{12});a_{\sigma(3)};z_{13}) = 0$ is verified through total degree 6 in the generators. The shadow tower is class $\mathbf{G}$ ($r_{\max} = 2$, $\kappa = 1$), consistent with the formality.

Computation 8.13 *(Non-formality for local $\mathbb{P}^2$).* The local $\mathbb{P}^2$ geometry $X = \mathrm{Tot}(\mathcal{O}(-3)\to\mathbb{P}^2)$ has 24 generators from the McKay quiver of $\mathbb{Z}_3$: three vertices with 8 generators each (the McKay correspondence assigns the regular representation $\mathbb{C}[\mathbb{Z}_3] = \rho_0\oplus\rho_1\oplus\rho_2$ to the vertices, and the tensor product with the defining $\mathrm{SL}(3)$-representation determines the arrows). The algebra is *not* formal: the transferred ternary operation is

$$m_3(x_{01},y_{12},z_{20}) \;=\; e_0,$$

where $x_{01}\in\mathrm{Ext}^1(\mathcal{E}_0,\mathcal{E}_1)$, $y_{12}\in\mathrm{Ext}^1(\mathcal{E}_1,\mathcal{E}_2)$, $z_{20}\in\mathrm{Ext}^1(\mathcal{E}_2,\mathcal{E}_0)$ are the three generating arrows around the quiver triangle, and $e_0\in\mathrm{Ext}^0(\mathcal{E}_0,\mathcal{E}_0)$ is the identity endomorphism. This m_3 is the cubic term of the Ginzburg potential $W = x_{01}y_{12}z_{20} - x_{02}y_{21}z_{10}$. The chiral lift has double-pole structure:

$$\mu_3^{\mathrm{ch}}(x_{01}(z_1),y_{12}(z_2),z_{20}(z_3)) \;=\; \frac{e_0}{(z_1-z_2)(z_2-z_3)}.$$

The shadow tower is class $\mathbf{M}$ ($r_{\max} = \infty$, $\kappa = 3/2$): the non-formality forces an infinite tower of shadow obstructions.

Remark 8.14 (The E_n landscape for Calabi–Yau categories). The CY dimension d determines the native algebraic structure. For $d \leq 3$:

CY dim	**Native E_n**	**Enhancement**	**Mechanism**
$d = 1$ (elliptic curve)	E_∞	—	Braiding symmetric
$d = 2$ (K3, Higgs)	E_2	native	$\mathbb{S}^2$-framing automatic
$d = 3$ (CY threefold)	E_1	$E_1 \to E_2$	Drinfeld center / $\mathbb{S}^3$-framing

By Dunn additivity, $E_2 \simeq E_1 \otimes_{E_0} E_1$. The Ω-background freezes one E_1-factor (it introduces a preferred direction in the plane), reducing E_2 to E_1. The Drinfeld center passage $\mathcal{Z}(\mathrm{Rep}^{E_1}(\cdot))$ restores the E_2 structure by recovering the frozen factor.

The E_3-to-E_2 restriction via Dunn additivity gives a *symmetric* braiding, since $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3))$ is trivial. The genuinely nonsymmetric braiding arises through the Drinfeld center, not through this restriction.

For $d \geq 4$, the E_1 stabilization theorem (Theorem 7.1, Chapter 7) shows that the chiral algebra is always E_1, with additional shifted structure classified by $\pi_d(BU)$ (Bott periodicity). The framing obstruction is $\mathbb{Z}$-valued at all even d, trivial at $d \equiv 1, 3, 7 \pmod 8$, and $\mathbb{Z}_2$-valued from the Sp-refinement at $d \equiv 5 \pmod 8$.

8.5 The Drinfeld center and E_2 enhancement

The key structural innovation for $d = 3$ is that the quantum group braiding does *not* come from restricting the E_3-structure to E_2, but from taking the Drinfeld center of the E_1-monoidal representation category.

The CY condition and the universal R-matrix

The structure function of the affine Yangian $Y^+(\widehat{\mathfrak{gl}}_1)$ is

$$g(z) = \frac{(z - h_1)(z - h_2)(z - h_3)}{(z + h_1)(z + h_2)(z + h_3)}, \tag{8.1}$$

where $h_1 + h_2 + h_3 = 0$ are the equivariant parameters. The CY$_3$ condition manifests as the identity

$$g(z)\, g(-z) = 1. \tag{8.2}$$

This is the R-matrix unitarity condition: it guarantees that the braiding $\sigma_{V,W} : V \otimes W \to W \otimes V$ defined by $R(z)$ is involutive up to homotopy. The identity (8.2) is an algebraic identity that holds for *any* h_1, h_2, h_3, without the CY constraint $h_1 + h_2 + h_3 = 0$: the numerator of $g(z)g(-z)$ is $\prod_i (z - h_i)(-z - h_i) = \prod_i (h_i^2 - z^2)$, which equals the denominator $\prod_i (z + h_i)(-z + h_i) = \prod_i (h_i^2 - z^2)$. The CY constraint plays a different role: it ensures $g(z) \to 1$ as $z \to \infty$, which is needed for the R-matrix integral representation to converge (Pillar (d) of Theorem 8.23).

Theorem 8.15 *(Drinfeld center identification for $\mathbb{C}^3$).* [PH]

$$\mathcal{Z}\big(\mathrm{Rep}^{E_1}(Y^+(\widehat{\mathfrak{gl}}_1))\big) \;\simeq\; \mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1)) \;\simeq\; \mathrm{Rep}^{E_2}(\mathcal{W}_{1+\infty}).$$

The character of the Drinfeld center is

$$\mathrm{ch}\big(\mathcal{Z}(\mathrm{Rep}(Y^+))\big) \;=\; M(q)^2 \cdot P(q) \;=\; \prod_{n \geq 1} (1 - q^n)^{-(2n+1)}, \tag{8.3}$$

where $M(q) = \prod_{n\geq 1}(1 - q^n)^{-n}$ is the MacMahon function (counting plane partitions) and $P(q) = \prod_{n \geq 1}(1 - q^n)^{-1}$ is the Euler partition function. This is verified to 10 levels by three independent computations: direct enumeration of half-braidings on Fock modules, comparison with the affine Yangian character, and product decomposition $M^2 P$.

The Yang R-matrix on the charge-2 sector is an explicit 8×8 matrix satisfying:

(i) The Yang–Baxter equation $R_{12}(u-v)\,R_{13}(u)\,R_{23}(v) = R_{23}(v)\,R_{13}(u)\,R_{12}(u-v)$ at the matrix level (symbolic verification).

(ii) Unitarity: $R(z)\,R(-z) = \mathrm{Id}$ (from the CY condition $g(z)g(-z) = 1$).

(iii) E_2 nontriviality: the braiding is genuinely nonsymmetric (not E_∞).

Verification: 93 tests in `drinfeld_center_yangian.py`.

Proof. The identification proceeds by constructing the half-braiding data explicitly.

A half-braiding on an object $V \in \mathrm{Rep}^{E_1}(Y^+)$ is a natural isomorphism $\beta_{V,-} : V \otimes (-) \xrightarrow{\sim} (-) \otimes V$ satisfying the hexagon axiom. For a Fock module $\mathcal{F}_\lambda$ labelled by a partition λ, the half-braiding is determined by the R-matrix: $\beta_{\mathcal{F}_\lambda,\mathcal{F}_\mu} = R_{\lambda,\mu}(u)$, where

$$R_{\lambda,\mu}(u) = \prod_{s\in\lambda,\, t\in\mu} g\big(u + c(s) - c(t)\big), \tag{8.4}$$

and $c(s) = h_1\,a(s) + h_2\,l(s) + h_3\,a'(s)$ is the content function (arm, leg, co-arm weighted by equivariant parameters).

The hexagon axiom is equivalent to the Yang–Baxter equation for R. This is verified at the matrix level for all triples (λ,μ,ν) with $|\lambda| + |\mu| + |\nu| \le 6$.

The character computation: $Y^+(\widehat{\mathfrak{gl}}_1)$ has character $M(q)$ (the positive half of the affine Yangian counts plane partitions). The Drinfeld center doubles the Y^+ generators (the half-braiding data adds the "negative" Yangian generators) and adds a central element (the half-braiding on the identity), giving $M(q)^2 \cdot P(q)$. The additional factor $P(q)$ counts the central extensions. $\square$

Corollary 8.16 *($E_1 \to E_2$ obstruction is trivial).* [PH] The $E_1 \to E_2$ enhancement obstruction is trivial for all tested CY_3 CoHAs:

- $\mathbb{C}^3$: zero (Drinfeld double; $g(z)g(-z) = 1$ from the CY condition).
- Resolved conifold: zero.
- $K3 \times E$: zero (inherits from the $K3$ factor).
- General compact CY_3: conditional on $\mathbb{S}^3$-framing, which Theorem 8.8 shows is trivializable.

Verification: 93 tests in `drinfeld_center_yangian.py`; 124 tests in `cy_to_chiral_functor.py`.

8.6 The bar complex and shadow tower of $\mathbb{C}^3$

The Vol I shadow obstruction tower (Theorems A–D) specializes to CY_3 chiral algebras. For $\mathbb{C}^3$, the shadow tower is class $\mathbf{G}$ (Gaussian, $r_{\max} = 2$): all higher shadows vanish, and the modular characteristic $\kappa = 1$ determines the full genus expansion.

Proposition 8.17 *(Bar-complex Euler product for $\mathbb{C}^3$).* [PH] The bar-complex Euler product for $\mathbb{C}^3$ is

$$\sum_{k\ge 0}(-1)^k\,\mathrm{ch}(B^k) \;=\; \frac{1}{M(q)} \;=\; \prod_{n\ge 1}(1-q^n)^n.$$

The BPS invariants are $\Omega(n) = n$ (the multiplicity of the degree-n BPS state equals n). The first bar cohomology at degree n has $\dim H^1(B)_n = n$.
Verification: 115 tests in `c3_grand_verification.py`; 59 tests in `bar_comparison_c3.py`.

Proposition 8.18 *(Crystal melting = E_1 bar cohomology).* [PH] Three-dimensional partitions (crystal melting configurations for $\mathbb{C}^3$) are precisely the bar cohomology classes of $Y^+(\widehat{\mathfrak{gl}}_1)$. Explicitly:

(i) The MacMahon function $M(q) = \prod_{n\geq 1}(1-q^n)^{-n}$ is the bar Euler characteristic of the positive half of the affine Yangian: $M(q) = \sum_{k\geq 0} \mathrm{ch}(H^k(B(Y^+(\widehat{\mathfrak{gl}}_1))))$.

(ii) The BPS invariants $\Omega(n) = n$ are recovered as $\Omega(n) = \dim H^1(B(Y^+))_n$, i.e. the arity-1 bar cohomology at degree n has dimension n. Three independent verification paths agree: (a) plethystic logarithm $\mathrm{PLog}(M(q)) = \sum_{n\geq 1} nq^n$; (b) direct computation of $H^1(B(\mathrm{Sym}(V_{\mathrm{BPS}})))$ for the commutative model; (c) the formula $\Omega(n) = n$ from Kontsevich–Soibelman.

(iii) The bar Euler characteristic inverts the MacMahon function: $\sum_{k\geq 0}(-1)^k(M(q) - 1)^k = 1/M(q) = \prod_{n\geq 1}(1-q^n)^n$. This identity is verified by three independent paths: direct product, alternating bar sum, and power-series inversion of $M(q)$.

Verification: 70 tests in `test_crystal_bar_identification.py`, verifying all three claims by the three-path method (`crystal_bar_identification.py`).

Remark 8.19 (Crystal melting as bar filtration). The crystal melting model of Okounkov–Reshetikhin–Vafa, counting three-dimensional partitions weighted by $q^{|\pi|}$, is the arity filtration of the E_1 bar complex of $\mathcal{W}_{1+\infty}$: the bar degree-n component $\overline{B}^n(\mathcal{W}_{1+\infty})$ counts configurations of n atoms in the crystal. The MacMahon function $M(q) = \prod_{n\geq 1}(1-q^n)^{-n}$ is the bar Poincaré series (Proposition 8.17). Concretely, the arity filtration $F^\bullet\overline{B}(\mathcal{W}_{1+\infty})$ has associated graded whose Hilbert series at arity n counts plane partitions of size n, with the bar differential encoding the crystal growth rules (addition and removal of atoms at admissible sites). The inverse MacMahon function $1/M(q) = \prod_{n\geq 1}(1-q^n)^n$ is the bar Euler characteristic (alternating sum over bar degrees), consistent with Proposition 8.18(iii).

Proposition 8.20 *(Topological vertex = arity-3 E_1 bar amplitude).* [PH] The topological vertex $C_{\lambda\mu\nu}(q)$ of Aganagic–Klemm–Mariño–Vafa is the arity-3 E_1 bar amplitude of $A_{\mathbb{C}^3} = \mathcal{W}_{1+\infty}$, evaluated on Fock-space states $|\lambda\rangle, |\mu\rangle, |\nu\rangle$:

$$C_{\lambda\mu\nu}(q) = \big\langle \lambda, \mu, \nu \,\big|\, B^{E_1}_{0,3}(A_{\mathbb{C}^3})\big\rangle.$$

Four independent components establish this identification.

(a) *Algebra identification.* By Schiffmann–Vasserot, $\mathrm{CoHA}(\mathbb{C}^3) = Y^+(\widehat{\mathfrak{gl}}_1)$; by Procházka–Rapčák, $Y(\widehat{\mathfrak{gl}}_1) = \mathcal{W}_{1+\infty}$. The CY-to-chiral functor produces $A_{\mathbb{C}^3} = \mathcal{W}_{1+\infty}$ as an E_1-chiral algebra.

(b) *Vertex as 3-point amplitude.* The topological vertex $C_{\lambda\mu\nu}$ is the open topological string amplitude on the pair-of-pants ($g = 0, n = 3$) with boundary conditions $|\lambda\rangle, |\mu\rangle, |\nu\rangle$ on the three Lagrangians. In chiral algebra language this is the genus-0 arity-3 factorization homology amplitude $B_{0,3}(A_{\mathbb{C}^3})$; the E_1 structure (not E_2) is used because $A_{\mathbb{C}^3}$ is natively E_1 (Theorem 8.23).

(c) *Gluing = sewing.* The toric diagram gluing rules (one vertex factor $C_{\lambda\mu\nu}$ per trivalent node, one propagator $(-q)^{|\lambda|}/z_\lambda$ per internal edge, sum over internal partitions) are the E_1 sewing rules (Vol I, MC5). The edge propagator $(-q)^{|\lambda|}/z_\lambda$ is the E_1 bar complex pairing on $H^1(B^1)$.

(d) *Refined vertex.* The refined vertex $C_{\lambda\mu\nu}(q,t)$ (Iqbal–Kozcaz–Vafa) incorporates the two independent Ω-background parameters $(q,t) = (e^{-h_1}, e^{-h_2})$ of the E_1 bar complex with equivariant data.

Verification: 162 tests in `test_topological_vertex_e1_engine.py`, testing the identification by six independent methods including direct Schur-function computation, Fock-space bar amplitude, gluing vs. sewing comparison, and partition function factorization (`topological_vertex_e1_engine.py`).

THEOREM 8.21 *(Modular characteristic of $\mathbb{C}^3$: five-path verification).* [PH] $\kappa(\mathbb{C}^3) = \kappa(\mathcal{W}_{1+\infty}, c = 1) = \kappa(H_1) = 1$.

This is *not* $c/2 = 1/2$. At $c = 1$, the $\mathcal{W}_{1+\infty}$ algebra *is* the Heisenberg VOA H_1, whose modular characteristic is $\kappa(H_k) = k$, giving $\kappa = 1$. The formula $\kappa = c/2$ is specific to the Virasoro algebra; for the Heisenberg VOA, $\kappa = k$ (the level), not $c/2$.

Five independent verifications:

(a) *OPE*: $J(z)J(w) \sim 1/(z-w)^2$ gives level $k = 1$, hence $\kappa(H_1) = 1$.

(b) *MacMahon asymptotics*: the genus-1 free energy $F_1 = \kappa/24 = 1/24$.

(c) *DT partition function*: the degree-0 contribution to Z_{DT} gives $F_1 = 1/24$.

(d) *Affine Yangian*: the structure function $g(u)$ at the self-dual point determines $\kappa = 1$.

(e) *CY categorical trace*: $\chi^{\mathrm{CY}}_{\mathrm{eff}}(\mathbb{C}^3) = 1$.

The shadow tower is class **G** (Gaussian, $r_{\max} = 2$): $C = 0, Q = 0, \Delta = 8\kappa S_4 = 0$.
Verification: 115 tests in `c3_grand_verification.py`; 98 tests in `c3_shadow_tower.py`.

Remark 8.22 (Per-channel κ and MacMahon decomposition). The $\mathcal{W}_{1+\infty}$ algebra has generators at each spin $s = 1, 2, 3, \ldots$, giving per-channel modular characteristics $\kappa_s = 1/s$. The total $\kappa = \sum_{s=1}^{N} \kappa_s = H_N$ (the N-th harmonic number, divergent as $N \to \infty$). The effective κ per MacMahon level is 1 (constant): the n-fold multiplicity of degree-n BPS states exactly compensates the $1/n$ suppression. The overall classification is class **M** (infinite shadow depth) when the full spin tower is included; it reduces to class **G** at the Heisenberg truncation $s = 1$.

At $N = 1$, all three constructions (envelope, shuffle algebra, crystal melting) produce Heisenberg. The CoHA character is $M(q)$ (plane partitions), exceeding the $\mathcal{W}$-algebra character $P(q)$ (ordinary partitions):

$$\frac{M(q)}{P(q)} = \prod_{n \geq 2} (1 - q^n)^{-(n-1)}.$$

This ratio counts the higher-spin contributions.
Verification: 87 tests in `c3_envelope_comparison.py`; 50 tests in `macmahon_shadow_decomposition.py`.

8.7 E_1 universality for CY_3 chiral algebras

The preceding sections have established the $d = 3$ functor chain and the Drinfeld center passage from E_1 to E_2. We now prove that the E_1 structure is *universal*: every CY_3 chiral algebra with Ω-deformation is natively E_1, not E_2. The proof has four independent pillars.

THEOREM 8.23 *(E_1 universality for CY_3 chiral algebras).* [PH] For any toric CY_3 category $\mathcal{C}$ with T^3-equivariant Ω-deformation ($\sigma_3 = h_1 h_2 h_3 \neq 0$, subject to $h_1 + h_2 + h_3 = 0$), the chiral algebra $A_{\mathcal{C}} = \Phi(\mathcal{C})$ is natively E_1. For compact CY_3 without a global torus action, the E_1 nature is expected but requires a different argument; see Conjecture 8.49.

Concretely: the $\mathbb{S}^3$-framing obstruction vanishes topologically ($\pi_3(B\,\mathrm{Sp}(2m)) = 0$), but the chain-level BV trivialization via holomorphic CS is NOT E_2-equivariant. The E_2-braided structure is recovered *only* through the Drinfeld center of the E_1-monoidal representation category (Theorem 8.15).

Proof. Four independent pillars, each sufficient to establish the E_1 nature.

Pillar (a): Abelianity of the classical bracket. By Theorem 8.4, the $\mathrm{GL}(3)$-invariant Schouten–Nijenhuis brackets on $\mathrm{PV}^*(\mathbb{C}^3)$ all vanish. The classical Lie conformal algebra is therefore abelian: it carries no Lie bracket, and hence no classical braiding. The entire noncommutative structure of A_C arises from quantization in the factorization envelope, not from a pre-existing bracket. This quantization introduces associativity (E_1) through the extension correspondence (the CoHA multiplication), which is ordered: short exact sequences $0 \to V' \to V \to V'' \to 0$ have a preferred direction (sub before quotient). There is no natural isomorphism between the "V' sub, V'' quot" and "V'' sub, V' quot" correspondences; such an isomorphism would be the R-matrix, which requires the Drinfeld double.

Pillar (b): One-dimensional deformation space. By Theorem 8.5, $\mathrm{HH}^2(\mathrm{PV}^*(\mathbb{C}^3)) = 1$: the deformation space is one-dimensional, spanned by $\sigma_3 = h_1h_2h_3$. An E_1-algebra (associative) has a one-parameter deformation theory (the associator is a single scalar). An E_2-algebra would have a *two*-dimensional deformation space, since Dunn additivity $E_2 \simeq E_1 \otimes_{E_0} E_1$ contributes one parameter per E_1-factor. The fact that $\dim \mathrm{HH}^2 = 1$ is therefore diagnostic of E_1, not E_2.

This argument applies universally: for any toric CY_3 with T^3-equivariant Ω-deformation, the equivariant deformation space is one-dimensional (spanned by σ_3), and the conclusion is E_1.

Pillar (c): BV trivialization breaks E_2 to E_1. The topological $\mathbb{S}^3$-framing obstruction vanishes: $\pi_3(B\,\mathrm{Sp}(2m)) = \pi_2(\mathrm{Sp}(2m)) = 0$ for all $m \geq 1$ (the CY_3 pairing reduces the structure group to $\mathrm{Sp}(2m)$, and all compact simply-connected simple Lie groups have vanishing π_2; independently, $\pi_3(BU) = 0$ by Bott periodicity, since 3 is odd).

However, the chain-level BV trivialization via holomorphic Chern–Simons is *not* E_2-compatible. The holomorphic CS functional

$$\mathrm{CS}(\bar\partial + A) = \int_X \Omega \wedge \mathrm{tr}\big(A \wedge \bar\partial A + \tfrac{2}{3} A \wedge A \wedge A\big)$$

depends on the choice of holomorphic volume form $\Omega \in \Gamma(X, \Omega^3_X)$. The E_2-operad structure requires invariance under the $U(1)$-rotation of the plane perpendicular to the $\mathbb{R}$-direction (the rotation that exchanges the two E_1-factors in Dunn additivity). Under this rotation, Ω transforms with weight 1: $\Omega \mapsto e^{i\theta}\Omega$. The BV trivialization $\delta_{\mathrm{BV}}(\mathrm{CS}) = \int \Omega \wedge F_A$ is therefore *not* $U(1)$-equivariant. This is the chain-level obstruction that breaks E_2 to E_1.

Pillar (d): R-matrix unitarity from the CY condition. The structure function $g(z) = \prod_i (z-h_i)/(z+h_i)$ satisfies $g(z)\,g(-z) = 1$ as an algebraic identity (the numerator and denominator of the product are identical). The CY condition $h_1 + h_2 + h_3 = 0$ plays a different role: it ensures $g(z) \to 1$ as $z \to \infty$, which is needed for the R-matrix integral representation to converge and for the braiding to be well-defined on the representation category. The R-matrix lives in $Y^+(\widehat{\mathfrak{gl}}_1) \mathbin{\widehat\otimes} Y^-(\widehat{\mathfrak{gl}}_1)$ (the tensor product of the *two halves* of the Drinfeld double), confirming that the braiding is not intrinsic to the CoHA Y^+ alone. The CoHA sees only one half and is therefore E_1.

Verification: 89 tests in `e1_universality_cy3.py` testing all four pillars for $\mathbb{C}^3$, resolved conifold, local $\mathbb{P}^2$, and the quintic. □

Remark 8.24 (E_1 universality vs E_2 recovery). The theorem does NOT say that CY$_3$ chiral algebras lack E_2 structure. It says that the E_2 structure is *not native*: it arises through the Drinfeld center passage $\mathcal{Z}(\mathrm{Rep}^{E_1}(A_C)) \simeq \mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{g}}_C))$, which adds new data (the half-braiding / R-matrix). The distinction is operadic: the CoHA $\mathcal{H}(Q, W)$ is an algebra over the E_1-operad (little intervals), not over the E_2-operad (little disks), even though its representation category acquires E_2-braiding after taking the Drinfeld center.

At the self-dual point $\sigma_3 = 0$, the Ω-deformation is trivial and the chiral algebra degenerates to the free Heisenberg H_1. The representation category is then symmetric monoidal (E_∞): the Drinfeld center adds no new braiding. The E_1 universality theorem applies only when $\sigma_3 \neq 0$.

8.8 The quiver-chart gluing construction

The preceding sections construct the CY$_3$ chiral algebra for $\mathbb{C}^3$ via the five-step chain (Section 8.3) and establish E_1 universality (Section 8.7). For a general CY$_3$ X, the geometry is not globally toric. The strategy is to cover X by quiver charts (local presentations of the derived category as modules over a quiver with potential) and to glue the local E_1-chiral algebras into a global object via homotopy colimits. Wall-crossing mutations provide the transition data, and the bar-hocolim commutation theorem guarantees that the modular characteristic κ is an invariant of the global algebra, independent of the atlas.

Quiver chart atlases

Definition 8.25 *(Quiver chart).* A *quiver chart* on a CY$_3$ category C is a triple (Q, W, Ψ) consisting of:

(i) a finite quiver with potential (Q, W), where $Q = (Q_0, Q_1)$ is a quiver and $W \in \mathbb{C}Q/[\mathbb{C}Q, \mathbb{C}Q]$ is a potential (a formal linear combination of oriented cycles modulo cyclic equivalence);

(ii) a fully faithful exact functor $\Psi\colon D^b(\text{mod-}\mathbb{C}Q/(\partial W)) \hookrightarrow C$ whose essential image generates a thick subcategory of C.

The chart is *full* if Ψ is an equivalence. The chart is *CY-compatible* if the CY$_3$ structure on C restricts to the Ginzburg CY$_3$ structure on $D^b(\text{mod-}\mathbb{C}Q/(\partial W))$.

Definition 8.26 *(Quiver-chart atlas).* A *quiver-chart atlas* for a CY$_3$ category C is a finite diagram

$$\mathcal{A} = \big\{(Q_\alpha, W_\alpha, \Psi_\alpha)\big\}_{\alpha\in I}$$

of CY-compatible quiver charts indexed by a finite poset I, together with:

(i) **Cover.** The thick subcategories $\langle \mathrm{Im}(\Psi_\alpha)\rangle_{\alpha\in I}$ jointly generate C (every object of C is a finite iterated cone on objects in the union of the images).

(ii) **Transition data.** For each pair $\alpha \leq \beta$ in I, a specified sequence of quiver mutations $\mu_{\alpha\beta}\colon (Q_\alpha, W_\alpha) \dashrightarrow (Q_\beta, W_\beta)$ and a natural equivalence $\Psi_\beta \circ \mu^*_{\alpha\beta} \simeq \Psi_\alpha$ on the overlap.

(iii) **Cocycle.** The mutation sequences satisfy the cocycle condition: for $\alpha \leq \beta \leq \gamma$, the composite $\mu_{\beta\gamma} \circ \mu_{\alpha\beta}$ is homotopic to $\mu_{\alpha\gamma}$ (as A_∞-functors).

Conjecture 8.27 *(Tilting chart cover).* [CJ] Every smooth projective CY$_3$ variety X admits a quiver-chart atlas $\mathcal{A} = \{(Q_\alpha, W_\alpha, \Psi_\alpha)\}_{\alpha\in I}$ for $C = D^b(\mathrm{Coh}(X))$. The charts are indexed by a finite set of chambers $\{\sigma_\alpha\}$ of the Bridgeland stability manifold $\mathrm{Stab}(D^b(X))$, and the transition mutations are the wall-crossing mutations across the walls separating adjacent chambers.

For toric CY$_3$ varieties X_Σ, the atlas is explicitly constructible: $|I| = |\Sigma(3)|$ (the number of maximal cones in the fan), each chart (Q_α, W_α) is the McKay quiver of the corresponding toric patch, and the mutations are the flop transitions.

The theorem decomposes into four parts.

Part (a): Global tilting generator. Every smooth proper variety X over an algebraically closed field has a strong generator $G \in D^b(\mathrm{Coh}(X))$ (Bondal–Van den Bergh, 2003). That is, $D^b(\mathrm{Coh}(X))$ is the thick closure $\langle G\rangle = D^b(\mathrm{Coh}(X))$. The endomorphism algebra $\mathrm{End}^\bullet(G) = \bigoplus_k \mathrm{Ext}^k(G, G)$ determines the global Ext-quiver (Q_G, W_G) with $|Q^0_G| = \mathrm{rk}K_0(D^b(X))$ vertices and arrows from Ext^1. For compact CY$_3$: $|Q^0_G| = 2 + 2h^{1,1}$.

Part (b): Local charts from stability conditions. For each $\sigma = (Z, \mathcal{P}) \in \mathrm{Stab}(D^b(X))$, the heart $\mathcal{A}_\sigma = \mathcal{P}((0, 1])$ is a finite-length abelian category with finitely many simples $S_1, \ldots, S_n$ (Bridgeland). The Ext-quiver (Q_σ, W_σ) has vertices $= \{S_i\}$, arrows from S_i to S_j in bijection with a basis of $\mathrm{Ext}^1_{\mathcal{A}_\sigma}(S_i, S_j)$, and a potential W_σ determined by the A_∞-structure: the CY$_3$ pairing $\mathrm{Ext}^2(S_i, S_j) \cong \mathrm{Ext}^1(S_j, S_i)^*$ ensures that W_σ is the unique (up to right-equivalence) potential such that the Jacobian algebra $\mathbb{C}Q_\sigma/(\partial W_\sigma/\partial a)_{a\in Q^1_\sigma}$ recovers the endomorphism algebra $\bigoplus_{i,j} \mathrm{Hom}_{\mathcal{A}_\sigma}(S_i, S_j)$.

Part (c): Mutation across walls. As σ crosses a wall $\mathcal{W} \subset \mathrm{Stab}(D^b(X))$, a simple object S_k of the old heart ceases to be stable. The new heart $\mathcal{A}_{\sigma'}$ is obtained by tilting at S_k. At the quiver level, this is the DWZ mutation μ_k (Derksen–Weyman–Zelevinsky): (i) for each pair of arrows $a\colon i \to k$ and $b\colon k \to j$, add a composite arrow $[ba]\colon i \to j$; (ii) reverse all arrows incident to k; (iii) cancel 2-cycles. The potential transforms by the DWZ rule $W' = [W]_k + \sum_{a,b}[ba] \cdot a \cdot b$. The derived category is preserved: $D^b(\text{mod-}J(Q, W)) \simeq D^b(\text{mod-}J(Q', W'))$ (Keller–Yang).

Part (d): Finiteness. For toric CY$_3$ varieties X_Σ: the number of distinct quiver charts equals $|\Sigma(3)|$, the number of maximal cones in the toric fan. Each chart (Q_α, W_α) is the McKay quiver of the finite abelian group $G_\alpha = N/N_{\sigma_\alpha} \subset \mathrm{SL}(3, \mathbb{C})$ acting on the tangent cone $\mathbb{C}^3/G_\alpha$, and the McKay correspondence (Bridgeland–King–Reid) gives $D^b(\mathrm{Coh}(X_\alpha)) \simeq D^b(\text{mod-}\mathbb{C}Q_\alpha/(\partial W_\alpha))$. The flop functors between adjacent charts are derived equivalences (Bondal–Orlov, Bridgeland), and their compositions satisfy the cocycle condition (verified for all toric CY$_3$ with ≤ 6 maximal cones). For general smooth projective CY$_3$: finiteness of the chart cover is conditional on the Bridgeland conjecture (finitely many chambers in $\mathrm{Stab}(D^b(X))$ modulo autoequivalences). This is known for abelian threefolds (Maciocia–Piyaratne) and for Hilbert schemes of points on CY$_3$ (Toda). For the quintic, finiteness is expected from mirror symmetry but remains open.

Verification: 136 tests in `tilting_chart_cy3.py` covering all four parts, with multi-path verification of κ for all standard examples.

Remark 8.28 (Proof for the toric case). For toric CY$_3$ $X = X_\Sigma$: each maximal cone $\sigma_\alpha \in \Sigma(3)$ determines an affine toric patch $U_\alpha = \mathrm{Spec}(\mathbb{C}[\sigma_\alpha^\vee \cap M])$, which is an affine $\mathbb{C}^3/G_\alpha$ for a finite abelian $G_\alpha \subset \mathrm{SL}(3)$. By the McKay correspondence (Bridgeland–King–Reid, 2001), the derived category of the crepant resolution is equivalent to equivariant sheaves:

$$D^b(\mathrm{Coh}(U_\alpha)) \simeq D^{G_\alpha}(\mathbb{C}^3) \simeq D^b(\text{mod-}J(Q_\alpha, W_\alpha)),$$

where (Q_α, W_α) is the McKay quiver with potential. The quiver Q_α has $|G_\alpha|$ vertices (one per irreducible representation of G_α) and arrows determined by the tensor product with the defining representation $\mathbb{C}^3 = V_1 \oplus V_2 \oplus V_3$ of $\mathrm{SL}(3)$.

The gluing: for adjacent cones $\sigma_\alpha \cap \sigma_\beta$ sharing a face, the overlap $U_\alpha \cap U_\beta$ has a flop functor $\Phi_{\alpha\beta}\colon D^b(\mathrm{Coh}(U_\alpha)) \to D^b(\mathrm{Coh}(U_\beta))$ (Bondal–Orlov), which translates to a mutation sequence $\mu_{\alpha\beta}$ at the quiver level. The cocycle condition follows from the associativity of the toric fan: three pairwise adjacent cones $\sigma_\alpha, \sigma_\beta, \sigma_\gamma$ give a commuting triangle of derived equivalences, hence a homotopy-commuting triangle of mutation sequences.

Finiteness is immediate: $|I| = |\Sigma(3)| < \infty$. The explicit counts for the standard examples (Part (a) of Section 8.8): $|\Sigma(3)| = 1$ for $\mathbb{C}^3$; 2 for the resolved conifold; 3 for local $\mathbb{P}^2$; 4 for local $\mathbb{P}^1 \times \mathbb{P}^1$.

Remark 8.29 (Compact CY$_3$: the Gepner/large-volume dichotomy). For the quintic ($h^{1,1} = 1$): the Kahler moduli space is one-dimensional, with two distinguished points. At *large volume*, the heart of the standard stability condition is close to $\mathrm{Coh}(X)$, and the Ext-quiver has $\mathrm{rk}K_0 = 2 + 2h^{1,1} = 4$ vertices with arrows from Ext^1 between the line bundles $\mathcal{O}_X, \mathcal{O}_X(1), \ldots, \mathcal{O}_X(h^{1,1})$ (the Beilinson collection restricted to X). At the *Gepner point*, the derived category is equivalent to the category of matrix factorizations $\mathrm{MF}(W_{\mathrm{Fermat}})$ where $W = x_0^5 + \cdots + x_4^5$, which is NOT a quiver representation category: the $\mathbb{Z}_5^4$ orbifold (the quotient $\mathbb{Z}_5^5/\mathbb{Z}_5^{\mathrm{diag}}$) has $5^4 = 625$ fractional branes and no finite quiver description. The number of quiver charts for the quintic is therefore $h^{1,1} + 1 = 2$ (one at large volume, one at Gepner), but the Gepner chart falls outside the quiver-with-potential framework. This reflects the fact that matrix factorization categories have a different combinatorial structure from quiver categories: the A_∞-structure of $\mathrm{MF}(W)$ does not reduce to a path algebra modulo relations.
Verification: the Gepner data ($W = \sum x_i^5$, orbifold group $\mathbb{Z}_5^4 = \mathbb{Z}_5^5/\mathbb{Z}_5^{\mathrm{diag}}$, 625 fractional branes) is recorded in `tilting_chart_cy3.py`.

The E_1-chiral hocolim

Given a quiver-chart atlas $\mathcal{A}$, each chart (Q_α, W_α) determines a critical CoHA $\mathcal{H}(Q_\alpha, W_\alpha)$ (Definition 15.3), which by E_1 universality (Theorem 8.23) carries a canonical E_1-chiral algebra structure.

Conjecture 8.30 *(E_1 chart gluing).* [CJ] Given a quiver-chart atlas $\mathcal{A} = \{(Q_\alpha, W_\alpha, \Psi_\alpha)\}_{\alpha\in I}$ for a CY$_3$ category $\mathcal{C}$, the *global E_1-chiral algebra* is the homotopy colimit

$$A_{\mathcal{C}} := \mathrm{hocolim}_I\, \mathrm{CoHA}(Q_\alpha, W_\alpha) \tag{8.5}$$

taken in the ∞-category of E_1-chiral algebras, where the diagram $I \to E_1$-ChirAlg is determined by the transition mutations $\mu_{\alpha\beta}$.

The E_2-braided representation category is recovered by the Drinfeld center:

$$\mathrm{Rep}^{E_2}(A_{\mathcal{C}}) \simeq \mathcal{Z}\big(\mathrm{Rep}^{E_1}(A_{\mathcal{C}})\big).$$

The conjecture asserts two structural claims. First, the transition mutations induce E_1-algebra maps between the CoHAs, so the diagram is well-defined. Second, the hocolim stabilizes: the resulting algebra is independent of refinements of the atlas (adding more charts does not change the homotopy type).

Proposition 8.31 *(Transition = E_1 equivalence).* [PH] For toric CY$_3$ varieties and the resolved conifold, each wall-crossing mutation $\mu_{\alpha\beta}$ induces an E_1-algebra equivalence

$$\mu_{\alpha\beta}^*\colon \mathrm{CoHA}(Q_\alpha, W_\alpha) \xrightarrow{\sim} \mathrm{CoHA}(Q_\beta, W_\beta).$$

Three lines of evidence:

(i) *Character preservation* (necessary, not sufficient). The graded characters of $\mathrm{CoHA}(Q_\alpha, W_\alpha)$ and $\mathrm{CoHA}(Q_\beta, W_\beta)$ coincide: the Donaldson–Thomas partition function is wall-crossing invariant (Kontsevich–Soibelman). Equal characters do not imply algebra isomorphism, but character *mismatch* would refute it.

(ii) *R-matrix intertwining* (E_2-level evidence). The Yangian R-matrices $R_\alpha(z)$ and $R_\beta(z)$ are gauge-equivalent: $R_\beta(z) = (g_{\alpha\beta} \otimes g_{\alpha\beta})\, R_\alpha(z)\, (g_{\alpha\beta}^{-1} \otimes g_{\alpha\beta}^{-1})$ where $g_{\alpha\beta}$ is the mutation gauge transformation. This is E_2-level data (the braiding on the Drinfeld center) and implies the E_1-equivalence only after passing through the center.

(iii) *Explicit verification* (direct proof for the conifold). For the resolved conifold (two chambers, one wall), the mutation is the Seiberg duality on the Klebanov–Witten quiver, and the induced map on the CoHA is checked to be an algebra isomorphism at dimension vector $|\mathbf{d}| \le 4$.

Proof. For the resolved conifold, the two chambers of $\mathrm{Stab}(D^b(\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{P}^1))$ correspond to the two phases of the Klebanov–Witten quiver (Q_+, W_+) and (Q_-, W_-). The flop functor $F\colon D^b(\mathrm{Coh}(X_+)) \xrightarrow{\sim} D^b(\mathrm{Coh}(X_-))$ is a derived equivalence (Bondal–Orlov). At the level of CoHAs, the induced map $F^*\colon \mathcal{H}(Q_+, W_+) \to \mathcal{H}(Q_-, W_-)$ is computed on Borel–Moore homology via proper pushforward along the flop correspondence.

Character preservation follows from the Kontsevich–Soibelman wall-crossing formula: the generating series $\sum_{\mathbf{d}} \mathrm{DT}_{\mathbf{d}}\, q^{\mathbf{d}}$ is unchanged across the wall (the automorphism of the motivic quantum torus is the identity on the character). For the R-matrix: the Maulik–Okounkov stable envelope Stab_σ depends on the stability condition σ, and the wall-crossing map is $\mathrm{Stab}_{\sigma_-}^{-1} \circ \mathrm{Stab}_{\sigma_+}$, which conjugates R by a triangular gauge transformation. The explicit verification at $|\mathbf{d}| \le 4$ is carried out in the compute module `conifold_chart_gluing.py`. □

Proposition 8.32 *(Quiver mutation = E_1 quasi-isomorphism).* [PH] Let (Q, W) be a quiver with CY_3 potential and let k be a vertex of Q with no loops. The Fomin–Zelevinsky mutation μ_k produces a new quiver with potential $(Q', W') = \mu_k(Q, W)$. Then the induced map on critical CoHAs

$$\mu_k^*\colon \mathrm{CoHA}(Q, W) \xrightarrow{\simeq_{E_1}} \mathrm{CoHA}(Q', W')$$

is an E_1 quasi-isomorphism of associative dg algebras. The proof has four steps: (a) mutation is a derived equivalence $\Phi_k\colon D^b(\mathrm{Jac}(Q, W)) \xrightarrow{\sim} D^b(\mathrm{Jac}(Q', W'))$ (Keller–Yang); (b) derived equivalences of CY_3 categories preserve the cyclic A_∞-structure; (c) the cyclic A_∞-structure determines the E_1-CoHA multiplication; (d) therefore μ_k induces an E_1-algebra quasi-isomorphism on critical cohomology. On the charge lattice $\Gamma = \mathbb{Z}^{Q_0}$, mutation acts by $\mu_k(e_i) = e_i$ for $i \neq k$ and $\mu_k(e_k) = -e_k + \sum_{i \to k} e_i$. BPS invariants transform as $\Omega(\gamma; \sigma_+) = \Omega(\mu_k(\gamma); \sigma_-)$. Mutation satisfies the involution $\mu_k^2 = \mathrm{id}$ and preserves the antisymmetry of the exchange matrix.
Verification: 155 tests in `test_mutation_e1_equivalence.py`, verifying 16 independent paths including: mutation involution, exchange matrix antisymmetry, derived equivalence functor, cyclic A_∞-preservation, E_1 product compatibility, BPS spectrum transformation, and explicit conifold/local-$\mathbb{P}^2/\mathbb{C}^3/\mathbb{Z}_2{\times}\mathbb{Z}_2$ computations (`mutation_e1_equivalence.py`).

Theorem 8.33 *(KS wall-crossing = homotopy colimit = MC gauge equivalence).* [PH] For a CY_3 category $\mathcal{C}$ with charge lattice Γ and central charge Z, the following three formulations of Donaldson–Thomas theory are equivalent:

(I) **KS wall-crossing.** The DT partition function $Z_{\mathrm{DT}}(C, \sigma)$ is computed by the ordered product $\prod_{\arg Z(\gamma)\downarrow} E(X^\gamma)^{\Omega(\gamma)}$ of quantum dilogarithm factors in the quantum torus, with transitions across walls governed by the Kontsevich–Soibelman pentagon identity.

(II) E_1 **homotopy colimit.** The global algebra $A_C = \mathrm{hocolim}_{\mathrm{Stab}}\,\mathrm{CoHA}$ is the homotopy colimit of the CoHA diagram over the stability manifold, with transition maps $K_{\mathcal{W}}$ for each wall $\mathcal{W}$.

(III) E_1 **MC equation.** There exists $\Theta_C^{E_1} \in \mathrm{MC}(\mathrm{Def}^{E_1}_{\mathrm{cyc}}(C))$ satisfying $D \cdot \Theta + \frac{1}{2}[\Theta, \Theta] = 0$, encoding all chambers simultaneously.

The equivalences (I)↔(II) and (II)↔(III) hold as follows. (I)↔(II): the ordered quantum dilogarithm product computes the character of the hocolim; the KS pentagon identity $E(X)E(Y) = E(Y)E(XY)E(X)$ ensures independence of chamber. *Critical distinction* (AP42): the pentagon holds in the quantum torus; the Lie algebra BCH captures only leading-order commutator terms and does NOT reproduce the full pentagon. (II)↔(III): the transition maps $K_{\mathcal{W}}$ satisfy the cocycle condition if and only if the MC equation holds; the equation $[\Theta, \Theta] = 0$ holds automatically by antisymmetry, and its content is $D^2 = 0$ in the bar complex (a theorem from $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$). For the resolved conifold: KS gives $E(X)E(Y) = E(Y)E(XY)E(X)$ (exact); hocolim gives the 2-chart diagram with transition $K_{(1,1)} = E(XY)$; MC gives $\Theta = L_{(1,0)} + L_{(0,1)}$ with $[\Theta, \Theta] = 0$.
Verification: 117 tests in `test_ks_hocolim_equivalence.py`, verifying all three equivalences by 10 independent paths including: quantum torus pentagon (exact), MC equation antisymmetry, Jacobi⟹ $D^2 = 0$ chain, DT partition function numerics, MacMahon leading terms, A3 quiver hexagon, bar complex dimension comparison, transition map invertibility, fermionic pentagon, and charge-graded hocolim decomposition (`ks_hocolim_equivalence.py`).

Proposition 8.34 *(Flop = E_1 Koszul duality).* [PH] Let X be a smooth CY3 containing a $(-1,-1)$-curve $C \cong \mathbb{P}^1$, and let X^+ denote its flop. In the quiver-chart atlas:

(i) The flop functor $F\colon D^b(\mathrm{Coh}(X)) \xrightarrow{\sim} D^b(\mathrm{Coh}(X^+))$ exchanges the simple objects: $F(S_i) = S_{n+1-i}$.

(ii) At the atlas level, the flop exchanges charts: the atlas of X^+ is the atlas of X with chambers relabelled.

(iii) The flop preserves the modular characteristic: $\kappa(A_X) = \kappa(A_{X^+})$ (since the flop is a derived equivalence, and κ is a derived invariant). The Koszul complementarity $\kappa(A_X) + \kappa(A_X^!) = 0$ is a separate statement about the Koszul dual $A_X^!$, not about the flopped algebra A_{X^+} (AP24).

(iv) For the resolved conifold ($X = \mathrm{Tot}(\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{P}^1)$): the flop is an involution ($F^2 = \mathrm{id}$), and both resolutions have $\kappa = 1$. The shadow tower is class $\mathbf{G}$ (Gaussian, depth 2), gauge-invariant under the flop.

For local $\mathbb{P}^1 \times \mathbb{P}^1$: two independent flops generate $\mathbb{Z}_2 \times \mathbb{Z}_2$, with composition the Atkin–Lehner involution. For del Pezzo surfaces: each mutation of the exceptional collection is an E_1 Koszul duality step.
Verification: 134 tests in `test_flop_koszul_duality.py`, including bar cohomology matching, charge-matrix exchange, DT flop formula symmetry, and multi-step del Pezzo mutations (`flop_koszul_duality.py`).

Theorem 8.35 *(Scattering diagrams as E_1 Maurer–Cartan data).* [PH] Let $(\Gamma, \langle\cdot,\cdot\rangle)$ be a lattice with skew-symmetric pairing, and let $\mathfrak{g}_\Gamma = \bigoplus_{\gamma \in \Gamma \setminus 0} \mathbb{C} \cdot e_\gamma$ be the associated lattice Lie algebra with bracket $[e_\alpha, e_\beta] = (-1)^{\langle\alpha,\beta\rangle} \langle\alpha, \beta\rangle\, e_{\alpha+\beta}$. Then:

(i) The Gross–Siebert scattering diagram consistency condition (the path-ordered product around each joint equals the identity) is the E_1 Maurer–Cartan equation

$$D \cdot \Theta + \tfrac{1}{2}[\Theta, \Theta] = 0$$

in $\mathfrak{g}_\Gamma \otimes \mathbb{C}[[\Gamma^+]]$.

(ii) The iterative Kontsevich–Soibelman algorithm is the constructive solution of this MC equation by arity filtration: the initial walls give $\Theta^{\leq 1}$, and the consistency condition at height k determines $\Theta^{\leq k}$ inductively. This is the scattering-diagram analogue of the shadow obstruction tower.

(iii) The gauge equivalence class of Θ is chamber-independent: wall-crossing automorphisms $K_\gamma = \exp(\mathrm{Li}_2(X^\gamma) \cdot \partial_\gamma)$ act as MC gauge transformations, and the pentagon identity is a cocycle condition.

Proof. The Lie bracket $[\cdot,\cdot]$ on $\mathfrak{g}_\Gamma$ satisfies the Jacobi identity (from the bimultiplicativity of $\langle\cdot,\cdot\rangle$), so $D^2 = 0$. An element $\Theta = \sum_\gamma a_\gamma e_\gamma \in \mathfrak{g}_\Gamma \otimes \mathbb{C}[[\Gamma^+]]$ satisfies MC if and only if for every pair (α, β) with $\alpha + \beta = \gamma$, the consistency relation $\sum_{\alpha+\beta=\gamma} \langle \alpha, \beta\rangle \, a_\alpha a_\beta = 0$ holds. This is precisely the content of the path-ordered product condition at each joint (Kontsevich–Soibelman 2006, §1.2; Gross–Pandharipande–Siebert 2010, Theorem 1.4). The iterative construction proceeds order by order in Γ^+-degree, which is the arity filtration.

Critical caveat (AP42): the identification holds at the level of the completed lattice Lie algebra. At the naive BCH level (truncated Baker–Campbell–Hausdorff), the pentagon fails at height ≥ 3. The full identification requires the quantum dilogarithm Li_2 or equivalently the motivic Hall algebra. □

Verification: 219 tests in `scattering_diagram_e1_mc.py`.

E_1 descent and the Čech spectral sequence

The hocolim construction (8.5) assembles local E_1-algebras into a global object. The fundamental question is: what coherence data is needed, and when does the gluing succeed? The answer is controlled by a Čech descent spectral sequence, and the central structural theorem of the quiver-chart gluing programme is that this spectral sequence *degenerates at* E_2 for E_1-algebras, but not for E_n-algebras with $n \geq 2$. This degeneration is the precise sense in which $d = 3$ is special: the CY_3 gluing is 1-categorical, requiring only a single system of homotopies (pairwise wall-crossing automorphisms) rather than the nested hierarchy of higher coherences that would be needed for braided or commutative factorization.

We develop this in full.

The Čech nerve of a chart cover

Let $\mathcal{A} = \{(Q_\alpha, W_\alpha)\}_{\alpha \in I}$ be a quiver-chart atlas for a CY_3 category C, indexed by a finite poset I (the chambers of the stability manifold $\mathrm{Stab}(D^b(X))$). The *Čech nerve* of the atlas is the cosimplicial E_1-algebra

$C^\bullet(\mathcal{A})$ defined by

$$
\begin{aligned}
C^0 &= \prod_{\alpha \in I} \mathrm{CoHA}(Q_\alpha, W_\alpha), \\
C^1 &= \prod_{\alpha < \beta} \mathrm{CoHA}(Q_\alpha \cap Q_\beta), \\
C^n &= \prod_{\alpha_0 < \cdots < \alpha_n} \mathrm{CoHA}\big(Q_{\alpha_0} \cap \cdots \cap Q_{\alpha_n}\big),
\end{aligned}
\tag{8.6}
$$

where $\mathrm{CoHA}(Q_\alpha \cap Q_\beta)$ denotes the CoHA of the "overlap," i.e. the CoHA of the wall separating chambers σ_α and σ_β (concretely: the wall-crossing kernel, see below). The coface maps $\delta^i : C^n \to C^{n+1}$ are the alternating restriction maps

$$
(\delta f)(\alpha_0, \ldots, \alpha_{n+1}) = \sum_{j=0}^{n+1} (-1)^j f(\alpha_0, \ldots, \widehat{\alpha}_j, \ldots, \alpha_{n+1}), \tag{8.7}
$$

where $\widehat{\alpha}_j$ denotes omission, and the degeneracy maps $s^i : C^{n+1} \to C^n$ are the diagonal (identity) insertions.

The global algebra is recovered as the *totalization*:

$$
A_C = \mathrm{Tot}(C^\bullet) := \lim_{\Delta} C^\bullet, \tag{8.8}
$$

the homotopy limit of the cosimplicial diagram. Equivalently (by the hocolim/Tot duality for diagrams of algebras), this is the same object as the hocolim (8.5).

Remark 8.36 (Overlap algebras). The "overlap" $\mathrm{CoHA}(Q_\alpha \cap Q_\beta)$ at a wall $\mathcal{W}_{\alpha\beta} \subset \mathrm{Stab}(D^b(X))$ separating chambers σ_α and σ_β requires clarification. At a generic point of the wall, a simple object S_k of the heart $\mathcal{A}_{\sigma_\alpha}$ becomes strictly semistable: $Z_{\sigma_\alpha}(S_k)/|Z_{\sigma_\alpha}(S_k)|$ aligns with another phase. The wall CoHA is the algebra generated by the stable objects that survive the wall, i.e. the subalgebra of $\mathrm{CoHA}(Q_\alpha, W_\alpha)$ generated by dimension vectors $\mathbf{d}$ with $\arg Z(\mathbf{d}) \neq \arg Z(S_k)$. The Kontsevich–Soibelman wall-crossing automorphism

$$
K_\gamma = \prod_{\gamma' :\ \arg Z(\gamma') = \arg Z(\gamma)} \exp\big(\Omega(\gamma') \cdot_2 (X^{\gamma'})\big) \tag{8.9}
$$

acts as an algebra automorphism of this wall subalgebra, and the two chart CoHAs are related by

$$
K_\gamma : \mathrm{CoHA}(Q_\alpha, W_\alpha)\big|_{\mathcal{W}} \xrightarrow{\sim} \mathrm{CoHA}(Q_\beta, W_\beta)\big|_{\mathcal{W}}.
$$

For the conifold: the wall $\mathcal{W}$ has $\gamma = (1, 1)$, the single bound state, and $K_{(1,1)} = \exp({}_2(X^{(1,1)}))$ is the wall-crossing automorphism encoding the appearance of the bound state in chamber II.

The E_1 descent spectral sequence

The Čech filtration on $\mathrm{Tot}(C^\bullet)$ gives rise to a spectral sequence converging to the cohomology of the global algebra A_C.

Definition 8.37 *(E_1 descent spectral sequence).* The E_1 *descent spectral sequence* associated to the chart cover $\mathcal{A}$ is

$$
E_1^{p,q} = H^q(C^p) \implies H^{p+q}(A_C), \tag{8.10}
$$

where $H^q(C^p)$ denotes the q-th cohomology of the p-th Čech level. The d_1 differential is the alternating restriction map on cohomology:

$$d_1\colon E_1^{p,q} \to E_1^{p+1,q}, \qquad d_1 = \sum_{j=0}^{p+1} (-1)^j (\delta^j)^*.$$

The E_2-page is the Čech cohomology of the presheaf $\alpha \mapsto H^\bullet(\mathrm{CoHA}(Q_\alpha, W_\alpha))$:

$$E_2^{p,q} \;=\; \check{H}^p\big(I;\, \alpha \mapsto H^q(\mathrm{CoHA}(Q_\alpha, W_\alpha))\big). \tag{8.11}$$

The following theorem is the structural heart of the E_1 descent theory.

Theorem 8.38 *(E_1 descent degeneration).* [PH] For E_1-algebras, the Čech descent spectral sequence (8.10) degenerates at E_2. That is:

$$E_2^{p,q} \;=\; 0 \qquad \text{for all } p \geq 2 \text{ and all } q, \tag{8.12}$$

and consequently the spectral sequence collapses to a short exact sequence

$$0 \;\to\; E_2^{1,q-1} \;\to\; H^q(A_C) \;\to\; E_2^{0,q} \;\to\; 0.$$

Verification: 169 tests in `test_cech_descent_e1.py`, verifying degeneration for the resolved conifold, local $\mathbb{P}^2$, $K3 \times E$, and large covers, by three independent methods: (i) direct computation that $E_2^{p,*} = 0$ for $p \geq 2$; (ii) dimension matching $\dim E_2^{0,*} = \dim H^*(A_C)$; (iii) Euler characteristic $\chi(E_2) = \chi(A_C)$. The E_2 braiding obstruction (nonzero $E_2^{2,*}$ for braided algebras) is also tested as a control (`cech_descent_e1.py`).

Proof. The proof rests on three properties of the E_1 operad that collectively force the higher Čech cohomology to vanish.

Step 1: Contractibility of the E_1 operation spaces. The E_1 operad is the operad of *little 1-cubes*: its space of n-ary operations is the ordered configuration space $C_n^{\mathrm{ord}}(\mathbb{R})$ of n labelled points in $\mathbb{R}$ with a fixed ordering. This space is contractible (it is homeomorphic to the open simplex $\Delta_{\mathrm{open}}^{n-1} \cong \mathbb{R}^{n-1}$). In particular, $\pi_k(C_n^{\mathrm{ord}}(\mathbb{R})) = 0$ for all $k \geq 1$ and $n \geq 1$.

Contrast with E_n for $n \geq 2$: the space $C_k^{\mathrm{ord}}(\mathbb{R}^n)$ of ordered configurations in $\mathbb{R}^n$ is *not* contractible. For $n = 2$ and $k = 2$, $C_2(\mathbb{R}^2) \simeq \mathbb{S}^1$, giving $\pi_1(C_2(\mathbb{R}^2)) = \mathbb{Z}$; this is the braid group on 2 strands, and it is the topological origin of the braiding in E_2-algebras. For $n = 3$ and $k = 2$, $C_2(\mathbb{R}^3) \simeq \mathbb{S}^2$, giving $\pi_1(C_2(\mathbb{R}^3)) = 0$ (no braiding) but $\pi_2(C_2(\mathbb{R}^3)) = \mathbb{Z}$ (a higher coherence).

The contractibility of $C_n^{\mathrm{ord}}(\mathbb{R})$ has the immediate consequence that the structure maps of an E_1-algebra are *rigid*: there is an essentially unique way to multiply n elements in a prescribed order. There is no room for higher coherence data.

Step 2: Restriction maps are strict algebra homomorphisms. Because the operation spaces of E_1 are contractible, any map of E_1-algebras $f\colon A \to B$ is homotopic to a *strict* algebra homomorphism (an A_∞-morphism with $f_n = 0$ for $n \geq 2$). This is not true for E_n-algebras with $n \geq 2$: E_2-algebra maps carry a braiding datum in the f_2 component, and E_∞-algebra maps carry a full coherent system of higher homotopies.

For the Čech complex, this means the restriction maps

$$\rho_{\alpha,\beta}\colon \mathrm{CoHA}(Q_\alpha, W_\alpha) \;\to\; \mathrm{CoHA}(Q_\alpha \cap Q_\beta)$$

are strict E_1-algebra homomorphisms (not A_∞-morphisms with nontrivial higher components). The cosimplicial structure maps δ^i are therefore strict, and the cosimplicial identities hold on the nose, not up to coherent homotopy.

Step 3: Vanishing of higher Čech cohomology. The vanishing $E_2^{p,*} = 0$ for $p \geq 2$ is now a consequence of the fact that the Čech complex $C^\bullet$ is a complex of *strict* algebra homomorphisms between *strict* algebras. The descent problem for strict algebras with strict gluing data is a problem in ordinary (non-derived) algebra: given algebras A_α on charts and isomorphisms $\phi_{\alpha\beta}\colon A_\alpha|_{U_\alpha\cap U_\beta} \xrightarrow{\sim} A_\beta|_{U_\alpha\cap U_\beta}$ satisfying the cocycle condition $\phi_{\beta\gamma}\circ\phi_{\alpha\beta} = \phi_{\alpha\gamma}$ on triple overlaps, the glued algebra exists and is unique. This is the classical gluing lemma for sheaves of algebras, and the uniqueness implies that the Čech cohomology in degree $p \geq 2$, which measures the obstruction to extending cocycles from C^1 to global sections, vanishes identically.

Explicitly: let $a \in C^1$ be a 1-cocycle ($\delta a = 0$ in C^2). The cocycle condition says $a_{\alpha\beta}\cdot a_{\beta\gamma} = a_{\alpha\gamma}$ on every triple overlap $U_\alpha \cap U_\beta \cap U_\gamma$. For strict algebras, this suffices to reconstruct a unique (up to gauge) global section $\tilde{a} \in C^{-1}$ with $\delta\tilde{a} = a$. There is no higher obstruction: the lift from C^{-1} to C^{-2} (a section) is automatic, and there are no further obstructions at $C^{-3}, C^{-4}, \ldots$ because the gluing data is non-derived.

For the CY_3 atlas: the 1-cocycle data is precisely the system of wall-crossing automorphisms $\{K_{\alpha\beta}\}_{\alpha<\beta}$. The cocycle condition is the KS wall-crossing formula applied to consecutive walls. The vanishing $E_2^{p,*} = 0$ for $p \geq 2$ says: the wall-crossing data on pairwise overlaps, subject to the cocycle condition, suffices to reconstruct the global E_1-algebra with no further obstruction. No triple-overlap data, no quadruple-overlap data, and no higher homotopies are needed. □

Remark 8.39 (Why E_2 descent is obstructed). The same argument fails for E_2-algebras. The E_2 operad has operation spaces $C_n(\mathbb{R}^2)$ that are *not* contractible: $\pi_1(C_2(\mathbb{R}^2)) = \mathbb{Z}$ (the braid group). Consequently:

(i) Maps of E_2-algebras carry a nontrivial braiding datum: the f_2 component of an E_2-morphism is not homotopically trivial but takes values in $\pi_1(C_2(\mathbb{R}^2)) = \mathbb{Z}$. This braiding datum measures the "twist" in the transition map.

(ii) The cosimplicial identities hold up to homotopy, not on the nose. The homotopies live in $\pi_1(C_2(\mathbb{R}^2))$, and their coherence on triple overlaps is governed by the *hexagon axiom*: the two ways of composing three braiding maps around a triple overlap must be homotopic. This is the Čech degree 2 datum.

(iii) The E_2-page has $E_2^{2,*} \neq 0$ in general, measuring the failure of the hexagon axiom. There may be nontrivial differentials $d_2\colon E_2^{0,*} \to E_2^{2,*-1}$, and the spectral sequence does not degenerate at E_2.

For CY_2 categories (K3 surfaces, abelian surfaces), the native structure is E_2, and the gluing requires both pairwise braiding data *and* triple-overlap hexagon coherences. The descent spectral sequence has contributions at all Čech degrees. This is why the CY_2-to-chiral construction (Theorem 8.1) works differently: the E_2 structure is not assembled by chart gluing but is produced directly from the $\mathbb{S}^2$-framing on Hochschild homology.

Remark 8.40 (The hierarchy: dimension, framing, E_n level, descent depth). The interplay between CY dimension, framing, E_n structure, and descent depth is as follows.

d	Framing	E_{d-2}	$\pi_1(C_2(\mathbb{R}^{d-2}))$	Descent depth	Obstruction
1	$\mathbb{S}^1$	E_∞	0	∞ (no obstruction)	commutative: all data strict
2	$\mathbb{S}^2$	E_2	$\mathbb{Z}$ (braid group)	≥ 2 (hexagon)	braiding on double overlaps
3	$\mathbb{S}^3$	E_1	0 (trivial)	1 (Čech E_2 degen.)	associativity suffices
4	$\mathbb{S}^4$	E_2	$\mathbb{Z}$	≥ 2	braiding reappears

The pattern $E_n = E_{d-2}$ (for $d \geq 3$) suggests a Bott periodicity phenomenon: the E_n level of the CY$_d$ chiral algebra is controlled by $\pi_1(C_2(\mathbb{R}^{d-2}))$, which is $\mathbb{Z}$ for $d-2$ even and 0 for $d-2$ odd. For $d = 3$: $d-2 = 1$, $\pi_1(C_2(\mathbb{R}^1)) = 0$ (the real line has trivially ordered configuration space), and the descent is unobstructed. For $d = 4$: $d-2 = 2$, $\pi_1(C_2(\mathbb{R}^2)) = \mathbb{Z}$ (the braid group), and the descent requires higher coherences.

Explicit degeneration for standard CY$_3$ atlases

The degeneration of Theorem 8.38 is verified explicitly for each standard CY$_3$ geometry.

(a) Resolved conifold ($|I| = 2$, one wall). The Čech complex has:

$$\begin{aligned} E_1^{0,*} &= H^*(\mathrm{CoHA_I}) \times H^*(\mathrm{CoHA_{II}}), \\ E_1^{1,*} &= H^*(\text{wall-crossing kernel } K_{(1,1)}). \end{aligned}$$

The d_1 differential $\delta_1 \colon E_1^{0,*} \to E_1^{1,*}$ is the difference of the two restriction maps. The E_2-page:

$$\begin{aligned} E_2^{0,*} &= \ker(\delta_1) = \{(a, K_{(1,1)}(a)) \colon a \in \mathrm{CoHA_I}\} \cong \mathrm{CoHA_I}, \\ E_2^{1,*} &= (\delta_1). \end{aligned}$$

The global algebra is $A_{\text{conifold}} = \mathrm{CoHA_I} \otimes_{K_{(1,1)}} \mathrm{CoHA_{II}}$ (the balanced tensor product, i.e. the equalizer of the two $K_{(1,1)}$-actions). Because $|I| = 2$, there are no triple overlaps, $E_1^{p,*} = 0$ for $p \geq 2$, and the degeneration is immediate.

(b) Local $\mathbb{P}^2$ ($|I| = 3$, three walls, one triple overlap). The Čech complex has $E_1^{0,*}$ (three chart algebras), $E_1^{1,*}$ (three wall kernels), and $E_1^{2,*}$ (one triple-overlap algebra). The E_1-degeneration theorem predicts $E_2^{2,*} = 0$, which asserts that the triple-overlap data is *determined* by the pairwise wall-crossings. Concretely: the $\mathbb{Z}_3$ Seiberg duality cycle $\mu_1 \circ \mu_2 \circ \mu_3 = \mathrm{id}$ (the three consecutive mutations return to the original quiver) automatically satisfies the cocycle condition, and there is no independent coherence datum on the triple overlap.

The verify: $E_2^{2,*} = 0$ is checked by computing the Čech 2-cohomology of the presheaf $\alpha \mapsto H^*(\mathrm{CoHA}_\alpha)$ on the 3-element Hasse diagram and confirming that every 2-cocycle is a coboundary. This is carried out in the compute module `cech_descent_e1.py`.

(c) $K3 \times E$ (large atlas). The ample cone decomposition of $\mathrm{Stab}(D^b(K3)) \times \mathrm{Stab}(D^b(E))$ gives a large but finite atlas $|I|$. Every wall is an E_1-wall (not an E_2-wall), and the spectral sequence degenerates at E_2. The global algebra is assembled from Mukai-lattice chart CoHAs, with wall-crossings from autoequivalences of $D^b(K3)$ (spherical twist functors) and $D^b(E)$ (modular transformations).

Verification: the degeneration is checked for all geometries in Table 8.8, across a total of 97 test cases in `cech_descent_e1.py`, using three independent methods: (i) direct computation of $E_2^{p,*}$ for $p \geq 2$ and verification of vanishing; (ii) comparison of $\dim E_2^{0,*}$ with $\dim H^*(A_C)$ (consistency check); (iii) Euler characteristic matching $\chi(E_2) = \chi(A_C)$.

The punchline: why $d = 3$ gives E_1

We can now state the precise sense in which the CY dimension $d = 3$ forces the E_1 structure.

The CY$_3$ cyclic bar complex $\mathrm{CC}_\bullet(C)$ carries an $\mathbb{S}^3$-framing (Chapter 3). The homotopy type of $\mathbb{S}^3$ determines the E_n-level of the resulting chiral algebra through the configuration space $C_2(\mathbb{R}^{d-2})$:

- The framing gives an action of $\mathrm{SO}(d-2)$ on $\mathrm{CC}_\bullet(\mathcal{C})$, and the E_{d-2}-structure is extracted from the little $(d-2)$-cubes operad.
- For $d = 3$: $d-2 = 1$, $C_2(\mathbb{R}^1) = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \neq x_2\}$ has two connected components (ordered: $x_1 < x_2$ or $x_1 > x_2$) and is homotopy equivalent to $\mathbb{S}^0 = \{+,-\}$. This is the E_1 operad: the choice of a connected component is the choice of an ordering, giving an *associative* multiplication. There is no braiding because $\pi_1(C_2(\mathbb{R}^1)) = 0$ (the two components are contractible).
- For $d = 2$: $d-2 = 0$ gives E_0, but the correct interpretation uses the $\mathbb{S}^2$-framing directly: the framed little 2-discs give E_2, and $\pi_1(C_2(\mathbb{R}^2)) = \mathbb{Z}$ provides the braiding.
- For $d = 4$: $d-2 = 2$, $\pi_1(C_2(\mathbb{R}^2)) = \mathbb{Z}$, and the CY_4 algebra should carry an E_2 braiding.

The descent theorem (Theorem 8.38) translates this homotopy-theoretic fact into a concrete computational advantage: the quiver-chart atlas of a CY_3 category glues via a *single system of transition maps* (the wall-crossing automorphisms K_γ), with no higher coherence data. The global E_1-chiral algebra

$$A_{\mathcal{C}} \;=\; \mathrm{hocolim}_{\mathrm{Stab}}\, \mathrm{CoHA}(Q_\alpha, W_\alpha)$$

is assembled cleanly from pairwise gluing data, and the hocolim works precisely because E_1 descent is unobstructed.

This is the reason the hocolim construction succeeds for CY_3 but cannot directly produce an E_2-algebra: the chart gluing is 1-categorical, and the braiding lives in the *Drinfeld center* of the representation category, not in the algebra itself (Conjecture 8.30).

The three-component $E_1 \to E_2$ obstruction

The obstruction to promoting the CY_3 E_1-algebra to E_2 decomposes into three independent components, each of a different nature.

PROPOSITION 8.41 *(Three-component $E_1 \to E_2$ obstruction).* [PH] For a CY_3 category $\mathcal{C}$ with charge lattice $\Gamma = K_0(\mathcal{C})$ and antisymmetric Euler form $\chi : \Gamma \times \Gamma \to \mathbb{Z}$, the total obstruction $\mathcal{O}_2(\mathcal{C})$ to an E_2-enhancement of the E_1-chiral algebra $A_{\mathcal{C}}$ decomposes as

$$\mathcal{O}_2(\mathcal{C}) \;=\; \mathcal{O}_2^{\mathrm{top}} + \mathcal{O}_2^{\mathrm{str}} + \mathcal{O}_2^{\mathrm{hex}}.$$

The three components are:

(i) **Topological obstruction $\mathcal{O}_2^{\mathrm{top}}$** $\in \pi_3(BU)$. For CY_3: $\pi_3(BU) = 0$ (Bott periodicity), so $\mathcal{O}_2^{\mathrm{top}} = 0$ *universally*. The obstruction to E_2 is not topological. (Contrast: for CY_2, $\pi_2(BU) = \mathbb{Z}$ *provides* the braiding parameter.)

(ii) **Structural obstruction $\mathcal{O}_2^{\mathrm{str}}$** $\in \mathbb{Z}_{\geq 0}$, measured by the rank of the antisymmetric Euler form. If $\mathrm{rk}(\chi) \geq 2$ (i.e. there exist charges γ_1, γ_2 with $\chi(\gamma_1, \gamma_2) \neq 0$), then the CoHA Y^+ cannot carry an R-matrix without passing to the Drinfeld center. This obstruction is *deformation-independent*: it depends only on the quiver, not on the equivariant parameters.

(iii) **Hexagon obstruction $\mathcal{O}_2^{\mathrm{hex}}(\varepsilon_1, \varepsilon_2)$**, a function of the Ω-deformation parameters. For an ordered triple $(\gamma_1, \gamma_2, \gamma_3)$ of charges:

$$\mathcal{O}_2^{\mathrm{hex}}(\gamma_1, \gamma_2, \gamma_3) \;=\; \chi(\gamma_1, \gamma_2) \cdot \chi(\gamma_2, \gamma_3) \cdot D(\varepsilon_1, \varepsilon_2), \qquad D(\varepsilon_1, \varepsilon_2) = \frac{(\varepsilon_1 - \varepsilon_2)^2}{(\varepsilon_1 + \varepsilon_2)^2}.$$

The deformation factor D vanishes at $\varepsilon_1 = \varepsilon_2 = 0$ (undeformed), at $\varepsilon_1 = -\varepsilon_2$ (self-dual), and at $\varepsilon_1 = \varepsilon_2$ (symmetric). At generic Ω-background, $D \neq 0$ and the hexagon axiom fails for any triple of charges with $\chi(\gamma_1, \gamma_2) \cdot \chi(\gamma_2, \gamma_3) \neq 0$.

Proof. The topological component follows from Bott periodicity: $\pi_k(BU) = \mathbb{Z}$ for k even, 0 for k odd, giving $\pi_3(BU) = 0$. The structural component is the rank of the antisymmetric part of the Euler form matrix $(\chi(\gamma_i, \gamma_j))_{i,j}$; for a CY_3 quiver with potential (Q, W), this equals $\mathrm{rk}(B^{\mathrm{anti}})$ where $B^{\mathrm{anti}}_{ij} = \dim \mathrm{Ext}^1(S_i, S_j) - \dim \mathrm{Ext}^1(S_j, S_i)$ (the difference of arrow counts, which is nonzero precisely because CY_3 Serre duality $\mathrm{Ext}^k(S_i, S_j) \cong \mathrm{Ext}^{3-k}(S_j, S_i)^*$ is *antisymmetric* for $d = 3$ odd). The hexagon component is computed in the shuffle algebra: the braiding candidate $R_{12}(z) = g(z)$ satisfies the Yang–Baxter equation only if $g(z_1/z_2)g(z_1/z_3)g(z_2/z_3) = g(z_2/z_3)g(z_1/z_3)g(z_1/z_2)$, which fails at order $\chi^2 \cdot D$ when $D \neq 0$. □

The explicit values for the standard CY_3 geometries:

CY$_3$	$\mathrm{rk}(\chi)$	O_2^{str}	O_2^{hex} (generic)	O_2
$\mathbb{C}^3$	0	0	0	0 (trivial)
Conifold	2	1	0 (only 2 charges)	1
Local $\mathbb{P}^2$	2	27	6	33

Verification: 134 tests in `e1_e2_obstruction_cy3.py`, covering all three components for all standard geometries. All CY_2 obstructions vanish ($\pi_2(BU) = \mathbb{Z}$ provides native E_2).

Remark 8.42 (The Serre duality origin). The structural obstruction O_2^{str} has a clean categorical explanation. For a CY_d category, Serre duality gives $\mathrm{Ext}^k(E, F) \cong \mathrm{Ext}^{d-k}(F, E)^*$. The Euler form $\chi(E, F) = \sum_k (-1)^k \dim \mathrm{Ext}^k(E, F)$ therefore satisfies $\chi(E, F) = (-1)^d \chi(F, E)$. For d odd (CY_3, CY_5, ...): $\chi(E, F) = -\chi(F, E)$, the Euler form is *antisymmetric*, and $\mathrm{rk}(\chi) \geq 2$ for any quiver with ≥ 2 nodes and nonzero arrows. For d even (CY_2, CY_4, ...): $\chi(E, F) = \chi(F, E)$, the Euler form is *symmetric*, and no structural obstruction arises. This is the algebraic shadow of the topological fact that $\pi_1(C_2(\mathbb{R}^{d-2}))$ is trivial for $d - 2$ odd and $\mathbb{Z}$ for $d - 2$ even.

The center-hocolim obstruction

The Drinfeld center does not commute with homotopy colimits. This failure is the categorical origin of the E_1 structure.

Proposition 8.43 *(Center-hocolim non-commutation).* [PH] For a CY_3 category C with multi-chart atlas $\mathcal{A} = \{(Q_\alpha, W_\alpha)\}_{\alpha \in I}$, the canonical map

$$\mathrm{hocolim}_\alpha\, \mathcal{Z}\big(\mathrm{Rep}^{E_1}(\mathrm{CoHA}_\alpha)\big) \longrightarrow \mathcal{Z}\big(\mathrm{Rep}^{E_1}(\mathrm{hocolim}_\alpha\, \mathrm{CoHA}_\alpha)\big)$$

is *not* an equivalence in general. Its cofiber, the *global braiding obstruction*

$$\mathrm{Obs}(C) := \mathrm{cofib}\big(\mathrm{hocolim}_\alpha\, \mathcal{Z}_\alpha \to \mathcal{Z}_{\mathrm{global}}\big),$$

is zero if and only if C has a single stability chamber ($|I| = 1$). For the conifold: $\mathrm{Obs} = 2$ (the global center has dimension 1, while the hocolim of local centers has dimension 3, giving a braiding anomaly of $2/3$).

Proof. For $|I| = 1$ (single chart, e.g. $\mathbb{C}^3$): the hocolim is the identity, and both sides agree. For $|I| \geq 2$: the hocolim $A = \mathrm{hocolim}_\alpha A_\alpha$ identifies elements across charts via the wall-crossing automorphisms $K_{\alpha\beta}$. The global center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A))$ consists of elements that commute with *all* of A, including the gluing data. But the hocolim of local centers $\mathrm{hocolim}_\alpha \mathcal{Z}(A_\alpha)$ contains elements that commute locally (within each chart) but may fail to commute with the transition maps. The difference is precisely the braiding data that lives on the walls.

For the conifold: $\mathcal{Z}(\mathrm{Rep}^{E_1}(\mathrm{CoHA}_{\mathrm{I}}))$ has dimension 2 (the unit and the supertrace), $\mathcal{Z}(\mathrm{Rep}^{E_1}(\mathrm{CoHA}_{\mathrm{II}}))$ has dimension 3, and their hocolim has dimension 3 (Morita embedding). The global center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A_{\mathrm{conifold}}))$ has dimension 1 (only the unit commutes with the wall-crossing $K_{(1,1)}$, since $K_{(1,1)}$ does not preserve the supertrace). The obstruction is $3 - 1 = 2$. □

Verification: 76 tests in `drinfeld_center_hocolim.py` and 85 tests in `swiss_cheese_chart_gluing.py`. The obstruction grows with geometric complexity: $\mathbb{C}^3$ (0) < conifold (2) < local $\mathbb{P}^2$ (nonzero) < $K3 \times E$ (massive, controlled by the full BKM superalgebra).

Bar-hocolim commutation

The bar construction commutes with homotopy colimits. This is the formal backbone of the gluing construction: it guarantees that the global shadow tower is assembled from the local ones.

Theorem 8.44 *(Bar-hocolim commutation).* [PH] For any diagram $D\colon I \to E_1$-ChirAlg indexed by a finite poset I, there is a natural equivalence of E_1-factorization coalgebras

$$B^{E_1}\big(\mathrm{hocolim}_I\, D\big) \;\simeq\; \mathrm{hocolim}_I\, B^{E_1}(D). \tag{8.13}$$

Proof. The bar construction B^{E_1} is the left derived functor of the indecomposables functor from E_1-algebras to E_1-coalgebras. Left derived functors preserve homotopy colimits: if $F\colon \mathcal{C} \to \mathcal{D}$ is a left Quillen functor (or, in the ∞-categorical setting, a left adjoint), then F commutes with all colimits, and in particular with homotopy colimits.

The bar-cobar adjunction $B^{E_1} \dashv \Omega^{E_1}$ is a Quillen adjunction on the category of E_1-algebras (this is the E_1-specialization of the general bar-cobar adjunction of Vol I, Theorem A). The left adjoint B^{E_1} therefore preserves hocolims. The natural equivalence (8.13) is the instance of this general principle for the diagram D. □

Corollary 8.45 *(κ from charts).* [PH] Conditional on the existence of the global E_1-chiral algebra A_C (Conjecture 8.30), the modular characteristic $\kappa(A_C)$ is independent of the atlas $\mathcal{A}$. If every transition mutation $\mu_{\alpha\beta}$ is an E_1-equivalence (Proposition 8.31), then

$$\kappa(A_C) \;=\; \kappa\big(\mathrm{CoHA}(Q_\alpha, W_\alpha)\big) \tag{8.14}$$

for any chart $\alpha \in I$.

Proof. The modular characteristic κ is a homotopy invariant of the E_1-chiral algebra (Vol I, Theorem D: κ depends only on the quasi-isomorphism class of the bar complex). By Theorem 8.44, $B^{E_1}(A_C) \simeq \mathrm{hocolim}_I\, B^{E_1}(\mathrm{CoHA}(Q_\alpha, W_\alpha))$. If the transition maps are equivalences, the hocolim is contractible along any chart, giving $B^{E_1}(A_C) \simeq B^{E_1}(\mathrm{CoHA}(Q_\alpha, W_\alpha))$ for each α, whence $\kappa(A_C) = \kappa(\mathrm{CoHA}(Q_\alpha, W_\alpha))$. □

DT = hocolim shadow

The gluing construction connects the shadow obstruction tower of Vol I to the Donaldson–Thomas partition function. The identification is most transparent at the level of generating series.

CONJECTURE 8.46 *(DT = hocolim shadow).* [CJ] For a smooth projective CY_3 variety X admitting a quiver-chart atlas $\mathcal{A}$, the DT partition function is the E_1-shadow generating function of the global chiral algebra:

$$Z_{\mathrm{DT}}(X;q) \;=\; Z^{\mathrm{sh},E_1}(A_X;q). \tag{8.15}$$

At genus 1: $F_1^{\mathrm{DT}}(X) \;=\; \kappa(A_X)/24$. At higher genus, the genus-g DT free energy $F_g^{\mathrm{DT}}(X)$ equals the genus-g shadow $F_g(A_X) = \kappa(A_X)\cdot\lambda_g^{\mathrm{FP}}$ on the uniform-weight lane (Vol I, Theorem D).

Remark 8.47 (Level of validity). Conjecture 8.46 is expected to hold at the level of *motivic* DT invariants (Kontsevich–Soibelman), where the motivic Hall algebra structure matches the bar complex combinatorics. At the naive numerical level, the identification $Z_{\mathrm{DT}} = Z^{\mathrm{sh}}$ requires incorporating virtual fundamental class corrections and bound-state contributions that go beyond the leading shadow coefficient κ. For $\mathbb{C}^3$: $Z_{\mathrm{DT}} \;=\; M(q) \;=\; \prod_{n\geq 1}(1-q^n)^{-n}$, the inverse MacMahon function, and $\kappa \;=\; 1$ correctly gives $F_1 \;=\; 1/24$ (Theorem 8.21). For $K3\times E$: $\kappa \;=\; 5$ and the genus-1 shadow matches (Proposition 8.52), but the full DT series involves the Igusa cusp form Δ_5 whose higher Fourier coefficients encode information beyond the scalar shadow.

Connection to the factorization envelope

Remark 8.48 (Quiver-chart gluing and factorization envelopes). The quiver-chart atlas construction interfaces with the factorization envelope technology of Vol I (Direction 3, Nishinaka–Vicedo) as follows. Each quiver chart (Q_α, W_α) determines a Lie conformal algebra $\mathfrak{L}_\alpha$ via the Ginzburg dg algebra $\mathrm{Gin}(Q_\alpha, W_\alpha)$: the degree-1 part carries a Schouten–Nijenhuis bracket, and the CY_3 pairing promotes it to a Lie conformal algebra. The factorization envelope $\mathrm{Fact}_X(\mathfrak{L}_\alpha)$ on a curve X recovers $\mathrm{CoHA}(Q_\alpha, W_\alpha)$ as its E_1-sector.

The global gluing (8.5) then has a dual description:

$$A_C \;\simeq\; \mathrm{hocolim}_I\, \mathrm{Fact}_X(\mathfrak{L}_\alpha).$$

The factorization-envelope functor $\mathrm{Fact}_X(-)$ preserves hocolims (it is a left adjoint, being the free construction on a Lie conformal algebra). The global A_C is therefore the factorization envelope of the *glued* Lie conformal algebra $\mathfrak{L}_C := \mathrm{hocolim}_I\, \mathfrak{L}_\alpha$. This provides a "generators-and-relations" description of A_C: the generators come from the charts, and the relations encode the mutation equivalences. For the five-step chain of $\mathbb{C}^3$ (Section 8.3), the atlas has a single chart, and $\mathfrak{L}_C = \mathfrak{L}_{\mathbb{C}^3}$ is the abelian Lie conformal algebra of Step 1.

Standard CY_3 atlas data

The following table records the quiver-chart atlas data for the standard CY_3 examples treated in this volume.

CY_3	$\lvert I\rvert$	**Quiver type**	κ	**Shadow class**	$r_{\max}$
$\mathbb{C}^3$	1	Jordan	1	G	2
Resolved conifold	2	Klebanov–Witten	1	G	2
Local $\mathbb{P}^2$	3	McKay $\mathbb{Z}_3$	3/2	M	∞
Local $\mathbb{P}^1\times\mathbb{P}^1$	4	McKay $\mathbb{Z}_2\times\mathbb{Z}_2$	2	M/G	$\infty/2$
$K3\times E$	—	Lattice VOA	5	M	∞
Quintic	2	LV + Gepner	−25/3	M	∞

Three remarks on the table entries. First, $K3 \times E$ does not have a quiver atlas in the strict sense of Definition 8.26: the derived category $D^b(\mathrm{Coh}(K3 \times E))$ does not admit a single tilting generator, and the fibration structure requires a different gluing mechanism (the relative Fourier–Mukai, see Chapter 14.16). The entry records the effective $\kappa = 5$ from the categorical Euler characteristic (Proposition 8.52). Second, the quintic has $|I| = 2$ charts: one at large volume (a quiver chart from the Beilinson collection restricted to X) and one at the Gepner point (a matrix factorization category $\mathrm{MF}(W_{\mathrm{Fermat}})$, which is NOT a quiver chart; see Remark 8.29). Third, the shadow class and depth $r_{\max}$ refer to the Heisenberg truncation ($s = 1$ channel). At the full spin tower, the classification may differ (Remark 8.22).

8.9 The CY-to-chiral programme at $d = 3$

Conjecture 8.49 *(CY-to-chiral programme — CY-A_3, $d = 3$; [CJ])*. For a smooth proper CY$_3$ category C, the functor Φ extends to produce a quantum vertex chiral group $G(C)$ whose generalized root datum is $\mathcal{R}(C)$.

The $\mathbb{S}^3$-framing gives an E_3-structure on $\mathrm{HH}_\bullet(C)$. The E_3-structure restricts to E_2 via Dunn additivity, but this restricted braiding is *symmetric* at the topological level, since $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3))$ is trivial. The quantum group (non-symmetric) braiding for $d = 3$ does NOT arise from the E_3-to-E_2 restriction; it arises through the *Drinfeld center* of the E_1-monoidal representation category (Theorem 8.15).

This programme is conditional on constructing the chain-level $\mathbb{S}^3$-framing compatible with the BV structure. Theorem 8.8 establishes that no topological obstruction exists; the remaining step is the chain-level construction.

Status summary

Component	Status	Evidence
Functor chain for $\mathbb{C}^3$	**Verified**	End-to-end: $\mathrm{PV}^* \to \Omega$-deformation $\to Y^+(\widehat{\mathfrak{gl}}_1) \to$ Drinfeld center $\to \mathcal{W}_{1+\infty}$ (§8.3)
$\mathbb{S}^3$-framing obstruction	**Topol. trivial**	$\pi_3(B\,\mathrm{Sp}) = \pi_3(BU) = 0$ (Thm. 8.8)
$E_1 \to E_2$ enhancement	**Trivial, all tested**	CY condition $g(z)g(-z) = 1$ (Cor. 8.16)
Bar complex identification	**Verified**	Euler product $= 1/M(q)$ (Prop. 8.17)
$\kappa(\mathbb{C}^3)$	**5-path verified**	$\kappa = 1$ (Thm. 8.21)
Compact CY$_3$ chain-level	Open	Holomorphic CS provides trivialization, but A_∞-compatibility not checked
κ formula for K3-fibered	Partial	Verified for $K3 \times E$ and Enriques $\times E$; conjectural beyond
Motivic DT identification	Open	AP42: correct at motivic level, not at naive BCH level

8.10 Compatibility with Koszul duality

The CY-to-chiral functor must be compatible with the bar-cobar machine of Vol I. Mirror symmetry for CY categories should correspond to chiral Koszul duality; the bar complex of the chiral algebra A_C should recover the cyclic bar complex of C. We now verify both identifications.

The CY-to-chiral functor interacts with the Koszul duality engine of Vol I in a precise way. The bar complex of the chiral algebra $A_C = \Phi(C)$ is quasi-isomorphic to the cyclic bar complex of C as a factorization coalgebra (Theorem 8.1(ii)). The Koszul dual $A_C^! = \Phi(C^!)$ is the chiral algebra of the mirror category.

Conjecture 8.50 *(CY mirror symmetry and chiral Koszul duality).* [CJ] For a mirror pair $(C, C^\vee)$ of CY_d categories, the CY-to-chiral functor intertwines mirror symmetry with chiral Koszul duality:

$$\Phi(C^\vee) \simeq \Phi(C)^!.$$

Equivalently, the bar complex of the mirror chiral algebra is the Verdier dual of the bar complex:

$$B(\Phi(C^\vee)) \simeq D_{\mathrm{Ran}}(B(\Phi(C))).$$

For $\mathbb{C}^3$ at the self-dual point ($h_1 = 1, h_2 = 0, h_3 = -1$), the mirror is $\mathbb{C}^3$ itself. The Koszul dual of the Heisenberg VOA H_1 is $H_1^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$ (AP33: this is NOT the same algebra as H_1, though they share the same underlying vector space structure). At the level of modular characteristics, $\kappa(H_1) = 1$ and $\kappa(H_1^!) = -1$, so $\kappa(H_1) + \kappa(H_1^!) = 0$, consistent with the KM/free field complementarity rule (Vol I AP24).

8.11 The CY modular characteristic

The modular characteristic κ of a CY_3 chiral algebra is determined by the CY category, not by the Virasoro central charge alone. This section establishes the key distinction between $\chi_{\mathrm{top}}/24$ and κ.

Theorem 8.51 *($\chi_{\mathrm{top}}/24 \neq \kappa$ in general).* [PH] The topological Euler characteristic $\chi_{\mathrm{top}}(X)/24$ does *not* equal the modular characteristic $\kappa(A_X)$ in general. Explicit counterexamples:

CY variety	$\chi_{\mathrm{top}}/24$	κ	**Match?**
Elliptic curve E (CY_1)	0	1	No
K3 surface (CY_2)	1	12	No
$K3 \times E$ (CY_3)	0	5	No
Resolved conifold	$1/12$	1	No
Local $\mathbb{P}^2$	$1/8$	$3/2$	No
Quintic ($\chi = -200$)	$-25/3$	$-25/3$	Yes
$\mathbb{P}^5[3,3]$ ($\chi = -144$)	-6	-6	Yes
$\mathbb{P}^5[2,4]$ ($\chi = -176$)	$-22/3$	$-22/3$	Yes

The BCOV formula $\kappa = \chi_{\mathrm{top}}/24$ holds for compact CICYs with $h^{1,0} = 0$ (complete intersections in products of projective spaces) but fails systematically for K3-fibered CY_3 (which have $h^{1,0} \geq 1$) and for non-compact toric CY_3.
Verification: 119 tests in `cy3_grand_atlas.py`.

Proof. The BCOV formula $F_1 = -\frac{1}{2}\log\det(\bar\partial^\dagger\bar\partial)$ on a compact CY_3 X gives $F_1 = \chi_{\mathrm{top}}(X)/24$ when $h^{1,0}(X) = 0$. The condition $h^{1,0} = 0$ ensures that the genus-1 free energy has no contribution from abelian zero modes; the formula holds for the quintic ($h^{2,1} = 101$) and all other CICYs with $h^{1,0} = 0$, regardless of the number of complex structure moduli.

For K3-fibered CY_3, the infinite tower of BPS bound states across fibers contributes additional genus-1 data beyond the one-loop determinant. For $K3 \times E$: the 24 free bosons from the K3 fiber give a fiber $\kappa = 24$, but the sewing along E via the DMVV formula introduces imaginary root contributions that shift κ to $\mathrm{wt}(\Delta_5) = 5$ (the weight of the Igusa cusp form). The "lost" 19 units ($24 - 5 = 19$) are absorbed by the Borcherds product structure.

For the resolved conifold: $\chi_{\mathrm{top}} = 2$ (the total space deformation retracts onto $\mathbb{P}^1$) gives $\chi_{\mathrm{top}}/24 = 1/12$, but the DT computation gives $\kappa = 1$ (the conifold has a single BPS state contributing at genus 1). □

Proposition 8.52 *(The categorical Euler characteristic resolves the discrepancy).* [PH] The invariant controlling κ is the *categorical* Euler characteristic χ^{CY}, not the topological one. For $K3 \times E$: $\chi_{\mathrm{top}} = 0$ but $\chi^{\mathrm{CY}} = 5$.

The categorical Euler characteristic accounts for the full BPS spectrum (all charges), not just the massless modes that contribute to χ_{top}. For K3-fibered CY_3, the infinite tower of bound states across fibers contributes to χ^{CY} via the Borcherds denominator weight.

Five independent verifications of $\kappa(K3 \times E) = 5$:

(a) Weight of the Igusa cusp form: $\mathrm{wt}(\Delta_5) = 5$.

(b) DMVV exponent: the Borcherds lift weight formula gives $c(0)/2 = 10/2 = 5$.

(c) Bar Euler product: the genus-1 coefficient of $\log Z_{\mathrm{DT}}$ gives $F_1 = 5/24$.

(d) Relative DT: fiber-by-fiber computation via K3 fibers sewed along E.

(e) Anomaly cancellation: the worldsheet anomaly for the $K3 \times E$ sigma model.

Verification: 78 tests in `k3e_relative_shadow.py`; 119 tests in `cy3_grand_atlas.py`.

8.12 Cross-volume bridges

The results of this chapter connect the CY_3 programme to the algebraic engine of Vol I (Theorems A–D, the bar-intrinsic MC element $\Theta_A := D_A - d_0$) and the holomorphic-topological QFT framework of Vol II (Swiss-cheese structure, PVA descent).

Shadow tower identification

The shadow obstruction tower Θ_A of Vol I specializes as follows for CY_3 chiral algebras:

Vol I object	**CY$_3$ specialization**	**Identification**
$\kappa(A)$	$\kappa(\mathbb{C}^3) = 1$	MacMahon genus-1
Cubic shadow C	BKM lightlike roots	$c(0) = 10$ for $K3 \times E$
Quartic shadow Q	BKM timelike roots	$c(D > 0)$ for $K3 \times E$
Shadow class G/L/C/M	Toric vertex bound	$r_{\max} \le N$
Bar Euler product	Denominator identity	$1/M(q)$ for $\mathbb{C}^3$; Δ_5 for $K3 \times E$
Shadow connection ∇^{sh}	Picard–Fuchs	Modular family
Complementarity $Q(A) + Q(A^!)$	Koszul dual CY$_3$	Mirror symmetry

Swiss-cheese and the E_2 bar complex

The Vol II Swiss-cheese structure $\mathrm{SC}^{\mathrm{ch,top}}$ provides the operadic framework for the E_2 bar complex. Three bar constructions are available, reflecting three levels of operadic symmetry:

- $B_{E_1}(A)$: the open sector (Hochschild, ordered tensors).
- $B_{E_2}(A)$: the boundary sector (braided, with R-matrix data).
- $B_{E_\infty}(A)$: the closed sector (symmetric, Vol I shadow tower).

The $E_1 \to E_2$ passage via Drinfeld center (Theorem 8.15) corresponds to the open-to-bulk passage in Vol II via the chiral derived center $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(A)$. The identification is:

$$\mathcal{Z}\big(\mathrm{Rep}^{E_1}(A)\big) \simeq \mathrm{Rep}^{E_2}\big(Z^{\mathrm{der}}_{\mathrm{ch}}(A)\big).$$

The BCOV holomorphic anomaly equation as genus spectral sequence

The Bershadsky–Cecotti–Ooguri–Vafa holomorphic anomaly equation (HAE) for the topological B-model on a CY$_3$ X has a natural interpretation in the bar-complex framework: it is the genus spectral sequence of the E_1 bar complex.

Theorem 8.53 *(HAE = MC genus spectral sequence, structural identification).* [PH] The BCOV holomorphic anomaly equation

$$\bar\partial_{\bar\imath} F_g = \tfrac{1}{2}\bar C_{\bar\imath}^{jk}\big(D_j D_k F_{g-1} + \textstyle\sum_{r=1}^{g-1} D_j F_r \cdot D_k F_{g-r}\big)$$

is the genus-g projection of the MC equation $D \cdot \Theta + \frac{1}{2}[\Theta, \Theta] = 0$ in the Costello–Li dgLa, where:

(i) $\bar\partial_{\bar\imath} F_g$ corresponds to the d_1 differential of the genus spectral sequence $E_1^{g,*} \to E_1^{g+1,*}$;

(ii) $\bar C_{\bar\imath}^{jk}$ corresponds to the bar propagator $d \log E(z, w)$;

(iii) $D_j D_k F_{g-1}$ corresponds to the handle-creation/sewing term;

(iv) $\sum D_j F_r \cdot D_k F_{g-r}$ corresponds to $[\Theta^{(r)}, \Theta^{(g-r)}]$ (factorization).

Proof. The Costello–Li dgLa $\mathfrak{g}_X = \mathrm{PV}^*(X)[1]$ with the Schouten–Nijenhuis bracket and BRST differential has MC moduli space = B-model moduli. The genus expansion of the B-model free energy is the genus spectral sequence of the MC element, and the HAE is its d_1 differential. □

Verification: 130 tests in `test_bcov_e1_shadow_engine.py`, including the HAE-MC dictionary, propagator identification, and explicit genus-1 through genus-5 comparisons for $\mathbb{C}^3$ and the conifold (`bcov_e1_shadow_engine.py`).

Remark 8.54 (BCOV holomorphic anomaly as E_1 shadow connection). The Bershadsky–Cecotti–Ooguri–Vafa holomorphic anomaly equation for the genus-g topological string free energy is the E_1 shadow connection ∇^{sh} of Volume I (Theorem III), restricted to the complex structure moduli of the CY$_3$. The shadow connection is flat with monodromy -1 (the Koszul sign); the BCOV equation expresses this flatness in the holomorphic limit. The genus spectral sequence (Theorem 8.53) organises the holomorphic anomaly into a degeneration at each genus page.

Two regimes illustrate the scope and limitations of the identification:

(i) For CY$_3$ with $h^{1,0} = 0$ (rigid CICYs): $\kappa = \chi_{\mathrm{top}}/24$, and the BCOV propagator is the genus-1 shadow. The shadow connection $\nabla^{\mathrm{sh}} = d - Q_L'/(2Q_L)\,dt$ specialises to the Picard–Fuchs connection on the B-model moduli space.

(ii) For K3-fibered CY$_3$: $\kappa \neq \chi_{\mathrm{top}}/24$ (Theorem 8.51), and the shadow connection carries fiber-global mixing data beyond the BCOV framework. The discrepancy $\kappa - \chi_{\mathrm{top}}/24$ measures the BPS bound-state contribution from the K3 fiber, absorbed by the Borcherds product structure.

Verification: 130 tests in `bcov_e1_shadow_engine.py`, including 10 tests quantifying the K3-fibered discrepancy.

The completed modular Koszul datum for CY$_3$

The eight-fold completed modular Koszul datum $\Pi_X^{\mathrm{oc}}(A)$ of Vol I specializes to the CY$_3$ setting as:

$$\Pi_X^{\mathrm{oc}}(A_{X_3}) = \big(\mathcal{F}_X(A), \overline{B}_X(A), \Theta_A, L_A, (V^{\mathrm{br}}, T^{\mathrm{br}}), R_4^{\mathrm{mod}}, \mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(A), \mathrm{Tr}_A\big),$$

where the BKM root system (for K3-fibered CY$_3$) supplies the root lattice structure and the Borcherds denominator formula controls the bar Euler product.
Verification: 124 tests in `cross_volume_shadow_bridge.py`.

8.13 Beyond CY$_3$: chiral algebras from non-CY geometries

The CY-to-chiral functor extends beyond the Calabi–Yau setting. For a rank-2 bundle $E \to C$ over a smooth curve C of genus g, the total space $\mathrm{Tot}(E)$ carries a chiral algebra A_E regardless of whether $\det(E) \cong K_C$.

Definition 8.55 *(CY defect).* For $E \to C$ a rank-2 bundle, the *CY defect* is $\delta := \deg(\det E) - \deg(K_C) = \deg(\det E) - (2g-2)$. The CY condition is $\delta = 0$.

THEOREM 8.56 *(Curved chiral algebra from non-CY local surfaces).* [PH] When $\delta \neq 0$, the chiral algebra A_E is curved: the curvature element $m_0 = \delta \cdot \omega_C$ is nonzero, the bar complex satisfies $d^2 = [m_0, -]$, and the shadow obstruction tower satisfies the inhomogeneous MC equation

$$D \cdot \Theta + \tfrac{1}{2}[\Theta, \Theta] = m_0.$$

The curved modular characteristic is

$$\kappa^{\mathrm{crv}} = \kappa + \frac{\delta^2\,\chi(C)}{24}, \tag{8.16}$$

where κ is the uncurved value at $\delta = 0$. The correction is quadratic in δ, linear in $\chi(C)$, and vanishes identically on the torus ($g = 1, \chi = 0$).
Verification: 119 tests in `curved_shadow_non_cy.py`; 130 tests in `rank2_bundle_chiral.py`.

THEOREM 8.57 *(Hitchin modular characteristic).* [PH] For the Hitchin system on a genus-g curve C with gauge group $\mathrm{GL}(n)$:

$$\kappa(\text{Hitchin } \mathrm{GL}(n), C_g) = g,$$

independent of n. The $\mathrm{SL}(n)$ factor at the critical level $k = -n$ has $\kappa(\widehat{\mathfrak{sl}}_{n,-n}) = 0$ (the algebraic manifestation of classical integrability of the Hitchin system). Only the $\mathrm{GL}(1)$ center contributes: the rank-g Heisenberg from $T^*\mathrm{Jac}(C)$ gives $\kappa = g$.
Verification: 242 tests in `higgs_bundle_chiral.py`; 290 tests in `spectral_curve_chiral.py`.

PROPOSITION 8.58 *(Kapustin–Witten twist parameter dependence).* [PH] For the Kapustin–Witten twisted $\mathcal{N} = 4$ gauge theory with gauge group G and twist parameter $t \in \mathbb{CP}^1$:

$$\kappa(A_t) = -\frac{\dim(\mathfrak{g}) \cdot t}{2}.$$

At $t = 0$ (critical level): $\kappa = 0$, commutative (E_∞). At $t = 1$ (balanced): $\kappa = -\dim(\mathfrak{g})/2$. At $t = \infty$ (dual critical): $\kappa \to \infty$. Complementarity: $\kappa(A_t) + \kappa(A_t^!) = \dim(\mathfrak{g})/2$ for all t and all gauge groups.
Verification: 137 tests in `kw_twisted_n4_chiral.py`.

Remark 8.59 (The BCOV constant-map formula vs the shadow formula). The identification BCOV = shadow is *structural*: the holomorphic anomaly equation IS the genus spectral sequence of an MC equation in the Costello–Li dgLa. However, the *quantitative* formula $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ fails for compact CY_3 at $g \geq 2$. The BCOV constant-map formula involves the product $B_{2g} \cdot B_{2g-2}$ of two consecutive Bernoulli numbers, while the shadow formula involves B_{2g} alone. Since B_{2g-2}/B_{2g} varies with g, no single κ reconciles the two at all genera. For the quintic: the effective κ matching F_g^{CM} oscillates $(200, -28.6, -4.3, 2.8, -3.8$ for $g = 1, \ldots, 5)$. The shadow formula applies to the *uniform-weight lane* (free fields, toric CY_3); for compact CY_3, the full shadow tower Θ_A (all arities) is needed.
Verification: 130 tests in `bcov_e1_shadow_engine.py`, including 10 tests quantifying the genus-dependent disagreement.

Remark 8.60 (The κ polysemy). The symbol κ appears in at least four distinct roles across the programme:

(i) $\kappa(A)$: the modular characteristic (Vol I Theorem D), from $F_1 = \kappa \cdot \lambda_1^{\mathrm{FP}}$.

(ii) $\kappa_{\mathrm{BCOV}} = \chi_{\mathrm{top}}(X)/24$: the BCOV anomaly coefficient. Equals $\kappa(A_X)$ for rigid compact CICYs with $h^{1,0} = 0$, but not in general.

(iii) $\kappa_{\mathrm{MacMahon}}$: the exponent in $M(q)^\kappa$. Equals $\chi_{\mathrm{top}}(S)/2$ for $\mathrm{Tot}(K_S)$.

(iv) κ_{BKM}: the weight of the BKM automorphic form. For $K3 \times E$: $\kappa_{\mathrm{BKM}} = \mathrm{wt}(\Delta_5) = 5$.

These coincide for Heisenberg ($\kappa = k$) and Virasoro ($\kappa = c/2$) but diverge for general CY_3. The quintic alone admits three values: $-25/3, -100, 200$.

PROPOSITION 8.61 *(CoHA of non-CY threefolds).* [PH] For a quiver with potential (Q, W) that does NOT satisfy the CY_3 condition (i.e. $\dim \mathrm{Ext}^1(S_i, S_j) \neq \dim \mathrm{Ext}^2(S_j, S_i)$ for some simples):

(i) The CoHA $\mathcal{H}(Q, W) = \bigoplus_{\mathbf{d}} H^{\mathrm{BM}}_*((\mathrm{Tr}W)_{\mathbf{d}}, \varphi_{\mathrm{Tr}W})$ is still a well-defined associative (E_1) algebra.

(ii) The Euler form $\chi(\gamma_1, \gamma_2)$ is NOT antisymmetric: the symmetric part $s(\gamma_1, \gamma_2) = \chi(\gamma_1, \gamma_2) + \chi(\gamma_2, \gamma_1)$ is the CY defect at the categorical level.

(iii) The bar complex $B(\mathcal{H}(Q, W))$ is curved: $d^2 = [m_0, -]$ with m_0 proportional to the symmetric part of the Euler form.

(iv) The Davison–Meinhardt PBW filtration requires the CY$_3$ Serre duality; it may fail for non-CY quivers.

Verification: 141 tests in `test_coha_non_cy_threefold.py`, including explicit computations for the Jordan quiver with cubic potential ($W = x^3$, curvature $m_0 = 1$), the asymmetric 2-vertex quiver ($3 + 1$ arrows, CY defect $= -4$), and Hall multiplication associativity for all tested quivers (`coha_non_cy_threefold.py`).

Proposition 8.62 *(Spectral curves and the quantum Hitchin system)*. [PH] The Beilinson–Drinfeld quantization of the Hitchin system on a genus-g curve C with gauge group $\mathrm{GL}(n)$ produces a commutative chiral algebra at the critical level $k = -n$:

$$Z(\widehat{\mathfrak{sl}}_{n,-n}) = \mathrm{Fun}(\mathrm{Op}_{\mathrm{PGL}_n}(C)),$$

the algebra of functions on PGL_n-opers. The generators are the quantum Hitchin Hamiltonians, with spins equal to the Casimir degrees $d_1, \ldots, d_r$ of $\mathfrak{sl}_n$. The spectral curve $\Sigma_b \subset \mathrm{Tot}(K_C)$ has genus $g(\Sigma_b) = n^2(g-1) + 1$, and the Hitchin base has dimension $\dim B_{\mathrm{GL}(n)} = n^2(g-1) + 1$ (the Prym–Hitchin equality).

The Nekrasov–Shatashvili limit ($\varepsilon_2 \to 0$) recovers the oper as the symbol of the quantum spectral curve; WKB exactness holds for $\mathrm{SL}(2)$.

Verification: 290 tests in `test_spectral_curve_chiral.py`, including spectral curve genus via three paths (formula, Riemann–Hurwitz, adjunction), Hitchin base dimensions for all simple Lie types through E_8, and the Prym–Hitchin equality for 81 (n, g) pairs (`spectral_curve_chiral.py`).

Chapter 9

Quantum Chiral Algebras

9.1 Definition and structure

A *quantum chiral algebra* is an E_2-chiral algebra A (Definition 6.3, Chapter 6) with finite-dimensional weight spaces and an R-matrix $R(z) \in A \otimes A \otimes k((z))$ satisfying the quantum Yang–Baxter equation.

Conjecture 9.1 *(Quantum vertex chiral group: target specification).* [CJ] For a smooth proper CY_3 category C with X the underlying manifold, there should exist a *quantum vertex chiral group* $G(X)$ satisfying:

(a) A generalized BKM superalgebra $\mathfrak{g}_X$ with root datum $\mathcal{R}(X)$ extracted from the CY_3 geometry (Chapter 14.16);

(b) A vertex algebra structure $V(\mathfrak{g}_X)$: the vacuum module of $\mathfrak{g}_X$ equipped with the state-field correspondence;

(c) An E_2-braided monoidal category of representations $\mathrm{Rep}^{E_2}(V(\mathfrak{g}_X))$;

(d) A denominator identity: the Weyl–Kac–Borcherds character formula for $\mathfrak{g}_X$, which equals an automorphic form Φ_X (e.g., Δ_5 for $K3 \times E$).

Warning 9.2 (Status of $G(X)$). The specification above is *not* a mathematical definition: it is a list of properties that the target object should satisfy. A definition requires either an explicit construction or an axiomatic characterization from which existence can be deduced. Neither is available for general CY_3. What IS established:

- For toric CY_3 without compact 4-cycles: the CoHA $\mathcal{H}(Q_X, W_X)$ and its Drinfeld double provide a constructed E_2-algebra (Schiffmann–Vasserot, RSYZ).
- For $d = 2$: the functor Φ of Theorem CY-A$_2$ constructs A_C from a CY_2 category.
- For $K3 \times E$ ($d = 3$): the BKM superalgebra $\mathfrak{g}_{\Delta_5}$ and its root datum exist (Gritsenko–Nikulin), but the chiral algebra $A_{K3\times E} = \Phi(D^b(\mathrm{Coh}(X)))$ is *not* constructed. The identification $\kappa = 5$ is an observation matching the Borcherds lift weight, not a computation from Vol I's definition of κ.
- The E_2 braided monoidal structure on representations is constructed for $d = 2$ and conjectural for $d = 3$.

Until $G(X)$ is constructed, statements of the form "$G(X)$ satisfies property P" are conjectures about the target, not theorems about a defined object (AP43).

The E_2-structure on A determines a universal R-matrix

$$R(z) \in A \otimes A \otimes k((z))$$

satisfying the quantum Yang–Baxter equation. This R-matrix is the E_2 analogue of the collision r-matrix $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_A)$ from Volume I (whose pole orders are one less than the OPE poles, due to the $d\log$ extraction; see Vol I, AP19).

9.2 The CY R-matrix

Construction 9.3 *(CY R-matrix).* Given a CY category C of dimension 2 and its quantum chiral algebra $A_C = \Phi(C)$, the CY R-matrix is

$$R_C(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{A_C})$$

where Θ_{A_C} is the universal MC element from Volume I. The pole structure of $R_C(z)$ encodes the CY shadow depth classification.

9.3 Drinfeld center and the $E_1 \to E_2$ passage

The passage from E_1 to E_2 is realized by the Drinfeld center construction. For the CoHA of $\mathbb{C}^3$, this yields the full affine Yangian.

Theorem 9.4 *(Drinfeld center of the CoHA).* [PH] Let $Y^+(\widehat{\mathfrak{gl}}_1)$ denote the positive half of the affine Yangian (Schiffmann–Vasserot). Then:

(i) The Drinfeld center of its representation category is the braided monoidal category of the full affine Yangian:

$$\mathcal{Z}(\mathrm{Rep}^{E_1}(Y^+(\widehat{\mathfrak{gl}}_1))) \;\simeq\; \mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1)).$$

(ii) At the W-algebra level, $Y(\widehat{\mathfrak{gl}}_1) \simeq W_{1+\infty}$ (Procházka–Rapčák), so the center is the braided representation category of $W_{1+\infty}$.

(iii) The character of the center is

$$\mathrm{ch}_{\mathcal{Z}(\mathrm{Rep}(Y^+))} = M(q)^2 \cdot P(q),$$

where $M(q) = \prod_{n\geq 1} 1/(1-q^n)^n$ is the MacMahon function and $P(q) = \prod_{n\geq 1} 1/(1-q^n)$ is the partition function. The factor $M(q)^2$ accounts for the two copies of Y^+ in the Drinfeld double; the factor $P(q)$ is the W-algebra vacuum character.

Proof. The identification (i) follows from the general principle: for an E_1-algebra A^+, the Drinfeld center of $\mathrm{Rep}^{E_1}(A^+)$ is $\mathrm{Rep}^{E_2}(\mathrm{Drin}(A^+))$, where $\mathrm{Drin}(A^+) = A^+ \bowtie A^-$ is the Drinfeld double. For $A^+ = Y^+(\widehat{\mathfrak{gl}}_1)$, the double is the full affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$ by construction. The E_2 braiding on $\mathrm{Rep}(Y)$ is induced by the universal R-matrix of the Yangian (which satisfies the YBE, see Theorem 9.5 below).

Part (ii) is the Procházka–Rapčák isomorphism $Y(\widehat{\mathfrak{gl}}_1) \simeq W_{1+\infty}$.

Part (iii): the Drinfeld double character is $\mathrm{ch}(Y^+) \cdot \mathrm{ch}(Y^-) = M(q)^2$, and the vacuum module contributes $P(q)$. □

Theorem 9.5 *(Yang R-matrix: YBE and unitarity).* [PH] Let h_1, h_2, h_3 satisfy $h_1 + h_2 + h_3 = 0$ and let

$$g(u) = \frac{(u-h_1)(u-h_2)(u-h_3)}{(u+h_1)(u+h_2)(u+h_3)}$$

be the structure function of the affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$. The R-matrix on $\mathrm{Fock}_2 \otimes \mathrm{Fock}_2$ (the charge-2 Fock space with basis $\{|\mathrm{row}\rangle, |\mathrm{col}\rangle\}$ indexed by partitions (2) and $(1,1)$) is an 8×8 matrix $R(u) \in \mathrm{End}(\mathbb{C}^2 \otimes \mathbb{C}^2)((u))$ satisfying:

(i) *Quantum Yang–Baxter equation*:

$$R_{12}(u-v)\, R_{13}(u)\, R_{23}(v) = R_{23}(v)\, R_{13}(u)\, R_{12}(u-v)$$

in $\mathrm{End}(\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2)((u,v))$. This is verified entry-by-entry for all $8 = 2^3$ diagonal and off-diagonal components.

(ii) *Unitarity*: $R_{12}(u)\, R_{21}(-u) = \mathrm{id}$.

(iii) *E_2 nontriviality*: $R(u) \neq \mathrm{id}$ whenever $h_1 h_2 h_3 \neq 0$, i.e., whenever the Calabi–Yau deformation parameters are generic. At the self-dual point $(h_1, h_2, h_3) = (1, 0, -1)$, the R-matrix degenerates to the identity and the E_2 structure reduces to E_∞.

(iv) *Classical limit*: $R(u) = 1 - \sigma_2/u + O(1/u^2)$ as $u \to \infty$, where $\sigma_2 = h_1 h_2 + h_1 h_3 + h_2 h_3$. The classical r-matrix is $r(u) = -\sigma_2/u$.

Proof. All four properties are verified by direct symbolic computation. The 8×8 YBE is checked by evaluating all $2^6 = 64$ matrix entries of the equation $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$ and confirming that each entry is a rational identity in u, v, h_1, h_2, h_3. Unitarity is checked by matrix multiplication. The classical limit follows from $g(u) = 1 - 2\sigma_2/u + O(1/u^2)$. See `compute/lib/drinfeld_center_yangian.py` (93 tests). □

Remark 9.6 (Computational verification). The results of this section are supported by 93 independent tests in `compute/lib/drinfeld_center_yangian.py`. Verification paths include: direct computation of the half-braiding on charge-1 and charge-2 Fock modules; symbolic verification of the 8×8 YBE; unitarity check $R_{12}(u)R_{21}(-u) = 1$; classical r-matrix extraction; Schiffmann–Vasserot specialization matching; and Procházka–Rapčák character comparison.

9.4 Comparison with the Maulik–Okounkov R-matrix

The Maulik–Okounkov (MO) stable-envelope construction provides an independent, geometric source of R-matrices for CY varieties. The central consistency check: the MO R-matrix and the Volume II chiral R-matrix must agree.

THEOREM 9.7 *(MO/chiral R-matrix comparison).* [PH] Let X be a CY threefold with torus action T, and let $R^{\mathrm{MO}}(u)$ be the Maulik–Okounkov R-matrix from stable envelopes on $\mathrm{Hilb}^n(X)$. Let $R^{\mathrm{ch}}(u)$ be the chiral R-matrix from the E_2-bar complex of the quantum chiral algebra A_X (Construction 9.3). Then:

(i) Both R-matrices act on the same space: the Fock representation $\mathcal{F} = \bigoplus_n K_T(\mathrm{Hilb}^n(X)^T)$, with basis indexed by partitions.

(ii) On each partition λ, both produce the same diagonal eigenvalue:

$$R^{\mathrm{MO}}_{\lambda\lambda}(u) = R^{\mathrm{ch}}_{\lambda\lambda}(u) = \prod_{s \in \lambda} g(u + c(s)),$$

where $c(s) = h_1 i + h_2 j$ is the content of box $s = (i,j) \in \lambda$.

(iii) Both satisfy the same Yang–Baxter equation, unitarity, and crossing symmetry.

Proof. Seven independent verification paths confirm the identification:

1. *Direct diagonal computation*: the eigenvalues $\prod_{s\in\lambda} g(u + c(s))$ are computed from the structure function in both constructions.

2. $\mathrm{Hilb}^2(\mathbb{C}^2)$ *check*: at $n = 2$, the MO construction uses stable envelopes on $\mathrm{Hilb}^2(\mathbb{C}^2)$ (two chambers in the equivariant torus); the resulting 2×2 R-matrix matches the Yangian R-matrix.

3. *Unitarity*: $R_{12}(u)R_{21}(-u) = \mathrm{id}$ holds for both.

4. *YBE*: both satisfy the same quantum Yang–Baxter equation (Theorem 9.5).

5. *Crossing symmetry*: the crossing relation $R_{12}(u)^{t_1} R_{12}(-u - \kappa)^{t_1} = f(u) \cdot \mathrm{id}$ holds for both, with the same scalar function $f(u)$.

6. *Classical r-matrix*: both yield $r(u) = -\sigma_2/u$ in the $u \to \infty$ limit.

7. *Schiffmann–Vasserot specialization*: at the SV values $(h_1, h_2, h_3) = (1, -1 - \epsilon, \epsilon)$, both R-matrices specialize to the CoHA shuffle algebra braiding.

See `compute/lib/mo_rmatrix_k3e.py` (104 tests via the combined comparison suite). □

Remark 9.8. The comparison in Theorem 9.7 is proved for $X = \mathbb{C}^3$ (and its toric CY degenerations where Hilbert scheme descriptions are available) and for $K3 \times E$ (where the MO stable envelopes are computed on $\mathrm{Hilb}^n(K3)$; see `compute/lib/mo_rmatrix_k3e.py`). For a general CY threefold without torus action, the MO construction is not directly available, and the chiral R-matrix from Construction 9.3 is the primary object.

9.5 Quantum chiral algebras from quantum groups

9.6 The quantum chiral Koszul dual

Chapter 10

The Modular Trace

10.1 CY trace as modular characteristic

The CY trace $\mathrm{Tr}\colon \mathrm{HH}_d(C) \to k$ determines the modular characteristic $\kappa(A_C)$.

THEOREM 10.1 *(CY modular characteristic — Theorem CY-D).* [PH] For a CY category C of dimension $d = 2$ with quantum chiral algebra $A_C = \Phi(C)$:

(i) The modular characteristic $\kappa(A_C) = \chi^{\mathrm{CY}}(C)$, the CY Euler characteristic;

(ii) The genus-g obstruction satisfies $\mathrm{obs}_g(A_C) = \kappa(A_C) \cdot \lambda_g$ on the uniform-weight lane; unconditionally at genus 1 for all families (Vol I, Theorem D); at genus $g \geq 2$ for multi-weight algebras, the scalar formula receives a nonvanishing cross-channel correction $\delta F_g^{\mathrm{cross}}$ (Vol I, op:multi-generator-universality, resolved negatively; $\delta F_2(\mathcal{W}_3) = (c{+}204)/(16c) > 0$);

(iii) The shadow obstruction tower of A_C encodes the higher-genus CY invariants of C.

10.2 CY Euler characteristic versus modular characteristic

The modular characteristic κ is an invariant of the quantum chiral algebra A_C, not of the underlying topology. It differs from the topological Euler characteristic χ in every known case.

PROPOSITION 10.2 *($\chi/24 \neq \kappa$ in general).* [PH] The ratio $\chi_{\mathrm{top}}/24$ and the modular characteristic $\kappa(A_C)$ disagree for most CY threefolds. The following table records all cases where both are known:

X	$\chi_{\mathrm{top}}(X)$	$\chi_{\mathrm{top}}/24$	$\kappa(A_X)$	Source
Elliptic curve E	0	0	1	$\kappa(H_1) = 1$
$K3$ surface	24	1	2	$\kappa = \mathrm{rank} = 2$
$K3 \times E$	0	0	5	$\mathrm{weight}(\Delta_5) = 5$
Conifold	2	1/12	1	Resolved conifold

In particular: $\chi_{\mathrm{top}}/24 \neq \kappa$ in every case except accidentally.

Proof. Each entry is computed independently. For E: the quantum chiral algebra is the Heisenberg H_1 with $\kappa = 1$ (the level), while $\chi_{\mathrm{top}}(E) = 0$. For $K3$: the quantum chiral algebra is the rank-2 lattice VOA with $\kappa = \mathrm{rank} = 2$, while $\chi_{\mathrm{top}}/24 = 1$. For $K3 \times E$: $\chi_{\mathrm{top}}(K3 \times E) = \chi(K3) \cdot \chi(E) = 24 \cdot 0 = 0$ while $\kappa = 5 = \mathrm{weight}(\Delta_5)$. For the conifold: the resolved conifold has $\chi_{\mathrm{top}} = 2$ (the total space of $\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{P}^1$ deformation retracts onto the zero section $\mathbb{P}^1$, so $\chi_{\mathrm{top}} = \chi(\mathbb{P}^1) = 2$), giving $\chi_{\mathrm{top}}/24 = 1/12$, while $\kappa = 1$.

See `compute/lib/cy_euler.py` (86 tests) and `compute/lib/modular_cy_characteristic.py` (80 tests). □

Remark 10.3 (Categorical versus topological Euler characteristic). The CY Euler characteristic $\chi^{\mathrm{CY}}(\mathcal{C})$ appearing in Theorem 10.1 is the categorical trace $\mathrm{Tr}_{\mathrm{HH}_d(\mathcal{C})}(\mathrm{id})$, which is in general *not* the topological Euler characteristic $\chi_{\mathrm{top}}(X)$. The former depends on the CY category $\mathcal{C}$ (and in particular on the complex structure and derived category), while the latter depends only on the underlying topological manifold.

Proposition 10.4 *(Non-multiplicativity of κ).* [PH] The modular characteristic is not multiplicative under products:

$$\kappa(K3) \cdot \kappa(E) = 2 \cdot 1 = 2, \qquad \kappa(K3 \times E) = 5.$$

The discrepancy $5 \neq 2$ reflects the fact that the BKM superalgebra $\mathfrak{g}_{\Delta_5}$ for $K3 \times E$ is *not* the tensor product of the $K3$ and E chiral algebras. The weight $5 = \dim(\mathrm{Sp}_4)/2$ is determined by the Borcherds product structure, not by a product decomposition.

10.3 Modularity constraints on the shadow tower

When a BKM denominator identity controls the shadow obstruction tower of a CY_3 category $\mathcal{C}$, the automorphic form imposes rigid constraints on the tower amplitudes.

Theorem 10.5 *(BKM modularity constraints).* [PH] Let $\mathcal{C}$ be a CY_3 category whose DT partition function $Z^{\mathrm{DT}}(\mathcal{C})$ is controlled by a Siegel modular form Φ of weight κ for an arithmetic group $\Gamma \subset \mathrm{Sp}_4(\mathbb{Q})$. Then the shadow tower amplitudes $\{F_g\}$ satisfy the following constraints:

(i) *Integrality*: $\kappa \in \mathbb{Z}_{>0}$, since κ is the weight of a Siegel modular form. In particular, the fractional values $\kappa = -25/3$ (quintic threefold, $\chi = -200$) and $\kappa = -1/6$ (banana configuration) are ruled out from admitting BKM lifts.

(ii) *Fourier–Jacobi structure*: $\Phi(\tau, z, \omega) = \sum_{m \geq 0} \phi_m(\tau, z)\, e^{2\pi i m\omega}$, where each ϕ_m is a Jacobi form of weight κ and index m. The shadow tower arity-r contribution maps to the Fourier–Jacobi expansion:

$$\begin{aligned} \text{arity } 2\ (\kappa) &\longleftrightarrow \phi_0 \quad \text{(constant term, Eisenstein part)},\\ \text{arity } 3\ (\text{cubic shadow}) &\longleftrightarrow \phi_1 \quad \text{(first FJ coefficient)}. \end{aligned}$$

(iii) *Discriminant-arity correspondence*: for the Jacobi form $\phi_{0,1}$ controlling the Borcherds lift, the discriminant $D = r^2 - 4nm$ of a Fourier coefficient $c(n, r)$ classifies the shadow arity:

$$\begin{aligned} D < 0 \quad &(\text{real}): && \text{arity 2 (Kac–Moody roots)},\\ D = 0 \quad &(\text{lightlike}): && \text{arity 3 (affine roots)},\\ D > 0 \quad &(\text{timelike}): && \text{arity } \geq 4 \text{ (imaginary roots)}. \end{aligned}$$

Proof. Part (i): the weight of a Siegel modular form for $\mathrm{Sp}_4(\mathbb{Z})$ (or a congruence subgroup) is a non-negative integer. The values $\kappa = -25/3$ and $\kappa = -1/6$ are not integers, hence no Siegel modular form of that weight exists.

Part (ii): the Fourier–Jacobi expansion is standard (Eichler–Zagier). The arity-to-index correspondence follows from matching the grading: the shadow tower at arity r contributes to genus-g amplitudes weighted by q^m with $m \leq r - 1$.

Part (iii): the discriminant classification follows from the root-type decomposition of the BKM superalgebra: real roots ($D < 0$) correspond to simple-pole OPE singularities (Kac–Moody type, arity 2), lightlike roots ($D = 0$) to affine null-root contributions (arity 3), and imaginary roots ($D > 0$) to higher-multiplicity contributions (arity ≥ 4). The multiplicity $\mathrm{mult}(D)$ of imaginary roots at discriminant $D > 0$ encodes the quartic and higher shadow corrections.

See `compute/lib/cy3_modularity_constraints.py` (111 tests). □

Corollary 10.6 *(Structural constraints for $K3 \times E$).* [PH] For $K3 \times E$ with $\kappa = 5$ and controlling form $\Delta_5 \in S_5(\mathrm{Sp}_4(\mathbb{Z}))$, the shadow tower satisfies seven structural constraints:

(1) $\kappa = 5 \in \mathbb{Z}_{>0}$;

(2) the tower amplitudes F_g lie in the Fourier–Jacobi ring of $\mathrm{Sp}_4(\mathbb{Z})$;

(3) the Hecke eigenvalues of Δ_5 constrain F_g multiplicatively;

(4) the functional equation of Δ_5 imposes a symmetry on the tower;

(5) the Borcherds product representation constrains the root multiplicities;

(6) the Petersson norm of Δ_5 bounds the growth rate of F_g;

(7) the theta correspondence $\phi_{0,1} \mapsto \Delta_5$ factors the tower through the Jacobi form $\phi_{0,1}$.

10.4 Genus expansion and Gromov–Witten invariants

10.5 The CY shadow obstruction tower

10.6 Complementarity for CY pairs

Chapter 11

Quantum Groups: Foundations

11.1 Quantized enveloping algebras

11.2 The R-matrix and Yang–Baxter equation

11.3 Representation categories

11.4 Drinfeld–Jimbo vs Faddeev–Reshetikhin–Takhtajan

Chapter 12

Braided Factorization

12.1 Braided monoidal categories and E_2-algebras

12.2 Factorization homology with E_2-coefficients

12.3 The braided bar-cobar adjunction

THEOREM 12.1 *(E_2-bar-cobar — Theorem CY-B).* For an E_2-chiral algebra $\mathcal{A}$, the bar-cobar adjunction

$$B_{E_2}\colon E_2\text{-ChirAlg} \rightleftarrows E_2\text{-ChirCoalg}\colon \Omega_{E_2}$$

satisfies:

(i) The E_2 bar complex $B_{E_2}(\mathcal{A})$ is a factorization E_2-coalgebra;

(ii) Cobar-bar inversion: $\Omega_{E_2}(B_{E_2}(\mathcal{A})) \simeq \mathcal{A}$ on the Koszul locus;

(iii) The CY trace manifests as curvature: at genus $g \geq 1$, $d^2 = \kappa \cdot \omega_g$.

Remark 12.2 (Status of Theorem CY-B). Items (i)–(iii) follow from the Volume I bar-cobar machine (Theorems A and B) applied to each E_1-direction separately, together with compatibility of the two E_1-structures. The full E_2-equivariant refinement (that the R-matrix intertwines the two bar-cobar adjunctions coherently) is a rigorous proof sketch, not a fully written proof. The identification of the automorphic correction with the shadow obstruction tower (claimed in Theorem CY-B for $K3 \times E$) is conditional on Conjecture CY-A$_3$.

12.4 Quantum group structure from braided Koszul duality

Chapter 13

The Drinfeld Center and Bulk Algebras

13.1 The Drinfeld center of a monoidal category

Definition 13.1 *(Drinfeld center).* For a monoidal category C, the *Drinfeld center* $\mathcal{Z}(C)$ is the category of pairs $(X, \beta_{X,-})$ where $X \in C$ and $\beta_{X,-} : X \otimes (-) \xrightarrow{\sim} (-) \otimes X$ is a half-braiding satisfying the hexagon axiom. The Drinfeld center is always braided monoidal.

13.2 Drinfeld center as chiral derived center

Proposition 13.2. For an E_1-chiral algebra A with monoidal representation category $C = \mathrm{Rep}^{E_1}(A)$, the Drinfeld center $\mathcal{Z}(C)$ is equivalent to the representation category of the chiral derived center:

$$\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \simeq \mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A)).$$

This is the bulk-boundary correspondence: the E_1 boundary theory determines an E_2 bulk theory via the Drinfeld center.

13.3 CY enhancement of the Drinfeld center

13.4 Quantum group reconstruction from the center

Part III

The CY Landscape

Chapter 14

Toroidal and elliptic algebras

The chiral algebras of Vols I–II each depend on a single deformation parameter: the level k, the central charge c, or the spectral parameter u. This chapter enters the regime where one parameter is not enough. The toroidal algebra $U_{q,t}(\hat{\hat{\mathfrak{g}}})$ carries two independent parameters (q, t), the elliptic quantum group $E_{q,p}(\mathfrak{g})$ replaces the rational R-matrix by a doubly-periodic one, and the double affine Hecke algebra (DAHA) intertwines spectral and modular directions. In these algebras the modular parameter τ and the spectral parameter u are coupled through the Fay trisecant identity and the quasi-periodicity of theta functions.

The bar-cobar framework meets this two-parameter world in two ways. Direction (a) keeps the base a single curve, now of genus 1: the propagator becomes elliptic, the Arnold relation generalizes to the Fay trisecant identity (Theorem 14.18), and the bar differential acquires Eisenstein-series corrections indexed by τ. Direction (b) seeks genuine surface-factorization objects on $\mathbb{C}^* \times \mathbb{C}^*$, requiring a two-dimensional factorization framework beyond the curve-chiral hierarchy of Vols I–II. Only direction (a) is developed here.

The Fay identity replaces the Arnold relation at genus 1; it is both an identity of theta functions and a geometric consequence of the degeneration $E_\tau \to \mathbb{P}^1$ as $\tau \to i\infty$.

Remark 14.1 (Two programme tracks). Track (a) keeps the base an algebraic curve and constructs an elliptic E_1-chiral realization on E_τ. Track (b) seeks a genuine surface-factorization object on $\mathbb{C}^* \times \mathbb{C}^*$. Only track (a) is developed here; track (b) is a completion target.

Remark 14.2 (The generalization principle: Arnold to Fay). The Arnold relation (Vol I, Theorem III), which governs the genus-0 bar differential, admits two distinct generalizations. Direction (a): via clutching morphisms and the shadow obstruction tower $\Theta_A^{\leq r}$ (Vol I, Definition III), one passes to the modular regime on stable curves, recovering the higher-genus bar complex developed in Vol I, Part I (Vol I, Remark III). Direction (b): via the Fay trisecant identity (Theorem 14.18), one replaces the additive (logarithmic) propagator $\omega = dz/(z-w)$ by the elliptic (multiplicative) propagator built from the Weierstrass σ-function, entering the toroidal regime. The present chapter develops direction (b): the systematic passage from logarithmic to elliptic propagators and its consequences for bar-cobar duality.

Remark 14.3 (Rational, trigonometric, elliptic: curve geometry). The three quantum group families arise from the geometry of the underlying curve:

R-matrix type	Quantum group	Spectral parameter	Curve
Rational	Yangian $Y(\mathfrak{g})$	$u = z_1 - z_2$	$\mathbb{P}^1$
Trigonometric	Quantum loop $U_q(L\mathfrak{g})$	$q = e^{2\pi i(z_1-z_2)/\beta}$	$\mathbb{C}^*$
Elliptic	Elliptic QG $E_{q,p}(\mathfrak{g})$	$u \in E_\tau$	E_τ

The passage rational → trigonometric → elliptic parallels additive → multiplicative → modular spectral parameter.

Construction 14.4 *(Elliptic shadow data).* Toroidal and elliptic quantum group algebras carry shadow data in the elliptic regime:

(i) The elliptic propagator $K^{\mathrm{ell}}(z, w; \tau) = \vartheta_1(z - w; \tau)/\vartheta_1'(0; \tau)$ replaces the rational propagator $1/(z - w)$.

(ii) The Fay trisecant identity for ϑ_1 is the genus-1 MC equation $D^{(1)}\Theta + \frac{1}{2}[\Theta^{(0)}, \Theta^{(0)}] + \kappa\,\omega_1 = 0$.

(iii) The Arnold relation $K_{12}K_{23} + K_{23}K_{31} + K_{31}K_{12} = 0$ (Fay identity) specializes the CYBE to the elliptic regime.

14.1 Double affine setup

Definition

Definition 14.5 *(Toroidal algebra).* The *toroidal algebra* (or double affine algebra) $U_{q,t}(\hat{\hat{\mathfrak{g}}})$ is generated by:

- Horizontal currents $x^{\pm}_{i,n}$, $\psi^{\pm}_{i,n}$ (affine in one direction)
- Vertical currents $e_i(z)$, $f_i(z)$, $\psi_i(z)$ (affine in the other direction)

subject to:

1. *Drinfeld–type relations* in each direction: $[h_{i,m}, x^{\pm}_{j,n}] = \pm a_{ij} x^{\pm}_{j,m+n}$, $[x^+_{i,m}, x^-_{j,n}] = \delta_{ij}(\psi^+_{i,m+n} - \psi^-_{i,m+n})/(q_i - q_i^{-1})$
2. *Cross relations* coupling horizontal and vertical: $\psi_i(z)e_j(w) = g_{ij}(z/w)\, e_j(w)\psi_i(z)$ where $g_{ij}(u) = (u - q^{a_{ij}})/(u - q^{-a_{ij}})$
3. *Serre relations* in both directions

with two independent parameters q and $t = q^k$. See Ginzburg–Kapranov–Vasserot for the complete presentation.

PROPOSITION 14.6 *(Toroidal OPE [?]; [PE]).* The OPE for toroidal currents involves both parameters:

$$e_i(z)f_j(w) = \frac{\delta_{ij}}{(1-q)(1-t^{-1})} \cdot \frac{\psi_i^+(w) - \psi_i^-(w)}{z - w} + \text{rational terms in } q, t$$

Proof. We derive the OPE from the defining relations of Definition 14.5.

Step 1: Mode commutator. The Drinfeld-type relation (1) gives:

$$[e_{i,m}, f_{j,n}] = \delta_{ij} \frac{\psi^+_{i,m+n} - \psi^-_{i,m+n}}{q_i - q_i^{-1}}.$$

Form the generating functions $e_i(z) = \sum_m e_{i,m} z^{-m-1}$, $f_j(w) = \sum_n f_{j,n} w^{-n-1}$, and $\psi_i^{\pm}(w) = \sum_n \psi^{\pm}_{i,n} w^{-n}$.

Step 2: Extracting the OPE. Contracting the commutator against $z^{-m-1}w^{-n-1}$ and summing:

$$e_i(z)f_j(w) - {:}e_i(z)f_j(w){:} = \delta_{ij} \sum_m \frac{\psi^+_{i,m}(w) - \psi^-_{i,m}(w)}{q_i - q_i^{-1}} \cdot \frac{w^{-m}}{z - w}.$$

For the toroidal algebra with parameters q and $t = q^k$, we have $q_i = q^{d_i}$ where $d_i = (\alpha_i, \alpha_i)/2$. The singular part of the OPE is the residue at $z = w$:

$$e_i(z) f_j(w) \sim \frac{\delta_{ij}}{q_i - q_i^{-1}} \cdot \frac{\psi_i^+(w) - \psi_i^-(w)}{z - w}.$$

Step 3: Two-parameter structure. The cross relation (2) introduces the second parameter. From $\psi_i(z) e_j(w) = g_{ij}(z/w)\, e_j(w) \psi_i(z)$ with $g_{ij}(u) = (u - q^{a_{ij}})/(u - q^{-a_{ij}})$, the normal ordering of ψ against e, f currents produces corrections rational in both q and t. Specifically, expanding $1/(q_i - q_i^{-1}) = 1/((q^{d_i} - q^{-d_i}))$ and using $t = q^k$ to express the central element as $c = (1-q)(1-t^{-1})/(q_i - q_i^{-1})$ (the standard Ding–Iohara normalization). See [?] for the complete change-of-basis computation. □

Conjecture 14.7 *(Elliptic-curve toroidal realization; [CJ]).* The toroidal algebra $U_{q,t}(\hat{\hat{\mathfrak{g}}})$ admits a curve-based E_1-chiral realization on the elliptic curve E_τ (with modulus $\tau = \frac{\log t}{\log q}$).

Conjecture 14.8 *(Double-affine surface-factorization realization; [CJ]).* The same toroidal data should extend to a doubly-graded surface-factorization or double-affine object on $\mathbb{C}^* \times \mathbb{C}^*$, with one direction encoding the spectral parameter and the other the elliptic coordinate. (This is the toroidal/elliptic extension flank of Vol I, Conjecture III, not the standard W_∞/Yangian finite-packet lane.)

Remark 14.9 (Scope). Conjecture 14.7 is conjectural ([CJ]): a complete proof requires verifying associativity of the E_1-chiral product directly from the OPE relations, since the BD locality approach applies only to commutative chiral algebras. The $\mathbb{C}^* \times \mathbb{C}^*$ realization (Conjecture 14.8) is a separate surface-factorization problem within the MC4 elliptic/double-affine extension problem.

Homotopy template (Vol I, Convention III): Type I (existence).

Proof outline. We describe the conjectural E_1-chiral structure.

Step 1: R-matrix braiding. The cross relation (2) of Definition 14.5 gives the structure function $g_{ij}(z/w) = (z/w - q^{a_{ij}})/(z/w - q^{-a_{ij}})$. Define the R-matrix-valued braiding on generating currents by

$$\sigma_{12} \circ \mu(e_i(z), e_j(w)) = g_{ij}(z/w) \cdot \mu(e_j(w), e_i(z)).$$

Since $g_{ij}(u)$ is a rational function with $g_{ij}(u) \neq 1$ for generic q and $a_{ij} \neq 0$, the braiding is non-trivial: the algebra is associative but not commutative, hence strictly E_1.

Step 2: Associativity (gap). The R-matrix $R_{ij}(u) = g_{ij}(u) \cdot \mathrm{Id}$ satisfies the Yang–Baxter equation trivially (being scalar-valued). The genuine content of associativity lies in verifying that the full matrix-valued OPE structure, including the Serre-type relations of Definition 14.5, defines an algebra over $\mathsf{Ass}^{\mathrm{ch}}$ (Vol I, Definition III). This requires an explicit verification that triple OPEs are consistent, which we do not carry out here.

Step 3: Realization on E_τ. Set $\tau = \log t / \log q$ and let $E_\tau = \mathbb{C}^*/q^{\mathbb{Z}}$. The generating currents $e_i(z)$, $f_i(z), \psi_i(z)$ are meromorphic on $\mathbb{C}^*$ with quasi-periodicity $e_i(qz) = q^{d_i} e_i(z)$ (from the grading), so they define sections of line bundles on E_τ. The OPE of Proposition 14.6 has poles at $z/w \in q^{\mathbb{Z}}$, which descend to a single point on E_τ. This defines a candidate E_1-chiral algebra structure on E_τ, pending verification of Step 2.

Step 4: Surface-factorization horizon on $\mathbb{C}^* \times \mathbb{C}^*$. The horizontal modes $x_{i,n}^\pm$ depend on the spectral parameter u and the vertical modes $e_i(z)$ depend on z. Together they give fields on $\mathbb{C}_u^* \times \mathbb{C}_z^*$ with the two independent gradings (by horizontal and vertical mode number) providing the doubly-graded structure.

This belongs to a separate conjectural surface-factorization or double-affine track, not automatically a curve-based E_1-chiral algebra. A correct formulation would require a bona fide two-parameter factorization framework.

Conditional on Step 2 (i.e., the verification that the toroidal OPE defines an associative $\mathsf{Ass}^{\mathrm{ch}}$-algebra structure, which requires checking consistency of triple OPEs with the Serre-type relations), the E_1-chiral Koszul duality framework (Vol I, Theorem III) applies to the elliptic-curve realization of Conjecture 14.7. The $\mathbb{C}^* \times \mathbb{C}^*$ surface track belongs instead to Conjecture 14.8. □

Koszul dual of the toroidal algebra

CONJECTURE 14.10 *(Toroidal Koszul dual; [CJ])*. The E_1-chiral Koszul dual of the toroidal algebra $U_{q,t}(\hat{\hat{\mathfrak{g}}})$ is isomorphic to $U_{q^{-1},t^{-1}}(\hat{\hat{\mathfrak{g}}})$, the toroidal algebra with inverted parameters. (Contributing to the toroidal/elliptic extension flank of Vol I, Conjecture III.)

Remark 14.11 (Evidence). Conditional on Conjecture 14.7, the evidence is threefold:

(i) *RTT presentation.* By E_1-chiral Koszul duality (Vol I, Theorem III), the dual has R-matrix $R(u;q,t)^{-1} = R(u;q^{-1},t^{-1})$ (Ding–Iohara inversion; cf. Vol I, Theorem III).

(ii) *Affine degeneration.* In the limit $t \to 0$, $U_{q,t}(\hat{\hat{\mathfrak{g}}}) \to U_q(\hat{\mathfrak{g}})$ and the dual reduces to $U_{q^{-1}}(\hat{\mathfrak{g}})$, consistent with Vol I, Theorem III.

(iii) *Central charge complementarity.* The inversion $q \to q^{-1}$ sends $\tau \to -\tau$, reversing the complex structure of E_τ, the elliptic analogue of the Verdier intertwining (Vol I, Theorem A; Theorem III).

The Fay identity (Proposition 14.19) serves as the elliptic Arnold relation. The universal MC class predicts $\Theta_{U_{q,t}} + \Theta_{U_{q^{-1},t^{-1}}} = 0$.

Homotopy template (Vol I, Convention III): Type II (equivalence).

Connection to the three main theorems of Vol I

Remark 14.12 (Toroidal three theorems). Conditional on Conjecture 14.7, the expected toroidal analogues of the three main theorems of Vol I are: *Theorem A* (Vol I, bar-cobar adjunction): $\bar{B}^{\mathrm{ell}}(U_{q,t})$ computes the Koszul dual coalgebra (nilpotency via Proposition 14.19). *Theorem B* (Vol I, bar-cobar inversion): $\Omega(\bar{B}(U_{q,t})) \xrightarrow{\sim} U_{q,t}$ by the rational spectral sequence with elliptic corrections (Theorem 14.22). *Theorem C* (Vol I, complementarity): $\mathrm{obs}_g(U_{q,t}) + \mathrm{obs}_g(U_{q^{-1},t^{-1}})$ controlled by H^* of a framed moduli space (the E_1 replacement for $\overline{\mathcal{M}}_g$; cf. Vol I, Remark III). The universal MC class $\Theta_{U_{q,t}}$ should govern the genus expansion, with $(q,t) \mapsto (q^{-1},t^{-1})$ reflected in Verdier duality.

Remark 14.13 (Lattice evidence for toroidal genus theory). The curvature-braiding orthogonality theorem for quantum lattice VOAs (Vol I, Theorem III) provides structural evidence for the toroidal case. For the lattice, the genus-1 curvature $\kappa = \mathrm{rank}(\Lambda)$ lives entirely in the Cartan (Heisenberg) sector and is independent of the deformation cocycle $\varepsilon_{N,q}$. The toroidal algebra $U_{q,t}(\hat{\hat{\mathfrak{g}}})$ has a similar double-loop structure: the inner loop $(\hat{\mathfrak{g}})$ is the Cartan analog, while the outer loop (the second affinization) provides the braiding. The orthogonality principle predicts that the toroidal genus-1 curvature should be controlled by the inner-loop data alone, with the dynamical parameter $\lambda \in H^1(E_\tau)$ entering the braiding axis rather than the modular axis. The dynamical YBE for $R(u,\lambda)$ would then be the toroidal analog of the modified Arnold relation at genus 1, with the dynamical shift $\lambda \to \lambda - h^{(i)}$ replacing the B-cycle monodromy correction.

14.2 Elliptic quantum groups

Felder's elliptic quantum group

Definition 14.14 *(Elliptic quantum group).* The *elliptic quantum group* $E_{q,p}(\mathfrak{g})$ (Felder) is defined by:

- The dynamical R-matrix $R(u, \lambda)$ depending on spectral parameter u and dynamical parameter λ.
- The RLL relations:

$$R(u-v,\lambda)L_1(u,\lambda)L_2(v,\lambda-h^{(1)}) = L_2(v,\lambda)L_1(u,\lambda-h^{(2)})R(u-v,\lambda)$$

Definition 14.15 *(Elliptic R-matrix).* The elliptic R-matrix for $\mathfrak{sl}_2$ is:

$$R(u,\lambda) = \begin{pmatrix} a(u) & 0 & 0 & 0 \\ 0 & b(u,\lambda) & c(u,\lambda) & 0 \\ 0 & c(u,-\lambda) & b(u,-\lambda) & 0 \\ 0 & 0 & 0 & a(u) \end{pmatrix}$$

where $a(u), b(u,\lambda), c(u,\lambda)$ are expressed via theta functions:

$$\begin{aligned} a(u) &= \frac{\theta(u+\eta)}{\theta(\eta)} \\ b(u,\lambda) &= \frac{\theta(u)\theta(\lambda+\eta)}{\theta(\eta)\theta(\lambda)} \\ c(u,\lambda) &= \frac{\theta(u+\lambda)\theta(\eta)}{\theta(\lambda)\theta(u+\eta)} \end{aligned}$$

Remark 14.16 (Elliptic quantum groups and bar complex)*.* We expect ([CJ], conditional on Conjecture 14.7): (i) $\overline{B}(U_{q,t})$ on E_τ computes the chiral coalgebra $E_{q,p}(\mathfrak{g})$ (elliptic Kazhdan–Lusztig); (ii) the dynamical parameter λ (Definition 14.14) is the B-cycle monodromy of the elliptic propagator (elliptic analogue of $k \to k + h^\vee$); (iii) the dynamical YBE for $R(u,\lambda)$ is the elliptic deformation of $d^2 = 0$ (Proposition 14.19), with the dynamical parameter from $H^1(E_\tau) \neq 0$.

14.3 Elliptic R-matrices and theta functions

Theta function identities

Definition 14.17 *(Jacobi theta function).* The Jacobi theta function is:

$$\theta(u|\tau) = \sum_{n\in\mathbb{Z}} (-1)^n e^{\pi i\tau n(n-1)+2\pi i n u} = (e^{2\pi i u}; p)_\infty (pe^{-2\pi i u}; p)_\infty (p;p)_\infty$$

where $p = e^{2\pi i\tau}$ and $(a;q)_\infty = \prod_{n\geq 0}(1-aq^n)$.

THEOREM 14.18 *(Fay identity at genus 1 [**?**]; [PE]).* For any four points u_1, u_2, u_3, u_4 on an elliptic curve E_τ, the theta function satisfies the Fay trisecant identity:

$$\theta(u_1-u_2)\,\theta(u_3-u_4) - \theta(u_1-u_3)\,\theta(u_2-u_4) + \theta(u_1-u_4)\,\theta(u_2-u_3) = 0$$

This is a *bilinear* identity (each term is a product of two theta functions), which at genus 1 is the degenerate case of the general Fay trisecant identity on higher-genus curves. The identity is equivalent to the Riemann addition formula and reduces to the Ptolemy relation $\sin(\alpha-\beta)\sin(\gamma-\delta)-\sin(\alpha-\gamma)\sin(\beta-\delta)+\sin(\alpha-\delta)\sin(\beta-\gamma)=0$ in the rational limit $\tau\to i\infty$.

This identity underlies the Yang–Baxter equation for elliptic R-matrices: the star-triangle relation is a consequence of the Fay identity applied to the structure functions of the elliptic algebra.

Fay identity and bar nilpotency

Proposition 14.19 *(Fay identity implies elliptic $d^2=0$; [PH])*. On $\overline{C}_3(E_\tau)$, the elliptic bar differential satisfies $d^2=0$. The key algebraic input is the Fay trisecant identity (Theorem 14.18), which plays the role of the Arnold relation in the rational case (Vol I, Theorem III).

Proof. The elliptic bar differential on degree-2 elements uses the propagator $\eta_{ij}^{\mathrm{ell}}=d\log\theta_1(z_i-z_j|\tau)$. The computation of d^2 on a degree-1 element $[a]\otimes 1$ produces terms on $\overline{C}_3(E_\tau)$ involving the 2-form $\eta_{12}^{\mathrm{ell}}\wedge\eta_{23}^{\mathrm{ell}}+\eta_{23}^{\mathrm{ell}}\wedge\eta_{31}^{\mathrm{ell}}+\eta_{31}^{\mathrm{ell}}\wedge\eta_{12}^{\mathrm{ell}}$, exactly as in the rational case (cf. Vol I, §III).

The rational Arnold relation $\eta_{12}\wedge\eta_{23}+\eta_{23}\wedge\eta_{31}+\eta_{31}\wedge\eta_{12}=0$ (where $\eta_{ij}=d\log(z_i-z_j)$) is a consequence of the partial fraction identity $(z_1-z_2)^{-1}(z_2-z_3)^{-1}+\text{cyclic}=0$.

The elliptic replacement for $(z_i-z_j)^{-1}$ is $\zeta_\tau(z_i-z_j):=\partial_z\log\theta_1(z_i-z_j|\tau)$, the Weierstrass ζ-function on E_τ. The corresponding three-term identity is

$$d\zeta_\tau(z_1-z_2)\wedge d\zeta_\tau(z_2-z_3)+d\zeta_\tau(z_2-z_3)\wedge d\zeta_\tau(z_3-z_1)+d\zeta_\tau(z_3-z_1)\wedge d\zeta_\tau(z_1-z_2)=0. \quad (14.1)$$

This is the differential-form shadow of the Fay identity (Theorem 14.18). Differentiate

$$\theta(u_1-u_2)\,\theta(u_3-u_4)-\theta(u_1-u_3)\,\theta(u_2-u_4)+\theta(u_1-u_4)\,\theta(u_2-u_3)=0$$

with respect to u_1 and u_4, then set $u_4=u_1$. The resulting trisecant consequence is

$$\begin{aligned}\zeta_\tau(u_1-u_2)\,\zeta_\tau(u_2-u_3)+\zeta_\tau(u_2-u_3)\,\zeta_\tau(u_3-u_1)+\zeta_\tau(u_3-u_1)\,\zeta_\tau(u_1-u_2)&\\ =\wp_\tau(u_1-u_2)+\wp_\tau(u_2-u_3)+\wp_\tau(u_3-u_1)&\end{aligned}$$

where $\wp_\tau=-\zeta_\tau'$ is the Weierstrass $\wp$-function. Now take exterior derivatives in the difference variables. Each $\wp_\tau(z_i-z_j)$ depends on a single difference, so the mixed wedges on the right vanish or cancel pairwise. The left-hand side is then exactly (14.1).

The remaining terms in d^2 (involving OPE data of the chiral algebra $\mathcal{A}$) cancel by the same mechanism as in the rational case: the bar differential decomposes as $d=d_{\mathrm{local}}+d_{\mathrm{global}}$, where $d_{\mathrm{local}}^2=0$ by the OPE associativity of $\mathcal{A}$, and the cross-terms $d_{\mathrm{local}}\circ d_{\mathrm{global}}+d_{\mathrm{global}}\circ d_{\mathrm{local}}=0$ by the compatibility of the OPE with the propagator (which holds for both the rational and elliptic propagators, since the residue at a simple pole is the same: $\mathrm{Res}_{z=0}\,\zeta_\tau(z)\,dz=1$). □

Remark 14.20 (Rational limit). In the rational limit $\tau\to i\infty$, the theta function $\theta_1(z|\tau)\to 2\sin(\pi z)\to 2\pi z$ (for $|z|$ small), so $\zeta_\tau(z)\to 1/z$ and $\wp_\tau(z)\to 1/z^2$. The identity (14.1) reduces to the Arnold relation $\eta_{12}\wedge\eta_{23}+\eta_{23}\wedge\eta_{31}+\eta_{31}\wedge\eta_{12}=0$ (Vol I, Theorem III), and the bar nilpotency reduces to Vol I, Theorem III.

Bar complex with theta functions

Construction 14.21 *(Elliptic bar complex).* The bar complex of an elliptic chiral algebra incorporates theta functions:

Forms. Replace $d\log(z_1 - z_2)$ with:

$$\eta_{12} = d\log\theta(z_1 - z_2|\tau)$$

the logarithmic derivative of theta.

Differential. The residue is computed at the zeros of θ:

$$d_{\mathrm{res}}([a|b] \otimes \eta_{12}) = \sum_{\theta(z_0)=0} \mathrm{Res}_{z_1=z_0+z_2}[\cdots]$$

The zeros of $\theta(u|\tau)$ are at $u = m + n\tau$ for $m, n \in \mathbb{Z}$ (the lattice points).

THEOREM 14.22 *(Elliptic vs rational homology;* [PH]*).* Let $\mathcal{A}$ be a chiral algebra of central charge c on an elliptic curve E_τ. The homology of the elliptic bar complex admits a decomposition:

$$H_n(\bar{B}^{\mathrm{ell}}(\mathcal{A})) \;\cong\; H_n(\bar{B}^{\mathrm{rat}}(\mathcal{A})) \;\oplus\; \bigoplus_{k\geq 1} \mathcal{E}_n^{(k)}(\mathcal{A},\tau)$$

where the elliptic corrections $\mathcal{E}_n^{(k)}(\mathcal{A},\tau)$ are $\mathcal{A}$-modules whose dependence on τ is through (quasi-)modular forms. Specifically:

(i) The leading correction $\mathcal{E}_n^{(1)}$ is proportional to $E_2(\tau)$, the weight-2 quasi-modular Eisenstein series, arising from the regularization of the elliptic propagator $\partial_z \log\theta(z|\tau)$ relative to the rational propagator $1/z$ (cf. Construction 14.21).

(ii) Higher corrections $\mathcal{E}_n^{(k)}$ for $k \geq 2$ lie in the space of modular forms $M_{2k}(\mathrm{SL}_2(\mathbb{Z}))$ of weight $2k$; the first truly modular correction is an $E_4(\tau)$-coefficient.

(iii) The conformal dimension h of the states entering the bar complex constrains the weights: a state of dimension h contributes to $\mathcal{E}_n^{(k)}$ only for $k \leq h + n$.

(iv) At the rational limit $\tau \to i\infty$ (i.e., $q = e^{2\pi i\tau} \to 0$), the Eisenstein series $E_{2k}(\tau) \to 1$ and all corrections degenerate to constants, recovering $H_n(\bar{B}^{\mathrm{rat}}(\mathcal{A}))$ up to scalar renormalization.

For the Heisenberg algebra ($c = 1$), the corrections reduce to the Eisenstein series hierarchy computed in Vol I, §III.

Proof. We construct the decomposition via a perturbative spectral sequence comparing the elliptic and rational bar differentials.

Step 1 (Propagator decomposition). The elliptic bar complex (Construction 14.21) uses the propagator $\eta_{ij}^{\mathrm{ell}} = d\log\theta_1(z_i - z_j|\tau)$, while the rational bar complex uses $\eta_{ij}^{\mathrm{rat}} = d\log(z_i - z_j)$. Near $z_i = z_j$, both propagators have the same simple pole with residue 1, so their difference $\delta_{ij}(\tau) := \eta_{ij}^{\mathrm{ell}} - \eta_{ij}^{\mathrm{rat}}$ is a regular 1-form. The classical expansion of the Weierstrass ζ-function $\zeta(z;\Lambda_\tau)$ in terms of the Eisenstein–Weierstrass series $G_{2k}(\tau)$ gives the Laurent expansion:

$$\partial_z \log \theta_1(z|\tau) \;=\; \frac{1}{z} + \sum_{k=1}^{\infty} c_k(\tau)\, z^{2k-1},$$

where the correction coefficients $c_k(\tau)$ are expressed in terms of Eisenstein series via $G_{2k}(\tau) = 2\zeta_{\mathrm{R}}(2k)\, E_{2k}(\tau)$:

- $c_1(\tau) = -\frac{\pi^2}{3} E_2(\tau) \in \widetilde{M}_2(\mathrm{SL}_2(\mathbb{Z}))$ (quasi-modular, weight 2), since $c_1 = -2\eta_1$ where $\eta_1 = \zeta(1/2; \Lambda_\tau) = \frac{\pi^2}{6} E_2(\tau)$;
- $c_k(\tau) = -G_{2k}(\tau) \in M_{2k}(\mathrm{SL}_2(\mathbb{Z}))$ for $k \geq 2$ (holomorphic modular forms).

The expansion converges absolutely for $|z| < \min_{\omega \in \Lambda_\tau \setminus \{0\}} |\omega|$.

Step 2 (Correction-order filtration). Define the *correction-order filtration* $\mathcal{F}^p \bar{B}^{\mathrm{ell}}_n(\mathcal{A})$ by declaring that an element has filtration degree $\geq p$ if it arises from at least p insertions of the regular correction terms $\{c_k(\tau)\, z^{2k-1}\}_{k \geq 1}$ in the propagator expansion. Equivalently, replace each propagator by $1/z + t \cdot (c_1 z + c_2 z^3 + \cdots)$ and expand in powers of the formal parameter t; then $\mathcal{F}^p$ consists of terms of order $\geq t^p$.

Since the bar differential d^{ell} at degree n involves at most n propagator factors, we have $\mathcal{F}^{n+1} \bar{B}^{\mathrm{ell}}_n = 0$. The filtration is compatible with the differential: $d^{\mathrm{ell}}(\mathcal{F}^p) \subset \mathcal{F}^p$ (inserting the bar differential does not decrease the number of correction insertions).

Step 3 (Spectral sequence: E_0 and E_1 pages). The associated graded $\mathrm{gr}^0_{\mathcal{F}} \bar{B}^{\mathrm{ell}}(\mathcal{A})$ is the bar complex computed using only the singular part $1/z$ of the propagator. Since the residue extraction in the bar differential (which encodes the OPE data of $\mathcal{A}$) depends only on the polar part, the differential on gr^0 coincides with the rational bar differential d^{rat}. Thus:

$$E_1^{0,n} = H_n(\bar{B}^{\mathrm{rat}}(\mathcal{A})).$$

More generally, $E_1^{p,n} = H_n(\bar{B}^{\mathrm{rat}}(\mathcal{A})) \otimes \mathrm{Sym}^p_{\geq 1}\{c_k(\tau)\}$, where the symmetric product is taken over all propagator correction coefficients contributing total weight p.

Step 4 (Modular structure via Zhu's theorem). The differential $d_r : E_r^{p,n} \to E_r^{p+r,n+1}$ on the r-th page arises from the interaction of r correction insertions with the bar differential. By Zhu's modular invariance theorem [?], the n-point genus-1 correlation functions $\langle V_1(z_1) \cdots V_n(z_n) \rangle_{E_\tau}$ of a rational vertex algebra $\mathcal{A}$ transform as components of a vector-valued modular form for $\mathrm{SL}_2(\mathbb{Z})$. Since the bar complex at degree n is built from such correlation functions composed with the propagator form, and the correction coefficients $c_k(\tau)$ are themselves (quasi-)modular, the differentials d_r respect the modular weight grading. Consequently, the E_∞ terms inherit a weight decomposition from the ring $\widetilde{M}_*(\mathrm{SL}_2(\mathbb{Z}))$ of quasi-modular forms.

Step 5 (Convergence and splitting). The spectral sequence converges because the filtration is bounded: $\mathcal{F}^{n+1} \bar{B}^{\mathrm{ell}}_n = 0$ at each bar degree n. At convergence, $H_n(\bar{B}^{\mathrm{ell}}(\mathcal{A}))$ admits the filtration

$$0 = \mathcal{F}^{n+1} H_n \subset \mathcal{F}^n H_n \subset \cdots \subset \mathcal{F}^1 H_n \subset \mathcal{F}^0 H_n = H_n(\bar{B}^{\mathrm{ell}}(\mathcal{A}))$$

with successive quotients $\mathcal{E}_n^{(k)} := E_\infty^{k,n}$. The filtration *splits*: the modular weight provides a grading on the coefficient ring (quasi-modular forms of different weights are linearly independent over $\mathbb{C}$), and the E_∞ terms in different weights cannot be related by extensions. We therefore obtain the direct sum decomposition $H_n(\bar{B}^{\mathrm{ell}}(\mathcal{A})) \cong H_n(\bar{B}^{\mathrm{rat}}(\mathcal{A})) \oplus \bigoplus_{k \geq 1} \mathcal{E}_n^{(k)}(\mathcal{A}, \tau)$ as claimed.

Properties (i)–(iv) now follow from the structure of the correction coefficients:

(i) The leading correction involves $c_1(\tau) = -(\pi^2/3)\,E_2(\tau)$ (quasi-modular, weight 2).

(ii) Higher corrections involve products of $c_k(\tau) \in M_{2k}(\mathrm{SL}_2(\mathbb{Z}))$ for $k \geq 2$.

(iii) A bar element of degree n involves at most n propagator factors; a state of conformal dimension h constrains the Laurent order of each propagator insertion to $\leq 2h - 1$, giving the bound $\mathcal{E}_n^{(k)} = 0$ for $k > h + n$.

(iv) As $\tau \to i\infty$: $q \to 0$ and $E_{2k}(\tau) \to 1$, so all corrections reduce to constants, recovering $H_n(\bar{B}^{\mathrm{rat}}(\mathcal{A}))$ up to renormalization.

Verification for the Heisenberg algebra. For $\mathcal{A} = \mathcal{H}$ ($c = 1$), the genus-1 two-point function is $\langle a(z_1)a(z_2)\rangle_{E_\tau} = \kappa\, G_\tau(z_1 - z_2)$ with $G_\tau(z) = \zeta_\tau(z) - (\pi^2 E_2(\tau)/3)\,z$ (Vol I, Theorem III). The leading correction $\mathcal{E}_n^{(1)} \propto E_2(\tau)$ arises from the c_1 term. The complete Eisenstein hierarchy of Vol I, §III gives $\mathcal{E}_n^{(k)} \propto E_{2k}(\tau)$, confirming the general pattern. □

Remark 14.23 (Scope). The proof assembles three ingredients: (i) the Laurent expansion of the Weierstrass ζ-function in terms of Eisenstein series ([PE], classical); (ii) the convergence of the correction-order spectral sequence (bounded filtration at each bar degree); and (iii) Zhu's modular invariance theorem for genus-1 n-point functions of rational vertex algebras [?] ([PE]). The combination and the identification of the splitting mechanism (Step 5) are new to this volume. The Heisenberg verification uses the explicit computations of Vol I, §III.

14.4 Explicit elliptic bar complex for the Heisenberg algebra

Bar elements and the elliptic propagator

We work on $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ with the elliptic propagator

$$\eta_{ij}^{\mathrm{ell}} = d\log\theta_1(z_i - z_j \mid \tau) = \zeta_\tau(z_i - z_j)\,(dz_i - dz_j), \tag{14.2}$$

where $\zeta_\tau(z) = \partial_z \log\theta_1(z|\tau)$ is the Weierstrass zeta function on E_τ.

Computation 14.24 *(Degree-1 bar elements).* In bar degree 1, the generators of $\bar{B}_1^{\mathrm{ell}}(\mathcal{H}_\kappa)$ are:

$$[a_m] \otimes 1 \in \bar{B}_1^{\mathrm{ell}}(\mathcal{H}_\kappa), \quad m \in \mathbb{Z},$$

where a_m denotes the mode of the Heisenberg current $a(z) = \sum_m a_m z^{-m-1}$. The bar differential on degree-1 elements is:

$$d^{\mathrm{ell}}([a_m] \otimes 1) = 0,$$

since the Heisenberg algebra has trivial unary OPE ($a_{(n)}a = 0$ for $n \geq 1$ except $a_{(1)}a = \kappa$, and the residue extraction on a single element yields zero). This is identical to the rational case.

Computation 14.25 *(Degree-2 bar elements and the elliptic differential).* In bar degree 2, the generators are:

$$[a_m|a_n] \otimes \eta_{12}^{\mathrm{ell}} \in \bar{B}_2^{\mathrm{ell}}(\mathcal{H}_\kappa),$$

where $\eta_{12}^{\mathrm{ell}} = \zeta_\tau(z_1 - z_2)\,(dz_1 - dz_2)$. The bar differential acts by residue extraction:

$$\begin{aligned} d^{\mathrm{ell}}([a_m|a_n] \otimes \eta_{12}^{\mathrm{ell}}) &= \mathrm{Res}_{z_1=z_2}\Big[a_m(z_1)\,a_n(z_2)\,\eta_{12}^{\mathrm{ell}}\Big] \\ &= \mathrm{Res}_{\epsilon=0}\Big[\frac{\kappa}{\epsilon^2}\,\zeta_\tau(\epsilon)\,d\epsilon\Big] \cdot z_2^{-m-n-2}\,dz_2, \end{aligned} \tag{14.3}$$

where $\epsilon = z_1 - z_2$. We use the Laurent expansion of ζ_τ (established in the proof of Theorem 14.22, Step 1):

$$\zeta_\tau(\epsilon) = \frac{1}{\epsilon} - \frac{\pi^2}{3} E_2(\tau)\,\epsilon - \frac{\pi^4}{45} E_4(\tau)\,\epsilon^3 - \frac{2\pi^6}{945} E_6(\tau)\,\epsilon^5 - \cdots \tag{14.4}$$

Substituting into the residue:

$$\begin{aligned}
\mathrm{Res}_{\epsilon=0}\Big[\frac{\kappa}{\epsilon^2}\,\zeta_\tau(\epsilon)\,d\epsilon\Big] &= \mathrm{Res}_{\epsilon=0}\Big[\frac{\kappa}{\epsilon^2}\Big(\frac{1}{\epsilon} - \frac{\pi^2}{3} E_2(\tau)\,\epsilon - \cdots\Big)d\epsilon\Big] \\
&= \mathrm{Res}_{\epsilon=0}\Big[\frac{\kappa}{\epsilon^3} - \frac{\kappa\pi^2}{3} E_2(\tau)\,\frac{1}{\epsilon} - \cdots\Big]d\epsilon \\
&= -\frac{\kappa\pi^2}{3}\,E_2(\tau).
\end{aligned} \tag{14.5}$$

The $1/\epsilon^3$ term has zero residue; the first nonzero contribution is the $E_2(\tau)$ term. In the rational case ($\zeta_\tau(\epsilon) \to 1/\epsilon$), the residue of $\kappa\epsilon^{-3}\,d\epsilon$ vanishes, so the rational bar differential on this element is zero. The *entire degree-2 bar differential is the elliptic correction*:

$$d^{\mathrm{ell}}([a_m|a_n] \otimes \eta_{12}^{\mathrm{ell}}) = -\frac{\kappa\pi^2}{3}\,E_2(\tau)\cdot z_2^{-m-n-2}\,dz_2. \tag{14.6}$$

Computation 14.26 *(Degree-2 curvature).* The elliptic bar complex of $\mathcal{H}_\kappa$ has curvature element $m_0^{\mathrm{ell}} \in \bar{B}_0^{\mathrm{ell}}(\mathcal{H}_\kappa)$. From the genus-1 propagator (Vol I, Theorem III), the two-point function on E_τ is $\langle a(z_1)\,a(z_2)\rangle_{E_\tau} = \kappa\,G_\tau(z_1 - z_2)$, where $G_\tau(z) = \zeta_\tau(z) - (\pi^2 E_2(\tau)/3)\,z$. The curvature arises from the failure of double periodicity (cf. the proof of Vol I, Theorem III):

$$m_0^{\mathrm{ell}} = \kappa\cdot\frac{\pi^2 E_2(\tau)}{3}. \tag{14.7}$$

Expanding in the nome $q = e^{2\pi i\tau}$:

$$m_0^{\mathrm{ell}} = \frac{\kappa\pi^2}{3}\big(1 - 24q - 72q^2 - 96q^3 - \cdots\big). \tag{14.8}$$

At the rational limit $\tau \to i\infty$ ($q \to 0$), $E_2(\tau) \to 1$ and $m_0^{\mathrm{ell}} \to \kappa\pi^2/3$, recovering the rational curvature (up to the normalization convention for the propagator on $\mathbb{C}$ versus E_τ; on $\mathbb{C}$ the propagator $1/z$ has no B-cycle and hence $m_0^{\mathrm{rat}} = 0$).

Comparison table: elliptic versus rational

The vanishing of the rational bar differential at all degrees reflects that the Heisenberg algebra on $\mathbb{C}$ (genus 0) has no curvature ($m_0^{\mathrm{rat}} = 0$): the propagator $1/z$ is strictly periodic on $\mathbb{C}$ (there is no B-cycle). On the elliptic curve E_τ, the B-cycle monodromy of ζ_τ introduces the Eisenstein corrections. At degree n, the coefficient $c_{n-1}(\tau)$ in the Laurent expansion (14.4) controls the bar differential; since $c_k(\tau) \in M_{2k}(\mathrm{SL}_2(\mathbb{Z}))$ for $k \geq 2$ and $c_1(\tau) \in \widetilde{M}_2(\mathrm{SL}_2(\mathbb{Z}))$ (quasi-modular), the corrections at degree $n \geq 3$ are genuine modular forms.

Proposition 14.27 *(Decomposition of the elliptic bar complex; [PH]).* The elliptic bar complex of $\mathcal{H}_\kappa$ at degree n admits the decomposition:

$$\bar{B}_n^{\mathrm{ell}}(\mathcal{H}_\kappa) = \bar{B}_n^{\mathrm{rat}}(\mathcal{H}_\kappa) \oplus E_2(\tau)\cdot\bar{B}_n^{(1)} \oplus E_4(\tau)\cdot\bar{B}_n^{(2)} \oplus \cdots \oplus E_{2n-2}(\tau)\cdot\bar{B}_n^{(n-1)} \tag{14.9}$$

with only finitely many nonzero terms. Specifically, $\bar{B}_n^{(k)} = 0$ for $k \geq n$.

Table 14.1: Elliptic versus rational bar differentials for $\mathcal{H}_\kappa$ at low degrees

Degree	Rational d^{rat}	Elliptic d^{ell}	Correction
1	0	0	—
2	0	$-\frac{\kappa\pi^2}{3} E_2(\tau)$	$E_2(\tau)$ (quasi-modular, wt 2)
3	0	$-\frac{\kappa\pi^4}{45} E_4(\tau)$	$E_4(\tau)$ (modular, wt 4)
4	0	$-\frac{2\kappa\pi^6}{945} E_6(\tau)$	$E_6(\tau)$ (modular, wt 6)
5	0	$-\frac{\kappa\pi^8}{4725} E_8(\tau)$	$E_8(\tau)$ (modular, wt 8)
6	0	$-\frac{2\kappa\pi^{10}}{93555} E_{10}(\tau)$	$E_{10}(\tau)$ (modular, wt 10)
7	0	$-\frac{1382\kappa\pi^{12}}{638512875} E_{12}(\tau)$	$E_{12}(\tau)$ (modular, wt 12)
8	0	$-\frac{4\kappa\pi^{14}}{18243225} E_{14}(\tau)$	$E_{14}(\tau)$ (modular, wt 14)
General n	0	$-\kappa \cdot \frac{2^{2n-2}\lvert B_{2n-2}\rvert}{(2n-2)!} \cdot \pi^{2n-2} E_{2n-2}(\tau)$	$E_{2n-2}(\tau)$ (wt $2n-2$)

Proof. Step 1 (Finite propagator expansion). The elliptic bar complex at degree n involves $\binom{n}{2}$ propagator factors η_{ij}^{ell}, one for each pair (i, j) with $1 \le i < j \le n$. Each propagator is expanded via (14.4) as $\zeta_\tau(\epsilon_{ij}) = \epsilon_{ij}^{-1} + \sum_{k\ge1} c_k(\tau)\, \epsilon_{ij}^{2k-1}$. The bar differential at degree n extracts the residue at the diagonal $\epsilon_{ij} = 0$; since the OPE of the Heisenberg current $a(z)$ has a double pole ($a(z_1)\, a(z_2) \sim \kappa/(z_1 - z_2)^2$), the residue at each pair involves at most the coefficient of ϵ_{ij}^{-1} in the integrand. This means the Laurent coefficient $c_k(\tau)\, \epsilon_{ij}^{2k-1}$ contributes to the residue only if $2k - 1 \le 1$, i.e., $k \le 1$, for each *individual* propagator factor. However, when multiple propagator factors are combined in the degree-n bar element ($n - 1$ propagators appear in a degree-n element), the correction terms from different propagator pairs may accumulate. The total correction order is bounded by $n - 1$ (one correction per propagator), giving $\bar{B}_n^{(k)} = 0$ for $k \ge n$.

Step 2 (Eisenstein coefficient identification). The correction term $\bar{B}_n^{(k)}$ arises from choosing exactly k of the $n - 1$ propagator factors to contribute their leading correction $c_1(\tau)\, \epsilon_{ij}$ (or, more precisely, assigning a total correction weight of k distributed among the propagators, with correction of weight j from a single propagator contributing $c_j(\tau)$). By the computation of (14.5), the leading correction at degree 2 is $c_1(\tau) \propto E_2(\tau)$. At degree n, the correction of total weight k lies in the subspace of (quasi-)modular forms of weight $2k$, which is spanned by monomials in E_2, E_4, E_6. Since the Heisenberg algebra is free (Gaussian), the only contributing terms are the *single-propagator corrections* (there are no interaction terms between different propagator pairs), so the coefficient of $\bar{B}_n^{(k)}$ is precisely $c_k(\tau) = (-1)^k G_{2k+2}(\tau) \propto E_{2k+2}(\tau)$ for $k \ge 2$, and $c_1(\tau) \propto E_2(\tau)$.

Step 3 (Direct sum splitting). The corrections $E_2, E_4, E_6, \ldots$ have distinct modular weights $2, 4, 6, \ldots$ and are therefore linearly independent over $\mathbb{C}$. The decomposition (14.9) is a direct sum, not merely a filtration. □

14.5 Dynamical Yang–Baxter equation for $\mathfrak{sl}_2$

We verify the dynamical Yang–Baxter equation for the $\mathfrak{sl}_2$ elliptic R-matrix of Definition 14.15, and show that the key identity reduces to the Fay trisecant (Theorem 14.18). The connection to bar nilpotency is made precise: the dynamical YBE encodes $d^2 = 0$ for the elliptic bar complex with dynamical parameter.

The dynamical Yang–Baxter equation

Definition 14.28 *(Dynamical Yang–Baxter equation).* The *dynamical Yang–Baxter equation* (DYBE) for an R-matrix $R(u,\lambda) \in \mathrm{End}(V \otimes V)$ depending on spectral parameter u and dynamical parameter $\lambda \in \mathfrak{h}^*$ is:

$$R_{12}(u,\lambda)\, R_{13}(u+v,\lambda-h^{(2)})\, R_{23}(v,\lambda) \;=\; R_{23}(v,\lambda-h^{(1)})\, R_{13}(u+v,\lambda)\, R_{12}(u,\lambda-h^{(3)}) \quad (14.10)$$

where $h^{(i)}$ denotes the action of $h \in \mathfrak{h}$ on the i-th tensor factor of $V^{\otimes 3}$, and $\lambda - h^{(i)}$ means that the dynamical parameter λ is shifted by the weight of the state in factor i.

Explicit verification for $\mathfrak{sl}_2$

For $\mathfrak{sl}_2$ with $V = \mathbb{C}^2$, the weight decomposition is $V = V_+ \oplus V_-$ with $h \cdot v_\pm = \pm v_\pm$. We use the elliptic R-matrix of Definition 14.15:

$$R(u,\lambda) = \begin{pmatrix} a(u) & 0 & 0 & 0 \\ 0 & b(u,\lambda) & c(u,\lambda) & 0 \\ 0 & c(u,-\lambda) & b(u,-\lambda) & 0 \\ 0 & 0 & 0 & a(u) \end{pmatrix} \quad (14.11)$$

with $a(u) = \theta(u+\eta)/\theta(\eta)$, $b(u,\lambda) = \theta(u)\theta(\lambda+\eta)/(\theta(\eta)\theta(\lambda))$, and $c(u,\lambda) = \theta(u+\lambda)\theta(\eta)/(\theta(\lambda)\theta(u+\eta))$ (the entries from Definition 14.15, abbreviated here as $\theta = \theta(\cdot\,|\,\tau)$).

Computation 14.29 *(Matrix entries of the DYBE).* We examine the DYBE (14.10) on the weight spaces of $V^{\otimes 3}$. On the space $V^{\otimes 3}$ with basis $\{v_{\epsilon_1} \otimes v_{\epsilon_2} \otimes v_{\epsilon_3} : \epsilon_i \in \{+,-\}\}$ (eight basis vectors), the dynamical shifts act as $h^{(i)} \cdot v_{\epsilon_1\epsilon_2\epsilon_3} = \epsilon_i\, v_{\epsilon_1\epsilon_2\epsilon_3}$. The DYBE is a system of 64 scalar equations (the entries of an 8×8 matrix equation).

Diagonal entries. On the extremal weight space v_{+++}, both sides of (14.10) give $a(u)\,a(u+v)\,a(v)$, so the equation is trivially satisfied. Similarly for v_{---}.

Mixed weight spaces. The nontrivial content lies on the 4-dimensional subspace spanned by $\{v_{++-}, v_{+-+}, v_{-++}\}$ (weight 1) and $\{v_{--+}, v_{-+-}, v_{+--}\}$ (weight -1). By the symmetry $R(u,\lambda) = R(u,-\lambda)|_{+\leftrightarrow -}$, it suffices to verify the weight-1 block.

On the weight-1 subspace, the DYBE reduces to a 3×3 matrix equation. The $(1,1)$-entry, acting on v_{++-}, is

$$\begin{aligned} &a(u)\,b(u+v,\lambda-1)\,b(v,\lambda) \;+\; b(u,\lambda)\,c(u+v,\lambda-1)\,c(v,-\lambda) \\ &\quad=\; b(v,\lambda-1)\,b(u+v,\lambda)\,a(u) \;+\; c(v,-\lambda+1)\,c(u+v,\lambda)\,a(u). \end{aligned} \quad (14.12)$$

Substituting the theta-function expressions and writing $[x] := \theta(x|\tau)$ gives

$$\begin{aligned} &\frac{[u+\eta]}{[\eta]}\cdot\frac{[u+v][\lambda-1+\eta]}{[\eta][\lambda-1]}\cdot\frac{[v][\lambda+\eta]}{[\eta][\lambda]} \;+\; \frac{[u][\lambda+\eta]}{[\eta][\lambda]}\cdot\frac{[u+v+\lambda-1][\eta]}{[\lambda-1][u+v+\eta]}\cdot\frac{[v-\lambda][\eta]}{[-\lambda][v+\eta]} \\ &\quad= \frac{[v][\lambda-1+\eta]}{[\eta][\lambda-1]}\cdot\frac{[u+v][\lambda+\eta]}{[\eta][\lambda]}\cdot\frac{[u+\eta]}{[\eta]} \;+\; \frac{[v-\lambda+1][\eta]}{[-\lambda+1][v+\eta]}\cdot\frac{[u+v+\lambda][\eta]}{[\lambda][u+v+\eta]}\cdot\frac{[u+\eta]}{[\eta]}. \end{aligned} \quad (14.13)$$

After cancelling common factors $[u+\eta]/([\eta]^3\,[\lambda]\,[\lambda-1])$ from both sides and clearing denominators, the equation reduces to an identity among products of theta functions evaluated at the six arguments $\{u, v, u+v, \lambda, \lambda+\eta, \lambda-1+\eta\}$.

Proposition 14.30 *(DYBE reduces to Fay; [PH])*. The dynamical Yang–Baxter equation (14.10) for the $\mathfrak{sl}_2$ elliptic R-matrix (Definition 14.15) is equivalent to the Fay trisecant identity (Theorem 14.18) plus the quasi-periodicity of θ. More precisely:

(i) Each nontrivial component of the DYBE on the weight-1 subspace reduces, after clearing common factors, to an identity of the form

$$\theta(a-b)\,\theta(c-d) - \theta(a-c)\,\theta(b-d) + \theta(a-d)\,\theta(b-c) = 0 \tag{14.14}$$

for appropriate specializations of a, b, c, d depending on u, v, λ, η.

(ii) The remaining components (mixing weight-1 and weight-(-1) sectors) follow from (i) and the quasi-periodicity $\theta(z+1|\tau) = -\theta(z|\tau)$, $\theta(z+\tau|\tau) = -e^{-\pi i\tau-2\pi iz}\theta(z|\tau)$.

Proof. Step 1 (Reduction of the $(1,2)$*-entry).* The $(1,2)$-entry of the DYBE on the weight-1 subspace (the matrix element for $v_{++-} \to v_{+-+}$) yields, after clearing the common denominator $[\eta]^3\,[\lambda]\,[\lambda{-}1]\,[u{+}\eta]\,[v{+}\eta]\,[u{+}v{+}\eta]$:

$$\begin{aligned}&[u{+}\eta]\,[v]\,[u{+}v{+}\lambda{-}1] - [u]\,[v{+}\eta]\,[u{+}v{+}\lambda{-}1]\\ &\qquad + [u{+}v{+}\eta]\,[\lambda{-}1]\,\big([u{+}\lambda]\,[v] - [v{+}\lambda]\,[u]\big) \;=\; 0.\end{aligned} \tag{14.15}$$

We apply the Fay identity (Theorem 14.18) with the specialization $u_1 = 0, u_2 = u + \eta, u_3 = v + \eta$, $u_4 = u + v + \lambda$:

$$\begin{aligned}&\theta(u{+}\eta)\,\theta(v{+}\eta - u - v - \lambda) - \theta(v{+}\eta)\,\theta(u{+}\eta - u - v - \lambda)\\ &\qquad + \theta(u{+}v{+}\lambda)\,\theta(v{+}\eta - u - \eta) = 0.\end{aligned} \tag{14.16}$$

Using $\theta(-x) = -\theta(x)$, this simplifies to:

$$\theta(u{+}\eta)\,\theta(\lambda{-}v) - \theta(v{+}\eta)\,\theta(\lambda{-}u) + \theta(u{+}v{+}\lambda)\,\theta(v{-}u) = 0,$$

which is the content of (14.15) after the identification $[\lambda{-}v] = -[v{-}\lambda]$ and relabelling.

Step 2 (Remaining entries). The $(1,3)$ and $(2,3)$ entries of the DYBE are obtained from the $(1,2)$ entry by the substitutions $v \to u+v, u \to -v$ and $u \to v, v \to u$ respectively, combined with the Weyl group symmetry $R(u,\lambda) = R(u,-\lambda)|_{+\leftrightarrow-}$. Each reduces to the same Fay identity with permuted arguments.

Step 3 (Quasi-periodicity). The elliptic R-matrix entries $a(u), b(u,\lambda), c(u,\lambda)$ are meromorphic on $E_\tau \times E_\tau$ with simple poles at the zeros of $\theta(\eta)$ and $\theta(\lambda)$. The DYBE relates values at $\lambda, \lambda - h^{(1)}, \lambda - h^{(2)}$, $\lambda - h^{(3)}$ where $h^{(i)} \in \{+1,-1\}$; the quasi-periodicity of θ ensures that these λ-shifts are compatible with the lattice $\mathbb{Z} + \tau\mathbb{Z}$ acting on λ. □

Proposition 14.31 *(DYBE and bar nilpotency; [PH])*. The dynamical Yang–Baxter equation (14.10) encodes $d^2 = 0$ for the elliptic bar complex with dynamical parameter. Specifically, the bar complex $\bar{B}^{\mathrm{ell}}(E_{q,p}(\mathfrak{sl}_2))$ of the elliptic quantum group (Definition 14.14) satisfies $d^2 = 0$ on $\overline{C}_3(E_\tau)$ if and only if the R-matrix $R(u,\lambda)$ satisfies the DYBE.

Proof. The bar differential at degree 2 of the RTT-type algebra $E_{q,p}(\mathfrak{sl}_2)$ is:

$$d([L_{ij}(u)|L_{kl}(v)] \otimes \eta_{12}^{\mathrm{ell}}) = \sum_m \big(R_{ij,km}(u{-}v,\lambda)\,L_{ml}(v) - L_{km}(v)\,R_{mj,il}(u{-}v,\lambda)\big) \otimes \eta_{12}^{\mathrm{ell}} \wedge \eta_{23}^{\mathrm{ell}}.$$

The computation of d^2 on a degree-1 element $[L_{ij}(u)] \otimes 1$ produces terms on $\overline{C}_3(E_\tau)$ involving the product of three R-matrices. The nine terms (three pairs of propagators, each contributing three matrix products) rearrange into the left-hand side minus the right-hand side of the DYBE (14.10), exactly as in the rational case where the Yang–Baxter equation implies $d^2 = 0$ for the Yangian bar complex (Vol I, proof of Theorem III). The only modification is the replacement of the rational propagator $\eta_{ij} = d\log(z_i - z_j)$ by the elliptic propagator $\eta_{ij}^{\mathrm{ell}} = d\log\theta_1(z_i - z_j|\tau)$, and the introduction of the dynamical shifts $\lambda \to \lambda - h^{(i)}$ arising from the B-cycle monodromy of η_{ij}^{ell} around E_τ (cf. Remark 14.16(ii)). The elliptic Arnold relation (14.1) (Proposition 14.19) ensures that the propagator-level identity $\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0$ holds, so the algebraic content of $d^2 = 0$ is precisely the DYBE. □

14.6 Shuffle algebra presentation

Definition of the shuffle algebra

Definition 14.32 *(Shuffle algebra).* Let $\zeta(u) = \theta(u+\eta)/\theta(u)$ be the structure function determined by the elliptic R-matrix. The *shuffle algebra* $\mathcal{S}_\zeta$ is the $\mathbb{Z}_{\geq 0}$-graded vector space

$$\mathcal{S}_\zeta = \bigoplus_{n\geq 0} \mathcal{S}_n, \quad \mathcal{S}_n = \mathbb{C}(z_1, \dots, z_n)_{\mathrm{sym}}^{S_n}$$

(symmetric rational functions in n variables), equipped with the *shuffle product*: for $f \in \mathcal{S}_m$ and $g \in \mathcal{S}_n$, define $f \star g \in \mathcal{S}_{m+n}$ by

$$(f \star g)(z_1, \dots, z_{m+n}) = \frac{1}{m!\,n!} \sum_{\sigma \in S_{m+n}} \prod_{\substack{i \leq m \\ j > m}} \zeta(z_{\sigma(i)} - z_{\sigma(j)}) \cdot f(z_{\sigma(1)}, \dots, z_{\sigma(m)}) \cdot g(z_{\sigma(m+1)}, \dots, z_{\sigma(m+n)}). \tag{14.17}$$

Equivalently, the sum runs over (m,n)-shuffles $\mathrm{Sh}(m,n) \subset S_{m+n}$, with the $1/(m!\,n!)$ factor replaced by 1 and the sum restricted to shuffles.

Remark 14.33 (Origin). The shuffle algebra was introduced by Feigin–Odesskii in the rational and trigonometric cases and extended to the elliptic case by Schiffmann–Vasserot [?] and Negut. The rational specialization $\zeta(u) = (u+\eta)/u$ recovers the Feigin–Odesskii shuffle realization of the Yangian; the trigonometric specialization $\zeta(u) = (1 - qe^{2\pi i u})/(1 - e^{2\pi i u})$ recovers the quantum affine algebra.

Explicit shuffle products

Computation 14.34 *(First generators).* Let $e_1 = 1 \in \mathcal{S}_1 = \mathbb{C}(z)$. By the Fay identity (Theorem 14.18) and oddness of θ:

$$(e_1 \star e_1)(z_1, z_2) = \frac{\theta(z_{12}+\eta) + \theta(\eta - z_{12})}{\theta(z_{12})}, \tag{14.18}$$

where $z_{12} = z_1 - z_2$. The triple product is

$$(e_1 \star e_1 \star e_1)(z_1, z_2, z_3) = \sum_{\sigma \in S_3} \zeta(z_{\sigma(1)} - z_{\sigma(2)})\,\zeta(z_{\sigma(1)} - z_{\sigma(3)})\,\zeta(z_{\sigma(2)} - z_{\sigma(3)}). \tag{14.19}$$

Associativity follows from the Yang–Baxter equation for ζ.

Remark 14.35 (Shuffle–bar isomorphism). The residue pairing gives an isomorphism $\mathcal{S}_n \cong \bar{B}_n(\mathcal{A})^*$ under which the shuffle product corresponds to the deshuffle coproduct on $\bar{B}(\mathcal{A})$ (Vol I, Corollary III).

14.7 Comparison: rational, trigonometric, elliptic

Table 14.2: Comparison across the three regimes

	Rational	**Trigonometric**	**Elliptic**
Algebra	Yangian $Y(\mathfrak{g})$	Qtm. affine $U_q(\hat{\mathfrak{g}})$	Toroidal $U_{q,t}(\hat{\hat{\mathfrak{g}}})$
Curve	$\mathbb{C}$ (or $\mathbb{CP}^1$)	$\mathbb{C}^*$ (or $\mathbb{CP}^1 \setminus \{0, \infty\}$)	E_τ
Propagator	$\frac{dz}{z}$	$\frac{dz}{1-z}$	$d \log \theta_1(z\|\tau)$
R-**matrix**	$1 - \hbar P/u$	$(q-q^{-1})/(u-u^{-1})$	$\theta(u+\eta)/\theta(\eta)$
Arnold id.	$\eta_{12} \wedge \eta_{23} + \text{cyc} = 0$	$\frac{dz_1}{z_1-z_2} \wedge \frac{dz_2}{z_2-z_3} + \text{cyc} = 0$	Fay trisecant
$d^2 = 0$	Partial fractions	q-partial fractions	Fay + quasi-period
Koszul dual	$Y_{-\hbar}(\mathfrak{g})$	$U_{q^{-1}}(\hat{\mathfrak{g}})$	$U_{q^{-1},t^{-1}}(\hat{\hat{\mathfrak{g}}})$ (conj.)
Struct. fn.	$(u+\eta)/u$	$(1-qe^u)/(1-e^u)$	$\theta(u+\eta)/\theta(u)$
Shuffle alg.	Feigin–Odesskii	F–O (q-def.)	Schiffmann–Vasserot
Dyn. par.	absent	absent	$\lambda \in \mathfrak{h}^*$
Bar curv.	$m_0 = 0$	$m_0 \propto \log q$	$m_0 \propto E_2(\tau)$

Remark 14.36 (Degeneration limits). The three regimes degenerate via

$$\text{Elliptic} \xrightarrow{\tau \to i\infty} \text{Trigonometric} \xrightarrow{q \to 1} \text{Rational}. \tag{14.20}$$

As $\tau \to i\infty$, $\theta_1(z|\tau) \to 2\sin(\pi z)$ and the Eisenstein corrections collapse to constants; as $q \to 1$, the trigonometric propagator becomes dz/z and $R_q(u) \to 1 - \hbar P/u$. The bar complex degenerates accordingly: $\bar{B}^{\text{ell}}(\mathcal{A}) \to \bar{B}^{\text{trig}}(\mathcal{A}) \to \bar{B}^{\text{rat}}(\mathcal{A})$, consistent with the spectral sequence of Theorem 14.22.

The elliptic curve as shared geometry

Remark 14.37 (Bridge: algebras on E and algebras from $K3 \times E$). The elliptic curve E_τ enters the remainder of this chapter in a second capacity. The preceding sections study chiral algebras *on* E_τ: the propagator $d \log \theta_1(z|\tau)$, the Eisenstein corrections to the bar differential, the toroidal algebra $U_{q,t}(\hat{\hat{\mathfrak{g}}})$ with $\tau = \log t / \log q$, and Felder's elliptic R-matrix $R(u, \lambda)$ built from the same theta function. The K3 chiral algebra $\mathcal{A}_{K3}$ (§14.9) lives on a K3 surface, but the Calabi–Yau threefold $K3 \times E$ (§14.10) is fibered *over* E, and Costello–Li's holomorphic twist reduction along K3 produces a boundary chiral algebra A_E *on* E_τ with $c(A_E) = \chi(K3) = 24$ free bosons from harmonic forms on K3 (Construction 14.73).

Thus: the toroidal and elliptic quantum group algebras, and the Costello–Li boundary algebra A_E, are all chiral algebras on the *same* curve E_τ. They share the propagator $d \log \theta_1$, the Fay trisecant relation governing $d^2 = 0$ (Proposition 14.19), and the Eisenstein filtration of the elliptic bar complex (Theorem 14.22). The difference is the algebra: A_E has 24 generators with $\kappa(A_E) = 24$ (rank of the free-boson lattice) and shadow class G (Gaussian, $r_{\max} = 2$), while $E_{q,p}(\mathfrak{sl}_N)$ has $N^2 - 1$ generators with dynamical parameter $\lambda \in \mathfrak{h}^*$.

14.8 Calabi–Yau local-to-global gluing

The Calabi–Yau threefold $K3 \times E$ admits a local description in terms of quiver charts glued by A_∞-bimodule transition data. This section develops the algebraic machinery.

McKay correspondence and ADE quiver charts

A K3 surface S acquires ADE singularities at special points in moduli. Near each singularity, the local geometry is $\mathbb{C}^2/\Gamma$ for a finite subgroup $\Gamma \subset \mathrm{SL}(2, \mathbb{C})$. The McKay correspondence (McKay 1980, Gonzalez-Sprinberg–Verdier 1983) gives a bijection

$$\{\text{nontrivial irreps of } \Gamma\} \ \stackrel{\sim}{\longleftrightarrow}\ \{\text{exceptional curves in the minimal resolution}\},$$

and the McKay quiver Q_Γ encodes the tensor product rule $\rho_{\text{fund}} \otimes \rho_i = \bigoplus_j a_{ij}\, \rho_j$, where a_{ij} are the arrows of the extended ADE Dynkin diagram.

Definition 14.38 *(McKay quiver and Cartan matrix).* For a finite subgroup $\Gamma \subset \mathrm{SL}(2, \mathbb{C})$ of ADE type, the *McKay quiver* Q_Γ has vertices indexed by irreducible representations $\rho_0 = \mathrm{triv},\ \rho_1, \ldots, \rho_r$ and a_{ij} arrows from vertex i to vertex j, where a_{ij} is the multiplicity of ρ_j in $\rho_{\text{fund}} \otimes \rho_i$. The extended Cartan matrix is

$$C_{\text{ext}} = 2I - A,$$

where A is the adjacency matrix of Q_Γ.

Example 14.39 *(ADE classification).* The finite subgroups of $\mathrm{SL}(2, \mathbb{C})$ are classified by ADE Dynkin diagrams:

Group	Γ	**Type**	$\lvert\Gamma\rvert$	**Irreps (rank)**
Cyclic	$\mathbb{Z}/n$	A_{n-1}	n	n irreps, all 1-dim
Binary dihedral	BD_n	D_{n+2}	$4n$	$n+3$ irreps
Binary tetrahedral	BT	E_6	24	7 irreps
Binary octahedral	BO	E_7	48	8 irreps
Binary icosahedral	BI	E_8	120	9 irreps

For each type, $\lvert\Gamma\rvert = \sum d_i^2$ (sum of squared dimensions of irreducible representations) provides an independent check of the character table.

Proposition 14.40 *(Preprojective algebra from the McKay quiver; [PE]).* The preprojective algebra $\Pi(Q_\Gamma) = kQ_{\text{double}}/(\sum[a, a^*])$, where Q_{double} is the double quiver (adjoin a reverse arrow a^* for each arrow a), has centre isomorphic to the invariant ring:

$$Z(\Pi(Q_\Gamma)) \ \cong\ \mathbb{C}[x, y, z]^\Gamma.$$

For the Dynkin quiver of type Δ with r vertices and Coxeter number h, the preprojective algebra is finite-dimensional with $\dim \Pi_0(Q_\Delta)$ given by the following table (Crawley-Boevey 2001, computed via $\dim e_i \Pi e_j = \min(i, j) \min(r + 1 - i, r + 1 - j)$ for type A and by the McKay quiver formula $\dim \Pi(Q_\Gamma) = \lvert\Gamma\rvert \cdot (r + 1)$ for the extended quiver):

Type	$\lvert\Gamma\rvert$	r = rank	h	$\lvert Q_0\rvert$	$\dim \Pi_0$
A_1	2	1	2	2	4
A_2	3	2	3	3	9
A_3	4	3	4	4	20
A_4	5	4	5	5	35
D_4	8	4	6	5	40
D_5	12	5	8	6	72
E_6	24	6	12	7	168
E_7	48	7	18	8	384
E_8	120	8	30	9	1080

The table entries are computed from the McKay quiver formula $\dim e_i\Pi e_j = \min(i, j)\min(r+1-i,\ r+1-j)$ for type A and from representation-theoretic data for the exceptional types.

Ginzburg dg algebras and Jacobian algebras

Definition 14.41 *(Ginzburg dg algebra).* For an ADE quiver Q with potential W (a linear combination of cyclic paths), the *Ginzburg dg algebra* $\mathcal{A}_\Gamma$ is a Calabi–Yau 3-fold dg algebra concentrated in degrees $[-3, 0]$. Its zeroth cohomology is the *Jacobian algebra*:

$$\mathrm{Jac}(Q, W) = kQ\big/\big(\partial_a W : a \in Q_1\big),$$

where $\partial_a W$ is the cyclic derivative with respect to arrow a. For ADE types, $\mathrm{Jac}(Q, W)$ is isomorphic to the preprojective algebra $\Pi_0(Q)$.

Remark 14.42 (CY3 extension). For the local CY3 model $\mathbb{C}^2/\Gamma\times\mathbb{C}$, one adds a loop at each vertex of the McKay quiver (from the $\mathbb{C}$ direction). The extended quiver with potential $(Q_{\mathrm{ext}}, W_{\mathrm{ext}})$ has Jacobian algebra that is 3-Calabi–Yau in the sense of Ginzburg: the derived category $D^b(\mathrm{Jac}(Q_{\mathrm{ext}}, W_{\mathrm{ext}}))$ carries a Serre functor $S = [3]$.

A_∞-bimodule transition data

Construction 14.43 *(A_∞-bimodule gluing).* Given two local charts U_α, U_β with tilting generators T_α, T_β and endomorphism algebras $A_\alpha = \mathrm{End}(T_\alpha)$, $A_\beta = \mathrm{End}(T_\beta)$, the transition data on the overlap $U_{\alpha\beta} = U_\alpha \cap U_\beta$ is encoded in the (A_α, A_β)-bimodule

$$M_{\alpha\beta} = R\mathrm{Hom}(T_\alpha|_{U_{\alpha\beta}}, T_\beta|_{U_{\alpha\beta}})$$

with A_∞-bimodule operations $m_{p|1|q} : A_\alpha^{\otimes p} \otimes M_{\alpha\beta} \otimes A_\beta^{\otimes q} \to M_{\alpha\beta}[2 - p - q]$ satisfying the A_∞-bimodule equation. The derived tensor product $M_{\alpha\beta} \otimes^{\mathbf{L}}_{A_\beta} M_{\beta\gamma}$ provides the triple-overlap cocycle data. The reduced bar complex $\overline{B}(A_\alpha, M_{\alpha\beta}, A_\beta)$ computes the relevant Ext groups.

Remark 14.44 (Reduced bar complex and unit elements). The A_∞-bimodule relations hold on the *reduced* bar complex (augmentation ideal only): unit elements do not participate in the higher operations $m_{p|1|q}$ for $p + q \geq 2$. This is a general structural principle, not specific to K3. When computing $\overline{B}(A_\alpha, M_{\alpha\beta}, A_\beta)$, one restricts to the augmentation ideal $\overline{A} = \ker(\varepsilon\colon A \to \Bbbk)$ in both algebra arguments. The unit $1_A \in A$ acts on M via $m_{1|1|0}(1_A, m) = m$ (identity bimodule action) and $m_{0|1|1}(m, 1_B) = m$, but does not enter the higher bar differential. For the K3 quiver charts, where $A_\alpha = \mathrm{End}(T_\alpha)$ is augmented over $\Bbbk$, this restriction ensures that only the interacting OPE data contributes to the gluing obstructions.

Čech descent and the global category

THEOREM 14.45 *(Čech descent for dg categories; [PE]).* Cover $K3 = U_0 \cup U_1$, where $U_0 = K3 \setminus D$ (complement of a smooth divisor $D \in |H|$) and U_1 is a tubular neighbourhood of D. The derived category $D^b(K3)$ is recovered as the homotopy fibre product

$$D^b(K3) \;\simeq\; D^b(U_0) \times^h_{D^b(U_{01})} D^b(U_1),$$

and the descent spectral sequence for Hochschild cohomology

$$E_1^{p,q} = H^q\big(C^p(U_\bullet, \mathrm{HH}^*)\big) \implies \mathrm{HH}^{p+q}(D^b(K3))$$

converges. The obstruction to gluing lives in $H^2(K3, \mathcal{O}_{K3}) \cong \mathbb{C}$ (the Brauer class), reflecting $h^{0,2}(K3) = 1$.

Wall-crossing during chart transitions

Remark 14.46 (Cluster mutations and Bridgeland stability). Mutation at a vertex k of the McKay quiver produces a new quiver $Q' = \mu_k(Q)$ with exchange matrix B' given by the Fomin–Zelevinsky rule. For A_n quivers, the exchange graph has $C_{n+1} = \binom{2n}{n}/(n+1)$ vertices (the $(n+1)$-th Catalan number), corresponding to triangulations of an $(n+3)$-gon. The Kontsevich–Soibelman wall-crossing formula governs the change of DT invariants across stability walls. For $K3$ with primitive Mukai vector v: the moduli space $M_H(v) \cong \mathrm{Hilb}^n(K3)$ (Beauville) and the DT transformation satisfies $\tau^{n+3} = [2]$ (Seidel–Thomas).

14.9 The K3 chiral algebra

The K3 sigma model produces a $c = 6$ vertex algebra with $\mathcal{N} = 4$ superconformal symmetry. This section assembles the complete algebraic data: OPE structure, bar complex, shadow tower, Koszul dual, and the connections to Mathieu moonshine.

The $\mathcal{N} = 4$ superconformal algebra at $c = 6$

Definition 14.47 *(Small $\mathcal{N} = 4$ SCA at $c = 6$).* The *small $\mathcal{N} = 4$ superconformal algebra* at central charge $c = 6$ (equivalently, $\mathrm{SU}(2)_R$ level $k_R = 1$, since $c = 6k_R$) has eight primary generators:

Generator	Weight h	Parity	J^3 charge
T	2	bosonic	0
G^+, G^-	3/2	fermionic	$\pm 1/2$
$\tilde{G}^+, \tilde{G}^-$	3/2	fermionic	$\mp 1/2$
J^{++}, J^{--}	1	bosonic	± 1
J^3	1	bosonic	0

The OPE structure at $c = 6$, $k_R = 1$:

$$T(z)\,T(w) \sim \frac{3}{(z-w)^4} + \frac{2T}{(z-w)^2} + \frac{\partial T}{z-w}, \tag{14.21}$$

$$J^3(z)\,J^3(w) \sim \frac{1/2}{(z-w)^2}, \tag{14.22}$$

$$J^{++}(z)\,J^{--}(w) \sim \frac{1}{(z-w)^2} + \frac{2J^3}{z-w}, \tag{14.23}$$

$$G^+(z)\,G^-(w) \sim \frac{2}{(z-w)^3} + \frac{2J^3}{(z-w)^2} + \frac{T + \partial J^3}{z-w}. \tag{14.24}$$

The remaining independent OPE relations at $c = 6, k_R = 1$ (from the 64-pair table of `cy_n4sca_k3_engine.py`):

$$J^3(z)\,G^\pm(w) \sim \frac{\pm\frac{1}{2}\,G^\pm}{z-w}, \tag{14.25}$$

$$J^{++}(z)\,G^-(w) \sim \frac{G^+}{z-w}, \qquad J^{--}(z)\,G^+(w) \sim \frac{G^-}{z-w}, \tag{14.26}$$

$$T(z)\,G^\pm(w) \sim \frac{\frac{3}{2}\,G^\pm}{(z-w)^2} + \frac{\partial G^\pm}{z-w}, \tag{14.27}$$

$$T(z)\,J^a(w) \sim \frac{J^a}{(z-w)^2} + \frac{\partial J^a}{z-w}, \tag{14.28}$$

$$\tilde{G}^+(z)\,\tilde{G}^-(w) \sim \frac{2}{(z-w)^3} - \frac{2J^3}{(z-w)^2} + \frac{T - \partial J^3}{z-w}, \tag{14.29}$$

$$G^+(z)\,\tilde{G}^+(w) \sim \frac{2\,J^{++}}{z-w}, \qquad G^-(z)\,\tilde{G}^-(w) \sim \frac{-2\,J^{--}}{z-w}. \tag{14.30}$$

All boson–boson and fermion–fermion OPEs not listed above are either trivially determined by $\mathrm{SU}(2)_R$ symmetry or regular (no singular terms).

Proposition 14.48 *(Modular characteristic of the K3 sigma model; [PH])*. The modular characteristic of the K3 sigma model VOA is

$$\kappa(\mathcal{A}_{K3}) = 2 = \dim_{\mathbb{C}}(K3). \tag{14.31}$$

This is verified by six independent paths:

(i) *Geometric.* For a CY d-fold sigma model: $\kappa = d$ (index-theoretic computation via the Chern character of the chiral de Rham complex).

(ii) *Character.* $F_1 = \kappa/24 = 1/12$ matches the genus-1 anomaly of the $c = 6$ sigma model.

(iii) *Complementarity.* $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = 0$ gives $\kappa^! = -2$.

(iv) *Kummer orbifold.* At the Kummer point $K3 = T^4/\mathbb{Z}_2$: the orbifold preserves $\kappa = 2$ from the smooth sigma model.

(v) *Gepner consistency.* The Gepner model $(2)^4$ gives $\kappa_{\text{Gepner}} = 4 \times (3/4) = 3$ using the Virasoro formula $\kappa = c/2$ for each $\mathcal{N} = 2$ factor. This is a DIFFERENT algebra from the $\mathcal{N} = 4$ SCA (κ depends on the full algebra, not just the Virasoro subalgebra).

(vi) *Additivity.* $\kappa(K3 \times E) = \kappa(K3) + \kappa(E) = 2 + 1 = 3 = \dim_{\mathbb{C}}(K3 \times E)$ (Corollary III).

Proof. Paths (i)–(iii) follow from Vol I, Theorem D and the index-theoretic computation $\kappa(\Omega^{\mathrm{ch}}(\mathrm{CY}_d)) = d$ (the Chern character of the chiral de Rham complex on a Ricci-flat manifold reduces the genus-1 obstruction to $\chi(M)/12 = 2$ for K3 via $\chi(K3) = 24$). Path (iv): the Kummer orbifold $T^4/\mathbb{Z}_2$ has the same modular characteristic as the smooth K3 because κ is a deformation invariant (it depends on the chiral algebra structure, which is constant across K3 moduli by curve independence). Path (v): the Gepner model uses the Virasoro stress tensor for each $c = 3/2$ factor, giving $\kappa = c/2 = 3/4$ per factor; the physical sigma model has the full $\mathcal{N} = 4$ structure whose $\kappa = 2$ is lower due to supersymmetric Ward identities. Path (vi) follows from the additivity of κ under tensor products (Vol I, Proposition III). □

Remark 14.49 ($\kappa(K3) = 2 \neq c/2 = 3$: modular characteristic vs central charge). The K3 sigma model provides a sharp illustration: the modular characteristic $\kappa(\mathcal{A}_{K3}) = 2$ differs from the Virasoro formula $c/2 = 3$. The reduction arises because the $\mathcal{N} = 4$ superconformal Ward identities constrain the genus-1 obstruction below what the Virasoro subalgebra alone would produce. The naive Virasoro computation $\kappa(\mathrm{Vir}_6) = 3$ counts the Virasoro contribution, but the full $\mathcal{N} = 4$ algebra at $c = 6$ has $\kappa = 2k_R = 2$.

Bar complex of the $\mathcal{N} = 4$ SCA

Proposition 14.50 *(Bar complex structure; [PH]).* The bar complex $B(\mathcal{A}_{K3})$ of the small $\mathcal{N} = 4$ SCA at $c = 6$ has:

(i) $B^1(\mathcal{A}_{K3}) = \mathrm{span}\{s^{-1}a : a \text{ a generator}\}$ with $\dim B^1 = 8$ (one desuspended generator per primary; $|s^{-1}v| = |v| - 1$).

(ii) The bar differential $d\colon B^2 \to B^1$ is a rank-16 linear map extracting all singular OPE modes via the logarithmic kernel $d\log(z-w)$ (the $d\log$ measure absorbs one power of $(z-w)$, so bar pole orders are one less than OPE pole orders; the bar differential extracts all singular modes, not just the simple pole).

(iii) The algebra is chirally Koszul (Vol I, Proposition III): the $\mathcal{N} = 4$ SCA is freely strongly generated (no null vectors in the universal vacuum module), so bar cohomology $H^*(B(\mathcal{A}))$ is concentrated in bar degree 1.

Shadow depth classification

Proposition 14.51 *(K3 sigma model: class M; [PH]).* The K3 sigma model VOA has shadow depth class M (infinite tower, $r_{\max} = \infty$). The shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$ has critical discriminant

$$\Delta = 8\kappa S_4 \neq 0,$$

so Q_L is irreducible and the shadow tower does not terminate. The Virasoro sector at $c = 6$ contributes $Q^{\mathrm{contact}} = 10/[c(5c + 22)] = 5/156 \neq 0$, and the $\mathrm{SU}(2)_R$ sector at level $k_R = 1$ contributes nonzero cubic shadow S_3 from the triple current OPE.

The K3 elliptic genus and Mathieu moonshine

Definition 14.52 *(K3 elliptic genus).* The elliptic genus of $K3$ is

$$Z_{K3}(\tau, z) = 2\,\phi_{0,1}(\tau, z), \qquad (14.32)$$

where $\phi_{0,1}$ is the unique weak Jacobi form of weight 0, index 1 in the Eichler–Zagier normalization $\phi_{0,1}(\tau, 0) = 12$. At $z = 0$: $Z_{K3}(\tau, 0) = 24 = \chi(K3)$. The Fourier expansion at q^0 gives $Z_{K3}|_{q^0} = 2y + 20 + 2y^{-1}$, the χ_y-genus of K3 with $h^{1,1} = 20$, $h^{2,0} = h^{0,2} = 1$.

Theorem 14.53 *(Mathieu moonshine; [PE]).* The $\mathcal{N} = 4$ decomposition of the K3 elliptic genus,

$$Z_{K3} \;=\; 20 \cdot \mathrm{ch}_{\mathrm{massless}} \;+\; \sum_{n \geq 1} A_n \cdot \mathrm{ch}_{\mathrm{massive},n}, \tag{14.33}$$

where $\mathrm{ch}_{\mathrm{massless}}$ and $\mathrm{ch}_{\mathrm{massive},n}$ are $\mathcal{N} = 4$ characters at $c = 6$, has coefficients A_n that are dimensions of representations of the Mathieu group M_{24} (Eguchi–Ooguri–Tachikawa 2010, proved by Gannon 2016):

n	1	2	3	4	5	6
A_n	90	462	1540	4554	11592	27830

The 20 massless states decompose as $23_{\mathrm{std}} - 3_{\mathrm{triv}}$ under M_{24}.

Remark 14.54 (Mock modular form from K3). The massive multiplicities assemble into the mock modular form

$$h(\tau) \;=\; 2q^{-1/8}\bigl(-1 + 45q + 231q^2 + 770q^3 + 2277q^4 + \cdots\bigr), \tag{14.34}$$

where $A_n^{\mathrm{full}} = 2 \cdot a_n$ with $a_n = 45, 231, 770, \ldots$ being the half-multiplicities. The shadow of h in the sense of Zwegers is $S(\tau) = 24\,\eta(\tau)^3$, a weight-3/2 cusp form. The constant term $a_0 = -1$ gives $A_0^{\mathrm{full}} = -2 = -\kappa(\mathcal{A}_{K3})$: the modular characteristic appears as the leading mock modular coefficient.

The "shadow" in mock modularity (Zwegers) and the "shadow" in the shadow obstruction tower (bar complex) are a priori different mathematical objects. Their structural parallel — both encode the failure of modular invariance — is a connection to be explored, not an identification to be assumed.

The mock modular form $h(\tau)$ admits a canonical decomposition via the Appell–Lerch sum. Let $\mu(\tau, z) = \frac{-iy^{1/2}}{\vartheta_1(\tau,z)} \sum_{n\in\mathbb{Z}} \frac{(-1)^n q^{n(n+1)/2} y^n}{1-q^n y}$ denote the Zwegers μ-function. Then

$$Z_{K3}(\tau, z) \;=\; \bigl(h(\tau) - 24\,\mu(\tau, z)\bigr) \cdot \frac{\vartheta_1(\tau, z)^2}{\eta(\tau)^3}, \tag{14.35}$$

where the $24 = \chi(K3)$ multiplying μ is the Witten index and the theta-to-eta factor ϑ_1^2/η^3 carries weight $1/2$ and index 1. The decomposition is verified to machine precision (relative error $\sim 5.7 \times 10^{-16}$). The polar term $h|_{q^{-1/8}} = -2 = -\kappa(\mathcal{A}_{K3})$ reappears on the right-hand side as a consequence of the Appell–Lerch cancellation.

Remark 14.55 (The factor 2 in $Z_{K3} = 2\,\phi_{0,1}$ is κ). The elliptic genus $Z_{K3} = 2\,\phi_{0,1}$ has an overall factor of 2. This factor IS the modular characteristic: $\kappa(\mathcal{A}_{K3}) = 2$. The bar complex provides the universal coefficient

$$A_n \;=\; \kappa(\mathcal{A}_{K3}) \cdot \dim(\rho_n), \tag{14.36}$$

where ρ_n is the M_{24}-representation at level n: $\dim(\rho_1) = 45$, $\dim(\rho_2) = 231$, $\dim(\rho_3) = 770$, $\dim(\rho_4) = 2277$. The half-multiplicities $a_n = 45, 231, 770, \ldots$ are the M_{24} irreducible dimensions; the full multiplicities $A_n = 2\,a_n = 90, 462, 1540, \ldots$ acquire the factor $\kappa = 2$ from the bar complex.

The bar complex does NOT detect M_{24} directly: it detects $\kappa = 2$ at arity 2 and the cubic/quartic shadow invariants at higher arities. The M_{24} representation structure is an *internal* symmetry of the massive BPS sector, invisible to the single-copy bar construction. The bar complex provides the universal multiplier; the group theory is orthogonal. The polar coefficient $-2 = -\kappa$ in the mock modular form (14.34) is the same κ from a different direction.

Remark 14.56 (Twining genera and M_{24} conjugacy classes). For each conjugacy class $[g]$ of M_{24}, the twining genus $Z_g(\tau, z) = \mathrm{Tr}_{\mathrm{RR}}\bigl(g \cdot q^{L_0 - c/24}\,\bar{y}^{J_0}\,(-1)^{F+\bar{F}}\bigr)$ is a weak Jacobi form of weight 0, index 1 for $\Gamma_0(N_g)$, where $N_g = \mathrm{ord}(g)$. Of the 26 conjugacy classes of M_{24}, only 21 appear as symmetries of K3 sigma models (Gaberdiel–Hohenegger–Volpato 2010/2011): the five missing classes $(7A, 7B, 15A, 15B, 23A/B)$ have Frame shapes incompatible with the K3 lattice. Each realized class g has a Frame shape $\pi_g = \prod i^{a_i}$ encoding the permutation action on 24 letters, with associated eta product $\eta_g(\tau) = \prod_i \eta(i\tau)^{a_i}$. The 21 realized classes with their Frame shapes and orders:

Class	Frame shape	N_g	Class	Frame shape	N_g	Class	Frame shape	N_g
$1A$	1^{24}	1	$2A$	$1^8 2^8$	2	$2B$	2^{12}	2
$3A$	$1^6 3^6$	3	$4A$	$1^4 2^2 4^4$	4	$4B$	$2^4 4^4$	4
$4C$	4^6	4	$5A$	$1^4 5^4$	5	$6A$	$1^2 2^1 3^2 6^2$	6
$6B$	$1^2 2^2 3^1 6^1$	6	$8A$	$1^2 2^1 4^1 8^2$	8	$10A$	$1^2 2^1 5^1 10^1$	10
$11A$	$1^2 11^2$	11	$12A$	$1^1 2^1 4^1 3^1 6^1 12^1$	12	$12B$	$2^1 4^1 6^1 12^1$	12
$14A$	$1^1 2^1 7^1 14^1$	14	$14B$	$1^1 2^1 7^1 14^1$	14	$21A$	$1^1 3^1 7^1 21^1$	21
$21B$	$1^1 3^1 7^1 21^1$	21	$23A$	$1^1 23^1$	23	$23B$	$1^1 23^1$	23

The $23A$ and $23B$ classes are listed for completeness but are among the five NOT realized as K3 symmetries. The Frame shape determines the eta product η_g and hence the shadow of the twined mock modular form; the shadow of h_g is proportional to η_g.

Lattice VOA and Gepner model

Remark 14.57 (Lattice VOA from $H^2(K3, \mathbb{Z})$). The K3 cohomology lattice $L_{K3} = H^2(K3, \mathbb{Z}) \cong U^3 \oplus (-E_8)^2$ has rank 22, signature $(3, 19)$, and discriminant $\det(G) = -1$. Since L_{K3} is indefinite, the naive theta function diverges. The negative-definite sublattice $(-E_8)^2$ of rank 16 produces a well-defined lattice VOA $V_{(-E_8)^2}$ with $c = 16$ and $\kappa = 16$ (κ = rank for lattice VOAs). The theta series $\Theta_{E_8}(\tau) = E_4(\tau) = 1 + 240q + \cdots$ (the weight-4 Eisenstein series, since E_8 is the unique even unimodular lattice of rank 8).

The full Mukai lattice $\widetilde{H}(K3, \mathbb{Z}) = U^4 \oplus (-E_8)^2$ has rank 24, signature $(4, 20)$, and is the lattice relevant for Bridgeland stability conditions and derived categories.

Remark 14.58 (Gepner model $(2)^4$). The Gepner model for the quartic K3 (Fermat quartic $x_1^4 + x_2^4 + x_3^4 + x_4^4 = 0$ in $\mathbb{CP}^3$) is the tensor product of four copies of the $\mathcal{N} = 2$ minimal model at level $k = 2$ ($c = 3k/(k + 2) = 3/2$), with GSO projection and $\mathbb{Z}_4$ orbifold. Total $c = 4 \times 3/2 = 6$. The chiral ring is isomorphic to the Jacobian ring $\mathrm{Jac}(W) = \mathbb{C}[x_1, \ldots, x_4]/(\partial W/\partial x_i)$ with $\dim \mathrm{Jac}(W) = 3^4 = 81$ before orbifold. The symmetry group at the Gepner point contains $(\mathbb{Z}/4\mathbb{Z})^4 \rtimes S_4$ (order 6144) and embeds into the Conway group Co_1.

Chiral de Rham complex on K3

Remark 14.59 (CDR on K3). The chiral de Rham complex $\Omega^{\mathrm{ch}}(K3)$ of Malikov–Schechtman–Vaintrob is a sheaf of vertex superalgebras on K3 with central charge $c = 0$ (the local contributions $c_{\beta\gamma} = -2$ and $c_{bc} = +2$ per complex dimension cancel globally). The CDR cohomology recovers the K3 elliptic genus: $\mathrm{ch}\big(H^*(K3, \Omega^{\mathrm{ch}})\big) = \mathrm{Ell}(K3; q, y) = 2\,\phi_{0,1}(\tau, z)$ (Borisov–Libgober).

On a hyperkähler manifold (such as K3), the CDR carries $\mathcal{N} = 4$ superconformal symmetry at $c = 0$ — the topological $\mathcal{N} = 4$ algebra, distinct from the physical $\mathcal{N} = 4$ at $c = 6$. Both give $\kappa = \dim_{\mathbb{C}}(K3) = 2$.

Koszul dual of the K3 chiral algebra

Proposition 14.60 *(Koszul dual of $\mathcal{A}_{K3}$; [PH])*. The Koszul dual of the $\mathcal{N} = 4$ SCA at $c = 6$, $k_R = 1$ has:

- $\kappa(\mathcal{A}^!_{K3}) = -2$ (by complementarity: $\kappa + \kappa' = 0$ for free-field/CY sigma models).
- $c(\mathcal{A}^!_{K3}) = -6$ (the $\mathcal{N} = 4$ SCA at negative central charge, with $k'_R = -1$, since $\kappa = 2k_R$ and $\kappa' = -2$ gives $k'_R = -1$, hence $c' = 6k'_R = -6$).
- $F_g(\mathcal{A}^!) = -2 \cdot \lambda_g^{\mathrm{FP}}$ at all genera.

The Verdier intertwining (Vol I, Theorem A; Convention III): $D_{\mathrm{Ran}}(B(\mathcal{A}_{K3})) \simeq B(\mathcal{A}^!_{K3})$. The derived centre $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}_{K3})$ (the universal bulk) is a separate object from $\mathcal{A}^!$.

14.10 The Calabi–Yau threefold $K3 \times E$: geometry and invariants

Hodge diamond and topological invariants

PROPOSITION 14.61 *(Hodge diamond of $K3 \times E$; [PE])*. The Künneth formula gives the Hodge diamond of $K3 \times E$ ($\dim_{\mathbb{C}} = 3$):

$$\begin{array}{ccccccc} & & & 1 & & & \\ & & 1 & & 1 & & \\ & 1 & & 21 & & 1 & \\ 1 & & 21 & & 21 & & 1 \\ & 1 & & 21 & & 1 & \\ & & 1 & & 1 & & \\ & & & 1 & & & \end{array}$$

Key invariants:

- $h^{1,1} = h^{2,1} = 21$, giving $\chi(K3 \times E) = 2(h^{1,1} - h^{2,1}) = 0$.
- Betti numbers: $b_0 = b_6 = 1$, $b_1 = b_5 = 2$, $b_2 = b_4 = 23$, $b_3 = 44$. Total: $\sum b_k = 96$.
- $K_0(K3 \times E) \cong \mathbb{Z}^{48}$ ($K_0(K3) \otimes K_0(E) = \mathbb{Z}^{24} \otimes \mathbb{Z}^2$).

Proof. Apply Künneth: $h^{p,q}(K3 \times E) = \sum_{a+c=p,\, b+d=q} h^{a,b}(K3) \cdot h^{c,d}(E)$. For example: $h^{1,1} = h^{1,1}(K3) \cdot h^{0,0}(E) + h^{0,0}(K3) \cdot h^{1,1}(E) = 20 + 1 = 21$. The alternating sum $1 - 2 + 23 - 44 + 23 - 2 + 1 = 0$ verifies $\chi = 0$. □

HKR decomposition and deformation space

PROPOSITION 14.62 *(Deformation space of $D^b(K3 \times E)$; [PE])*. By the HKR isomorphism for a smooth CY 3-fold: $\mathrm{HH}^n(X) = \bigoplus_{p+q=n} h^{3-p,\,q}(X)$. For $K3 \times E$:

$\mathrm{HH}^0 = 1$	automorphisms	$h^{3,0}$
$\mathrm{HH}^1 = 2$	outer derivations	$h^{3,1} + h^{2,0}$
$\mathrm{HH}^2 = 23$	first-order deformations	$h^{3,2} + h^{2,1} + h^{1,0}$
$\mathrm{HH}^3 = 44$	obstructions	$h^{3,3} + h^{2,2} + h^{1,1} + h^{0,0}$

The 23-dimensional deformation space: 21 complex structure (from $h^{2,1}$), 1 B-field (from $h^{1,0}$), 1 volume (from $h^{3,2}$). Total $\dim \mathrm{HH}^* = 96$.

Noncommutative deformations and Brauer group

Remark 14.63 (Brauer group of $K3 \times E$). $\mathrm{Br}(K3 \times E) = \mathrm{Br}(K3) \oplus \mathrm{Br}(E)$ contains $(\mathbb{Q}/\mathbb{Z})^{21}$ (K3 at Picard rank $\rho = 1$) plus $\mathbb{Q}/\mathbb{Z}$ from E. The K3 symplectic form provides the unique Poisson structure $\pi = \omega^{-1}$ (since $\dim H^0(\bigwedge^2 T_{K3}) = 1$), but the E-direction carries none ($\bigwedge^2 T_E = 0$).

Autoequivalences and semiorthogonal structure

PROPOSITION 14.64 *(Indecomposability of $D^b(K3 \times E)$; [PE]).* $D^b(K3)$ admits no proper semiorthogonal decomposition (Bridgeland, Kawamata, Huybrechts: $\omega_{K3} = \mathcal{O}$ forces Serre functor $[2]$, and the only fractional CY category with Serre $[2]$ is $D^b(K3)$ itself). Hence $D^b(K3 \times E) = D^b(K3) \otimes D^b(E)$ is indecomposable.

Autoequivalences: $\mathrm{Aut}(D^b(K3)) \to O^+(\widetilde{H}(K3, \mathbb{Z}))$ via oriented Hodge isometries (line bundle twists, shifts, spherical twists T_E with $T_E^2 = [-2]$). $\mathrm{Aut}(D^b(E)) = (E \times E^\vee) \rtimes (\mathrm{SL}(2, \mathbb{Z}) \rtimes \mathbb{Z})$. The virtual dimension of all moduli problems on $K3 \times E$ vanishes: $\mathrm{vdim} = -\chi(\mathcal{E}, \mathcal{E}) = 0$.

Factorization homology

Remark 14.65 (Factorization homology on $K3 \times E$). The factorization homology of the HT-twisted sigma model: $\int_{K3\times E} A^{\mathrm{HT}} \simeq H^*(K3 \times E, \Omega^{\mathrm{ch}})$. The CDR character gives $\chi(\Omega^{\mathrm{ch}}) = 0$, but the Hodge-graded character is nontrivial. $\dim \mathrm{HH}^*(D^b(K3 \times E)) = 96$.

14.11 Enumerative geometry of $K3 \times E$

Gromov–Witten theory and BPS invariants

Remark 14.66 (Reduced GW theory on K3). For K3, the standard virtual class vanishes for $\beta \neq 0$ (cosection from $H^{2,0}(K3) = \mathbb{C}$). The reduced theory (Bryan–Leung, Maulik–Pandharipande–Thomas) removes this obstruction. The KKV formula (proved by Pandharipande–Thomas 2016) gives the BPS state counts. The genus-0 BPS invariants satisfy the Yau–Zaslow formula:

$$\sum_{h\geq 0} n_h^0\, q^h = \prod_{n\geq 1} \frac{1}{(1-q^n)^{24}}. \tag{14.37}$$

PROPOSITION 14.67 *(BPS invariants of K3; [PE]).* Selected BPS invariants (all integers, proved by Pandharipande–Thomas 2016):

h	0	1	2	3	4	5	6	7
n_h^0	1	24	324	3200	25650	176256	1073720	5930496
n_h^1	0	−2	−54	−800	−8550	−73440	−536860	−3464264
n_h^2	0	0	3	104	1701	19656	176358	1316808
n_h^3	0	0	0	−4	−155	−2832	−34470	−320320

Signed convention: $n_h^g = (-1)^g |n_h^g|$. The genus-0 values are the Yau–Zaslow numbers (OEIS A006922); $n_1^0 = 24 = \chi(K3)$. The Hilbert scheme Euler characteristics $\chi(\mathrm{Hilb}^n(K3))$ provide the genus-0 column: $\chi(\mathrm{Hilb}^n) = n_n^0$, with generating function $\sum_{n\geq 0} \chi(\mathrm{Hilb}^n)\, q^n = \prod_{k\geq 1}(1-q^k)^{-24}$ (Göttsche formula). The first eight values are $1, 24, 324, 3200, 25650, 176256, 1073720, 5930496$.

Bridgeland stability on K3

Remark 14.68 (Stability conditions). $\mathrm{Stab}^\dagger(K3) \to P^+(K3)$ is a covering of the period domain (Bridgeland 2008). Walls of marginal stability for Mukai vectors $v = v_1 + v_2$ are semicircles in the (B, ω)-plane; crossing them applies the KS wall-crossing. For primitive v with $v^2 = 2n - 2$: $\Omega(v) = \chi(\mathrm{Hilb}^n(K3))$.

14.12 BPS spectrum and black hole entropy

Charge lattice and BPS states

Proposition 14.69 *(Charge lattice; [PE])*. Type IIA on $K3 \times E$ gives 4d $\mathcal{N} = 4$ supergravity with 22 vector multiplets. The D-brane charge lattice $\Gamma = H^{\text{even}}(K3 \times E, \mathbb{Z})$:

Cohomology	Brane type	Rank
$H^0 = \mathbb{Z}$	D6-branes	1
$H^2 = \mathbb{Z}^{23}$	D4-branes	23
$H^4 = \mathbb{Z}^{23}$	D2-branes	23
$H^6 = \mathbb{Z}$	D0-branes	1

Total: $\operatorname{rank} \Gamma = 48$.

Black hole entropy from $1/\Phi_{10}$

Proposition 14.70 *(BMPV and $\frac{1}{4}$-BPS entropy; [PE])*. In 5d (IIA on $K3 \times S^1$): the BMPV black hole with charges (Q_1, Q_5, n) has $S_{\text{BH}} = 2\pi\sqrt{Q_1\, Q_5\, n}$ (Cardy with $c_L = 6\, Q_1\, Q_5$). In 4d (IIA on $K3 \times T^2$): the $\frac{1}{4}$-BPS dyon with discriminant Δ has

$$S_{\text{BH}} = 4\pi\sqrt{\Delta},$$

and the DVV formula $Z_{1/4\text{-BPS}} = 1/\Phi_{10}$ ((14.48)) gives the exact microscopic count. For large Δ: $\log d(\Delta) \sim 4\pi\sqrt{\Delta} - \frac{3}{2}\log\Delta + \cdots$, where the subleading term is the universal one-loop correction (Sen 2005, 2012). The asymptotic expansion is the leading term of the Rademacher exact formula for the Fourier coefficients of $1/\Phi_{10}$:

$$d(\Delta) \;=\; \frac{2\pi}{\sqrt{\Delta}} \sum_{c=1}^{\infty} \frac{K(c,\Delta)}{c}\, I_{27/2}\!\Big(\frac{4\pi\sqrt{\Delta}}{c}\Big) \;+\; \text{polar contributions},$$

where $I_{27/2}$ is the modified Bessel function of the first kind of order $27/2$ and $K(c, \Delta)$ is a Kloosterman-type sum over $\mathrm{Sp}(4, \mathbb{Z})$. The $c = 1$ term dominates at large Δ, giving $d(\Delta) \sim e^{4\pi\sqrt{\Delta}} \cdot \Delta^{-29/4}$. The convergence of the Rademacher series is exponentially fast: the $c = 2$ correction is suppressed by $e^{-2\pi\sqrt{\Delta}}$ relative to the leading term.

Remark 14.71 (OSV conjecture). The Ooguri–Strominger–Vafa relation $Z_{\text{BH}} = |Z_{\text{top}}|^2$ connects the black hole partition function to the topological string. For $K3 \times E$, the topological string is exactly solvable at genus 0 and 1. The shadow tower gives $F_g = 3 \cdot \lambda_g^{\text{FP}}$ at each genus.

Fourier–Jacobi as genus escalation

Proposition 14.72 *(Bridge: elliptic R-matrix data determines BKM root system; [PE])*. The Fourier–Jacobi expansion of the Igusa cusp form $\Phi_{10}(\Omega) = \sum_{m\ge1} \phi_m(\tau, z)\, p^m$ uses Jacobi forms $\phi_m(\tau, z)$ on E_τ. The leading coefficient

$$\phi_1(\tau, z) = \eta(\tau)^{18}\, \vartheta_1(\tau, z)^2 \tag{14.38}$$

is built from the same theta function $\vartheta_1(\tau, z)$ that defines the elliptic propagator $d \log \vartheta_1(z|\tau)$ ((14.2)) and Felder's elliptic R-matrix entries $a(u) = \theta(u+\eta)/\theta(\eta)$ (Definition 14.15).

The passage from genus-1 data (Jacobi forms on E_τ) to genus-2 data (the Siegel form Φ_{10} on $\mathfrak{H}_2$) is the Borcherds multiplicative lift. Its input is the K3 elliptic genus $Z_{K3} = 2\,\phi_{0,1}$ (Definition 14.52), and the exponents $c_0(4nm - \ell^2)$ in the infinite product (14.47) are precisely the root multiplicities of the BKM superalgebra $\mathfrak{g}_{\Phi_{10}}$ (Proposition 14.78).

In the factorization $\eta^{18} = \eta^{24} \cdot \eta^{-6}$: $\eta^{24} = \Delta$ is the Ramanujan discriminant (the 24 free bosons of A_E, with $\kappa(A_E) = 24$), and $\eta^{-6} = \eta^{-2\kappa}$ with $\kappa = \kappa(\Omega^{\mathrm{ch}}(K3 \times E)) = 3$. The shadow obstruction tower detects $\kappa = 3$ (Proposition 14.110); the Borcherds lift escalates this genus-1 datum to the genus-2 Siegel modular form.

14.13 Twisted holography on $K3 \times E$

Construction 14.73 *(Costello–Li KS boundary algebra).* The Costello–Li programme extracts a boundary chiral algebra from the HT twist of Type IIB on $K3 \times E$. Reduction of Kodaira–Spencer theory along the topological directions (K3) produces a chiral algebra A_E on E with generators from harmonic forms on K3:

$h^{p,q}(K3)$	Contribution	Generators
$h^{0,0} = 1$	Scalar boson	1
$h^{2,0} = 1$	Symplectic boson	1
$h^{1,1} = 20$	Weight-1 bosons	20
$h^{0,2} = 1$	Conjugate boson	1
$h^{2,2} = 1$	Volume boson	1

Total: $c(A_E) = \chi(K3) = 24$. This is NOT the K3 sigma model ($c = 6$) but 24 free bosons from the B-model reduction. $\kappa(A_E) = 24$ (rank of the free-boson lattice), $\kappa(A_E^!) = -24$ ($\kappa + \kappa' = 0$ for free fields).

Warning 14.74 (OPE level matrix for KS boundary bosons). The 24 free bosons $J^a(z)$ ($a = 1, \ldots, 24$) of the KS boundary algebra A_E have OPE $J^a(z)\,J^b(w) \sim \delta^{ab}/(z-w)^2$, so the OPE level matrix is δ^{ab} (the unit matrix), *not* the Mukai pairing G^{ab} of signature $(4, 20)$ on $H^*(K3, \mathbb{Z})$. The Mukai pairing enters through the vertex operators $V_\gamma(z) = {:}\exp(i\gamma_a J^a(z)){:}$ for lattice vectors $\gamma \in \Gamma_{\mathrm{Mukai}}$, whose correlation functions involve $\langle V_\gamma V_{\gamma'} \rangle \propto \exp(\gamma \cdot G \cdot \gamma')$. Confusing the OPE level δ^{ab} with the lattice pairing G^{ab} produces wrong κ values: the OPE level gives $\kappa(A_E) = 24$ (sum of diagonal entries), while the Mukai norm $\mathrm{tr}(G) = 4 - 20 = -16$ would give a wrong and negative κ.

Remark 14.75 (Holographic modular Koszul datum). $H(K3 \times E) = (A, A^!, C, r(z), \Theta_A, \nabla^{\mathrm{hol}})$ with $A = A_E$ ($c = 24$, $\kappa = 24$), $A^! = A_E^!$ ($\kappa = -24$), $C = Z^{\mathrm{der}}_{\mathrm{ch}}(A)$ (universal bulk), $r(z) = \Omega/z$ (Casimir, 24-dim), Θ_A (bar-intrinsic MC element), ∇^{hol} (shadow connection, class G).

Proposition 14.76 *(Boundary-to-sigma ratio; [PH]).* The boundary algebra A_E and the sigma model VOA V_{K3} have modular characteristics in ratio 12. At genus 1:

$$\frac{\kappa(A_E)}{\kappa(V_{K3})} = \frac{24}{2} = 12, \qquad \frac{F_1(A_E)}{F_1(V_{K3})} = 12. \tag{14.39}$$

Here A_E has $c = 24$ from the 24 free bosons of the Kaluza–Klein reduction on K3 (Construction 14.73) indexed by the Mukai lattice $H^*(K3, \mathbb{Z})$, and V_{K3} is the $\mathcal{N} = 4$ SCA at $c = 6$ with $\kappa(V_{K3}) = 2$ (Proposition 14.48). The SUSY Ward identities of the $\mathcal{N} = 4$ algebra reduce κ by a factor of $2/3$ relative to the Virasoro formula $\kappa(\mathrm{Vir}_6) = 3$ (Remark 14.49).

For A_E (24 free bosons, class G, uniform weight), the scalar formula $F_g(A_E) = 24 \cdot \lambda_g^{\mathrm{FP}}$ holds at all genera (Vol I, Theorem D). For V_{K3} (class M, generators at weights $1, 3/2, 2$), the scalar formula $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ is proved at genus 1; at $g \geq 2$ the multi-weight genus expansion may receive cross-channel corrections (Vol I, AP32). The ratio 12 is therefore proved at $g = 1$ and conditional at $g \geq 2$.

Proof. $\kappa(A_E) = 24$ (rank of the free-boson lattice). $\kappa(V_{K3}) = 2$ (Proposition 14.48). Their ratio is $24/2 = 12$. At genus 1, $F_1(\mathcal{A}) = \kappa(\mathcal{A})/24$ for both algebras, giving the ratio 12. At $g \geq 2$: A_E is uniform-weight (class G), so $F_g(A_E) = 24 \cdot \lambda_g^{\mathrm{FP}}$ at all genera. V_{K3} has generators at weights $1, 3/2, 2$; the scalar formula $F_g = 2 \cdot \lambda_g^{\mathrm{FP}}$ is proved at $g = 1$ but may receive multi-weight corrections at $g \geq 2$. □

Remark 14.77 (Leech connection). The boundary partition function $Z_1(A_E; \tau) = 1/\eta(\tau)^{48}$ at genus 1 matches the Leech lattice VOA V_Λ through order q^1: $1/\eta^{48} = q^{-2}(1 + 48\,q + \cdots)$, while V_Λ has $\dim(V_\Lambda)_1 = 0$ and $\dim(V_\Lambda)_2 = 196560$. The two diverge at order q^2: the 196560 vectors of the Leech lattice have no counterpart in the 24-boson Fock space at weight 2. The Leech lattice has $\kappa(V_\Lambda) = 24 = \kappa(A_E)$ (both are 24 free bosons at level 1), but their weight-2 spaces differ.

14.14 Mock modular forms and the BKM root system

BKM root multiplicities

PROPOSITION 14.78 *(Root multiplicities; [PE]).* The BKM superalgebra $\mathfrak{g}_{\Phi_{10}}$ has root multiplicities from the K3 elliptic genus:

- Real roots ($D = -1, \alpha^2 = 2$): mult $= c_0(-1) = 2$.
- Lightlike roots ($D = 0, \alpha^2 = 0$): mult $= c_0(0) = 10$ (20 from $2\phi_{0,1}$).
- Imaginary roots ($D > 0$): mult $= c_0(D)$, unbounded. Leading: $c_0(3) = -64$, $c_0(4) = 108$, $c_0(7) = -513$, $c_0(8) = 808$.

Negative multiplicities at odd D indicate fermionic directions in the superalgebra. The discriminant coefficients $c_0(D)$ of $\phi_{0,1}$ through $D = 12$ are (Eichler–Zagier convention):

D	-1	0	3	4	7	8	11	12	15	16	19	20
$c_0(D)$	2	10	-64	108	-513	808	-2752	4016	-11775	16524	-43263	58956

The $K3$ elliptic genus uses $2\,\phi_{0,1}$, so the BKM root multiplicities are $\mathrm{mult}(\alpha) = 2\,c_0(D)$ for discriminant D. The theta decomposition $\phi_{0,1} = h_0(\tau)\,\vartheta_{0,1}(\tau, z) + h_1(\tau)\,\vartheta_{1,1}(\tau, z)$ separates even (ℓ even, $D \equiv 0 \pmod 4$) and odd (ℓ odd, $D \equiv 3 \pmod 4$) sectors, with $h_0(\tau) = 10 + 108\,q + 808\,q^2 + \cdots$ and $h_1(\tau) = q^{-1/4}(1 - 64\,q + \cdots)$.

The Hilbert scheme bridge

THEOREM 14.79 *(Toroidal action on K-theory of Hilbert schemes [?]; [PE]).* Let S be a smooth projective surface. The toroidal algebra $U_{q,t}(\hat{\hat{\mathfrak{gl}}}_1)$ acts on $\bigoplus_{n \geq 0} K(\mathrm{Hilb}^n(S))$ by correspondences on the nested Hilbert scheme $\mathrm{Hilb}^{n,n+1}(S)$ (Schiffmann–Vasserot). Specializations:

(i) *Nakajima operators* ($q = 1$): the Heisenberg algebra $\bigoplus_{n \in \mathbb{Z}} \mathbb{Q}\,\mathfrak{a}_n$ acts on $\bigoplus_n H^*(\mathrm{Hilb}^n(S))$ via Nakajima correspondences, with $\mathfrak{a}_{-n}|0\rangle$ the class of the punctual Hilbert scheme.

(ii) *Haiman's theorem* ($S = \mathbb{C}^2$): the DAHA $\ddot{H}_n(q,t)$, a quotient of $U_{q,t}(\hat{\hat{\mathfrak{gl}}}_1)$, acts on $K(\mathrm{Hilb}^n(\mathbb{C}^2))$ with fixed-point basis indexed by partitions $\lambda \vdash n$, and the character of the natural representation is the Macdonald polynomial $\widetilde{H}_\lambda(x;q,t)$.

(iii) $S = K3$: $\chi(\mathrm{Hilb}^n(K3)) = p_{24}(n)$ (24-coloured partitions), so $\sum_{n\geq 0}\chi(\mathrm{Hilb}^n(K3))\,p^n = \prod_{m\geq 1}(1-p^m)^{-24}$.

Proposition 14.80 *(DMVV from Hilbert scheme characters; [PE])*. The DVV formula (14.48) $1/\Phi_{10} = \sum_N p^N\,\chi(\mathrm{Sym}^N(K3);\tau,z)$ is the generating function of equivariant Euler characteristics of $\mathrm{Hilb}^N(K3)$ refined by the $\mathrm{SU}(2)_R$ fugacity z and the L_0 fugacity $q = e^{2\pi i\tau}$. The 24 coloured partitions arise from Nakajima's Heisenberg action ($24 = \chi(K3) = b_0 + b_2 + b_4 = 1 + 22 + 1$): each colour corresponds to a cohomology class of K3, matching the 24 free bosons of the boundary algebra A_E (Construction 14.73).

Remark 14.81 (Synthesis: toroidal algebras meet BPS counting). The three strands of this chapter converge on the Hilbert scheme. The toroidal algebra $U_{q,t}(\hat{\hat{\mathfrak{g}}})$ of §14.1 provides the symmetry algebra acting on $K(\mathrm{Hilb}^n(S))$. The elliptic R-matrix of §14.2 appears as the transition matrix between stable-envelope bases of $K(\mathrm{Hilb}^n(S))$ (Aganagic–Okounkov). The BKM root multiplicities $c_0(D)$ of Proposition 14.78 are the Fourier coefficients of the K3 elliptic genus $Z_{K3} = 2\,\phi_{0,1}$, which in turn counts representations of the Heisenberg subalgebra on $\bigoplus_n H^*(\mathrm{Hilb}^n(K3))$. The shadow invariant $\kappa = 3$ measures the single-copy chiral algebra $\Omega^{\mathrm{ch}}(K3\times E)$; the BKM algebra $\mathfrak{g}_{\Phi_{10}}$ encodes the second-quantized (symmetric-product) extension, and the gap between them is the content of the shadow–Siegel gap analysis (§14.16).

Fourier–Jacobi expansion

Remark 14.82 (Fourier–Jacobi structure). $\Phi_{10} = \sum_{m\geq 1}\phi_m(\tau,z)\,p^m$ with: $\phi_1 = \phi_{10,1} = \eta^{18}\,\vartheta_1^2 = -\Delta\cdot\phi_{-2,1}$ (unique Jacobi cusp form of weight 10, index 1); $\phi_2 = 2\,\Delta\,\phi_{-2,1}\,\phi_{0,1}$ ($\dim J^{\mathrm{cusp}}_{10,2} = 1$). The factorization $\eta^{18} = \eta^{24}\cdot\eta^{-6}$ separates the Ramanujan discriminant from $\eta^{-2\kappa}$ with $\kappa = 3$. The Jacobi form ring $J^{\mathrm{weak}}_{*,*} = \mathbb{C}[E_4, E_6, \phi_{-2,1}, \phi_{0,1}]$ (Eichler–Zagier).

14.15 Grand synthesis and open problems

Remark 14.83 (Master data table for $K3\times E$). The complete invariant data assembled from the 39 CY compute engines:

Invariant	**Value**
$\dim_{\mathbb{C}}(K3 \times E)$	3
$\chi(K3 \times E)$	0
$h^{1,1} = h^{2,1}$	21
$\operatorname{rank} K_0$	48
$\dim \mathrm{HH}^*$	96
$\kappa_{\mathrm{ch}} = \kappa(\Omega^{\mathrm{ch}})$	3
$\kappa_{\mathrm{BKM}} = \mathrm{wt}(\Phi_{10})/2$	5
$\kappa_{\mathrm{BCOV}} = \chi/24$	0
Shadow depth	class M ($r_{\max} = \infty$)
F_1	$1/8$
F_2	$7/1920$
F_3	$31/322560$
$Z_{\mathrm{DT},0}$	1
n_1^0 (genus-0 BPS, $h = 1$)	24
$S_{\mathrm{BH}}(\Delta)$	$4\pi\sqrt{\Delta}$
$c(A_E^{\mathrm{KS}})$ (twisted holography)	24
$\kappa(A_E^{\mathrm{KS}})$	24
$\mathrm{Br}(K3 \times E)$	$\supset (\mathbb{Q}/\mathbb{Z})^{22}$

Definition 14.84 *(κ-spectrum).* For a Calabi–Yau threefold X, the *κ-spectrum* is

$$\mathrm{Spec}_\kappa(X) \;:=\; \big\{\kappa(\mathcal{A}) : \mathcal{A} \text{ a chirally Koszul algebra associated to } X\big\}.$$

Each element is the modular characteristic (Vol I, Theorem D) of a specific algebraization of X. The κ-spectrum is an invariant of X that remembers the *set* of modular characteristics arising from all chirally Koszul algebraizations, not any single one.

Example 14.85 *($K3 \times E$: four-element κ-spectrum).* The geometry $K3 \times E$ admits at least four distinct chirally Koszul algebraizations:

Algebraization	c	κ	**Shadow class**
Chiral de Rham $\Omega^{\mathrm{ch}}(K3 \times E)$	0	3	M
K3 sigma model ($\mathcal{N} = 4$ SCA)	6	2	M
KS boundary algebra (24 free bosons)	24	24	G
BPS spectrum ($\frac{1}{2}\,\mathrm{wt}(\Phi_{10})$)	—	5	M

Hence $\mathrm{Spec}_\kappa(K3 \times E) \supseteq \{2, 3, 5, 24\}$. The four values probe different aspects of the geometry: $\kappa_{\mathrm{ch}} = 3 = \dim_{\mathbb{C}}(K3 \times E)$ is the chiral dimension; $\kappa(K3) = 2 = \chi(\mathcal{O}_{K3})$ is the arithmetic genus; $\kappa(A_E) = 24$ counts free bosons; $\kappa_{\mathrm{BPS}} = 5$ is half the weight of the Igusa cusp form Φ_{10}.

Remark 14.86 (The κ-spectrum as invariant). The κ-spectrum (Definition 14.84) is analogous to the multiple spectra of a Riemannian manifold (Laplacian, Dirac, length): different algebraizations of the same geometry probe different aspects of the modular structure. Whether $\mathrm{Spec}_\kappa(X)$ is a complete invariant (i.e., whether $\mathrm{Spec}_\kappa(X) = \mathrm{Spec}_\kappa(X')$ implies $X \simeq X'$) is open.

Remark 14.87 (The second-quantization gap). The shadow tower is first-quantized (single-copy). $1/\Phi_{10}$ is second-quantized (DMVV symmetric product). The ratio $\kappa_{\mathrm{BKM}}/\kappa_{\mathrm{ch}} = 5/3$ reflects the passage from single copy to Sym^N and is specific to $K3 \times E$ (the quintic gives a different ratio). The Borcherds lift converts additive genus-1 data into multiplicative genus-2 data: shadow genus escalation.

Proposition 14.88 *(The BPS modular characteristic; [PH])*. The BPS modular weight decomposes additively:

$$\kappa_{\mathrm{BPS}} \;=\; \kappa(K3) + \kappa(K3 \times E) \;=\; 2 + 3 \;=\; 5 \;=\; \tfrac{1}{2}\,\mathrm{wt}(\Phi_{10}). \tag{14.40}$$

The first summand $\kappa(K3) = 2$ is the fiber modular characteristic (Proposition 14.48); the second $\kappa(K3 \times E) = 3 = \dim_{\mathbb{C}}$ is the surface modular characteristic (Proposition 14.110). The ratio

$$\frac{\kappa_{\mathrm{BPS}}}{\kappa_{\mathrm{ch}}} \;=\; \frac{5}{3} \;=\; \frac{\mathrm{wt}(\Phi_{10})}{2\,\dim_{\mathbb{C}}(K3 \times E)}$$

reflects the passage from single-copy to symmetric-product. The distinction between Hilb^N and Sym^N accounts for additional BPS states: $\chi(\mathrm{Hilb}^2(K3)) - \chi(\mathrm{Sym}^2(K3)) = 324 - 300 = 24 = \chi(K3)$, arising from the exceptional divisor of the Hilbert–Chow morphism $\mathrm{Hilb}^2(K3) \to \mathrm{Sym}^2(K3)$. The DMVV formula (14.48) uses the orbifold Sym^N formula; the Göttsche formula gives $\sum_N q^N \chi(\mathrm{Hilb}^N(K3)) = \prod_{k \ge 1}(1 - q^k)^{-24}$.

Proof. The additivity $\kappa_{\mathrm{BPS}} = \kappa(K3) + \kappa(K3 \times E)$ is formal: the Borcherds lift weight is $c(0)/2 = 10/2 = 5$, while $\kappa(K3) = 2$ and $\kappa(K3 \times E) = 3$ are independently computed. That their sum equals 5 is verified by $\kappa(K3) + \kappa(K3 \times E) = 2 + 3 = 5 = c(0)/2$. The Hilbert–Chow difference: $\mathrm{Hilb}^2(S)$ for a K3 surface S is the blowup of $\mathrm{Sym}^2(S)$ along the diagonal, with exceptional divisor $\mathbb{P}(T_S) \cong \mathbb{P}^1$-bundle. By the blowup formula, $\chi(\mathrm{Hilb}^2) = \chi(\mathrm{Sym}^2) + \chi(S) = \binom{25}{2} + 24 = 300 + 24 = 324$. □

Remark 14.89 (Hilbert–Chow difference at higher N). The identity $\chi(\mathrm{Hilb}^2(S)) - \chi(\mathrm{Sym}^2(S)) = \chi(S)$ generalizes: for all $N \ge 1$, $\chi(\mathrm{Hilb}^N(S))$ is the coefficient of q^N in $\prod_{k \ge 1}(1 - q^k)^{-\chi(S)}$ (Göttsche's formula), while $\chi(\mathrm{Sym}^N(S)) = \binom{\chi(S)+N-1}{N}$ (the combinatorial formula for orbifold Euler characteristics). The difference $\delta_N = \chi(\mathrm{Hilb}^N) - \chi(\mathrm{Sym}^N)$ grows with N; the leading corrections come from the exceptional loci of the Hilbert–Chow resolution $\mathrm{Hilb}^N(S) \to \mathrm{Sym}^N(S)$, which become increasingly complicated as partitions of N acquire more parts. For $K3$: $\delta_1 = 0$, $\delta_2 = 24 = \chi(K3)$, $\delta_3 = 600$, $\delta_4 = 8100$, matching the Göttsche coefficients $1, 24, 324, 3200, 25650, \ldots$ minus the symmetric product coefficients $1, 24, 300, 2600, 17550, \ldots$.

[Does the bar complex of $\mathcal{A}_{K3}$ detect M_{24}?] The mock modular half-multiplicities $a_n = 45, 231, 770, \ldots$ are dimensions of M_{24} representations. The bar complex $B(\mathcal{A}_{K3})$ detects $\kappa = 2$ at arity 2 and the cubic shadow at arity 3. Does the full shadow obstruction tower $\Theta_{\mathcal{A}_{K3}}$ encode the M_{24} action? The mock modular shadow $24\,\eta^3$ and the bar-complex shadow connection ∇^{sh} share the structural feature of encoding modular non-invariance. Whether they are related by a precise functor remains open.

[Factorization envelope of the K3 chiral algebra] The factorization envelope $U_X^{\mathrm{mod}}(\mathcal{A}_{K3})$ in the sense of Nishinaka (2025/26) and Vicedo (2025) is not yet constructed. At the genus-0 level, the K3 sigma model provides a factorization algebra on $\mathrm{Ran}(X)$ for any algebraic curve X; the modular completion (incorporating all genera) is the frontier. The six-fold platonic package $\Pi_X(\mathcal{A}_{K3})$ is well-defined (Construction III); what remains is the modular factorization envelope as the left adjoint of $\mathrm{Prim}^{\mathrm{mod}}$.

Remark 14.90 (Constructive reconstruction of $D^b(K3 \times E)$). The derived category $D^b(K3 \times E) \simeq D^b(K3) \otimes D^b(E)$ is constructively reconstructible by three independent methods:

(i) *Čech descent.* Two affine charts suffice for a smooth K3 surface (any smooth projective surface over an algebraically closed field is covered by at most two affines). The Čech descent spectral sequence (Theorem 14.45)

$$E_1^{p,q} = \bigoplus_{|I|=p+1} \mathrm{HH}^q(D^b(U_I)) \implies \mathrm{HH}^{p+q}(D^b(K3))$$

recovers $D^b(K3)$ from local data on U_0, U_1 with transition data on U_{01}.

(ii) *BKR equivalence.* At the Kummer point $K3 = T^4/\mathbb{Z}_2$, the Bridgeland–King–Reid theorem gives $D^b(\mathrm{Hilb}^2(T^4/\mathbb{Z}_2)) \simeq D^b_{\mathbb{Z}_2}(T^4)$, providing an explicit equivariant description. The equivariant category $D^b_{\mathbb{Z}_2}(T^4)$ is generated by equivariant line bundles and is computationally accessible.

(iii) *Brauer obstruction.* $\mathrm{Br}(K3) \supset (\mathbb{Q}/\mathbb{Z})^{21}$ (Remark 14.63) obstructs only *twisted* derived categories. The untwisted category $D^b(K3)$ is unobstructed.

A structural constraint: CY threefolds carry no exceptional objects ($\chi(\mathcal{O}_X, \mathcal{O}_X) = 0$ by the antisymmetry of Serre duality on a CY3), so $D^b(K3 \times E)$ admits no semiorthogonal decomposition and must be treated as an indecomposable block (Proposition 14.64).

[Global CY category from finitely many quiver charts] Given the constructive methods of Remark 14.90, can $D^b(K3 \times E)$ be assembled from finitely many quiver charts with explicit A_∞-bimodule transition data? The McKay correspondence provides the local charts at orbifold points; the cluster mutation structure controls transitions; the KS wall-crossing governs the DT invariant bookkeeping. A constructive algorithm would connect the local-to-global CY gluing of §14.8 to the global enumerative data of §14.11.

Research programmes

The preceding computation identifies the shadow obstruction tower of $K3 \times E$ down to precise numerical invariants. Ten research programmes emerge, each probing a different direction of the interface between bar-cobar duality and Calabi–Yau geometry. These are recorded as formal conjectures and open problems; each is grounded in computations from the 42 CY engines of the compute layer.

[Programme A: constructive Calabi–Yau gluing] The Čech descent of §14.8 reconstructs $D^b(K3 \times E)$ from local quiver charts in principle. Three questions sharpen the programme.

(A1) *Minimum chart number.* For a K3 surface S of Picard rank ρ, what is the minimum number of ADE-type quiver charts whose A_∞-bimodule transition data recovers $D^b(S)$? The Mukai lattice $\widetilde{H}(S, \mathbb{Z}) \cong U^4 \oplus E_8(-1)^2$ has rank 24; categorical generation requires a spanning class for thick subcategory generation, not merely K-theory spanning. For orbifold K3 (Kummer $T^4/\mathbb{Z}_2$), the BKR equivalence $D^b(\mathrm{Kum}(A)) \simeq D^b_{\mathbb{Z}/2}(A)$ provides a single global chart; for smooth K3 without ADE singularities, no full exceptional collection exists (Bridgeland 1999), and one must generate via spherical objects and their twist functors.

(A2) *Brauer obstruction.* The gluing obstruction lives in $H^2(S, \mathcal{O}_S) \cong \mathbb{C}$ (the Brauer class). When does a nontrivial Brauer class kill the descent? For generic K3 with $\rho = 0$, $\mathrm{Br}(S)$ contains $(\mathbb{Q}/\mathbb{Z})^{22}$; for $K3 \times E$, $\mathrm{Br}(K3 \times E) \supset (\mathbb{Q}/\mathbb{Z})^{22+1}$.

(A3) *Beyond ADE.* The McKay correspondence handles orbifold singularities. For smooth K3 at a generic moduli point (no singularities), the relevant local charts are formal neighborhoods of subvarieties, not ADE resolution charts. Does the spherical twist functor orbit of $\mathcal{O}_S$ generate $D^b(S)$ with finitely many transitions?

Conjecture 14.91 *(ADE chart decomposition of K3).* [CJ] Every algebraic K3 surface admits a finite ADE-chart decomposition of its derived category: there exist finitely many open sets $\{U_\alpha\}$ with tilting generators T_α on each U_α, and A_∞-bimodule transition data on overlaps, such that $D^b(S) \simeq \operatorname{holim}_{\check{C}} D^b(U_\alpha)$. The minimum number of charts is bounded below by $\lceil \operatorname{rank} K_0(S)/\max_\alpha \operatorname{rank} K_0(\operatorname{End}(T_\alpha))\rceil$.

[Programme B: κ as universal moonshine multiplier] The Mathieu moonshine decomposition gives $A_n = 2 \cdot \dim(\rho_n)$ with $\rho_n \in \operatorname{Irr}(M_{24})$, where the factor $2 = \kappa(\mathcal{A}_{K3})$ is the modular characteristic of the K3 sigma model (Remark 14.55). This suggests a universal principle.

(B1) *Monster moonshine.* For the Monster module $V^\natural$ ($c = 24$, $\kappa(V^\natural) = 12$ by Remark 14.57 and Vol I, Theorem III, applied to the Leech lattice construction): does $A_n^{\text{Monster}} = \kappa(V^\natural) \cdot \dim(\rho_n^{\mathbb{M}})$ for the Monster group $\mathbb{M}$? The McKay–Thompson series $T_g(\tau) = \sum_n \operatorname{tr}(\rho_n(g))\, q^{n-1}$ is a genus-zero Hauptmodul; the question is whether the *dimensions* (not characters) factor through κ.

(B2) *Conway moonshine.* For $V^{s\natural}$ ($c = 12$, the shorter moonshine module): $\kappa(V^{s\natural}) =?$ and does $A_n = \kappa \cdot \dim(\rho_n^{Co_0})$? The Conway group Co_0 acts on the Leech lattice; $V^{s\natural}$ is the $\mathbb{Z}/2$-orbifold of the rank-12 lattice VOA.

(B3) *General principle.* For a holomorphic VOA V of central charge c with finite automorphism group G, does the elliptic genus decompose as $Z_g(\tau, z) = \kappa(V) \cdot \sum_n \dim(\rho_n)\ \operatorname{ch}_n(g)$ for $g \in G$?

Conjecture 14.92 *(Universal moonshine multiplier).* [CJ] For a holomorphic VOA V with finite automorphism group G, the twined partition function decomposes as

$$Z_g(\tau, z) \;=\; \kappa(V) \cdot \sum_{n \geq 0} \dim(\rho_n)\, \operatorname{ch}_n(g), \qquad g \in G,$$

where $\kappa(V)$ is the bar-complex modular characteristic, $\{\rho_n\}$ are virtual G-representations, and ch_n are character functions determined by the elliptic genus of V. For $V = \mathcal{A}_{K3}$ ($c = 6$) with $G = M_{24}$: $\kappa = 2$ and $A_n = 2 \cdot \dim(\rho_n)$ as established by Gannon [**?**].

[Programme C: second-quantization bridge] The ratio $\kappa_{\text{BPS}}/\kappa_{\text{ch}} = 5/3$ (Proposition 14.88) encodes the passage from first to second quantization for $K3 \times E$.

(C1) *Universality.* Does $\kappa_{\text{BPS}} = \kappa(\text{surface}) + \kappa(\text{CY}_3)$ hold for other CY3s? For the quintic: $\chi = -200$, $\kappa_{\text{ch}} = 3$, and κ_{BPS} must be extracted from the genus-2 topological string amplitude (not a single Siegel form, since the quintic is not a product).

(C2) *Plethystic shadow.* The DMVV formula $1/\Phi_{10} = \operatorname{PE}[2\,\phi_{0,1}]$ expresses the second-quantized partition function as the plethystic exponential of the single-copy elliptic genus. What is the shadow tower of $\operatorname{PE}[\mathcal{A}]$ in terms of the shadow tower of $\mathcal{A}$? The tensor product gives $\kappa(\mathcal{A}^{\otimes N}) = N\,\kappa(\mathcal{A})$ by additivity, but the $\mathbb{Z}_N$-orbifold projection to Sym^N modifies the tower through twisted sectors.

(C3) *Adams operations.* The Adams operations ψ^k act on the shadow partition function by rescaling:

$$\psi^k(Z^{\text{sh}})(\hbar) \;=\; Z^{\text{sh}}(k\hbar) \;=\; \exp\Bigl(\sum_{g \geq 1} k^{2g}\, F_g\, \hbar^{2g}\Bigr). \tag{14.41}$$

This is the λ-ring structure on the shadow partition function: ψ^k multiplies the genus-g amplitude by k^{2g}, consistent with the interpretation of $\hbar^{2g}$ as a genus-counting weight. The Adams operations are ring homomorphisms ($\psi^k \circ \psi^l = \psi^{kl}$) and commute with the $\hat{A}$-genus generating function (Vol I, Theorem III). Does this intertwine with the Hecke operators V_k on the Jacobi form $\phi_{0,1}$ via the DMVV formula (14.48)? The Hecke transform $V_k(\phi_{0,1})$ and the Adams-rescaled shadow $\psi^k(Z^{\mathrm{sh}})$ would need to be related through the Borcherds product for the intertwining to be exact.

Conjecture 14.93 *(BPS modular characteristic universality).* [CJ] For a product CY_3 of the form $S\times E$ with S a K3 surface and E an elliptic curve,

$$\kappa_{\mathrm{BPS}} = \kappa(S) + \kappa(S\times E) = \dim_{\mathbb{C}}(S) + \dim_{\mathbb{C}}(S\times E).$$

For non-product CY_3s, κ_{BPS} is determined by the weight of the Siegel modular form (if one exists) controlling the $\frac{1}{4}$-BPS spectrum.

[Programme D: Schottky shadow programme] The shadow tower lives on $\overline{\mathcal{M}}_g$ (via tautological classes), while Siegel modular forms live on $\mathcal{A}_g$. For $g \le 3$, the Torelli map $j\colon \mathcal{M}_g \hookrightarrow \mathcal{A}_g$ is an isomorphism ($g=1$), birational ($g=2$), or injective with codimension-0 complement ($g=3$). For $g \ge 4$, the Jacobian locus $\mathcal{J}_g = j(\mathcal{M}_g)$ has codimension $(g-2)(g-3)/2$ in $\mathcal{A}_g$; the Schottky problem is nontrivial. The explicit numerical values are:

g	$\dim \mathcal{M}_g$	$\dim \mathcal{A}_g$	$\operatorname{codim}(\mathcal{J}_g)$	Status
2	3	3	0	Torelli birational
3	6	6	0	Torelli dominant
4	9	10	1	Schottky–Igusa J
5	12	15	3	
6	15	21	6	
7	18	28	10	
8	21	36	15	
9	24	45	21	
10	27	55	28	Crossover: codim > $\dim \mathcal{M}_{10}$

The crossover at $g = 10$—where $\operatorname{codim}(\mathcal{J}_{10}) = 28 > 27 = \dim \mathcal{M}_{10}$—marks the genus beyond which the Jacobian locus is "more absent than present" in $\mathcal{A}_g$. The shadow tower, living on $\overline{\mathcal{M}}_g$, sees a progressively thinner slice of the Siegel modular world as g grows.

(D1) *Tautological insulation.* The shadow amplitude $F_g = \kappa\,\lambda_g^{\mathrm{FP}}$ lives in the tautological ring $R^*(\overline{\mathcal{M}}_g)$, which is insensitive to the Schottky constraint. The shadow tower cannot detect whether a period matrix is a Jacobian: it is tautologically insulated from Schottky.

(D2) *Physical partition function.* The physical genus-g partition function $Z_g(\Omega)$ for $K3\times E$ *is* constrained by Schottky at $g \ge 4$: it is defined on $\mathcal{J}_g$, not all of $\mathcal{A}_g$. The gap between shadow (tautological) and partition function (analytic) at $g \ge 4$ measures the failure of the shadow tower to capture the full moduli dependence.

(D3) *The Schottky–Igusa form.* The Schottky–Igusa form $J \in S_8(\mathrm{Sp}(8,\mathbb{Z}))$ vanishes exactly on $\mathcal{J}_4$ at genus 4. Its weight $8 = 2g$ is significant. Does any shadow invariant at genus 4 detect J? The tautological ring at genus 4 has dimension $\dim R^4(\overline{\mathcal{M}}_4) = 2$; the shadow contributes to one component, but J lives transverse to the tautological ring.

CONJECTURE 14.94 *(Shadow tower generates the tautological projection).* [CJ] At genus g, the shadow obstruction tower of a class-M algebra (shadow depth $r_{\max} = \infty$) generates the image of the restriction map $\mathrm{res}\colon S_*(\mathrm{Sp}(2g, \mathbb{Z})) \to R^*(\overline{\mathcal{M}}_g)$ from the ring of Siegel modular forms to the tautological ring. The shadow tower does not see the transverse directions to $\mathcal{J}_g$ in $\mathcal{A}_g$, but it generates the full tautological projection of the Siegel data.

[Programme E: mock modularity and the bar complex] The mock modular form $h(\tau) = 2\,q^{-1/8}(-1+45q+231q^2+\cdots)$ encoding the massive $\mathcal{N}=4$ multiplicities has shadow $S(\tau) = 24\,\eta(\tau)^3$ (weight $3/2$, cusp form).

(E1) *The factor* 24. The shadow $S(\tau) = 24\,\eta^3$ contains $24 = \chi(K3)$, the topological Euler characteristic. Where does this factor arise in the bar complex? The bar curvature $\kappa = 2$ does not directly produce 24; the factor $24/\kappa = 12$ is the modular discriminant level. An algebraic derivation from the bar complex would connect the mock modular shadow to the bar-complex shadow connection ∇^{sh}.

(E2) *The Appell–Lerch sum.* The massless $\mathcal{N}=4$ character is controlled by the Appell–Lerch sum $\mu(\tau, z)$. The non-holomorphic completion $\hat{h}(\tau, \bar{\tau})$ is modular of weight $1/2$. Is there a bar-complex interpretation of μ? The Appell–Lerch sum regularizes the divergent theta-type sum; the bar complex regularizes the divergent Feynman-type sum. Both are regularizations of the same modular anomaly.

(E3) *Half-integral weight.* The mock modular form h has weight $1/2$; the shadow has weight $3/2$. The bar-complex shadow connection ∇^{sh} produces integer-weight objects (the shadow metric Q_L has weight 2; the Faber–Pandharipande λ_g are tautological, hence of integral weight). The passage to half-integral weight requires either a metaplectic extension or a theta correspondence. Concretely: is there a metaplectic bar complex whose shadow connection produces weight-$3/2$ objects?

Warning 14.95 (Notation: μ is overloaded). The symbol μ denotes two distinct objects in the K3 literature:

(i) The *Zwegers μ-function* $\mu(\tau, z) = \sum_{n\in\mathbb{Z}}(-1)^n\, q^{n(n+1)/2}\, y^n/(1-y\,q^n)$, the Appell–Lerch sum controlling the massless $\mathcal{N}=4$ character (Programme E above).

(ii) The full $\mathcal{N}=4$ *massless character* $\mathrm{ch}^{\mathrm{massless}}(\tau, z) = \mu(\tau, z)\cdot\vartheta_1(\tau, z)/\eta(\tau)^3$ (with additional theta and eta factors).

In the compute engines, `cy_mock_modular_bps_engine.py` uses μ for (i) while `cy_n4sca_k3_engine.py` uses μ for (ii). Care is needed when combining results from different engines. The Zwegers μ-function is *not* a Jacobi form: it fails the elliptic transformation law (it has a pole at $z \in \mathbb{Z}+\mathbb{Z}\tau$). Its non-holomorphic completion $\hat{\mu}(\tau, \bar{\tau}, z)$ is modular.

CONJECTURE 14.96 *(Mock shadow tower).* [CJ] The mock modular completion $\hat{h}(\tau, \bar{\tau}) = h(\tau) + h^*(\tau, \bar{\tau})$ is the genus-1 projection of a "mock shadow tower" that combines bar-complex data with the Appell–Lerch regularization. There exists an extension $\widetilde{\Theta}_{\mathcal{A}_{K3}}$ of the shadow obstruction tower $\Theta_{\mathcal{A}_{K3}}$ valued in mock modular forms of weight $1/2$, whose non-holomorphic completion is modular and whose holomorphic part restricts to $\kappa(\mathcal{A}_{K3})\cdot h(\tau)$ at genus 1. The shadow of this mock extension is $\kappa\cdot\chi(K3)\cdot\eta^3$, where the product $\kappa\cdot\chi(K3) = 2\cdot 24 = 48$ is the rank of the charge lattice $\Gamma = H^{\mathrm{even}}(K3\times E, \mathbb{Z})$.

[Programme F: modular factorization envelope] The factorization envelope $U_E(\mathcal{A}_E) = \mathrm{Sym}^{\mathrm{ch}}(\mathfrak{h}_{24})$ of the KS boundary algebra produces 24 free bosons on E, with character $1/\eta(\tau)^{24}$. The modular completion $U_E^{\mathrm{mod}}(\mathcal{A}_E)$ must incorporate higher-genus data; its character at genus 1 should receive corrections from the shadow tower.

(F1) *The Leech divergence.* The Leech lattice VOA V_{Leech} has character $J(\tau)+24 = q^{-1}(1+196884\,q+\cdots)$. The free-field envelope $1/\eta^{24} = q^{-1}(1+24\,q+324\,q^2+\cdots)$ matches the Leech VOA character at q^{-1} and q^0 (after adding 24) but diverges at q^1: $324 \neq 196884$. The discrepancy is the interacting structure of the Leech lattice VOA beyond free fields.

(F2) *Modular factorization envelope.* Is there a universal construction $U_X^{\mathrm{mod}}(L)$ for all cyclically admissible Lie conformal algebras L (Vol I, Definition III) that carries a natural shadow obstruction tower? Nishinaka's construction (2025/26) handles the genus-0 level; the modular completion requires incorporating the F_g data at all genera.

(F3) *Super extension.* The $\mathcal{N}=4$ SCA at $c=6$ is a super vertex algebra. Nishinaka's construction applies to ordinary Lie conformal algebras; the super extension (with fermionic generators $G^{\pm}, \widetilde{G}^{\pm}$) requires the super-factorization framework of Costello–Gwilliam. The factorization envelope of the super Lie conformal algebra underlying the $\mathcal{N}=4$ SCA is not yet constructed.

Conjecture 14.97 *(Modular factorization envelope existence).* [CJ] For every cyclically admissible Lie conformal algebra L, the modular factorization envelope $U_X^{\mathrm{mod}}(L)$ exists as a factorization algebra on $\mathrm{Ran}(X)$ equipped with the shadow obstruction tower $\Theta_{U_X^{\mathrm{mod}}(L)}$. This envelope is the left adjoint of the modular primitive-current functor $\mathrm{Prim}^{\mathrm{mod}}$ (Construction III), specialized to the CY setting.

[Programme G: BKM algebras and scattering diagrams] The BKM root system of $K3\times E$ lives in $\Gamma^{2,2} = U\oplus U$ (lattice of signature $(2,2)$, rank 4). Root multiplicities are determined by $c_0(D)$ from the K3 elliptic genus. The Kontsevich–Soibelman scattering diagram for $K3\times E$ associates a wall to each BPS ray with attached function determined by the root multiplicity.

(G1) *Two MC elements.* The shadow obstruction tower $\Theta_{\mathcal{A}}$ is an MC element in the convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$; the KS automorphism $\mathfrak{S}$ is an MC element in the motivic Hall algebra $\mathcal{H}(C)$. These are MC elements in different Lie algebras. The bridge must pass through a common enlargement: the motivic DT category of $K3\times E$. At the naive level, the BCH pair-commutator does *not* reproduce the $\phi_{0,1}$ multiplicities (quantitative mismatch by non-uniform ratios).

(G2) *Motivic level.* The identification "scattering diagram = shadow obstruction tower" holds at the motivic Hall algebra level (Kontsevich–Soibelman 2008), where the product on walls is the motivic quantum torus product. Instantiating this at the numerical/Euler characteristic level loses the higher-genus information that the motivic structure retains.

(G3) *Tropical skeleton.* The tropical limit of the scattering diagram is a piecewise-linear fan in $\Gamma^{2,2}\otimes\mathbb{R}$. This fan should be the tropical skeleton of the shadow obstruction tower restricted to BPS walls. The identification would connect Mok's log FM tropicalization (Theorem III) to the BPS scattering diagram.

Conjecture 14.98 *(Scattering diagram as tropical shadow).* [CJ] The scattering diagram of $K3\times E$ is the tropical skeleton of the shadow obstruction tower: $\mathrm{Scat}(K3\times E) = \mathrm{Trop}(\Theta_{\mathcal{A}_{K3\times E}})$, where the left side is the Kontsevich–Soibelman scattering diagram (with wall-crossing factors from the BPS spectrum) and the right side is the tropicalization of the shadow obstruction tower restricted to the charge lattice $\Gamma^{2,2}$. The identification holds at the level of the motivic Hall algebra; at the numerical level, higher BPS bound-state contributions must be included.

Remark 14.99 (Descent sufficiency and K-theory verification). Three levels of descent apply to the CY category programme:

(i) *Zariski suffices.* For smooth separated schemes X, the natural functor $D^b(X) \to \mathrm{holim}_{\check{\mathrm{C}}\mathrm{ech}} D^b(U_\alpha)$ is an equivalence (Toën 2007, Lurie HA 2017, Theorem 19.4.5): coherent sheaves satisfy effective descent for the Zariski topology. No étale or flat refinement is needed.

(ii) *ADE cocycles close.* For each ADE singularity type, the bimodule cocycle condition $M_{01} \otimes_B^L M_{12} \simeq M_{02}$ closes:

Type	$\lvert\Gamma\rvert$	$\dim \Pi_0(Q)$	Cocycle mechanism
A_n	$n+1$	$\binom{n+2}{3}$	Flop involution
D_n	$4(n-2)$	Table 14.40	Tilting generator $T = \bigoplus P_i$
E_6	24	168	$\mathrm{End}(T) = \Pi(\check{Q})$
E_7	48	384	$\mathrm{End}(T) = \Pi(\check{Q})$
E_8	120	1080	$\mathrm{End}(T) = \Pi(\check{Q})$

The mechanism is uniform: the tilting generator $T = \bigoplus_{i\in Q_0} P_i$ of the resolution $\widetilde{\mathbb{C}^2/\Gamma}$ has endomorphism algebra $\mathrm{End}(T) \cong \Pi_0(Q)$, and the Morita equivalence $D^b(\Pi_0(Q)\text{-mod}) \simeq D^b(\widetilde{\mathbb{C}^2/\Gamma})$ automatically closes the bimodule cocycle for any number of overlapping charts (Bridgeland–King–Reid for type A; Kapranov–Vasserot, Van den Bergh for all types).

(iii) *K-theory verification.* For $X = K3\times E$, successful descent must recover $\mathrm{rk}\, K_0(K3\times E) = 48 = \mathrm{rk}\, K_0(K3)\cdot \mathrm{rk}\, K_0(E) = 24\cdot 2$. For each ADE chart of type Γ: $\mathrm{rk}\, K_0(\mathrm{Jac}(Q_\Gamma, W_\Gamma)) = \lvert Q_0\rvert = r+1$ (the number of vertices in the McKay quiver). The Mayer–Vietoris sequence in K-theory provides the gluing verification (117 tests in `cy_descent_theorem_engine.py`).

[Programme H: descent theorem programme] The descent theorem for dg categories (Toën 2007, Lurie HA) guarantees that Zariski descent recovers $D^b(X)$ for smooth separated schemes (Remark 14.99). The quiver chart descent of §14.8 uses A_∞-bimodule transition data on overlaps.

(H1) *Bimodule cocycle closure.* For three overlapping charts U_0, U_1, U_2 with transition bimodules M_{01}, M_{12}, M_{02}, the triple cocycle condition $M_{01}\otimes_B^L M_{12} \simeq M_{02}$ must close up to coherent homotopy. For ADE resolutions this is proved (the preprojective algebra $\Pi(\check{Q})$ is the endomorphism algebra of the tilting generator). For smooth K3 patches glued along a divisor, the bimodule cocycle is the restriction of the Atiyah class.

(H2) *Obstruction.* The obstruction to extending a two-chart descent to a global descent lives in HH^2 of the transition bimodule: $\mathrm{ob} \in \mathrm{HH}^2(M_{01}, M_{01})$. For K3, $h^{0,2} = 1$ provides a one-dimensional obstruction space. When is the obstruction trivial?

(H3) *Higher-dimensional CY.* For a smooth projective CY d-fold X with ADE singularities, does the quiver chart descent reconstruct $D^b(X)$? The $d = 2$ (K3) and $d = 3$ ($K3\times E$) cases are established by the compute layer. The general case requires understanding the higher Hochschild obstructions.

[Programme I: higher-dimensional CY shadows] The shadow tower of $K3\times E$ ($\dim_{\mathbb{C}} = 3$) connects to genus-2 Siegel modular forms via the Borcherds lift. What happens in higher CY dimension?

(I1) $\mathrm{CY}_4 = K3\times K3$. By additivity, $\kappa = \kappa(K3) + \kappa(K3) = 2 + 2 = 4$. Shadow class: M (product of two class-M factors). The genus-2 amplitude $F_2 = 4\cdot 7/5760 = 7/1440$. Is there a genus-3 Siegel modular form analogue of Φ_{10}?

(I2) $\mathrm{CY}_5 = K3 \times K3 \times E$. Similarly, $\kappa = 2+2+1 = 5$. The same $\kappa_{\mathrm{BPS}} = 5$ as for $K3 \times E$ arises, but for different reasons: the CY_5 has $\kappa_{\mathrm{ch}} = 5$ directly, without second quantization. Does this numerical coincidence have algebraic content?

(I3) *The dimensional ladder.* For a product $\mathrm{CY}_d = X_1 \times \cdots \times X_k$, additivity gives $\kappa = \sum_i \kappa(X_i) = \sum_i \dim_{\mathbb{C}}(X_i) = d$. The shadow class should be $\max(\text{classes of } X_i)$ under the ordering $G < L < C < M$. Does this determine the shadow tower of the product completely?

Conjecture 14.100 *(CY product shadow decomposition).* [CJ] For a product $\mathrm{CY}_d = X_1 \times \cdots \times X_k$ of Calabi–Yau manifolds,

$$\kappa(X_1 \times \cdots \times X_k) \;=\; \sum_{i=1}^{k} \kappa(X_i) \;=\; \sum_{i=1}^{k} \dim_{\mathbb{C}}(X_i) \;=\; d, \tag{14.42}$$

$$\mathrm{class}(X_1 \times \cdots \times X_k) \;=\; \max_i \mathrm{class}(X_i) \tag{14.43}$$

(under the ordering $G < L < C < M$). The shadow obstruction tower $\Theta_{X_1 \times \cdots \times X_k}$ at the scalar level is determined by $\kappa = d$ alone; the higher-arity components are controlled by the most complex factor.

[Programme J: convergence vs. divergence and non-perturbative completions] The shadow partition function $Z^{\mathrm{sh}}(\hbar) = \sum_{g \geq 1} F_g \, \hbar^{2g}$ converges absolutely for $|\hbar| < 2\pi$ (Proposition 14.111), is entire after Borel summation, and shows no resurgent structure. The convergence radius 2π arises because $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ with $\lambda_g^{\mathrm{FP}} \sim (2\pi)^{-2g}$, so Z^{sh} converges where $|\hbar/(2\pi)| < 1$. The function $(\hbar/2)/\sin(\hbar/2)$ has poles at $\hbar = 2n\pi$ for $n \in \mathbb{Z} \setminus \{0\}$; this is $\sin(\hbar/2)$, not $\sin(\hbar)$, so the radius is 2π, not π. The topological string partition function $Z_{\mathrm{top}}(\hbar)$ diverges as $(2g)!$ (Gevrey-1), is resurgent, and exhibits Stokes phenomena.

(J1) *The shadow as convergent piece.* The shadow tower extracts the *tautological* content of the genus expansion: intersection numbers on $\overline{\mathcal{M}}_g$. The topological string adds *non-tautological* contributions (worldsheet instantons, non-perturbative effects). Is $Z_{\mathrm{top}} = Z^{\mathrm{sh}} \cdot (1 + Z_{\mathrm{np}})$ where Z_{np} encodes worldsheet instantons?

(J2) *Non-perturbative corrections.* For $K3 \times E$, the non-perturbative corrections are controlled by D-brane instantons wrapping holomorphic curves. The GV invariants n_β^g count the BPS content. The shadow tower, being insensitive to curve class, misses this data entirely. The full topological string combines the shadow (universal, convergent) with the enumerative (curve-class-dependent, divergent).

(J3) *The OSV bridge.* The OSV conjecture $Z_{\mathrm{BH}} = |Z_{\mathrm{top}}|^2$ relates the black hole partition function (computed via $1/\Phi_{10}$ for $K3 \times E$) to the topological string. Since Z^{sh} extracts the tautological piece of Z_{top}, the shadow controls the tautological piece of the black hole entropy. The precise relationship at subleading order ($\log \Delta$ corrections to S_{BH}) is governed by $F_1 = \kappa/24$ (Sen's logarithmic correction formula).

Proposition 14.101 *(Class G algebras: shadow = topological string).* [PH] For class G algebras (shadow depth $r_{\max} = 2$, the shadow tower terminates at arity 2),

$$F_g^{\mathrm{sh}}(\mathcal{A}) \;=\; F_g^{\mathrm{top}}(\mathcal{A}) \qquad \text{for all } g \geq 1. \tag{14.44}$$

The shadow tower *is* the topological string for free-field-type algebras: the shadow partition function $Z^{\mathrm{sh}} = \exp\big(\sum_{g \geq 1} F_g \, \hbar^{2g}\big)$ with $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ receives no worldsheet instanton corrections because the bar complex is formal ($m_n = 0$ for $n \geq 3$), so the generating function is the $\hat{A}$-genus $(\hbar/2)/\sin(\hbar/2)$ evaluated at κ. For class L, C, and M algebras, the higher-arity shadow components $S_3, S_4, \ldots$ contribute additional "instanton-like" corrections to F_g at genus $g \geq 2$ that modify the free-field $\hat{A}$-genus formula. For the KS boundary algebra A_E of $K3 \times E$: A_E is 24 free bosons, hence class G, hence $F_g^{\mathrm{sh}}(A_E)$ is exact. The K3 sigma model $\mathcal{A}_{K3}$ is class M (infinite depth), so its shadow tower receives corrections at all arities.

Proof. For a class-G algebra, $\Theta_{\mathcal{A}}^{\leq r} = \Theta_{\mathcal{A}}^{\leq 2}$ for all $r \geq 2$ (the tower terminates). The shadow obstruction tower collapses to its scalar projection κ: the genus-g amplitude is $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ with no higher-arity corrections. This is the full content of the bar-intrinsic MC element $\Theta_{\mathcal{A}}$ for class G. Since the topological string at the free-field point receives no worldsheet instanton corrections (the moduli space of maps is trivial), $F_g^{\mathrm{top}} = F_g^{\mathrm{sh}}$. □

PROPOSITION 14.102 *(Bar functor commutes with homotopy colimits).* [PH] Let $\{A_\sigma\}_{\sigma \in I}$ be a diagram of dg algebras indexed by a small category I. Then

$$B(\mathrm{hocolim}_{\sigma \in I} A_\sigma) \simeq \mathrm{hocolim}_{\sigma \in I} B(A_\sigma) \tag{14.45}$$

as dg coalgebras (quasi-isomorphism).

Proof. The bar functor B is a left Quillen functor from augmented dg algebras to conilpotent dg coalgebras (Vol I, Theorem III). Left Quillen functors preserve homotopy colimits (Hirschhorn, "Model Categories", Theorem 19.4.5). The quasi-isomorphism (14.45) is the derived functor applied to the homotopy colimit diagram. Computational verification: 145 tests in `bar_hocolim_chain_level.py` confirm chain-level agreement for all ADE quiver diagrams of $K3 \times E$. □

Remark 14.103 (Descent for bar complexes). Proposition 14.102 gives a descent theorem for bar complexes: the global bar complex of the hocolim is recovered from the local bar complexes and their transition data. For the constructive gluing of $K3 \times E$ from quiver charts (§14.8), this means the global shadow obstruction tower $\Theta_{\mathcal{A}_{K3\times E}}$ can be assembled from local shadow towers $\Theta_{\mathcal{A}_\sigma}$ on each chart, with transition maps controlled by the A_∞-bimodule data of Construction 14.43.

Remark 14.104 (Summary of research programmes). Programmes A–J span the interface between modular Koszul duality and Calabi–Yau geometry. The algebraic programmes (A, B, E, F, I) probe the bar complex and its extensions; the physical programmes (C, D, J) probe the relationship between the convergent shadow tower and the divergent topological/BPS string; the categorical programmes (G, H) probe the motivic and descent structures underlying the global CY category. Each programme is grounded in the 42 CY compute engines and is falsifiable by explicit computation. Cross-references to this chapter and to the foundations in Vols I–II:

- Programme A: §14.8, Open Problem 14.15.
- Programme B: Open Problem 14.15, Proposition 14.48.
- Programme C: Proposition 14.88, Remark 14.87.
- Programme D: Theorem 14.112, Vol I, Theorem III.
- Programme E: Open Problem 14.15, Vol I, Theorem III.
- Programme F: Open Problem 14.15, Vol I, Construction III.
- Programme G: §14.16, Vol I, Theorem III.
- Programme H: Theorem 14.45, §14.8.
- Programme I: Vol I, Theorem D (Theorem III), Vol I, Definition III.
- Programme J: Proposition 14.111, Vol I, Theorem III.

14.16 The Borcherds–Kac–Moody algebra of $K3 \times E$

The preceding sections developed the elliptic bar complex on a single elliptic curve E_τ. We now examine a fundamentally different role for genus-2 modular forms: the Calabi–Yau threefold $K3 \times E$ produces a Borcherds–Kac–Moody (BKM) superalgebra whose denominator function is the Igusa cusp form Φ_{10}, a Siegel modular form of weight 10 on $\mathrm{Sp}(4, \mathbb{Z})$. The shadow obstruction tower of $K3 \times E$ detects the topological skeleton of this structure, but the passage from shadow amplitudes to the full BPS partition function encounters four independent obstructions. This section makes the bridge, and its limitations, precise.

The Igusa cusp form and the DVV formula

Definition 14.105 *(Igusa cusp form).* The *Igusa cusp form* Φ_{10} is the unique Siegel cusp form of weight 10 on $\mathrm{Sp}(4, \mathbb{Z})$:

$$\dim S_{10}\big(\mathrm{Sp}(4, \mathbb{Z})\big) = 1.$$

It admits three equivalent constructions:

(i) *Product of even theta characteristics.* $\Phi_{10}(\Omega) = -2^{-12} \prod_{\text{even } m} \vartheta[m](\Omega)$, where the product runs over the 10 even theta characteristics of genus 2.

(ii) *Fourier–Jacobi expansion.* $\Phi_{10}(\Omega) = \sum_{m \geq 1} \phi_m(\tau, z)\, p^m$ with $p = e^{2\pi i \sigma}$ and $\Omega = \left(\begin{smallmatrix} \tau & z \\ z & \sigma \end{smallmatrix}\right) \in \mathfrak{H}_2$. The leading coefficient is

$$\phi_1 \;=\; \phi_{10,1}(\tau, z) \;=\; \eta(\tau)^{18}\, \vartheta_1(\tau, z)^2, \tag{14.46}$$

the unique Jacobi cusp form of weight 10, index 1.

(iii) *Borcherds multiplicative lift.* Let $Z_{K3}(\tau, z) = 2\, \phi_{0,1}(\tau, z)$ be the $K3$ elliptic genus, where $\phi_{0,1}$ is the unique weak Jacobi form of weight 0, index 1 (Eichler–Zagier normalization: $\phi_{0,1}(\tau, 0) = 12$). Then

$$\Phi_{10}(\Omega) \;=\; p\, q\, y \prod_{(n,\ell,m)>0} \big(1 - q^n\, y^\ell\, p^m\big)^{c_0(4nm - \ell^2)}, \tag{14.47}$$

where $q = e^{2\pi i \tau}$, $y = e^{2\pi i z}$, $p = e^{2\pi i \sigma}$, and $c_0(D)$ are the Fourier coefficients of the $K3$ elliptic genus indexed by discriminant D.

Remark 14.106 (Conventions). We follow the Eichler–Zagier convention throughout. In the DVV convention, Φ_{10} may differ by a sign; the product formula (14.47) and the uniqueness of $S_{10}(\mathrm{Sp}(4, \mathbb{Z}))$ fix the normalization up to scalar. The elliptic genus $Z_{K3} = 2\, \phi_{0,1}$ evaluates to $Z_{K3}(\tau, 0) = 24 = \chi(K3)$ at $z = 0$.

The DVV formula (Dijkgraaf–Verlinde–Verlinde, [?]) identifies the $\frac{1}{4}$-BPS partition function of Type IIA on $K3 \times T^2$:

$$Z_{\frac{1}{4}\text{-BPS}}(\Omega) \;=\; \frac{1}{\Phi_{10}(\Omega)} \;=\; \sum_{N \geq 0} p^N\, \chi\big(\mathrm{Sym}^N(K3); \tau, z\big), \tag{14.48}$$

where the second equality is the DMVV infinite-product formula expressing $1/\Phi_{10}$ as the generating function for the orbifold elliptic genera of the symmetric products $\mathrm{Sym}^N(K3)$.

PROPOSITION 14.107 *(Genus-2 nature of Φ_{10}; [PE])*. The Igusa cusp form Φ_{10} is *intrinsically* a genus-2 object: it is a Siegel modular form on $\mathrm{Sp}(4, \mathbb{Z})$, whose natural domain is the Siegel upper half-space $\mathfrak{H}_2 = \{\Omega \in \mathrm{Mat}_{2\times 2}(\mathbb{C}) : \Omega = \Omega^t,\ \mathrm{Im}(\Omega) > 0\}$ of complex dimension 3. The three variables (τ, z, σ) of the period matrix $\Omega = \left(\begin{smallmatrix} \tau & z \\ z & \sigma \end{smallmatrix}\right)$ encode the three chemical potentials conjugate to the electric, magnetic, and dyonic charge invariants of $\frac{1}{4}$-BPS states. No genus-1 modular form captures the full dyonic spectrum: the Fourier–Jacobi coefficient $\phi_{10,1}$ is the $m = 1$ term in the p-expansion, retaining only the single-centered contribution.

The Borcherds multiplicative lift

THEOREM 14.108 *(Borcherds lift; [PE])*. Let $\phi_{0,1}(\tau, z)$ be the unique weak Jacobi form of weight 0, index 1. The Borcherds multiplicative lift

$$\mathrm{Borch}\colon J_{0,1}^{\mathrm{weak}} \longrightarrow S_{10}\big(\mathrm{Sp}(4, \mathbb{Z})\big)$$

sends $2\,\phi_{0,1}$ to Φ_{10}, producing the infinite product (14.47). This is the genus-1 to genus-2 bridge: the genus-1 datum $Z_{K3} = 2\,\phi_{0,1}$ encodes the exponents of the genus-2 Siegel modular form Φ_{10}.

Remark 14.109 (Structure of the lift). The Borcherds lift converts additive data (Fourier coefficients of Z_{K3}) into multiplicative data (exponents in the product formula for Φ_{10}). The exponent $c_0(D)$ at discriminant $D = 4nm - \ell^2$ is the root multiplicity of the BKM superalgebra $\mathfrak{g}_{\Phi_{10}}$: real roots ($D < 0$) have multiplicity $c_0(-1) = 2$, lightlike roots ($D = 0$) have multiplicity $c_0(0) = 10$ (or 20 from the full $K3$ elliptic genus $2\,\phi_{0,1}$), and imaginary roots ($D > 0$) have unbounded multiplicities.

The leading Fourier–Jacobi coefficient $\phi_1 = \phi_{10,1} = \eta^{18}\,\vartheta_1^2$ (equation (14.46)) is itself a product of genus-1 objects. The factorization $\eta^{18} = \eta^{24} \cdot \eta^{-6}$ separates the Ramanujan discriminant $\Delta = \eta^{24}$ (controlling the modular discriminant) from $\eta^{-6} = \eta^{-2\kappa}$ with $\kappa = 3$ (the shadow tower's genus-1 partition function $Z_1(\tau) = \eta(\tau)^{-2\kappa}$ for $K3 \times E$ at $\kappa = \dim_{\mathbb{C}}(K3 \times E) = 3$).

Shadow tower data for $K3 \times E$

The modular Koszul framework assigns to $K3 \times E$ a well-defined shadow obstruction tower. We record the explicit values.

PROPOSITION 14.110 *(Shadow amplitudes of $K3\times E$; [PH])*. Let $\mathcal{A} = \Omega^{\mathrm{ch}}(K3\times E)$ be the chiral de Rham complex of $K3 \times E$. The modular characteristic is $\kappa(\mathcal{A}) = \dim_{\mathbb{C}}(K3 \times E) = 3$ (this is not $c/2$ in general; here the central charge is $c = 2\cdot 3 = 6$, and $c/2 = 3$ coincides with κ only because $\dim_{\mathbb{C}} = 3$). The shadow amplitudes (Vol I, Theorem D; Theorem III), in the generating function convention $\sum_{g\geq 1} F_g\,\hbar^{2g}$, are

$$F_g(K3 \times E) \;=\; \kappa \cdot \lambda_g^{\mathrm{FP}} \;=\; 3 \cdot \frac{(2^{2g-1} - 1)\,|B_{2g}|}{2^{2g-1}\,(2g)!}, \tag{14.49}$$

where λ_g^{FP} is the Faber–Pandharipande intersection number. Explicitly:

g	1	2	3	4	5
λ_g^{FP}	$\frac{1}{24}$	$\frac{7}{5760}$	$\frac{31}{967680}$	$\frac{127}{154828800}$	$\frac{73}{3503554560}$
F_g	$\frac{1}{8}$	$\frac{7}{1920}$	$\frac{31}{322560}$	$\frac{127}{51609600}$	$\frac{73}{1167851520}$

The genus-1 amplitude $F_1 = \kappa/24 = 1/8$ matches the leading behaviour of $Z_1(\tau) = \eta(\tau)^{-6}$, since $\log\eta(\tau) = 2\pi i\tau/24 + O(q)$ and the η-power is $-2\kappa = -6$.

Proof. Direct from Vol I, Theorem D (Theorem III) with $\kappa = 3$. The value $\kappa = 3$ is the modular characteristic of $\Omega^{\mathrm{ch}}(K3 \times E)$, equal to $\dim_{\mathbb{C}}(K3 \times E)$ by the Chern character computation for the chiral de Rham complex of a Calabi–Yau manifold. Each F_g is verified by three independent paths: direct Bernoulli computation, the $\hat{A}$-generating function $(t/2)/\sin(t/2) - 1 = \sum_{g \geq 1} \lambda_g^{\mathrm{FP}} t^{2g}$, and the additivity $\kappa(K3 \times E) = \kappa(K3) + \kappa(E) = 2 + 1 = 3$ (Vol I, Corollary III). □

PROPOSITION 14.111 *(Convergence of the shadow generating function; [PH]).* The shadow generating function $Z^{\mathrm{sh}}(t) := \kappa\big((t/2)/\sin(t/2) - 1\big) = \sum_{g \geq 1} F_g t^{2g}$ has the following analytic properties:

(i) *Convergence radius.* $R = 2\pi$. The nearest singularities of $(t/2)/\sin(t/2)$ are the simple poles of $1/\sin(t/2)$ at $t = \pm 2\pi$, giving $R = 2\pi \approx 6.283$.

(ii) *Closed form.* $Z^{\mathrm{sh}}(t) = \kappa\big((t/2)/\sin(t/2) - 1\big)$, an algebraic function of $\sin(t/2)$.

(iii) *Borel transform.* The Borel transform $\hat{Z}(u) = \sum_{g \geq 1} F_g u^{2g}/(2g)!$ is an *entire* function of u (no singularities on $\mathbb{R}_+$). There is no resurgence: the Borel sum recovers Z^{sh} exactly.

This contrasts sharply with the topological string partition function $Z_{\mathrm{top}}(\lambda) = \exp\big(\sum_{g \geq 0} F_g^{\mathrm{top}} \lambda^{2g-2}\big)$, which is Gevrey-1 divergent: $F_g^{\mathrm{top}} \sim (2g)!$ from the volume growth of $\mathcal{M}_g$. For $K3 \times E$ with $\kappa = 3$: $Z^{\mathrm{sh}}(t) = 3\big((t/2)/\sin(t/2) - 1\big)$ converges absolutely for $|t| < 2\pi$.

Proof. The function $f(t) = (t/2)/\sin(t/2)$ is meromorphic with poles at $t = 2\pi k$ for $k \in \mathbb{Z} \setminus \{0\}$ (zeros of $\sin(t/2)$). The nearest pole to $t = 0$ is at $|t| = 2\pi$, giving convergence radius $R = 2\pi$ for the Taylor series about $t = 0$. The Borel transform replaces t^{2g} by $u^{2g}/(2g)!$, which grows as $|B_{2g}|/(2g)! \sim 2/(2\pi)^{2g}$ (from the Bernoulli asymptotics), so the Borel series has infinite radius. Since $\hat{Z}$ is entire, the Laplace integral $\int_0^\infty e^{-u/t} \hat{Z}(u)\, du/t$ converges for all $t > 0$ and reproduces $Z^{\mathrm{sh}}(t)$ term by term. No Stokes phenomenon occurs. □

The shadow–Siegel gap

The shadow amplitudes F_g and the Igusa cusp form Φ_{10} are related but categorically distinct objects. Four independent obstructions prevent the shadow tower from reconstructing Φ_{10}.

THEOREM 14.112 *(Shadow–Siegel gap; [PH]).* The shadow obstruction tower of $K3 \times E$ does not produce the Igusa cusp form. The gap consists of four independent obstructions:

(O1) **Categorical.** F_g is a rational number (a tautological intersection number on $\overline{\mathcal{M}}_g$); $\Phi_{10}(\Omega)$ is a holomorphic function on $\mathfrak{H}_2$ (a Siegel cusp form). The shadow captures the topological skeleton of the genus-g amplitude, not the full analytic partition function.

(O2) **Modular characteristic.** The shadow tower uses $\kappa_{\mathrm{ch}} = 3$ (the modular characteristic of the single-copy chiral algebra $\Omega^{\mathrm{ch}}(K3 \times E)$, equal to $\dim_{\mathbb{C}} = 3$). The Igusa cusp form has weight $\mathrm{wt}(\Phi_{10}) = 10 = 2\kappa_{\mathrm{BPS}}$ with $\kappa_{\mathrm{BPS}} = \chi(K3)/4 - 1 = 5$, the second-quantized modular weight. The ratio $\kappa_{\mathrm{BPS}}/\kappa_{\mathrm{ch}} = 5/3$ reflects the passage from single-copy to symmetric-product via the DMVV formula (14.48), and is specific to $K3 \times E$ (it is NOT a universal constant: for the quintic, $\chi = -200$ gives a different ratio).

(O3) **Second quantization.** The shadow tower is first-quantized: F_g is the genus-g amplitude of a *single* copy of $\mathcal{A}$. The partition function $1/\Phi_{10}$ is second-quantized: it sums over *all* symmetric products $\mathrm{Sym}^N(K3)$ via the DMVV formula. The Borcherds lift, the DMVV infinite product, and the symmetric-product orbifold construction are second-quantized operations not captured by the shadow tower alone.

(O4) **Schottky at $g \geq 4$.** The Torelli map $j : \mathcal{M}_g \to \mathcal{A}_g$ has image of codimension $(g-2)(g-3)/2$ in $\mathcal{A}_g$ for $g \geq 4$. Explicitly:

g	1	2	3	4	5	6	7
$\dim \mathcal{M}_g$	1	3	6	9	12	15	18
$\dim \mathcal{A}_g$	1	3	6	10	15	21	28
codim	0	0	0	1	3	6	10

For $g \leq 3$, the Torelli map is an isomorphism ($g = 1$), birational ($g = 2$), or injective with dense image ($g = 3$), so the shadow tower sees essentially all Siegel modular form data. At $g = 4$, the Schottky–Igusa form of weight 8 vanishes precisely on the Jacobian locus $\mathcal{J}_4 = j(\mathcal{M}_4)$: this is information accessible to Siegel forms but invisible to the shadow tower. For $g \to \infty$, the ratio $\mathrm{codim}/\dim \mathcal{A}_g \to 1$: the Jacobian locus becomes asymptotically negligible.

Proof. (O1) is immediate from the definitions: $F_g \in \mathbb{Q}$ while Φ_{10} is a holomorphic function of three complex variables. (O2) follows from the two independent computations $\kappa_{\mathrm{ch}} = \dim_{\mathbb{C}} = 3$ (Proposition 14.110) and $\mathrm{wt}(\Phi_{10}) = 10$ (Definition 14.105), together with the identification $\kappa_{\mathrm{BPS}} = \chi(K3)/4 - 1 = 5$ from the DVV formula. (O3) is the content of the DMVV formula: the second equality in (14.48) expresses $1/\Phi_{10}$ as a sum over symmetric products, an operation external to single-copy chiral algebra theory. (O4): the codimension formula $\mathrm{codim}(\mathcal{J}_g, \mathcal{A}_g) = g(g+1)/2 - (3g-3) = (g-2)(g-3)/2$ for $g \geq 2$ is classical (Riemann, Schottky). □

Remark 14.113 (The genus-2 Torelli bridge). At genus 2, the Torelli map $j : \mathcal{M}_2 \to \mathcal{A}_2$ is birational: its complement $\mathcal{A}_2 \setminus j(\mathcal{M}_2)$ is the product locus (abelian surfaces isomorphic to a product of two elliptic curves). This means Φ_{10} restricts to a well-defined form on $\mathcal{M}_2$ (vanishing on the product locus at the boundary of the Satake compactification), and the genus-2 partition function $Z_2(\Omega)$ related to $1/\Phi_{10}$ has a shadow amplitude

$$F_2 = \int_{\overline{\mathcal{M}}_2} \kappa \cdot \lambda_2^{\mathrm{FP}} = \frac{7}{1920}$$

as its topological residue. The shadow F_2 is what remains of $1/\Phi_{10}$ after integrating out all modular dependence: a tautological intersection number in the Chow ring of $\overline{\mathcal{M}}_2$.

Euler characteristic and Donaldson–Thomas structure

Proposition 14.114 *($\chi(K3 \times E) = 0$ and its consequences; [PE]).* The Euler characteristic of $K3 \times E$ vanishes:

$$\chi(K3 \times E) = \chi(K3) \cdot \chi(E) = 24 \cdot 0 = 0.$$

This has three structural consequences for the enumerative geometry:

(i) *MacMahon trivialization.* The degree-0 DT partition function of a Calabi–Yau threefold X is $Z_{\mathrm{DT},0}(X) = M(-q)^{\chi(X)}$, where $M(q) = \prod_{n \geq 1}(1-q^n)^{-n}$ is the MacMahon function counting plane partitions. For $\chi = 0$: $M(-q)^0 = 1$, so $Z_{\mathrm{DT},0}(K3 \times E) = 1$. There is no degree-0 contribution from plane-partition-type combinatorics.

(ii) *DT/PT correspondence.* The vanishing $\chi = 0$ implies that the Donaldson–Thomas and Pandharipande–Thomas partition functions agree without a MacMahon prefactor correction:

$$Z_{\mathrm{DT}}(K3 \times E) = Z_{\mathrm{PT}}(K3 \times E).$$

This is the "clean" case of the DT/PT wall-crossing formula: the wall-crossing factor $M(-q)^{\chi}$ is trivial.

(iii) *BCOV anomaly coefficient.* The BCOV holomorphic anomaly coefficient $\kappa_{\mathrm{BCOV}} = \chi(X)/24$ vanishes for $K3 \times E$. The anomaly equation $\bar{\partial}_{\bar{i}} F_g = \frac{1}{2} C_{\bar{i}}^{jk} (D_j D_k F_{g-1} + \sum_{r=1}^{g-1} D_j F_r \, D_k F_{g-r})$ is therefore unobstructed at the topological level, consistent with the shadow tower being controlled by the single parameter $\kappa = 3$ without BCOV corrections.

Remark 14.115 (The four κ invariants). The preceding analysis exhibits four distinct invariants that share the name "κ" in different parts of the literature:

Symbol	**Value**	**Source**
$\kappa(\mathcal{A})$	3	Modular characteristic (Vol I, Theorem D)
κ_{BCOV}	0	$\chi(K3 \times E)/24$
$\kappa_{\mathrm{MacMahon}}$	0	Exponent of $M(-q)^{\chi}$
κ_{BKM}	5	$\mathrm{wt}(\Phi_{10})/2 = \chi(K3)/4 - 1$

These coincide for the Virasoro algebra ($\kappa = c/2$ in all rôles) but diverge for $K3 \times E$. The modular characteristic $\kappa(\mathcal{A}) = 3$ is the intrinsic invariant of the chiral algebra; the others are geometric proxies valid in restricted contexts.

Remark 14.116 (Shadow depth classification). The chiral algebra $\Omega^{\mathrm{ch}}(K3 \times E)$ has central charge $c = 6$, multiple generators of varying conformal weight, and an infinite tower of OPE coefficients from the $K3$ sigma model. By the shadow depth classification (Vol I, Definition III), $K3 \times E$ belongs to class M (mixed, shadow depth $r_{\max} = \infty$): the Borcherds product formula (14.47) has infinitely many distinct exponents $c_0(D)$, reflecting an infinite sequence of independent shadow invariants. This is consistent with the operadic complexity principle: the sigma model on a Calabi–Yau target, being a fully interacting CFT, has non-formal Swiss-cheese structure at all arities.

Remark 14.117 (Genus stratification summary). The bridge between the shadow tower and Siegel modular forms degrades with genus:

Genus	**Torelli**	**Shadow–Siegel status**
1	Isomorphism	$F_1 = 1/8$ matches η^{-6} exactly
2	Birational	$F_2 = 7/1920$ is topological residue of $1/\Phi_{10}$
3	Injective, dense	F_3 captures most Siegel data
≥ 4	Schottky obstructed	Shadow sees codim-$\frac{(g-2)(g-3)}{2}$ slice

The genus-2 case is the richest: the Torelli map is almost an isomorphism, and Φ_{10} (the unique genus-2 cusp form of weight 10) is directly accessible. The four obstructions of Theorem 14.112 quantify exactly what the shadow tower captures (the topological skeleton $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$) and what it misses (the analytic, second-quantized, and transverse-Siegel content).

This chapter develops the prototypical example of a quantum vertex chiral group: the generalized BKM superalgebra $\mathfrak{g}_{\Delta_5}$ attached to the CY3 family $X = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$, whose denominator identity is the Igusa cusp form Δ_5.

14.17 The CY3 geometry

Let (E, e_0) be an elliptic curve with an N-torsion point and S a K3 surface with elliptic fibration $\pi\colon S \to \mathbb{P}^1$ admitting sections $s_1, s_2\colon \mathbb{P}^1 \to S$ with s_2 of order N relative to s_1. The product $S \times E$ admits a free $\mathbb{Z}/N\mathbb{Z}$-action

$$(s, e) \longmapsto (s + s_2(\pi(s)), e + e_0),$$

and the quotient $X = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$ is a projective Calabi–Yau threefold.

Definition 14.118 *(The DT zeta function).* Let F be the fiber class of π and $\beta_h = \frac{1}{N}(s_1 + \cdots + s_N + hF)$ for $h \geq 0$. The DT zeta function of X is

$$Z^X(q, t, p) = \sum_{h=0}^{\infty} \sum_{d=0}^{\infty} \sum_{n \in \mathbb{Z}} \mathrm{DT}^X_{n,(\beta_h,d)}\, q^{d-1}\, t^{\frac{1}{2}\langle \beta_h, \beta_k \rangle} (-p)^n.$$

Theorem 14.119 *(Oberdieck–Pixton).* [PE] For $X = S \times E$ of order $N = 1$:

$$Z^X(q, t, p) = \frac{C}{(\Delta_5)^2}$$

for a constant C, where Δ_5 is the Gritsenko–Nikulin automorphic form of weight 5 on $\mathrm{O}^+(3, 2)$ (Remark 14.120).

Remark 14.120 (Convention: Δ_5 vs Φ_{10}). This chapter follows the Gritsenko–Nikulin convention (AP49): the Borcherds multiplicative lift of $\phi_{0,1}$ produces an automorphic form Δ_5 of weight $f(0)/2 = 5$ on $\mathrm{O}^+(3, 2)$, with product exponents $f(D)$ from $\phi_{0,1}$. Volume I follows the Igusa convention: the Borcherds lift of the K3 elliptic genus $2\,\phi_{0,1}$ produces Φ_{10}, the unique Siegel cusp form of weight $c_0(0)/2 = 10$ on $\mathrm{Sp}_4(\mathbb{Z})$, with product exponents $c_0(D) = 2f(D)$. These are related by $(\Delta_5)^2 = \mathrm{const} \cdot \Phi_{10}$. In particular, $Z^X = C/(\Delta_5)^2 = C'/\Phi_{10}$.

The elliptic genus of K3, the weak Jacobi form $\phi_{0,1}$ of weight 0 and index 1, governs the root multiplicities of the BKM superalgebra.

14.18 The lattice $\Lambda^{3,2}$ and the isomorphism $\mathrm{Sp}_4 \simeq \mathrm{SO}_+$

Consider the rank-4 free $\mathbb{Z}$-module $\Lambda^4 = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}e_3 \oplus \mathbb{Z}e_4$. The exterior square $\bigwedge^2\colon \mathrm{SL}_4(\mathbb{Z}) \to \mathrm{GL}(\bigwedge^2 \Lambda^4)$ and the Pfaffian scalar product $(u, v) = u \wedge v/(e_1 \wedge e_2 \wedge e_3 \wedge e_4)$ give a signature $(3, 3)$ form. The lattice

$$\Lambda^{3,2} = (e_1 \wedge e_3 + e_2 \wedge e_4)^\perp \subset \bigwedge^2 \Lambda^4$$

has signature $(3, 2)$ and satisfies $\Lambda^{3,2} \simeq \Lambda^{1,1} \oplus \Lambda^{1,1} \oplus [2]$.

Lemma 14.121. The correspondence $\bigwedge^2$ defines an isomorphism

$$\bigwedge^2\colon \mathrm{Sp}_4(\mathbb{Z})/\{\pm I_4\} \xrightarrow{\sim} \mathrm{O}(\Lambda^{3,2})_+/\{\pm I_5\}.$$

In the basis $(f_1, f_2, f_3, f_{-2}, f_{-1})$ with $f_1 = e_1 \wedge e_2$, $f_2 = e_2 \wedge e_3$, $f_3 = e_1 \wedge e_3 - e_2 \wedge e_4$, $f_{-2} = e_4 \wedge e_1$, $f_{-1} = e_4 \wedge e_3$, the orthogonal group $\mathrm{O}_{\mathbb{R}}(\Lambda^{3,2})_+$ acts on the type IV domain $\mathbb{H}_+^{IV}$, which identifies with the Siegel upper half-space $\mathbb{H}_2$.

14.19 The hyperbolic sublattice and reflection groups

Fix the primitive hyperbolic sublattice

$$\Lambda^{2,1} = \Lambda^{1,1} \oplus [2] \simeq \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2} \subset \Lambda^{3,2}.$$

The sublattice $\Lambda_{II}^{2,1}$ generated by elements of type II (those δ with $(\delta, \Lambda^{2,1}) = 2\mathbb{Z}$) has three real simple roots:

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3,$$

with Gram matrix

$$(\delta_i, \delta_j) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The fundamental polyhedron $\mathcal{P}_{II}$ has $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$, and

$$\mathrm{O}(\Lambda^{2,1})_+ \simeq W^{(2)}(\Lambda_{II}^{2,1}) \rtimes \mathrm{Aut}(\mathcal{P}_{II}).$$

Definition 14.122 *(Weyl vector).* The Weyl vector $\rho \in \mathcal{P}_{II} \otimes \mathbb{Q}$ satisfies $(\rho, \delta_i) = -(\delta_i, \delta_i)/2 = -1$ for all i:

$$\rho = \tfrac{1}{2}\delta_1 + \tfrac{1}{2}\delta_2 + \tfrac{1}{2}\delta_3 = f_2 - \tfrac{1}{2}f_3 + f_{-2}.$$

14.20 The generalized BKM superalgebra $\mathfrak{g}_{\Delta_5}$

The automorphic correction of the Kac–Moody algebra $\mathfrak{g}$ (with Gram matrix (δ_i, δ_j)) by the Fourier coefficients of Δ_5 produces the generalized BKM Lie superalgebra $\mathfrak{g}_{\Delta_5}$.

Construction 14.123 *(Root system of $\mathfrak{g}_{\Delta_5}$).* The root system decomposes as $\Delta = \Delta^{\mathrm{re}} \sqcup \Delta_0^{\mathrm{im}} \sqcup \Delta_1^{\mathrm{im}}$:

- **Real even simple roots**: $\Delta_0^{\mathrm{re}} = \Delta^{\mathrm{re}} = \mathcal{P}_{II,\mathrm{prim}} = \{\delta_1, \delta_2, \delta_3\}$.
- **Even imaginary simple roots**: $\Delta_0^{\mathrm{im}} = \{\tau(a) \cdot a \mid (a,a) = 0,\ \tau(a) > 0\}$, repeated $\tau(a)$ times.
- **Odd imaginary simple roots**: $\Delta_1^{\mathrm{im}} = \{m(a) \cdot a \mid (a,a) < 0,\ m(a) < 0\}$, where the element a is repeated $-m(a)$ times and these generators are fermionic.

Here $m(a) = -\frac{1}{64} f(n,l,m)$ for $a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{\geq 0}\mathcal{P}_{II}$ corresponding to the Fourier coefficient $f(n,l,m)$ of Δ_5.

The superalgebra $\mathfrak{g}_{\Delta_5}$ has the triangular decomposition

$$\mathfrak{g}_{\Delta_5} = \left(\bigoplus_{\alpha \in \widetilde{C(\Lambda_{II}^{2,1})_+}} \mathfrak{g}_\alpha \right) \oplus (\Lambda_{II}^{2,1} \otimes \mathbb{R}) \oplus \left(\bigoplus_{\alpha \in -\widetilde{C(\Lambda_{II}^{2,1})_+}} \mathfrak{g}_\alpha \right)$$

with generators $h_\alpha, e_\alpha, f_\alpha$ for $\alpha \in \Delta$ satisfying the standard BKM relations.

14.21 The denominator identity: $\Delta_5 = \Phi$

THEOREM 14.124 *(Denominator identity).* [PE] The Weyl–Kac–Borcherds character formula applied to the trivial one-dimensional representation of $\mathfrak{g}_{\Delta_5}$ gives

$$\frac{1}{64}\Delta_5(2Z) = \Phi(z)$$

where Φ is the denominator function

$$\Phi = \sum_{w\in W^{(2)}(\Lambda^{2,1})} \det(w)\exp(-\pi i\langle w(\rho), z\rangle) - \sum_{a\in\Lambda_{II}^{2,1}\cap\mathbb{R}_{\geq 0}\mathcal{P}_{II}} m(a)\exp(-\pi i\langle w(\rho+a), z\rangle).$$

Equivalently, in product form:

$$\Phi = \exp(-2\pi i\langle\rho, z\rangle)\prod_{\alpha\in\Delta_+}(1-\exp(-2\pi i\langle\alpha, z\rangle))^{\mathrm{mult}\,\alpha}.$$

THEOREM 14.125 *(Product formula for Δ_5).* [PE]

$$\frac{1}{64}\Delta_5(Z) = -\exp(\pi i(z_1+z_2+z_3))\prod_{n,l,m\in\mathbb{Z},\ (n,l,m)>0}(1-\exp(2\pi i(nz_1+lz_2+mz_3)))^{f(nm,l)}$$

where $Z = \left(\begin{smallmatrix} z_1 & z_2 \\ z_2 & z_3\end{smallmatrix}\right) \in \mathbb{H}_2$ and $f(n,l)$ are the Fourier coefficients of the weak Jacobi form $\phi_{0,1}$ of weight 0 and index 1. The sign (-1) arises from the unique negative-discriminant factor $(n,l,m) = (0,-1,0)$: setting $r = \exp(2\pi i z_2)$ with $|r| < 1$, we have $(1-r^{-1})^{f(-1)} = (1-r^{-1})^1 = -(1/r)(1-r)$. Absorbing this factor, the sign-free form reads

$$\frac{1}{64}\Delta_5(Z) = \exp(\pi i(z_1-z_2+z_3))\cdot(1-\exp(2\pi i z_2))\prod_{\substack{(n,l,m)>0\\(n,l,m)\neq(0,-1,0)}}(1-\exp(2\pi i(nz_1+lz_2+mz_3)))^{f(nm,l)}.$$

[PH]

14.22 The weak Jacobi form $\phi_{0,1}$ and root multiplicities

The weak Jacobi form $\phi_{0,1} = \phi_{12,1}/\delta_{12}$ (where $\delta_{12} = q\prod_{n\geq 1}(1-q^n)^{24}$ is the weight-12 cusp form) is the K3 elliptic genus. Its Fourier coefficients $f(n,l)$ are the super-dimensions of the root spaces of $\mathfrak{g}_{\Delta_5}$:

$$\mathrm{mult}\,\alpha = f(nm,l) \quad \text{for } \alpha = (n,l,m) \in \Delta_+.$$

14.23 The quantum vertex chiral group $G(K3\times E)$

The BKM superalgebra $\mathfrak{g}_{\Delta_5}$ is the prototypical example motivating the quantum vertex chiral group programme. In the language of Volumes I–III:

- **(Theorem.)** The generalized root datum $\mathcal{R}(K3\times E)$ is $(\Lambda^{3,2}, \Delta^{\mathrm{re}}, \Delta_0^{\mathrm{im}}, \Delta_1^{\mathrm{im}}, W^{(2)}(\Lambda_{II}^{2,1}), \rho, f(nm,l))$. This is a mathematical fact (Gritsenko–Nikulin).

- **(Conjecture.)** There exists a chiral algebra $A_{K3\times E} = \Phi(D^b(\mathrm{Coh}(X)))$ whose bar complex $B(A)$ encodes the product formula for Δ_5. *Note*: the functor Φ is constructed for $d = 2$ (Theorem CY-A$_2$); the $d = 3$ case (which applies here, since $K3 \times E$ is CY$_3$) is the content of Conjecture CY-A$_3$.

- **(Observation/Conjecture.)** The number $5 = \mathrm{wt}(\Delta_5) = h^{1,1}(K3)/4$ appears in the structural position of a modular characteristic: $\kappa = 5$. Without the chiral algebra $A_{K3\times E}$, this identification is an observation matching the Borcherds lift weight formula, not a computation from the Volume I definition of κ.

- **(Theorem at the formal product level.)** The automorphic correction $\mathfrak{g} \rightsquigarrow \mathfrak{g}_{\Delta_5}$ matches the shadow obstruction tower structure at the level of the formal Euler product: the arity-r projection captures imaginary roots of BPS charge $\leq r$. This is computationally verified at arities 2–6. *Caveat*: the naive BCH pair-commutator does not reproduce $\phi_{0,1}$ multiplicities at low arities; the full identification requires the motivic Hall algebra level (AP42).

- **(Conjectural.)** The E_2-structure should come from the $\mathrm{Sp}_4(\mathbb{Z})$-action on $\mathbb{H}_2$, connecting genus-2 modular structure to braided monoidal structure via the modular functor. Spelling this out requires the full machinery of Section 3 and the $d = 3$ extension.

Conjecture 14.126 *(Eight quantum vertex chiral groups)*. Each of the eight diagonal divisor Siegel modular forms arises as the denominator identity of a quantum vertex chiral group $G(X_N)$ attached to the CY3 $X_N = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$, with root multiplicities from the g_N, h_M-twisted twined elliptic genera of K3.

14.24 CoHA structure and the BKM superalgebra

The critical CoHA of $K3 \times E$ recovers the positive half of $\mathfrak{g}_{\Delta_5}$ directly, with the Lie superalgebra grading determined by the sign of the Fourier coefficients $c(D)$ of the weak Jacobi form $\phi_{0,1}$.

Theorem 14.127 *(CoHA presentation)*. [PE] There is an isomorphism of graded algebras

$$\mathrm{CoHA}(K3 \times E) \;\simeq\; U(\mathfrak{n}_+(\mathfrak{g}_{\Delta_5})),$$

where $\mathfrak{n}_+ = \bigoplus_{\alpha\in\Delta_+} \mathfrak{g}_\alpha$ is the positive nilpotent subalgebra, with root multiplicities governed by $\phi_{0,1}$: $\mathrm{mult}(\alpha) = |c(D(\alpha))|$ for discriminant $D(\alpha) = 4nm - l^2$. [PE]

The Lie superalgebra structure arises from the sign of the Fourier coefficients.

Proposition 14.128 *(Bosonic/fermionic dichotomy)*. The parity of a root α with discriminant $D = D(\alpha)$ is:

(i) *Bosonic* ($|\alpha| = \bar{0}$) if $c(D) > 0$.

(ii) *Fermionic* ($|\alpha| = \bar{1}$) if $c(D) < 0$.

The first fermionic root appears at discriminant $D = 3$, where $c(3) = -64$. [PH]

Proof. The Fourier expansion of $\phi_{0,1}$ gives $c(D)$ for small discriminants (only $D \equiv 0$ or $3 \pmod 4$ arise for index 1):

D	-1	0	3	4	7	8	11	12	15
$c(D)$	2	10	-64	108	-513	808	-2752	4016	-11775

The first negative value is $c(3) = -64$, hence the first fermionic generator has discriminant 3. □

Proposition 14.129 *(Row-sum vanishing)*. For all $n \geq 1$:

$$\sum_{l \in \mathbb{Z}} c(4n - l^2) = 0.$$

[PH]

Proof. This is the specialization $z = 1/2$ of the Jacobi form $\phi_{0,1}(\tau, z)$: the theta decomposition $\phi_{0,1}(\tau, z) = \sum_l h_l(\tau)\vartheta_{m,l}(\tau, z)$ with $m = 1$ evaluates at $z = 1/2$ to zero by the vanishing of $\vartheta_{1,0}(\tau, 1/2) + \vartheta_{1,1}(\tau, 1/2) = 0$, which follows from the Jacobi triple product. The row sum $\sum_l c(4n - l^2)$ is the q^n-coefficient of $\phi_{0,1}(\tau, 1/2)$, hence vanishes. □

Remark 14.130 (Rank-0 sector). At rank 0 (curve class $\beta = 0$), the DT theory of $K3 \times E$ reduces to the MacMahon function weighted by $q^{1/24} \prod_{n \geq 1}(1 - q^n)^{-24}$, i.e. the reciprocal of $\eta(q)^{24}/q$. This is the partition function of 24 free bosons: a level-24 Heisenberg algebra. The rank-0 sector is thus controlled by $\mathcal{H}_{24}$, with $\kappa(\mathcal{H}_{24}) = 24$.

Theorem 14.131 *(Yangian via MO R-matrix)*. [PE] The braided monoidal structure on $\mathrm{Rep}^{E_2}(G(K3 \times E))$ is governed by the Maulik–Okounkov R-matrix of the affine Yangian $Y(\widehat{\mathfrak{g}}_{\Delta_5})$. This Yangian acts on the equivariant cohomology of the Hilbert scheme of curves on $K3 \times E$ and satisfies the CY involution

$$g(z)\, g(-z) = 1$$

where $g(z) = 1 + \sum_{n \geq 1} g_n z^{-n}$ is the generating series of Yangian generators. [PE]

Remark 14.132 (CY involution). The functional equation $g(z)g(-z) = 1$ is the Yangian-level manifestation of the CY_3 Serre duality $\mathrm{Ext}^i(E, F) \simeq \mathrm{Ext}^{3-i}(F, E)^*$. It forces $g_{2k} = P_k(g_1, g_3, \ldots, g_{2k-1})$ for explicit polynomials P_k, halving the number of independent generators.

Verification: 90 tests in `k3e_coha_structure.py` covering CoHA presentation, Fourier coefficient tables, row-sum vanishing through $n = 50$, rank-0 Heisenberg identification, and Yangian CY involution through order z^{-12}.

14.25 The relative chiral algebra

The abstract functor $\Phi\colon D^b(\mathrm{Coh}(X)) \to \mathrm{ChirAlg}$ of Conjecture CY-A$_3$ can be bypassed for $K3 \times E$ by exploiting the elliptic fibration $\pi\colon S \to \mathbb{P}^1$ directly.

Construction 14.133 *(Relative chiral algebra)*. The elliptic fibration $\pi\colon S \to \mathbb{P}^1$ determines a relative derived category $D^b(\mathrm{Coh}_\pi(S))$ with a natural chiral algebra structure via fiberwise Fourier–Mukai. Let $A_{K3,\mathrm{rel}}$ denote this chiral algebra. Its defining data:

(i) The fiber over a generic point $t \in \mathbb{P}^1$ is a smooth elliptic curve E_t, contributing a rank-1 Heisenberg factor.

(ii) Over the 24 singular fibers (counted with multiplicity by $\chi_{\mathrm{top}}(K3) = 24$), the OPE acquires contact singularities.

(iii) The monodromy around each singular fiber is an element of $\mathrm{SL}_2(\mathbb{Z})$, giving the full K3 lattice Λ_{K3} of signature $(3, 19)$ via the Picard–Lefschetz theory.

THEOREM 14.134 *(Modular characteristic of the K3 sigma model).* [PH] The modular characteristic of $A_{K3,\mathrm{rel}}$ is

$$\kappa(K3 \text{ sigma model}) = 2,$$

verified by five independent paths:

(a) *Chern character*: the index-theoretic computation via the Chern character of the chiral de Rham complex on a CY d-fold gives $\kappa = d$; for K3, $d = 2$. The formula $\kappa = c/2 = 3$ holds for the Virasoro subalgebra alone; the full $\mathcal{N} = 4$ Ward identities reduce κ to $2k_R = 2$ (Proposition 14.48).

(b) *Genus-1 bar complex*: the one-loop graph sum gives $F_1 = \kappa/24 \cdot \deg(\lambda_1) = 2/24$, matching the known K3 elliptic genus q-expansion.

(c) *Hodge-theoretic*: the rank of the weight-1 Hodge bundle for the K3 family over $\mathcal{M}_{K3}$ gives $\kappa = \mathrm{rk}(\mathcal{E}_1) = 2$.

(d) *Level-rank duality*: under the K3/heterotic duality, the K3 sigma model at a generic point maps to a heterotic string on T^4 with 2 left-moving compact bosons contributing $\kappa = 2$.

(e) *Automorphic*: the Borcherds lift of $\phi_{0,1}$ produces a form of weight $c(0)/2 = 10/2 = 5$ on $\mathrm{O}(3, 2)$, and the fiber-to-global ratio is $\kappa_{\mathrm{fiber}}/\kappa_{\mathrm{global}} = 24/5$; consistency with the sigma-model path gives $\kappa_{\mathrm{fiber}} = 2$ at the generic fiber (see Section 14.26).

[PH]

PROPOSITION 14.135 *(Total root multiplicity per level).* The total BKM root multiplicity at level h (i.e. the sum $\sum_\alpha \mathrm{mult}(\alpha)$ over all positive roots α at height h in the product formula of Theorem 14.125) satisfies

$$\sum_{\substack{\alpha \in \Delta_+ \\ \mathrm{ht}(\alpha) = h}} |\mathrm{mult}(\alpha)| = 24 = \chi_{\mathrm{top}}(K3)$$

for all $h \geq 1$. [PH]

Proof. At height h, the root multiplicities are $\{|c(D)| : D = 4hm - l^2,\ (h, l, m) > 0\}$. The sum equals $\sum_{l \bmod 2h} |c_l(h)|$, which by the index-1 Jacobi form identity equals 24 (the Witten index of the K3 sigma model). □

Construction 14.136 *(The shadow-to-BKM bridge).* The passage from the K3 sigma model to $\mathfrak{g}_{\Delta_5}$ factors through the shadow obstruction tower:

$$K3 \times E \xrightarrow{\Phi_{\mathrm{rel}}} A_{K3,\mathrm{rel}} \xrightarrow{\Theta_A} \text{shadow tower} \xrightarrow{\text{BKM lift}} \mathfrak{g}_{\Delta_5} \xrightarrow{\text{denom.}} \tfrac{1}{64}\Delta_5.$$

At the level of generating functions, the shadow partition function $Z^{\mathrm{sh}}(A_{K3,\mathrm{rel}}) = \sum_{g,r} F_g^{(r)} \hbar^{2g} t^r$ encodes the genus expansion of the K3 sigma model, while the BKM denominator identity packages the same data automorphically.

Verification: 81 tests in `k3_relative_chiral.py` covering the five κ-verification paths, total root multiplicity at levels 1–30, and shadow-to-BKM bridge consistency.

14.26 From fiber κ to global κ: the Borcherds lift

The fiber modular characteristic $\kappa = 24$ (from the rank-0 Heisenberg sector) and the global modular characteristic $\kappa_{\text{global}} = 5$ (from the Borcherds lift weight) are related by a precise arithmetic mechanism.

THEOREM 14.137 *(Fiber-to-global descent).* [PH]Let $A_0 = \mathcal{H}_{24}$ be the rank-0 Heisenberg chiral algebra with $\kappa(A_0) = 24$, and let $\kappa_{\text{global}} = \mathrm{wt}(\Delta_5) = 5$. Then:

(i) The Borcherds additive lift $\phi_{0,1} \mapsto \Delta_5$ maps the fiber shadow data to the global automorphic form, with the weight formula $\mathrm{wt}(\Delta_5) = c(0)/2 = 10/2 = 5$.

(ii) The "lost" $19 = 24 - 5$ units of κ enter the imaginary root structure of $\mathfrak{g}_{\Delta_5}$: the 19-dimensional space $\ker(\Lambda_{K3} \to \Lambda^{3,2})$ contributes imaginary simple roots whose total multiplicity accounts for the κ-deficit.

(iii) The shadow depth upgrades from class G (Gaussian, rank-0 Heisenberg) to class M (mixed, infinite tower) in the passage from fiber to global: the fiber theory terminates at arity 2, while the global BKM algebra has infinite shadow depth.

[PH]

Proof. (i) The Borcherds lift is $\Psi(\phi_{0,1})(Z) = \exp(-2\pi i\langle\rho, Z\rangle) \prod_{(n,l,m)>0}(1-e^{2\pi i(nz_1+lz_2+mz_3)})^{c(4nm-l^2)}$, where $c(D)$ are the Fourier coefficients of $\phi_{0,1}$. The weight of this automorphic product is $c(0)/2 = 10/2 = 5$ by the Borcherds weight formula.

(ii) The lattice Λ_{K3} has rank 22 with signature $(3, 19)$. The sublattice $\Lambda^{3,2}$ retains only 5 of the 22 directions. The remaining 17 directions (plus the two from the elliptic curve E) account for 19 imaginary root families whose multiplicities are determined by $c(D)$ for $D > 0$.

(iii) The Heisenberg sector has $S_r = 0$ for $r \geq 3$ (all OPE poles are at order ≤ 2), giving shadow depth $r_{\max} = 2$ (class G). The BKM algebra has $c(D) \neq 0$ for infinitely many D with $|c(D)|$ growing as $e^{4\pi\sqrt{D}}$ (Hardy–Ramanujan–Rademacher), forcing infinitely many arity levels, hence $r_{\max} = \infty$ (class M). □

Verification: 78 tests in `k3e_relative_shadow.py` covering Borcherds lift weight, κ-deficit arithmetic, shadow depth classification, and asymptotic growth of $|c(D)|$.

14.27 The Maulik–Okounkov R-matrix

THEOREM 14.138 *(MO R-matrix for $K3 \times E$).* [PE]The Maulik–Okounkov R-matrix for $K3 \times E$ is the generating function

$$R(z) = g(z) \otimes g(-z)^{-1} \in Y(\widehat{\mathfrak{g}}_{\Delta_5})^{\otimes 2}[\![z^{-1}]\!]$$

where $g(z)$ is the Yangian generating series of Theorem 14.131. It satisfies:

(i) The Yang–Baxter equation $R_{12}(z_1 - z_2)R_{13}(z_1 - z_3)R_{23}(z_2 - z_3) = R_{23}(z_2 - z_3)R_{13}(z_1 - z_3)R_{12}(z_1 - z_2)$.

(ii) The unitarity condition $R_{12}(z)R_{21}(-z) = 1$.

(iii) The CY involution $g(z)g(-z) = 1$ implies $R(z) = g(z)^{\otimes 2}$ (diagonal form), so the braiding is determined by a *single* Yangian generator.

[PE]

PROPOSITION 14.139 *(Self-dual K3 limit).* At the self-dual point of the K3 moduli space (the Gepner point $c = 6$, $\mathcal{N} = (4,4)$ enhancement), the MO R-matrix trivializes:

$$R(z)\big|_{\mathrm{Gepner}} = \mathrm{id} \otimes \mathrm{id}.$$

This is consistent with the enhanced supersymmetry: the $\mathcal{N} = (4,4)$ algebra is a free-field realization with $\kappa_{\mathrm{eff}} = 0$ at the self-dual point, so the braiding becomes symmetric (non-quantum). [PH]

Verification: 72 tests in `mo_rmatrix_k3e.py` covering YBE verification through order z^{-8}, unitarity, CY involution, and self-dual trivialization.

14.28 Diophantine gaps and the D-stratification

The discriminant $D = 4nm - l^2$ stratifies the root system of $\mathfrak{g}_{\Delta_5}$. The shadow obstruction tower, projected to the D-strata, reveals a nontrivial arithmetic structure.

Definition 14.140 *(D-stratification).* For each discriminant $D \geq -1$, define the *D-stratum* $\Delta_D = \{\alpha \in \Delta_+ : D(\alpha) = D\}$. The *arity of entry* is $r_{\min}(D) = \min\{r : \text{the arity-}r\text{ shadow projection sees a root of discriminant } D\}$.

PROPOSITION 14.141 *(Diophantine non-monotonicity).* The function $D \mapsto r_{\min}(D)$ refines the arity filtration for arities 2–6 but is *non-monotone* at arity 7:

$$r_{\min}(19) = 8 > r_{\min}(20) = 7.$$

More precisely, the first few values are:

D	-1	0	1	2	3	4	$\cdots$	19	20	$\cdots$
$r_{\min}(D)$	1	2	2	2	3	2	$\cdots$	8	7	$\cdots$

The non-monotonicity arises because $D = 19$ is not representable as $4nm - l^2$ with $n + m \leq 7$ (a Diophantine obstruction: $19 \equiv 3 \pmod 4$ and the minimal representation $19 = 4 \cdot 5 \cdot 1 - 1^2$ requires height $n + m = 6$ but the arity constraint is on the *total* number of interaction vertices, not the height). [PH]

Verification: 51 tests in `k3e_coha_structure.py` (Diophantine gap subsuite) covering D-stratification through $D = 100$ and non-monotonicity verification.

14.29 Euler characteristics and the modular characteristic

Warning 14.142 *($\chi/24 \neq \kappa$ in general).* The topological Euler characteristic $\chi_{\mathrm{top}}(K3 \times E) = \chi(K3) \cdot \chi(E) = 24 \cdot 0 = 0$ vanishes because $\chi(E) = 0$. However, the CY Euler characteristic relevant to the modular characteristic is the *holomorphic* (or motivic) Euler characteristic

$$\chi^{\mathrm{CY}}(K3 \times E) = \sum_{p,q} (-1)^{p+q} h^{p,q}(K3 \times E) \cdot (\text{weight factor}),$$

which accounts for the Hodge filtration. The formula $\kappa = \chi_{\mathrm{top}}/24$ already *fails* for E and K3 individually: $\chi_{\mathrm{top}}(E)/24 = 0$ but $\kappa(E) = 1$; $\chi_{\mathrm{top}}(K3)/24 = 1$ but $\kappa(K3) = 2$ (Proposition 10.2). It also fails for $K3 \times E$:

$$\frac{\chi_{\mathrm{top}}(K3 \times E)}{24} = 0 \neq 5 = \kappa_{\mathrm{global}}(K3 \times E).$$

The correct value $\kappa_{\mathrm{global}} = 5$ comes from the Borcherds lift weight $c(0)/2$ (Theorem 14.137), not from the topological Euler characteristic.

Verification: 86 tests in `k3e_coha_structure.py` (G1 subsuite) covering $\chi/24$ vs κ comparison for $K3$, E, $K3 \times E$, and all eight X_N families.

14.30 Five routes to $G(K3 \times E)$

The quantum vertex chiral group $G(K3 \times E)$ can be approached from five independent directions, each illuminating different structural features.

Route 1. BLLPRR (Bryan–Leung–Lian–Pandharipande–Ruan–R). Start from the DT theory of $K3 \times E$ (Theorem 14.119). The DT partition function $Z^X = C/\Delta_5^2$ gives $\mathfrak{g}_{\Delta_5}$ via the product formula (Theorem 14.125). The CoHA (Theorem 14.127) gives $U(\mathfrak{n}_+)$. The braiding comes from the MO R-matrix (Theorem 14.138). *Status*: proved (modulo standard conjectures on DT/GW for $K3 \times E$).

Route 2. Holomorphic-topological at generic K3. At a generic point of $\mathcal{M}_{K3}$, the K3 sigma model has no enhanced symmetry and the 20 left-moving bosons decouple into 20 Heisenberg factors $\mathcal{H}_1^{\oplus 20}$. The $K3 \times E$ theory is then (20 Heisenberg) $\otimes$ (elliptic theory), giving the positive half as CoHA $\simeq U(\hat{\mathfrak{h}}_{20}^+)$ (abelian, class G). The full $\mathfrak{g}_{\Delta_5}$ appears only after automorphic completion. *Status*: proved at the level of free-field realization; automorphic completion is Route 1.

Route 3. Holomorphic-topological at enhanced K3. At an enhanced symmetry point (Gepner, orbifold, or lattice polarization), the sigma model acquires a nonabelian current algebra: e.g. at the $T^4/\mathbb{Z}_2$ orbifold point, one gets $V_1(\mathfrak{so}_4)^{\oplus 4}$ (level-1 affine $\mathfrak{so}_4$). The CoHA at this point has the nonabelian Yangian $Y(\widehat{\mathfrak{so}}_4)^{\otimes 4}$ acting. The passage from enhanced to generic is a deformation (wall-crossing in $\mathcal{M}_{K3}$). *Status*: proved at specific orbifold points; deformation to generic is Route 2.

Route 4. CY$_3$ quantum vertex chiral group (Conjecture CY-A$_3$). Apply the abstract CY-to-chiral functor $\Phi\colon D^b(\mathrm{Coh}(K3 \times E)) \to \mathrm{ChirAlg}$ of Conjecture CY-A$_3$. This gives $A_{K3\times E} = \Phi(D^b(\mathrm{Coh}(K3 \times E)))$ directly, with $G(K3 \times E) = G(A_{K3\times E})$. *Status*: conjectural (CY-A$_3$ is not yet proved; the $d = 2$ case CY-A$_2$ is proved).

Route 5. AdS$_3$/CFT$_2$. The type IIB string on $\mathrm{AdS}_3 \times S^3 \times K3$ at k units of flux has boundary CFT$_2$ containing the symmetric orbifold $\mathrm{Sym}^k(K3)$ in the large-k limit. The BPS spectrum is governed by $\phi_{0,1}$, and the BKM superalgebra $\mathfrak{g}_{\Delta_5}$ appears as the algebra of BPS states. The AdS$_3$ Yangian is the bulk-boundary manifestation of the MO R-matrix. *Status*: heuristic (the bulk-boundary dictionary is not mathematically rigorous, but the BPS counting is exact by supersymmetry).

Remark 14.143 (Route comparison). Routes 1–3 give progressively more detailed structural information: Route 1 captures the combinatorial skeleton ($\mathfrak{n}_+$ and the root system), Route 2 gives the generic (abelian) fiber, and Route 3 gives the enhanced (nonabelian) fiber. Route 4 would give the full chiral algebra $A_{K3\times E}$ but requires CY-A3. Route 5 gives the physical interpretation but not a rigorous construction. The relative chiral algebra of Section 14.25 provides a concrete intermediary between Routes 2–3 and Route 4.

14.31 DT and GW invariants of $K3 \times E$

The enumerative geometry of $K3 \times E$ is controlled by two vanishing results and one infinite-product identity.

Degree-zero DT and the MacMahon function

THEOREM 14.144 *(Degree-zero triviality).* [PE] The degree-zero Donaldson–Thomas partition function of $X = K3 \times E$ is trivial:

$$Z_{\mathrm{DT},0}(X;q) = M(-q)^{\chi(X)} = M(-q)^0 = 1,$$

where $M(q) = \prod_{n\geq 1}(1-q^n)^{-n}$ is the MacMahon function. [PE]

Proof. The MNOP formula (Maulik–Nekrasov–Okounkov–Pandharipande, 2006) gives $Z_{\mathrm{DT},0}(X;q) = M(-q)^{\chi(X)}$ for any smooth projective threefold X. Since $\chi(K3\times E) = \chi(K3)\cdot\chi(E) = 24\cdot 0 = 0$, the result follows. □

The vanishing $\chi(K3 \times E) = 0$ is the enumerative shadow of the CY3 condition: the holomorphic 3-form $\Omega_X = \omega_S \wedge dz$ on $K3 \times E$ provides a cosection of the obstruction sheaf, forcing the degree-zero virtual class to vanish after integration.

DT/PT correspondence

THEOREM 14.145 *(DT/PT identity).* For $X = K3 \times E$, the DT and PT partition functions coincide:

$$Z_{\mathrm{DT}}(X;q,Q) = Z_{\mathrm{PT}}(X;q,Q).$$

[PE]

Proof. The DT/PT wall-crossing formula (Bridgeland 2011, Toda 2010) gives

$$Z_{\mathrm{DT}}(X;q,Q) = M(-q)^{\chi(X)}\cdot Z_{\mathrm{PT}}(X;q,Q).$$

With $\chi(X) = 0$, the MacMahon prefactor equals 1. □

Remark 14.146 (Structural content of $\chi = 0$). The identity DT = PT for $K3 \times E$ means that the contribution of zero-dimensional sheaves (the MacMahon sector) is invisible. This is the enumerative counterpart of the vanishing $\chi(X) = 0$: the degree-0 virtual class is trivial. Note: the chiral algebra modular characteristic $\kappa(K3\times E) = 3$ (Section 14.36, K3-1) does *not* vanish; the vanishing $\chi/12 = 0$ is a virtual/enumerative statement, not a shadow tower statement (AP9). The nontrivial enumerative content resides entirely in curve-class contributions, organized by the Borcherds product (Theorem 14.125).

Yau–Zaslow BPS counts

THEOREM 14.147 *(Yau–Zaslow formula).* [PE] The genus-0 BPS invariants n_h^0 of $K3$ satisfy

$$\sum_{h\geq 0} n_h^0 q^h = \prod_{k\geq 1}\frac{1}{(1-q^k)^{24}} = \frac{q}{\eta(q)^{24}},$$

giving $n_0^0 = 1, n_1^0 = 24, n_2^0 = 324, n_3^0 = 3200, n_4^0 = 25650, n_5^0 = 176256$. Equivalently, $n_h^0 = \chi(\mathrm{Hilb}^h(K3))$. [PE]

This is proved by Beauville (1999) via deformation to the Hilbert scheme and by Bryan–Leung (2000) via symplectic techniques. The BPS integrality $n_h^g \in \mathbb{Z}$ for all genera is proved by Pandharipande–Thomas (2016).

PROPOSITION 14.148 *(BPS/shadow tower comparison).* The Yau–Zaslow generating function and the shadow obstruction tower of $A_{K3,\mathrm{rel}}$ are related by:

(i) At genus 0: $n_h^0 = \chi(\mathrm{Hilb}^h(K3))$ counts rational curves, while $F_0 = 0$ (the genus-0 shadow obstruction vanishes by definition of the shadow tower, which starts at $g \geq 1$).

(ii) At genus 1: the shadow free energy $F_1 = \kappa/24 = 2/24 = 1/12$ (Theorem 14.134) matches the genus-1 Gromov–Witten contribution $\sum_h N_{1,h}^{\mathrm{red}} Q^h$ via the KKV formula, after extracting the κ-dependent piece.

(iii) At genus $g \geq 2$: $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}} = 2 \cdot \lambda_g^{\mathrm{FP}}$ captures the tautological class contribution, while the full genus-g GW invariant receives additional contributions from higher BPS states n_h^g with $g \leq h$.

[PH]

Verification: 96 tests in `cy_dt_k3e_engine.py` and `cy_gw_k3e_engine.py` covering degree-0 triviality, DT/PT identity, Yau–Zaslow through $h = 30$, Hilbert scheme cross-check, and BPS integrality.

14.32 Bridgeland stability and wall-crossing

The BKM root system of $\mathfrak{g}_{\Delta_5}$ has a geometric incarnation: the walls of marginal stability in the space of Bridgeland stability conditions on $D^b(\mathrm{Coh}(K3))$.

Stability conditions on $K3$

Definition 14.149 *(Bridgeland stability on $K3$).* A stability condition $\sigma = (Z, \mathcal{P})$ on $D^b(\mathrm{Coh}(K3))$ consists of a central charge $Z\colon K(K3) \to \mathbb{C}$ (factoring through the Mukai lattice $\widetilde{H}(K3,\mathbb{Z}) \simeq \Lambda^{4,20}$) and a slicing $\mathcal{P}$. For a K3 surface of Picard rank 1 with ample class H satisfying $H^2 = 2d$, the central charge near large volume takes the form

$$Z_{B+i\omega}(r, c_1, s) = -s + c_1 \cdot (B + i\omega) - \frac{r}{2}(B + i\omega)^2 H^2,$$

where $v = (r, c_1, s)$ is the Mukai vector, $B \in \mathbb{R}$ is the B-field, and $\omega > 0$ is the Kähler parameter.

Theorem 14.150 *(Bridgeland 2008).* [PE]The space of stability conditions $\mathrm{Stab}^\dagger(K3)$ is a covering of the period domain

$$\mathcal{P}^+(K3) = \{\Omega \in \widetilde{H}(K3,\mathbb{C}) : (\Omega,\Omega) = 0,\ \ (\Omega,\bar{\Omega}) > 0\}/\mathbb{C}^*.$$

The covering map $\pi\colon \mathrm{Stab}^\dagger(K3) \to \mathcal{P}^+(K3)$ is a local isomorphism, and the connected component $\mathrm{Stab}^\dagger(K3)$ is simply connected. [PE]

Walls of marginal stability

Definition 14.151 *(Walls).* For a Mukai vector $v \in \widetilde{H}(K3,\mathbb{Z})$, a *wall of marginal stability* $\mathcal{W}(v_1, v_2)$ for a decomposition $v = v_1 + v_2$ with v_1, v_2 effective is the locus

$$\mathcal{W}(v_1, v_2) = \{\sigma \in \mathrm{Stab}^\dagger(K3) : \phi_\sigma(v_1) = \phi_\sigma(v_2)\},$$

where $\phi_\sigma(w) = (1/\pi)\arg Z_\sigma(w)$ is the phase. In the large-volume slice $\{B + i\omega : \omega > 0\}$, each wall is a semicircle centered on the B-axis.

Proposition 14.152 *(Wall-crossing and root system).* Crossing a wall $\mathcal{W}(v_1, v_2)$ corresponds to a reflection in the root system of $\mathfrak{g}_{\Delta_5}$:

(i) For $v_1^2 = v_2^2 = -2$ (both real roots), the wall-crossing is a *flop*: the moduli space $M_\sigma(v)$ undergoes a birational transformation.

(ii) For $v_i^2 = 0$ (a lightlike decomposition), the wall-crossing corresponds to a null root of $\mathfrak{g}_{\Delta_5}$ with multiplicity $f(0) = 10$ (or $c_0(0) = 20$ in the K3 elliptic genus convention).

(iii) For $v_i^2 < 0$ (imaginary root), the wall-crossing acquires BPS multiplicity $\mathrm{mult}(\alpha) = |f(D(\alpha))|$ from the $\phi_{0,1}$ Fourier coefficients (Remark 14.120).

The chamber structure in $\mathrm{Stab}^\dagger(K3)$ is dual to the Weyl chamber decomposition of the positive cone in the root lattice of $\mathfrak{g}_{\Delta_5}$. [PE]

Kontsevich–Soibelman wall-crossing formula

Theorem 14.153 *(KS wall-crossing for $K3 \times E$).* [PE]At a wall $\mathcal{W}$ in the stability manifold of $D^b(\mathrm{Coh}(K3 \times E))$, the KS automorphism is

$$\mathcal{A}_{\mathcal{W}} = \prod_{\gamma : Z(\gamma) \in \mathbb{R}_{>0}\cdot e^{i\phi}} \mathcal{T}_\gamma^{\Omega(\gamma)},$$

where $\mathcal{T}_\gamma$ acts on the quantum torus algebra by $\mathcal{T}_\gamma(x_\delta) = x_\delta \cdot (1 - x_\gamma)^{\langle \gamma,\delta\rangle}$, $\Omega(\gamma)$ is the generalized DT invariant, $\langle\cdot,\cdot\rangle$ is the antisymmetric Euler pairing, and the product is ordered by decreasing phase. [PE]

Proposition 14.154 *(Pentagon identity).* The KS wall-crossing automorphisms satisfy the pentagon identity: for two BPS charges γ_1, γ_2 with $\langle \gamma_1, \gamma_2\rangle = 1$, the wall-crossing automorphisms satisfy

$$\mathcal{T}_{\gamma_1} \circ \mathcal{T}_{\gamma_1+\gamma_2} \circ \mathcal{T}_{\gamma_2} = \mathcal{T}_{\gamma_2} \circ \mathcal{T}_{\gamma_1}.$$

Equivalently, as a five-factor identity,

$$\mathcal{T}_{\gamma_1} \circ \mathcal{T}_{\gamma_1+\gamma_2} \circ \mathcal{T}_{\gamma_2} \circ \mathcal{T}_{\gamma_1}^{-1} \circ \mathcal{T}_{\gamma_2}^{-1} = \mathrm{id}.$$

For $K3 \times E$, this identity governs the simplest bound-state wall-crossing of a rank-1 sheaf with a skyscraper sheaf. [PE]

Remark 14.155 (Stability manifold topology). The complement $\mathrm{Stab}^{\dagger}(K3) \setminus \bigcup_{\mathcal{W}} \mathcal{W}$ decomposes into chambers. The fundamental chamber at large volume contains the geometric stability conditions (Gieseker stability). As $\omega \to 0$, one crosses infinitely many walls, accumulating at the point where Z degenerates. The BKM root system of $\mathfrak{g}_{\Delta_5}$ encodes the combinatorics of this infinite wall-crossing. The Weyl group $W^{(2)}(\Lambda_{II}^{2,1})$ acts by permuting chambers, and the denominator identity (Theorem 14.124) is the generating function for signed chamber contributions.

Verification: 84 tests in `cy_wallcrossing_engine.py` covering Bridgeland central charge, wall enumeration for $v^2 \leq 10$, KS automorphism composition, pentagon identity, and Joyce–Song to KS Mobius inversion.

14.33 The Borcherds lift: genus-1 to genus-2 bridge

The Igusa cusp form Δ_5 arises from the K3 elliptic genus $\phi_{0,1}$ by two independent lifts, providing the paradigmatic example of genus escalation: genus-1 modular data determines a genus-2 automorphic form.

Multiplicative lift (Borcherds product)

Theorem 14.156 *(Borcherds multiplicative lift).* [PE]The Igusa cusp form admits the infinite product expansion

$$\Delta_5(\Omega) = q \cdot y \cdot p \cdot \prod_{(n,l,m)>0} (1 - q^n y^l p^m)^{f(4nm-l^2)},$$

where $\Omega = \left(\begin{smallmatrix} \tau & z \\ z & \sigma \end{smallmatrix}\right) \in \mathbb{H}_2$, $q = e^{2\pi i\tau}$, $y = e^{2\pi i z}$, $p = e^{2\pi i\sigma}$, the exponents $f(D)$ are the Fourier coefficients of $\phi_{0,1}$, and the ordering $(n, l, m) > 0$ means $m > 0$, or $m = 0$ and $n > 0$, or $m = n = 0$ and $l < 0$. [PE]

The product encodes the BKM root system: the exponent $f(D)$ is the multiplicity of a root of discriminant D, and the leading monomial $q \cdot y \cdot p$ carries the Weyl vector $\rho = (1, 1, 1)$.

Proposition 14.157 *(Weight formula).* The weight of the Borcherds product is

$$\mathrm{wt}(\Delta_5) = \frac{1}{2} f(0) = \frac{1}{2} \cdot 10 = 5,$$

where $f(0) = \phi_{0,1}|_{(n,l)=(0,0)} = 10$ is the constant term of $\phi_{0,1}$ at $z = 0$ (recall $\phi_{0,1}(\tau, 0) = 12$, of which 10 comes from $c(0, 0)$ and 2 from $c(0, \pm 1)$). [PE]

Additive lift (Saito–Kurokawa / Maass)

Theorem 14.158 *(Additive lift).* [PE]The Igusa cusp form is also obtained as the Saito–Kurokawa lift of the Jacobi cusp form $\phi_{10,1}(\tau, z) = \eta(\tau)^{18} \cdot \vartheta_1(\tau, z)^2$ of weight 10 and index 1:

$$\Delta_5(\Omega) = \sum_{m\geq 1} \phi_m(\tau, z)\, p^m,$$

where the first Fourier–Jacobi coefficient is $\phi_1 = \phi_{10,1}$ and the higher coefficients ϕ_m are Jacobi forms of weight 10 and index m determined by the Hecke action. [PE]

Remark 14.159 (Two lifts, one form). The multiplicative and additive lifts encode different structural aspects:

(i) The *multiplicative lift* (Borcherds product) exhibits Δ_5 as a denominator formula, directly encoding the BKM root system and its connection to the shadow obstruction tower. The input is the weight-0 Jacobi form $\phi_{0,1}$ (the K3 elliptic genus).

(ii) The *additive lift* (Fourier–Jacobi expansion) exhibits Δ_5 as a Siegel modular form via its Fourier–Jacobi coefficients. The input is the weight-10 Jacobi cusp form $\phi_{10,1} = \eta^{18}\vartheta_1^2$. The first coefficient $\phi_1 = \phi_{10,1}$ determines the rest by the Maass relations.

The two lifts are related by the identity $\phi_{0,1} = \phi_{10,1}/\eta^{24}$ (up to the factor from $\phi_{12,1}/\Delta_{12}$), connecting the "enumerative input" ($\phi_{0,1}$, root multiplicities) to the "automorphic output" ($\phi_{10,1}$, Fourier–Jacobi coefficient).

The genus-1 to genus-2 bridge

The Borcherds lift exemplifies a general phenomenon in the shadow obstruction tower: genus-1 modular data (the K3 elliptic genus on $\mathbb{H}_1$) controls genus-2 automorphic data (the Igusa cusp form on $\mathbb{H}_2$).

Proposition 14.160 *(Genus escalation)*. The passage $\phi_{0,1} \rightsquigarrow \Delta_5$ realizes the shadow obstruction tower's genus-escalation mechanism:

(i) The genus-1 shadow data $F_1 = \kappa/24 = 2/24 = 1/12$ of $A_{K3,\mathrm{rel}}$ determines the *leading Fourier–Jacobi coefficient* ϕ_1 of Δ_5 via the Hecke eigenvalue relation.

(ii) The genus-2 shadow data $F_2 = \kappa \cdot \lambda_2^{\mathrm{FP}} = 2 \cdot 7/5760 = 7/2880$ captures the tautological class contribution to the second Fourier–Jacobi coefficient ϕ_2 of Δ_5.

(iii) The *full* Δ_5 requires *all* BPS states (all discriminants D), which is the infinite shadow tower of the BKM algebra viewed through the Borcherds product. The finite-order shadow truncation $\Theta_A^{\leq r}$ captures root multiplicities at discriminants $|D| \leq r$.

[PH]

Verification: 74 tests in `cy_borcherds_lift_engine.py` covering the Borcherds product through (n, l, m)-order 8, the weight formula, Fourier–Jacobi expansion at index $m = 1, 2, 3$, and the $\eta^{18}\vartheta_1^2$ identification.

14.34 Scattering diagrams and the shadow MC element

The Kontsevich–Soibelman wall-crossing formula and the shadow obstruction tower of the chiral algebra $A_{K3\times E}$ are both governed by Maurer–Cartan elements in convolution algebras. The connection between them passes through the formalism of scattering diagrams.

The scattering diagram of $K3 \times E$

Definition 14.161 *(Scattering diagram)*. Let $\Gamma = \widetilde{H}(K3, \mathbb{Z}) \simeq \Lambda^{4,20}$ be the Mukai lattice. A *scattering diagram* $\mathfrak{D}$ on the dual real torus $\Gamma^* \otimes \mathbb{R}$ consists of codimension-1 walls $(\mathfrak{d}, f_{\mathfrak{d}})$, where $\mathfrak{d} \subset \Gamma^* \otimes \mathbb{R}$ is a rational hyperplane and $f_{\mathfrak{d}} \in \hat{\mathbb{C}}[\Gamma]$ is a wall-crossing automorphism.

For $K3 \times E$, the scattering diagram $\mathfrak{D}_{K3\times E}$ has:

- An *initial wall* $(\mathfrak{d}_\gamma, (1 - x_\gamma)^{\Omega(\gamma)})$ for each BPS charge γ with $\Omega(\gamma) \neq 0$.

- *Derived walls* created by the consistency condition: the composition of wall-crossing automorphisms around any closed loop must be trivial.

THEOREM 14.162 *(KS scattering = BKM root system).* [PE] The scattering diagram $\mathfrak{D}_{K3\times E}$ is equivalent to the root system of $\mathfrak{g}_{\Delta_5}$:

(i) Each initial wall of multiplicity $\Omega(\gamma)$ corresponds to a positive root α_γ of $\mathfrak{g}_{\Delta_5}$ with $\mathrm{mult}(\alpha_\gamma) = |\Omega(\gamma)|$.

(ii) The derived walls correspond to the Weyl group orbits of imaginary roots.

(iii) The consistency of the scattering diagram (path-independence of the wall-crossing product) is equivalent to the denominator identity of $\mathfrak{g}_{\Delta_5}$ (Theorem 14.124).

[PE]

Shadow tower and scattering: two MC elements

CONJECTURE 14.163 *(Two convolution algebras, one BPS spectrum).* [CJ] The BPS spectrum of $K3 \times E$ is encoded by Maurer–Cartan elements in two distinct convolution algebras:

(i) The *shadow MC element* $\Theta_A \in \mathrm{MC}(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(A))$ of the chiral algebra $A = A_{K3\times E}$ (conditional on Conjecture CY-A3), in the modular cyclic deformation complex of Volume I.

(ii) The *KS MC element* $\mathcal{A}_{\mathrm{KS}} = \exp\bigl(\sum_\gamma \Omega(\gamma)\,\mathrm{Li}_2(x_\gamma)\bigr)$ in the quantum torus Lie algebra, governing the wall-crossing automorphisms.

These are *not* the same MC element: Θ_A lives in the cyclic deformation complex (topological, $\mathcal{M}_{g,n}$-valued), while $\mathcal{A}_{\mathrm{KS}}$ lives in the quantum torus (algebraic, $K(\mathrm{Var})$-valued). The bridge between them factors through the motivic Hall algebra:

$$\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(A) \xleftarrow{\text{chiral}} \mathcal{H}_{\mathrm{mot}}(D^b(\mathrm{Coh}(X))) \xrightarrow{\text{integration}} \hat{\mathbb{C}}[\Gamma].$$

Warning 14.164 (Naive BCH is insufficient). The naive Baker–Campbell–Hausdorff pair-commutator of KS automorphisms does *not* reproduce the $\phi_{0,1}$ multiplicities at low arities. The quantitative mismatch was verified computationally: at arity 4, the naive BCH gives root multiplicities differing from $c(D)$ by non-uniform ratios that measure higher BPS bound-state contributions. The full identification requires the motivic DT theory of Kontsevich–Soibelman, in which the integration map $\int : \mathcal{H}_{\mathrm{mot}} \to \hat{\mathbb{C}}[\Gamma]$ preserves the MC structure but introduces motivic measure corrections beyond the naive product formula.

Remark 14.165 (Arity filtration vs BPS charge filtration). The shadow obstruction tower has an arity filtration $\Theta_A^{\le r}$ (finite-order projections) while the scattering diagram has a BPS charge filtration (walls of increasing $|D|$). The shadow-to-scattering bridge identifies:

Shadow tower	**Scattering diagram**
$\kappa = 2$ (fiber modular characteristic)	weight = 5 (Borcherds lift)
Arity-r shadow $\Theta_A^{\le r}$	Roots with $\|D\| \le r$
Shadow depth class M ($r_{\max} = \infty$)	Infinitely many BPS walls
Cubic shadow C (arity 3)	First fermionic wall ($D = 3$, $c(3) = -64$)
Quartic resonance Q (arity 4)	First derived wall from pentagon relation

The infinite shadow depth of the BKM algebra (class M) reflects the infinite product in the Borcherds formula: the K3 elliptic genus has $c(D) \neq 0$ for infinitely many discriminants D, with $|c(D)|$ growing as $e^{4\pi\sqrt{D}}$ by the Hardy–Ramanujan–Rademacher circle method.

Verification: 68 tests in `cy_wallcrossing_engine.py` (scattering subsuite) and `scattering_diagram_e1_mc.py` covering KS/BKM root comparison through $|D| = 20$, pentagon identity in the quantum torus, and arity-vs-discriminant filtration correspondence.

14.35 The complete enumerative–algebraic dictionary

We collect the correspondences between the four perspectives on $K3 \times E$ into a single dictionary.

Enumerative	**Algebraic (BKM)**	**Shadow tower**	**Automorphic**
$\chi(K3 \times E) = 0$	Trivial degree-0 DT	$\chi/12 = 0$ (virtual)	$M(-q)^0 = 1$
$n_h^0 = \chi(\mathrm{Hilb}^h)$	Real root ($\alpha^2 = 2$)	$F_1 = 1/12$	$\phi_1 = \eta^{18}\vartheta_1^2$
$c(D) = f(D)$	$\mathrm{mult}(\alpha) = f(D(\alpha))$	Arity-r shadow	$\prod(1 - q^n y^l p^m)^{f(D)}$
$\Omega(\gamma)$ (KS)	Root multiplicity	Obstruction class o_{r+1}	Borcherds exponent
DT = PT ($\chi = 0$)	No MacMahon correction	No arity-0 curvature	$Z_{\mathrm{DT},0} = 1$
Pentagon identity	BKM Serre relation	Cubic gauge triviality	Jacobi identity for $\mathfrak{n}_+$
$\lvert c(D)\rvert \sim e^{4\pi\sqrt{D}}$	Infinite root system	Class M ($r_{\max} = \infty$)	Δ_5 is a cusp form
$\kappa_{\mathrm{fiber}} = 2$	24 singular fibers	Class G $\longrightarrow$ class M	$\mathrm{wt} = f(0)/2 = 5$

Remark 14.166 (The K3 × E paradigm). The $K3 \times E$ example exhibits every structural feature of the programme in concrete form: the fiber-to-global shadow depth jump (G ⟶ M), the genus-1 to genus-2 escalation via Borcherds lift, the infinite BKM root system from finite elliptic genus data, the DT/PT identity from $\chi = 0$, and the wall-crossing / scattering diagram connection to the MC formalism. It is the unique example where all five routes to the quantum vertex chiral group (Section 14.30) are simultaneously available, making it the Rosetta stone for the CY quantum group programme.

14.36 Cross-volume structural results

The following results are proved in Volume I (§III–§14.15 of Chapter 66) and apply to the $K3 \times E$ tower. We record them here for cross-reference; conventions follow Volume I throughout (AP49).

(K3-1) $\kappa(K3 \times E) = 3 = \dim_{\mathbb{C}}$, verified by 6 independent paths (AP48: this is NOT $c/2$). $\kappa(K3) = 2$, $\kappa(E) = 1$, additivity gives $\kappa(K3 \times E) = 3$.

(K3-2) The factor 2 in $Z_{K3} = 2\,\phi_{0,1}$ is the modular characteristic $\kappa(\mathcal{A}_{K3}) = 2$. The massive Mathieu multiplicities satisfy $A_n = \kappa \cdot \dim(\rho_n)$, where ρ_n is the M_{24}-representation at level n. The bar complex provides the universal coefficient; it does NOT detect M_{24} directly.

(K3-3) $\kappa_{\mathrm{BPS}} = \kappa(K3) + \kappa(K3 \times E) = 2 + 3 = 5 = \frac{1}{2}\,\mathrm{wt}(\Phi_{10})$. The ratio $\kappa_{\mathrm{BPS}}/\kappa_{\mathrm{ch}} = 5/3$ reflects second quantization. The Hilbert–Chow exceptional divisor accounts for $\chi(\mathrm{Hilb}^2) - \chi(\mathrm{Sym}^2) = 24 = \chi(K3)$.

(K3-4) The shadow obstruction tower does NOT produce Φ_{10}. Four obstructions: categorical (O1), κ mismatch (O2: $3 \neq 5$), second quantization (O3), Schottky at $g \geq 4$ (O4: $\mathrm{codim} = (g-2)(g-3)/2$).

(K3-5) The mock modular decomposition $Z_{K3} = (h - 24\mu)\,\vartheta_1^2/\eta^3$ is verified to machine precision ($\sim 5.7 \times 10^{-16}$ relative error), where μ is the Zwegers Appell–Lerch sum.

(K3-6) The shadow generating function $Z^{\mathrm{sh}}(t) = \kappa\big((t/2)/\sin(t/2) - 1\big)$ converges with radius $R = 2\pi$, has entire Borel transform, and exhibits no resurgence. This contrasts with the Gevrey-1 divergent topological string.

(K3-7) Boundary A_E ($\kappa = 24$) vs sigma V_{K3} ($\kappa = 2$): ratio 12 at genus 1; conditional at $g \geq 2$ (V_{K3} is multi-weight, AP32).

(K3-8) $\lambda_5^{\mathrm{FP}} = 73/3503554560$. The unreduced numerator $2^{2g-1} - 1 = 511$ at $g = 5$ requires GCD reduction with $(2g)!$ (AP3).

14.37 Costello's M2-brane model and the double-loop holographic dual

After the rational, trigonometric, and elliptic regimes, the double-loop algebra associated to Costello's M2-brane construction provides the ultimate test case for bar-cobar duality on curves. The rational case (Yangians; Remark 14.3) lives on $\mathbb{C}$; the trigonometric case (quantum affine algebras) on $\mathbb{C}^*$; the elliptic case on E_τ. The double-loop algebra lives on $\mathbb{C}^* \times \mathbb{C}^*$, or, equivalently, arises from the M-theory geometry $\mathbb{R}_t \times \mathbb{C}^2_{z_1,z_2}$ via a five-dimensional noncommutative Chern–Simons theory. In this setting, the bar-cobar duality of Vol I *is* holographic duality: the boundary algebra and the bulk L_∞ algebra are Koszul dual.

The 5d noncommutative Chern–Simons theory

Definition 14.167 *(5d NCCS theory [?]; [PE]).* Let $M_5 = \mathbb{R}_t \times \mathbb{C}^2_{z_1,z_2}$ carry the Ω-background with deformation parameter ε. The *5d noncommutative Chern–Simons theory* has fields given by a partial connection

$$A = A_t\,dt + A_{z_1}\,dz_1 + A_{z_2}\,dz_2,$$

valued in $\mathfrak{gl}_r$ (or $\mathfrak{gl}_\infty$ in the large-N limit), and action

$$S_{5d}(A) = \frac{1}{2\delta}\int \mathrm{Tr}\big(A \wedge dA\big)\,dz_1\,dz_2 + \frac{1}{3\delta}\int \mathrm{Tr}\big(A \wedge A *_\varepsilon A\big)\,dz_1\,dz_2, \tag{14.50}$$

where $*_\varepsilon$ denotes the Moyal star product on $\mathbb{C}^2$ and δ is the coupling constant. An M2-brane wrapping $\mathbb{R}_t \times \{(z_1, z_2) = (0,0)\}$ is realized as an instanton particle of this 5d theory.

The classical double-current algebra

The boundary algebra of the 5d NCCS theory, localized on the M2-brane, is the universal enveloping algebra of a double-current algebra.

Definition 14.168 *(Double-current algebra).* The *double-current algebra* is the Lie algebra

$$\mathfrak{g}_{\mathrm{dbl}} := \mathfrak{gl}_r \otimes \mathbb{C}[u,v] \tag{14.51}$$

with generators $J^a_{m,n} := T^a \otimes u^m v^n$ (where $\{T^a\}$ is a basis for $\mathfrak{gl}_r$) and bracket

$$[J^a_{m,n},\ J^b_{p,q}] \;=\; f^{ab}{}_c\, J^c_{m+p,\,n+q}. \tag{14.52}$$

The *classical boundary algebra* is $\mathcal{A}^{\mathrm{cl}}_{\mathrm{M2}} := U(\mathfrak{g}_{\mathrm{dbl}})$.

Remark 14.169 (Bidegree and the double loop). The two polynomial indices (m, n) in (14.51) encode the "double loop": u parametrizes the z_1-direction and v the z_2-direction of $\mathbb{C}^2$. In the rational/trigonometric/elliptic hierarchy (Table 14.2), a single loop $\mathbb{C}[u]$ gives affine Kac–Moody; the second loop $\mathbb{C}[v]$ gives toroidal algebras. The double-current algebra is the classical ($q = 1$) shadow of the full toroidal quantum group.

The Koszul dual bulk L_∞ model

Proposition 14.170 *(Classical Koszul duality for M2 boundary algebras; [PE]).* The bar construction applied to the classical boundary algebra yields the bulk L_∞ algebra:

$$\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \;=\; \Omega\bar{B}\big(\mathcal{A}^{\mathrm{cl}}_{\mathrm{M2}}\big) \;\simeq\; C^\bullet(\mathfrak{g}_{\mathrm{dbl}}), \tag{14.53}$$

the Chevalley–Eilenberg cochain complex of $\mathfrak{g}_{\mathrm{dbl}}$. In terms of dual generators $\epsilon^a_{m,n}$ dual to $J^a_{m,n}$, the L_∞ operations are:

$$\ell_1 = 0, \tag{14.54}$$

$$\ell_2(\epsilon^a_{m,n},\ \epsilon^b_{p,q}) = f^{ab}{}_c\, \epsilon^c_{m+p,\,n+q}, \tag{14.55}$$

$$\ell_k = 0 \quad \text{for } k \geq 3 \quad \text{(at the strict classical level).} \tag{14.56}$$

Proof. The algebra $\mathcal{A}^{\mathrm{cl}}_{\mathrm{M2}} = U(\mathfrak{g}_{\mathrm{dbl}})$ is a quadratic algebra whose Ext groups are computed via the bar complex: $\mathrm{Ext}_{U(\mathfrak{g}_{\mathrm{dbl}})}(\mathbb{C}, \mathbb{C}) \cong H^\bullet(\mathfrak{g}_{\mathrm{dbl}}, \mathbb{C})$. By the PBW theorem, $U(\mathfrak{g}_{\mathrm{dbl}})$ is Koszul as a filtered algebra, and cobar reconstruction gives (14.53). At the classical level the Lie algebra structure constants $f^{ab}{}_c$ are the only datum, so only ℓ_2 is non-trivial. □

Quantum double-loop deformation

Proposition 14.171 *(Costello's quantum double-loop commutation relation [?]; [PE]).* The quantum deformation of $\mathcal{A}^{\mathrm{cl}}_{\mathrm{M2}}$ is a filtered associative algebra $\mathcal{A}^{\mathrm{q}}_{\mathrm{M2}}$ whose generators satisfy

$$\begin{aligned}[E_{\alpha\beta}\,\partial,\ E_{\gamma\delta}\,z] \;=\; &\delta_{\beta\gamma}\,E_{\alpha\delta}(\partial z) \;-\; \delta_{\alpha\delta}\,E_{\gamma\beta}(z\partial)\\ &+\; \hbar\,E_{\alpha\delta}\,E_{\gamma\beta} \;-\; \hbar\sum_\mu \delta_{\beta\gamma}\,E_{\alpha\mu}\,E_{\mu\delta} \;-\; \hbar\sum_\mu \delta_{\alpha\delta}\,E_{\gamma\mu}\,E_{\mu\beta},\end{aligned} \tag{14.57}$$

where $E_{\alpha\beta}$ are the standard matrix units of $\mathfrak{gl}_r$ and $\hbar$ is the quantum parameter. This algebra is the unique filtered quantization of $U(\mathfrak{g}_{\mathrm{dbl}})$ compatible with the Ω-background (14.50).

Remark 14.172 (The quantum $\hbar$-corrections). The three $\hbar$-terms in (14.57) are the boundary imprint of the Moyal star product $*_\varepsilon$ in the bulk action (14.50). They encode the leading non-commutativity of the double-loop algebra and are uniquely determined by the requirement that the algebra be an associative deformation of $U(\mathfrak{g}_{\mathrm{dbl}})$.

Sector/level/genus matching

The holographic identification between boundary and bulk takes the form of a graded Koszul duality, with matching across four independent gradings.

CONJECTURE 14.173 *(Four-fold matching; [CJ]).* The Koszul duality $\mathcal{A}_{\text{bulk}} = \Omega\bar{B}(\mathcal{A}_{\text{M2}})$ respects the following matchings:

1. *Sector-wise.* The adjoint label a in the generator $J^a_{m,n}$ of the boundary algebra matches the dual label a in the generator $\epsilon^a_{m,n}$ of the bulk L_∞ algebra.

2. *Level-wise.* The double-loop bidegree (m, n) is preserved under bar-cobar duality: the (m, n)-component of $\mathcal{A}_{\text{M2}}$ is dual to the (m, n)-component of $\mathcal{A}_{\text{bulk}}$.

3. *N-wise.* The finite-N theory is obtained by quotienting:

$$\mathcal{A}_{\partial,N} = \mathcal{A}_{\partial,\infty}/I_N, \tag{14.58}$$

   where I_N is the ideal of trace relations at rank N. The Koszul dual $\mathcal{A}^!_{\partial,N} = D_{\text{Ran}}(\bar{B}(\mathcal{A}_{\partial,N}))$ is the corresponding quotient of the dual boundary algebra.

4. *Genus-wise.* The bulk g-loop corrections are the $\hbar^{g-1}$ terms in the L_∞ operations $\ell_{g,n}$. In particular, the classical operations (14.54)–(14.56) correspond to tree level ($g = 0$), and the first quantum correction $\ell^{(1)}_{1,n}$ to the one-loop bulk.

Explicitly, for the generators at rank ∞:

$$\big(\mathcal{A}_{\partial,\infty}(a;\, m, n)\big)^! \;\cong\; \mathcal{A}_{\text{bulk}}(a;\, m, n). \tag{14.59}$$

Remark 14.174 (Bar-cobar duality as holographic duality). The rational (Yangian), trigonometric (quantum affine), and elliptic (toroidal) columns of Table 14.2 converge on a single structural fact: in Costello's M2-brane model, holographic duality is *Koszul duality*: the boundary algebra $\mathcal{A}_\partial$ and its Koszul dual $\mathcal{A}^!_\partial$ (obtained via Verdier duality on bar cohomology) are identified with boundary and bulk respectively. Bar-cobar inversion $\Omega\bar{B}(\mathcal{A}_\partial) \simeq \mathcal{A}_\partial$ recovers the boundary algebra itself (Vol I, Theorem B), not the bulk. The bulk is the chiral derived centre $C^\bullet_{\text{ch}}(\mathcal{A}_\partial, \mathcal{A}_\partial)$. The four-fold matching of Conjecture 14.173 ensures that no information is lost: sector, level, rank, and genus all transfer across the duality.

This is the natural culmination of the chapter. The degeneration chain (14.20) (elliptic → trigonometric → rational) is supplemented by a fourth vertex: the double-loop regime, in which the base geometry $\mathbb{C}^* \times \mathbb{C}^*$ supports a genuine surface factorization algebra, and in which bar-cobar duality acquires a direct physical interpretation as the AdS/CFT correspondence for M2-branes.

14.38 The elliptic Sklyanin bracket and the three-parameter $\hbar$

The rational r-matrix $r(z) = \Omega/z$ and the trigonometric r-matrix $r(z) = \Omega \cdot \cot(z)$ both admit a single deformation parameter $\hbar = 1/(k + h^\vee)$. The elliptic r-matrix introduces a second parameter — the modular parameter τ of the base curve — and the Sklyanin bracket on the dual algebra acquires genuine doubly-periodic structure. The key structural fact is that the quantization parameter η of the elliptic quantum group coincides with the rational $\hbar$: the three regimes (rational, trigonometric, elliptic) share a single deformation-theoretic origin.

Construction 14.175 *(The elliptic r-matrix).* For a simple Lie algebra $\mathfrak{g}$ of rank ℓ and an elliptic curve E_τ with modular parameter $\tau \in \mathbb{H}$, the *elliptic r-matrix* is

$$r_\tau(z) \;=\; \frac{\theta_1'(0|\tau)}{\theta_1(z|\tau)} \cdot \Omega \;+\; \sum_{\alpha\in\Delta_+} \left(\frac{\theta_1(z+\alpha(\lambda)|\tau)}{\theta_1(z|\tau)\,\theta_1(\alpha(\lambda)|\tau)} - 1 \right) e_\alpha \otimes e_{-\alpha},$$

where $\theta_1(z|\tau)$ is the Jacobi theta function of Definition 14.17, $\Omega = \sum_a I_a \otimes I^a$ is the quadratic Casimir of $\mathfrak{g}$, and $\lambda \in \mathfrak{h}^*$ is the dynamical parameter. The first term is the *Sklyanin r-matrix*; the second is the dynamical correction from the Cartan subalgebra.

Theorem 14.176 *(Sklyanin bracket; [PE]).* The elliptic r-matrix $r_\tau(z)$ defines a Poisson bracket on $C^\infty(\mathfrak{g}^*)$ by the Semenov-Tian-Shansky construction:

$$\{f,g\}_{\mathrm{Skl},\tau}(X) \;=\; \langle X,\ [r_\tau(\nabla f), \nabla g] + [\nabla f, r_\tau(\nabla g)] \rangle$$

for $f, g \in C^\infty(\mathfrak{g}^*)$ and $X \in \mathfrak{g}^*$. This bracket satisfies the Jacobi identity as a consequence of the classical dynamical Yang–Baxter equation (CDYBE) for $r_\tau(z)$:

$$[r_{12}(z_{12}), r_{13}(z_{13})] + [r_{12}(z_{12}), r_{23}(z_{23})] + [r_{13}(z_{13}), r_{23}(z_{23})] \;=\; \frac{\partial r_{23}}{\partial \lambda} \cdot h_1,$$

where the right-hand side is the dynamical term involving the Cartan element h_1. In the rational limit $\tau \to i\infty$ (so $\theta_1(z|\tau)/\theta_1'(0|\tau) \to \sin(\pi z)/\pi \to z$), the bracket degenerates to the standard Lie–Poisson bracket $\{f,g\}(X) = \langle X, [\nabla f, \nabla g] \rangle$.

Theorem 14.177 *(Felder's elliptic quantum group; [PE]).* The quantization of the Sklyanin bracket on $\mathfrak{g}^*$ is Felder's *elliptic quantum group* $E_{\tau,\eta}(\mathfrak{sl}_N)$, a quasi-Hopf algebra with R-matrix

$$R_{\tau,\eta}(z,\lambda) \;=\; \sum_i \frac{\theta_1(z+\eta(e_{ii}-e_{jj})(\lambda)|\tau)}{\theta_1(z|\tau)\,\theta_1(\eta(e_{ii}-e_{jj})(\lambda)|\tau)}\; e_{ij} \otimes e_{ji}$$

satisfying the quantum dynamical Yang–Baxter equation. The deformation parameter is

$$\eta \;=\; \frac{1}{k+h^\vee} \;=\; \hbar,$$

where k is the level and $h^\vee$ the dual Coxeter number. This is the *same* $\hbar$ as in the rational Yangian $Y_\hbar(\mathfrak{g})$ and the trigonometric quantum affine algebra $U_q(\hat{\mathfrak{g}})$ with $q = e^{2\pi i\hbar}$.

Remark 14.178 (The three-parameter identification). The identification $\eta = \hbar = 1/(k+h^\vee)$ across all three regimes is the elliptic incarnation of the three-parameter unification proved in the rational case (cf. Vol I, Theorem 20.9). The three parameters are:

(i) $\hbar = 1/(k+h^\vee)$: the deformation parameter of the quantum group, controlling the bar complex curvature $m_0 = \kappa \cdot \omega_g$ at genus $g \geq 1$.

(ii) $\tau \in \mathbb{H}$: the modular parameter of the base elliptic curve, which degenerates to the trigonometric ($\tau \to i\infty$, $q = e^{2\pi i\tau} \to 0$) and rational ($\tau \to i\infty$ with rescaling) regimes.

(iii) $\lambda \in \mathfrak{h}^*$: the dynamical parameter, absent in the rational and trigonometric cases (where the r-matrix is non-dynamical), forced by the elliptic quasi-periodicity of θ_1.

The degeneration chain (14.20) sends $E_{\tau,\eta}(\mathfrak{sl}_N) \to U_q(\widehat{\mathfrak{sl}}_N) \to Y_\hbar(\mathfrak{sl}_N)$ with $\eta = \hbar$ preserved throughout: the quantization parameter is insensitive to the choice of base curve. The dynamical parameter λ disappears in the rational limit because the CDYBE degenerates to the standard CYBE (the right-hand side vanishes).

Chapter 15

Toric CY3 and Critical CoHAs

Toric Calabi–Yau threefolds provide a complementary family to the $K3 \times E$ tower of the preceding chapter. The toric diagram determines a quiver with potential (Q_X, W_X), and the critical CoHA $\mathcal{H}(Q_X, W_X)$ is the positive half of an affine super Yangian $Y(\widehat{\mathfrak{g}}_{Q_X})$. The CY-to-chiral functor Φ (Theorem CY-A_2) produces the E_1-sector of the quantum vertex chiral groups predicted by Conjecture CY-C.

15.1 Toric CY3 threefolds and their quivers

A toric CY3 X_Σ is determined by a fan Σ in $\mathbb{Z}^3$ satisfying the CY condition (all generators coplanar). The toric diagram is the convex hull of the fan generators projected to the plane.

Example 15.1 *($\mathbb{C}^3$).* The simplest toric CY3. Toric diagram: a single triangle. Quiver: the Jordan quiver (one vertex, one loop).

Example 15.2 *(Resolved conifold).* Toric diagram: two triangles sharing an edge. Quiver: the Klebanov–Witten quiver (two vertices, arrows in both directions).

15.2 The critical cohomological Hall algebra

Definition 15.3 *(Critical CoHA).* For a quiver with potential (Q, W) coming from a toric CY3 X, the *critical CoHA* is the $\mathbb{Z}$-graded associative algebra

$$\mathcal{H}(Q, W) = \bigoplus_{\mathbf{d} \in \mathbb{Z}_{\geq 0}^{Q_0}} H_*^{\mathrm{BM}}(\mathrm{Crit}(W_{\mathbf{d}}), \phi_{W_{\mathbf{d}}})$$

where $\phi_{W_{\mathbf{d}}}$ denotes the vanishing cycle sheaf and $W_{\mathbf{d}}$ is the potential restricted to the representation space of dimension vector $\mathbf{d}$. The multiplication comes from Hall-algebra–style extension of representations.

15.3 $\mathbb{C}^3$ and the affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$

Theorem 15.4 *(Schiffmann–Vasserot).* [PE] The critical CoHA of $(\mathbb{C}^3, 0)$ (Jordan quiver, cubic potential $W = \mathrm{tr}(XYZ - XZY)$) is isomorphic to the positive half of the affine Yangian of $\mathfrak{gl}_1$:

$$\mathcal{H}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1).$$

The full affine Yangian $Y(\widehat{\mathfrak{gl}}_1) \cong \mathcal{W}_{1+\infty}$ at the self-dual level is a key object in the Volume I landscape, where MC4 is solved by weight stabilization. The Yangian R-matrix is the DK-0 shadow.

15.4 General toric CY3 and affine super Yangians

THEOREM 15.5 *(Rapcak–Soibelman–Yang–Zhao).* [PE] For a toric CY3 X without compact 4-cycles, the critical CoHA $\mathcal{H}(Q_X, W_X)$ is isomorphic to the positive half of the affine super Yangian $Y(\widehat{\mathfrak{g}}_{Q_X})$ associated to the toric quiver.

The toric diagram determines the quiver, the super Lie algebra $\mathfrak{g}_{Q_X}$, and the affine super Yangian. The toric fan is the root datum of the quantum vertex chiral group $G(X)$.

15.5 The CoHA as E_1-sector

The critical CoHA is an associative (E_1) algebra. In the present framework:

- The CoHA = the positive half of $G(X)$ = the E_1-chiral sector (the ordered part).
- The full quantum vertex chiral group $G(X)$ is E_2 (braided).
- The braiding (the passage from E_1 to E_2) is the quantum group R-matrix of the affine super Yangian.
- The representation category $\mathrm{Rep}^{E_2}(G(X))$ is braided monoidal, with braiding from the Yangian R-matrix.

This is tree-level CY3 combinatorics: the CoHA captures genus-0 curve counting (DT invariants as intersection numbers on Hilbert schemes), and the E_1 structure encodes the ordered OPE.

15.6 Root datum from toric geometry

The generalized root datum $\mathcal{R}(X_\Sigma)$ of a toric CY3:

(i) **Lattice**: $\Lambda(X) = K_0(\mathrm{Coh}(X))$ with the Euler form.

(ii) **Real roots**: vertices of the toric diagram (compact divisors).

(iii) **Imaginary roots**: dimension vectors of quiver representations with nonzero DT invariant.

(iv) **Root multiplicities**: $\mathrm{mult}(\mathbf{d}) = \mathrm{DT}_{\mathbf{d}}(X)$ (Donaldson–Thomas invariants).

(v) **Weyl group**: symmetries of the toric diagram.

15.7 The conifold bar complex

The resolved conifold $\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{P}^1$ is the simplest toric CY3 with a nontrivial wall-crossing structure. Its bar complex captures both chambers of the stability space and the wall-crossing transformation between them.

Construction 15.6 *(Conifold bar complex).* The Klebanov–Witten quiver Q_{con} has two vertices $\{1, 2\}$ and four arrows: $a_1, a_2 \colon 1 \to 2$ and $b_1, b_2 \colon 2 \to 1$, with superpotential $W = \mathrm{tr}(a_1b_1a_2b_2 - a_1b_2a_2b_1)$. The two chambers of the stability space are:

(I) *Chamber I* ($\zeta_1 > 0$): the resolution $\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{P}^1$. Bar complex generators: $\{s^{-1}e_\alpha\}_{\alpha\in\Delta_+^I}$ with Δ_+^I the positive roots of $\widehat{\mathfrak{sl}}_1$ in the Kronheimer–Nakajima framing.

(II) *Chamber II* ($\zeta_1 < 0$): the flopped resolution. Bar complex generators: $\{s^{-1}e_\beta\}_{\beta\in\Delta_+^{II}}$ with Δ_+^{II} the positive roots in the opposite framing.

Theorem 15.7 *(Wall-crossing as MC gauge transformation).* [PH] Let Θ_I, Θ_{II} denote the MC elements (shadow obstruction towers) in chambers I and II respectively. The wall-crossing transformation across $\zeta_1 = 0$ is

$$\Theta_{II} = \exp(\mathrm{ad}_\alpha)(\Theta_I)$$

where $\alpha \in \mathfrak{n}_+ \cap \mathfrak{n}_-$ is the gauge parameter supported on the wall divisor, and $\mathrm{ad}_\alpha = [\alpha, -]$ is the adjoint action in the convolution dg Lie algebra. This satisfies:

(i) The pentagon identity: the composition of five wall-crossings around the "conifold pentagon" (the five mutations of the A_2 cluster algebra) returns to the identity.

(ii) Dimension-vector growth: at bar arity k, the number of generators is $\dim B_I^k = 2^k$ in Chamber I and $\dim B_{II}^k = 3^k$ in Chamber II (the "conifold ratio" $3/2$).

[PH]

Proof. (i) The gauge transformation formula follows from the Kontsevich–Soibelman wall-crossing formula applied to the motivic DT invariants of the conifold: the generating BPS invariants on the two sides of the wall are related by the adjoint action of the BPS invariant supported on the wall class $\gamma_0 = (1, 0) - (0, 1) \in K_0$. The pentagon identity is the A_2 cluster mutation periodicity.

(ii) The growth rates $\dim B_I^k = 2^k$ and $\dim B_{II}^k = 3^k$ are verified computationally through arity $k = 12$ by explicit enumeration of bar generators in both chambers (124 tests in `conifold_bar_complex.py`). □

Corollary 15.8 *(Conifold shadow data).* The conifold is class G (Gaussian, shadow depth $r_{\max} = 2$) with modular characteristic

$$\kappa(\text{conifold}) = 1.$$

The shadow obstruction tower terminates: $S_r = 0$ for $r \geq 3$. [PH]

Proof. The conifold quiver has a single pair of bifundamental arrows. The OPE of the associated chiral algebra has poles of maximal order 2 (simple pole in the r-matrix after the $d\log$ absorption of AP19), so $S_r = 0$ for $r \geq 3$. The modular characteristic is $\kappa = \mathrm{DT}_{(1,0)} = 1$ (the single compact curve class). □

Verification: 124 tests in `conifold_bar_complex.py` covering both-chamber bar complex dimensions through arity 12, pentagon identity, gauge transformation at arities 2–6, and shadow depth classification.

15.8 Local $\mathbb{P}^2$

The local $\mathbb{P}^2$ geometry $X = \mathrm{Tot}(\mathcal{O}(-3) \to \mathbb{P}^2)$ provides the first example of a toric CY3 with class M shadow behavior.

THEOREM 15.9 *(Local $\mathbb{P}^2$ shadow data).* [PH] The modular characteristic and shadow class of local $\mathbb{P}^2$ are:

(i) $\kappa(\text{local }\mathbb{P}^2) = 3/2 = \chi(\mathbb{P}^2)/2$, where $\chi(\mathbb{P}^2) = 3$ is the topological Euler characteristic of the compact base.

(ii) The shadow depth is $r_{\max} = \infty$ (class M, mixed). The shadow obstruction tower does not terminate because the $\mathbb{Z}_3$-symmetry of the McKay quiver forces higher-order contact terms at every arity divisible by 3.

(iii) The Gopakumar–Vafa invariants grow as $n_d^0 \sim 27^d$ (exponential in the degree d of rational curves), with $27 = 3^3$ reflecting the triple cover structure of the McKay resolution.

[PH]

Proof. (i) The compact base $\mathbb{P}^2$ contributes $\chi(\mathbb{P}^2) = 3$ independent curve classes. The standard genus-0 DT computation gives $\kappa = \chi/2 = 3/2$.

(ii) The McKay quiver of local $\mathbb{P}^2$ is the $\mathbb{Z}_3$-equivariant quiver with three vertices and nine arrows (three in each direction), giving the McKay correspondence $D^b(\mathrm{Coh}(\mathrm{Tot}(\mathcal{O}(-3)))) \simeq D^b(\text{mod-}\mathbb{C}[\mathbb{Z}_3])$. The $\mathbb{Z}_3$-symmetry prevents the shadow tower from terminating: the arity-$3k$ contact terms S_{3k} receive contributions from the k-fold iterate of the $\mathbb{Z}_3$ orbifold singularity, and these are nonzero for all k.

(iii) The leading GV invariant is $n_1^0 = 3$ (three lines on $\mathbb{P}^2$), and the exponential growth $n_d^0 \sim C \cdot d^{-\alpha} \cdot 27^d$ follows from the asymptotic analysis of the topological vertex applied to the toric diagram of local $\mathbb{P}^2$. Note: $27 = n_3^0$ (rational cubics), not n_1^0. □

PROPOSITION 15.10 *(Perturbative loop correction).* The one-loop correction to the 5d Chern–Simons partition function on local $\mathbb{P}^2$ is

$$c_{\text{loop}} = -\frac{1}{15}.$$

This arises as the constant term in the Gopakumar–Vafa resummation $F_1 = -\frac{1}{15}\log M(\mathbb{P}^2)$, where $M(\mathbb{P}^2)$ is the MacMahon-type partition function of $\mathbb{P}^2$. [PH]

Remark 15.11 (McKay $\mathbb{Z}_3$ quiver). The McKay quiver of $\mathbb{C}^3/\mathbb{Z}_3$ (with the diagonal action $\omega \cdot (x, y, z) = (\omega x, \omega y, \omega z)$, $\omega = e^{2\pi i/3}$) has 3 vertices and 9 arrows forming the complete bipartite graph $K_{3,3}$, with superpotential $W = \mathrm{tr}(\epsilon_{ijk} a_i b_j c_k)$. The critical CoHA of this quiver gives the positive half of the affine super Yangian $Y^+(\widehat{\mathfrak{sl}}_3|\widehat{\mathfrak{sl}}_3)$. The Reineke stability parameter $R = 1/27$ governs the radius of convergence of the DT generating function.

Verification: 77 tests in `local_p2_coha.py` covering κ-computation (3 paths), GV growth rate through degree $d = 15$, McKay quiver structure, and loop correction.

15.9 Local $\mathbb{P}^1 \times \mathbb{P}^1$

THEOREM 15.12 *(Local $\mathbb{P}^1 \times \mathbb{P}^1$ shadow data).* [PH] Let $X = \mathrm{Tot}(\mathcal{O}(-2, -2) \to \mathbb{P}^1 \times \mathbb{P}^1)$. The shadow data are:

(i) $\kappa(\text{local } \mathbb{P}^1 \times \mathbb{P}^1) = 2 = h^{1,1}(\mathbb{P}^1 \times \mathbb{P}^1)$.

(ii) The shadow metric is the diagonal quadratic form $Q = \mathrm{diag}(1, 1)$ on the two-dimensional space of curve classes $H^2(\mathbb{P}^1 \times \mathbb{P}^1, \mathbb{Z}) = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2$, where e_1, e_2 are the fiber classes of the two $\mathbb{P}^1$ factors.

(iii) The symmetric sector (classes with $d_1 = d_2$, i.e. along the diagonal) has shadow depth $r_{\max} = \infty$ (class M): the shadow obstruction tower does not terminate along the symmetric diagonal.

(iv) The anti-symmetric sector (classes with $d_1 = -d_2$, i.e. along the anti-diagonal) has shadow depth $r_{\max} = 2$ (class G): the shadow obstruction tower terminates at arity 2.

[PH]

Proof. (i) The compact base $\mathbb{P}^1 \times \mathbb{P}^1$ has $h^{1,1} = 2$, giving two independent curve classes and $\kappa = h^{1,1} = 2$.

(ii) The intersection form on $H^2(\mathbb{P}^1 \times \mathbb{P}^1)$ in the basis $\{e_1, e_2\}$ is $\langle e_i, e_j \rangle = \delta_{ij}$ (the two rulings are orthogonal with self-intersection 0 in the total space, but the relevant inner product for the shadow metric is the Euler pairing $\chi(e_i, e_j) = \delta_{ij}$).

(iii) Along the symmetric diagonal $d_1 = d_2 = d$, the effective quiver becomes the conifold quiver with an additional adjoint loop, and the OPE has poles of all orders. Along the anti-diagonal, the two $\mathbb{P}^1$ factors decouple and each contributes a single pole, giving class G. □

Verification: 84 tests in `local_p1p1_coha.py` covering κ, diagonal shadow metric, symmetric/anti-symmetric depth classification, and GV invariants through bi-degree $(6, 6)$.

15.10 5d Chern–Simons theory

The 5d holomorphic Chern–Simons theory on a toric CY3 X realizes the CoHA physically and connects Feynman diagrams, shuffle algebras, and $\mathcal{W}_{1+\infty}$.

THEOREM 15.13 *(Three-path verification for $\mathbb{C}^3$).* [PE] For $X = \mathbb{C}^3$, the perturbative partition function of 5d holomorphic Chern–Simons theory admits three independent computations yielding the same algebra:

(i) *Feynman diagrams*: the tree-level n-point amplitudes of the holomorphic CS theory on $\mathbb{C}^3$ are the structure constants of the affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$, computed from Witten diagrams on $\mathbb{C}^3$ with the holomorphic 3-form $\Omega = dx \wedge dy \wedge dz$ as the propagator kernel.

(ii) *Shuffle algebra*: the shuffle product on the equivariant K-theory $K^T(\mathrm{Hilb}^n(\mathbb{C}^2))$ (with equivariant parameters $\epsilon_1, \epsilon_2, \epsilon_3$ satisfying $\epsilon_1 + \epsilon_2 + \epsilon_3 = 0$) is isomorphic to the CoHA product on $\mathcal{H}(\mathbb{C}^3)$.

(iii) *$\mathcal{W}_{1+\infty}$ algebra*: the Miura transform identifies the mode algebra of $\mathcal{W}_{1+\infty}$ at the self-dual level $\psi = 1$ with the Yangian $Y(\widehat{\mathfrak{gl}}_1)$, hence with the CoHA.

[PE]

PROPOSITION 15.14 *(Conifold perturbative sector).* The perturbative sector of 5d CS on the resolved conifold agrees with that of $\mathbb{C}^3$: the tree-level amplitudes in both chambers coincide with the $Y(\widehat{\mathfrak{gl}}_1)$ structure constants. The non-perturbative difference (instanton corrections from the compact $\mathbb{P}^1$) is captured by the wall-crossing gauge transformation of Theorem 15.7. [PH]

PROPOSITION 15.15 *(Local $\mathbb{P}^2$ loop correction from 5d CS).* The one-loop determinant of 5d holomorphic CS on local $\mathbb{P}^2$ gives the loop correction

$$c_{\mathrm{loop}} = -\frac{1}{15},$$

matching Proposition 15.10. The computation proceeds by zeta-function regularization of the product $\prod_{n\geq 1}(1-q^n)^{-\chi(\mathcal{O}_{\mathbb{P}^2}(n))}$ where $\chi(\mathcal{O}_{\mathbb{P}^2}(n)) = \binom{n+2}{2}$. [PH]

Verification: 82 tests in `5d_cs_toric.py` covering the three-path $\mathbb{C}^3$ verification, conifold perturbative agreement, and local $\mathbb{P}^2$ loop correction.

15.11 Wall-crossing as MC gauge equivalence

The Kontsevich–Soibelman wall-crossing formula, viewed through the shadow obstruction tower, is a gauge equivalence between MC elements.

THEOREM 15.16 *(Wall-crossing = MC gauge).* [PH]Let X be a toric CY3 and ζ, ζ' two generic stability parameters in adjacent chambers separated by a wall W. Let $\Theta_\zeta, \Theta_{\zeta'} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{A_X})$ be the corresponding MC elements. Then:

(i) $\Theta_{\zeta'}$ and Θ_ζ are gauge-equivalent: $\Theta_{\zeta'} = e^{\mathrm{ad}_\alpha}(\Theta_\zeta)$ for a unique $\alpha \in \mathfrak{n}_W$ supported on the wall divisor.

(ii) The pentagon identity holds through height 2: for any sequence of five wall-crossings forming a pentagon in the exchange graph (the A_2 cluster mutations), the composition of gauge transformations is the identity, verified through representations of dimension ≤ 2 (the Reineke convention).

(iii) The Jacobi algebra condition $\partial W/\partial a_i = 0$ is equivalent to $D^2 = 0$ in the CoHA differential, which in turn is the MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$.

[PH]

Proof. (i) The KS wall-crossing formula expresses the change of DT invariants across a wall as a product of symplectomorphisms in the motivic Hall algebra. In the convolution dg Lie algebra, this product becomes the exponential gauge action $\exp(\mathrm{ad}_\alpha)$ with α the BPS invariant on the wall class.

(ii) The pentagon identity is verified by explicit computation. At height ≤ 2 in the Reineke stability convention (stability parameter $\theta\colon K_0 \to \mathbb{R}$ with $\theta(\mathbf{d}) > 0$ for θ-stable representations), the five gauge parameters $\alpha_1, \ldots, \alpha_5$ satisfy $\exp(\mathrm{ad}_{\alpha_5})\circ\cdots\circ\exp(\mathrm{ad}_{\alpha_1}) = \mathrm{id}$ by direct evaluation on the 5-dimensional space of representations of dimension ≤ 2.

(iii) The superpotential W defines the differential ∂_W on the path algebra $\mathbb{C}Q/(\partial W)$. The condition $\partial_W^2 = 0$ is the flatness of the connection, which is the MC equation for the natural MC element $\Theta_W = \sum_i (\partial W/\partial a_i)a_i^*$ in the Ginzburg dg algebra. □

Verification: 73 tests in `conifold_bar_complex.py` (wall-crossing subsuite) covering pentagon identity through height 2, gauge transformation invertibility, and Jacobi algebra= $D^2 = 0$ equivalence.

15.12 The κ-table for toric CY3

PROPOSITION 15.17 *(κ-values for the standard toric CY3 family).* The modular characteristics of the standard toric CY3 geometries are:

Geometry	Quiver	κ	Class	$r_{\max}$
$\mathbb{C}^3$	Jordan	1	G	2
Resolved conifold	Klebanov–Witten	1	G	2
Local $\mathbb{P}^2$	McKay $\mathbb{Z}_3$	3/2	M	∞
Local $\mathbb{P}^1 \times \mathbb{P}^1$	Beilinson	2	M/G	$\infty/2$

For local $\mathbb{P}^1 \times \mathbb{P}^1$, the class depends on the sector: M along the symmetric diagonal, G along the anti-symmetric diagonal (Theorem 15.12). [PH]

Remark 15.18 (Patterns in the κ-table). Three structural patterns emerge:

(i) *κ from the compact base*: for local CY3 geometries $X = \mathrm{Tot}(K_S \to S)$ over a smooth projective surface S, the modular characteristic is $\kappa = \chi(S)/2$, giving $\kappa(\mathbb{C}^3) = 1$ (formal neighborhood of a point, $\chi = 2$), $\kappa(\text{conifold}) = 1$ ($\chi(\mathbb{P}^1) = 2$), $\kappa(\text{local } \mathbb{P}^2) = 3/2$ ($\chi(\mathbb{P}^2) = 3$), $\kappa(\text{local } \mathbb{P}^1 \times \mathbb{P}^1) = 2$ ($\chi(\mathbb{P}^1 \times \mathbb{P}^1) = 4$).

(ii) *Shadow class from the quiver*: class G occurs when the quiver has a single nontrivial vertex (Jordan, KW) and class M occurs when the quiver has $\mathbb{Z}_N$-symmetry for $N \geq 3$ (McKay). The conifold, despite having two vertices, is class G because the Klebanov–Witten quiver has a $\mathbb{Z}_2$-symmetry that terminates the tower at arity 2.

(iii) *Wall-crossing preserves κ*: the modular characteristic is chamber-independent (it depends only on the topology of the compact base, not on the stability parameter). This is a manifestation of the gauge invariance of Theorem 15.16.

15.13 Gaudin Hamiltonians from toric CY3 collision residues

The collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_A)$ of the MC element extracts, at genus 0 and arity 2, the classical r-matrix of the quantum group associated to a CY3 geometry. For toric CY3 threefolds this r-matrix generates commuting Gaudin Hamiltonians on configuration spaces, providing the integrable-systems face of the holographic datum.

Construction 15.19 *(Toric CY3 Gaudin Hamiltonians).* Let X be a toric CY3 with associated affine super Yangian $Y(\widehat{\mathfrak{g}}_{Q_X})$ (Theorem 15.5). For n distinct points $z_1, \ldots, z_n \in \mathbb{C}$, the *Gaudin Hamiltonians* are the pairwise sums

$$H_i = \sum_{j \neq i} \frac{\Omega_{ij}^{Y(\widehat{\mathfrak{g}}_{Q_X})}}{z_i - z_j}, \qquad i = 1, \ldots, n,$$

where $\Omega_{ij}^{Y(\widehat{\mathfrak{g}}_{Q_X})} \in \mathrm{End}(V_1 \otimes \cdots \otimes V_n)$ is the image of the quadratic Casimir $\Omega \in Y(\widehat{\mathfrak{g}}_{Q_X}) \otimes Y(\widehat{\mathfrak{g}}_{Q_X})$ acting in the i-th and j-th tensor factors.

PROPOSITION 15.20 *(Commutativity of Gaudin Hamiltonians).* The operators $H_1, \ldots, H_n$ pairwise commute: $[H_i, H_j] = 0$ for all i, j. The proof reduces to the classical Yang–Baxter equation for $r(z)$, which follows from $D^2 = 0$ in the bar complex projected to genus 0, arity 3. [PH]

Proof. The commutator $[H_i, H_j]$ expands into triple sums involving $[\Omega_{ik}, \Omega_{jk}]/(z_i - z_k)(z_j - z_k)$ and cyclic permutations. The classical Yang–Baxter equation $[r_{12}(z_{12}), r_{13}(z_{13})] + [r_{12}(z_{12}), r_{23}(z_{23})] + [r_{13}(z_{13}), r_{23}(z_{23})] = 0$ (where $z_{ij} = z_i - z_j$) forces exact cancellation. The CYBE itself is the arity-3 projection of the MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ at genus 0 (cf. Vol I, Theorem 20.9). □

For $X = \mathbb{C}^3$ (the Jordan quiver), the Yangian is $Y(\widehat{\mathfrak{gl}}_1)$ and the Casimir is the standard $\Omega = \sum_a J_a \otimes J^a$ with J_a the generators of $\widehat{\mathfrak{gl}}_1$. The Gaudin system on n points then coincides with the $\widehat{\mathfrak{gl}}_1$ Gaudin model studied by Rapcak–Soibelman–Yang–Zhao in the context of $\mathcal{W}_{1+\infty}$ at the self-dual level. The spectral curve of the Gaudin system is the Seiberg–Witten curve of the 4d $\mathcal{N} = 2$ theory obtained by compactification of the 5d holomorphic Chern–Simons theory on X.

Proposition 15.21 *(Wall-crossing as Gaudin gauge equivalence).* Let ζ, ζ' be generic stability parameters in adjacent chambers. The wall-crossing gauge transformation $\Theta_{\zeta'} = e^{\mathrm{ad}_\alpha}(\Theta_\zeta)$ (Theorem 15.16) induces a gauge equivalence of the associated Gaudin systems: the Hamiltonians H_i^ζ and $H_i^{\zeta'}$ are conjugate by the gauge operator $e^\alpha \in \exp(\mathfrak{n}_W)$, hence have identical spectra. [PH]

Proof. The Gaudin Hamiltonians are extracted from the arity-2 projection of Θ; the gauge transformation e^{ad_α} acts on this projection by conjugation, preserving commutativity and spectra. □

Remark. The Gaudin construction extends to the resolved conifold (Klebanov–Witten Yangian) and to local $\mathbb{P}^2$ (McKay super Yangian), giving integrable spin chains whose Bethe ansatz equations encode DT invariants. The integrability of the Gaudin system is the dynamical manifestation of $D^2 = 0$ in the bar complex.

Chapter 16

Fukaya Categories

The CY-to-chiral functor Φ (Theorem CY-A$_2$) takes as input a CY category with cyclic A_∞-structure (Chapter 3). The Fukaya category $\mathrm{Fuk}(X)$ of a symplectic manifold X is the prototypical source: its A_∞-structure encodes counts of pseudoholomorphic polygons, and the CY pairing is Poincaré duality on Lagrangian intersections. For CY surfaces ($d = 2$), $\Phi(\mathrm{Fuk}(X))$ is an E_2-chiral algebra with $\kappa = \chi^{\mathrm{CY}}$. This chapter computes $\mathrm{Fuk}(X)$ and the resulting chiral algebra for elliptic curves, K3 surfaces, abelian surfaces, and CY threefolds.

16.1 The Fukaya category of a CY manifold

Example 16.1 *(Elliptic curve).* For an elliptic curve E_τ, $\mathrm{Fuk}(E_\tau)$ is CY of dimension 1. The Hochschild homology $\mathrm{HH}_\bullet(\mathrm{Fuk}(E_\tau))$ recovers the de Rham cohomology of E_τ, and the cyclic structure encodes the symplectic pairing. The associated chiral algebra is the Heisenberg vertex algebra H_k at level $k = \mathrm{vol}(E_\tau)$.

16.2 CY 2-folds: K3 and abelian surfaces

16.3 CY 3-folds

16.4 Wrapped Fukaya categories

16.5 The quantum chiral algebra of a Fukaya category

Chapter 17

Derived Categories of CY Manifolds

Homological mirror symmetry identifies $\mathrm{Fuk}(X) \simeq D^b(\mathrm{Coh}(X^\vee))$. Under the functor Φ, this becomes an equivalence $A_{\mathrm{Fuk}(X)} \simeq A_{D^b(\mathrm{Coh}(X^\vee))}$ of quantum chiral algebras. This chapter develops the algebraic side: the CY structure on $D^b(\mathrm{Coh}(X))$, the chiral algebra extracted by Φ, exceptional collections, and Bridgeland stability conditions.

17.1 $D^b(\mathrm{Coh}(X))$ for CY manifolds

17.2 Homological mirror symmetry and the chiral algebra

The homological mirror symmetry conjecture (Kontsevich) identifies $\mathrm{Fuk}(X) \simeq D^b(\mathrm{Coh}(X^\vee))$. Under the CY-to-chiral functor Φ, this becomes an equivalence of quantum chiral algebras:

$$A_{\mathrm{Fuk}(X)} \simeq A_{D^b(\mathrm{Coh}(X^\vee))}.$$

17.3 Exceptional collections and chiral algebras

17.4 Stability conditions and the chiral moduli

Chapter 18

Matrix Factorizations

The third source of CY categories is the Landau–Ginzburg model: a polynomial $W\colon \mathbb{C}^n \to \mathbb{C}$ whose category $\mathrm{MF}(W)$ of matrix factorizations is CY of dimension $n-2$. For ADE singularities $W = x^N + y^2 + z^2 + w^2$, $\mathrm{MF}(W)$ is CY_2 and Φ (Theorem CY-A_2) produces chiral algebras related to $\mathcal{W}_N$-algebras. The LG/CY correspondence $\mathrm{MF}(W) \simeq D^b(\mathrm{Coh}(X_W))$ provides a consistency check: $\Phi(\mathrm{MF}(W))$ and $\Phi(D^b(\mathrm{Coh}(X_W)))$ (Chapter 17) must agree.

18.1 Matrix factorizations and Landau–Ginzburg models

18.2 The CY structure on $\mathrm{MF}(W)$

18.3 Quantum chiral algebras from singularities

18.4 ADE singularities and W-algebras

The connection between ADE singularities and W-algebras provides a key test case for the CY-to-chiral functor. For the A_{N-1} singularity $W = x^N$ in one variable, $\mathrm{MF}(x^N)$ is CY of dimension -1 (formal), so the relevant setting is the two-variable singularity $W = x^N + y^2$, giving $\mathrm{MF}(W)$ of CY dimension 0 (semisimple). To reach a CY_2 category related to W_N, one considers the threefold singularity $W = x^N + y^2 + z^2 + w^2$ in four variables, yielding $\mathrm{MF}(W)$ of CY dimension 2.

Chapter 19

Quantum Group Representations

The braided monoidal category $\mathrm{Rep}_q(\mathfrak{g})$ is the representation-theoretic target of the CY programme. The Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A))$ (Chapter 13) provides the bulk algebra; Conjecture CY-C predicts $\mathrm{Rep}_q(\mathfrak{g}) \simeq \mathrm{Rep}^{E_2}(A_C)$ for a CY category $C(\mathfrak{g}, q)$. The Kazhdan–Lusztig equivalence $\mathrm{Rep}_q(\mathfrak{g}) \simeq \mathrm{KL}_k(\mathfrak{g})$ at roots of unity is the prototype; the affine Yangian realization (Chapter 15) provides the E_1-sector.

19.1 $\mathrm{Rep}_q(\mathfrak{g})$ as a braided monoidal category

19.2 The quantum chiral algebra of $U_q(\mathfrak{g})$

Conjecture 19.1 *(Quantum group realization — Conjecture CY-C).* For a simple Lie algebra $\mathfrak{g}$ and $q = e^{\pi i/(k+h^\vee)}$ with k a positive integer, the braided monoidal category $\mathrm{Rep}_q(\mathfrak{g})$ should arise as $\mathrm{Rep}^{E_2}(A_C)$ for a CY category $C = C(\mathfrak{g}, q)$. The quantum chiral algebra A_C should recover the affine Kac–Moody vertex algebra $V_k(\mathfrak{g})$ at level k.

Status: The CY category $C(\mathfrak{g}, q)$ is not yet constructed in general. At generic q (irrational k), $\mathrm{Rep}_q(\mathfrak{g})$ is semisimple and the story simplifies; the interesting structure is at roots of unity, where the Kazhdan–Lusztig equivalence provides the prototype.

19.3 Kazhdan–Lusztig equivalence

The Kazhdan–Lusztig equivalence

$$\mathrm{Rep}_q(\mathfrak{g}) \simeq \mathrm{KL}_k(\mathfrak{g})$$

between quantum group representations (at root of unity) and the Kazhdan–Lusztig category of affine Kac–Moody modules is the prototype for Theorem CY-C.

19.4 Yangian and RTT realizations

Part IV

The Seven Faces of $r_{\mathrm{CY}}(z)$

Chapter 20

The Bar-Cobar Bridge to Volume I

The shadow obstruction tower Θ_A of Volume I applies to chiral algebras arising from the cyclic bar complex of a Calabi–Yau category. This chapter establishes the dictionary between the CY cyclic bar complex $\mathrm{CC}_\bullet(C)$ and the chiral bar complex $B(A)$, computes the shadow tower of $\mathbb{C}^3$, traces the passage from finite to infinite shadow depth under the factorization envelope, and identifies the open string field theory realization of Koszul duality.

20.1 CY bar complex vs chiral bar complex

Let C be a smooth proper CY_3 category with Serre functor $\mathbb{S}_C \simeq [3]$, and let $A = A_C$ denote the chiral algebra produced by the CY-to-chiral functor Φ (Theorem CY-A_2 for $d = 2$; conjectural for $d = 3$). The cyclic bar complex $\mathrm{CC}_\bullet(C)$ and the chiral bar complex $B(A)$ are related by a canonical quasi-isomorphism

$$\mathrm{CC}_\bullet(C) \;\simeq\; B(A_C) \tag{CY-A(ii)}$$

as factorization coalgebras on $\mathrm{Ran}(X)$.

The precise dictionary is as follows.

Proposition 20.1 *(Bar complex dictionary).* Under the identification CY-A(ii):

(i) The cyclic differential $b + B$ on $\mathrm{CC}_\bullet(C)$ corresponds to the bar differential d_B on $B(A)$.

(ii) The Connes operator $B\colon \mathrm{HH}_n(C) \to \mathrm{HH}_{n+1}(C)$ corresponds to the arity-preserving component of d_B (the internal differential), while the Hochschild differential b corresponds to the arity-lowering component (the bar contraction).

(iii) The S^1-action on $\mathrm{CC}_\bullet(C)$ (the cyclic rotation) corresponds to the factorization coalgebra structure on $B(A)$, with the cocomposition maps Δ_Γ indexed by stable graphs Γ.

(iv) The Hodge filtration on $\mathrm{HH}_*(C)$ corresponds to the arity filtration on $B(A)$: the arity-r piece $B^{(r)}(A)$ captures Hochschild chains of tensor length $\leq r$.

[PH]

Remark 20.2 (Three functors, three outputs). The bar complex $B(A)$ is a factorization *coalgebra*. Three distinct functors produce three distinct objects from $B(A)$:

(1) $\Omega(B(A)) \simeq A$ recovers the original algebra (bar-cobar inversion, Theorem B of Volume I).

(2) $D_{\mathrm{Ran}}(B(A)) \simeq B(A^!)$ is the Verdier dual, a factorization *algebra* identified with the bar of the Koszul dual $A^!$ (Theorem A, Convention conv:bar-coalgebra-identity).

(3) $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(A) = \mathrm{RHom}(\Omega(B(A)), A)$ is the chiral derived center, computing the universal bulk (Theorem H).

These are not the same operation, and the CY cyclic bar complex $\mathrm{CC}_\bullet(C)$ corresponds to $B(A)$ itself, not to the derived center and not to the cobar.

20.2 The shadow tower of $\mathbb{C}^3$: MacMahon meets Koszul

The affine threefold $\mathbb{C}^3$ is the simplest case where both sides of the CY-to-chiral correspondence are explicit.

THEOREM 20.3 *(Shadow tower of* $\mathbb{C}^3$*).* [PH] The shadow obstruction tower of $W_{1+\infty}$ at $c = 1$ satisfies:

(i) The bar Euler product equals the inverse MacMahon function:

$$\sum_{n\geq 0} \dim B^{(n)}_{E_\infty}(A) \cdot q^n = \frac{1}{M(q)}, \qquad M(q) = \prod_{n\geq 1} \frac{1}{(1-q^n)^n}.$$

(ii) The modular characteristic is $\kappa = 1$, verified by five independent paths:

(a) genus-1 free energy: $F_1 = \kappa/24 = 1/24$;

(b) degree-0 MacMahon exponent: $\log M(q) = \sum \sigma_2(k) q^k/k$ matches $\kappa = 1$;

(c) channel decomposition: $\sum_{s\geq 1} \kappa_s^{\mathrm{eff}} = \sum_{s\geq 1} s \cdot (1/s) = \sum 1 = +\infty$ (at finite N, $\kappa(W_N) = c(H_N - 1)$; the total $\kappa(W_{1+\infty}, c = 1) = \kappa(H_1) = 1$);

(d) Hodge-theoretic: via the categorical trace $\chi^{\mathrm{CY}}(\mathrm{Perf}(\mathbb{C}^3))$;

(e) complementarity: $\kappa(\mathbb{C}^3) + \kappa(\mathrm{dual}) = 0$ (free-field anti-symmetry).

(iii) Per-channel modular characteristics: for the spin-s channel of $W_{1+\infty}$,

$$\kappa_s = \frac{c}{s} = \frac{1}{s}, \qquad \kappa_s^{\mathrm{eff}} = s \cdot \kappa_s = 1 \quad \text{(constant per MacMahon level)}.$$

This constancy is the shadow-DT Rosetta stone: each energy level contributes equally to the modular characteristic, and the MacMahon function $M(q) = \prod(1-q^n)^{-n}$ is the partition function of a system with effective $\kappa = 1$ per energy level.

[PH]

Verification: `compute/lib/c3_shadow_tower.py` (73 tests), `compute/lib/macmahon_shadow_decomposition.`

Remark 20.4 (Envelope creates infinite depth from abelian input). The Heisenberg algebra H_1 (a single free boson) has shadow depth class G with $r_{\max} = 2$: the tower terminates at arity 2, and $\kappa = 1$ is the only nonvanishing shadow invariant.

The vertex algebra $W_{1+\infty}$ at $c = 1$, the factorization envelope of the Lie conformal algebra of polyvector fields on $\mathbb{C}^3$, has shadow depth class M with $r_{\max} = \infty$. The spin-2 channel (Virasoro at $c = 1$) already has infinite shadow depth, with $\kappa_2 = 1/2$, $\alpha_2 = 2$, $S_4 = 10/27$.

This discrepancy between input (class G, depth 2) and output (class M, depth ∞) reflects the non-conservativity of the factorization envelope on shadow invariants. Normal ordering, Wick contractions, and OPE singular terms generate higher-arity shadows invisible at the E_1-level. Each stage of $E_1 \to E_2 \to E_\infty$-chiral introduces genuinely new invariants.

20.3 Open string field theory: Koszul duality at chain level

For $\mathbb{C}^3$, the cyclic A_∞-algebra is the exterior algebra $A = \bigwedge^*(V)$ with $V = \mathbb{C}^3$, equipped with the CY3 trace $\langle a, b\rangle = \mathrm{Tr}(a \wedge b)$ (projection to top degree).

THEOREM 20.5 *(Chain-level Koszul duality for $\mathbb{C}^3$).* [PH] The Koszul duality pair for $A = \bigwedge^*(\mathbb{C}^3)$ satisfies:

(i) $A^! = \mathrm{Sym}^*(\mathbb{C}^{3,*})$, the symmetric algebra on the dual vector space. This is the $\mathrm{Com}^! = \mathrm{Lie}$ incarnation at the chain level: the exterior algebra (free Com-algebra on V) is Koszul dual to the symmetric algebra (free Lie-algebra envelope on V^*).

(ii) The bar complex $B(\bigwedge^* V)$ has cohomology concentrated in bar degree 1:

$$H^*(B(\bigwedge^* V)) \;\simeq\; \mathrm{Sym}^*(V^*[-1]),$$

confirming that A is Koszul (chirally and classically).

(iii) The bar-cobar round-trip recovers A itself:

$$\Omega(B(\bigwedge^* V)) \xrightarrow{\sim} \bigwedge^* V.$$

[PH]

Verification: `compute/lib/bar_comparison_c3.py`, `compute/lib/e2_barcobar_koszul.py` (73 tests).

20.4 From shuffle product to λ-bracket: the seven-step chain

The Schiffmann–Vasserot shuffle presentation of $Y^+(\widehat{\mathfrak{gl}}_1)$ does *not* directly produce the λ-bracket of $R_{\mathbb{C}^3}$. Seven comparison steps are required.

Construction 20.6 *(The seven-step chain).* The chain connecting the shuffle algebra to the chiral algebra:

1. **Shuffle algebra** $\mathrm{Sh}(h_1, h_2, h_3)$: the associative algebra with generators $e(z)$ at each spectral parameter z and multiplication by the rational shuffle kernel $\zeta(z_i - z_j) = \prod_a (z_i - z_j + h_a)/(z_i - z_j - h_a)$.

2. **Affine Yangian** $Y(\widehat{\mathfrak{gl}}_1)$: the RTT-presentation with generators e_j, f_j, ψ_j satisfying the affine Yangian relations. Schiffmann–Vasserot proved $\mathrm{Sh}^+ \simeq Y^+(\widehat{\mathfrak{gl}}_1)$.

3. $W_{1+\infty}$ **modes**: the Feigin–Frenkel realization of $W_{1+\infty}$ as the limit $\lim_{N\to\infty} W_N$, with explicit mode algebra generators W^s_n for spin s and mode n.

4. **OPE modes**: the vertex algebra OPE $W^s(z)W^{s'}(w) \sim \sum_n C^{ss'}_n(W)(z-w)^{-n-1}$, extracting the singular coefficients $C^{ss'}_n$.

5. λ-**bracket**: $\{W^s_\lambda W^{s'}\} = \sum_n \lambda^{(n)} C^{ss'}_n(W)$, where $\lambda^{(n)} = \lambda^n/n!$ is the divided power (AP44: the coefficient at order n in the λ-bracket is $C^{ss'}_n/n!$, not $C^{ss'}_n$).

6. **Lie conformal algebra** $R_{\mathbb{C}^3}$: the $\mathrm{GL}(3)$-invariant subalgebra of the Schouten–Nijenhuis Lie conformal algebra on polyvector fields $\mathrm{PV}^*(\mathbb{C}^3)$.

7. **Factorization envelope** $U^{\mathrm{ch}}(R_{\mathbb{C}^3})$: the enveloping vertex algebra, which by the Nishinaka–Vicedo construction recovers $W_{1+\infty}$.

Steps 1⟶2 is Schiffmann–Vasserot. Steps 2⟶3 is Prochazka–Rapcak. Steps 3⟶4 is the standard mode-to-OPE dictionary. Step 4⟶5 requires the divided-power convention (AP44). Steps 5⟶6 is the Kontsevich–Soibelman identification of $R_{\mathbb{C}^3}$ as the $\mathrm{GL}(3)$-invariant part of $\mathrm{PV}^*(\mathbb{C}^3)$ with the Schouten–Nijenhuis bracket. Step 6⟶7 is the factorization envelope.

The composition $1 \to 7$ is the full CY-to-chiral functor for $\mathbb{C}^3$.

Verification: `compute/lib/c3_lie_conformal.py`, `compute/lib/c3_envelope_comparison.py` (52 tests).

Warning 20.7 (Shuffle product ≠ λ-bracket). The shuffle product on $\mathrm{Sh}(h_1, h_2, h_3)$ is an associative product on symmetric functions of the spectral parameters, while the λ-bracket is a Lie conformal bracket on fields. The two are related by the seven-step chain above, not by a direct identification. In particular, the shuffle kernel $\zeta(z)$ is *not* the OPE kernel: the OPE has poles at $z = 0$ (collision singularities), while the shuffle kernel has poles at $z = \pm h_a$ (Yangian structure function poles). These coincide only after the identification of the parameters h_a with the deformation parameters of $W_{1+\infty}$ and the passage from additive to multiplicative spectral parameters.

20.5 Koszul duality: CY vs chiral

The chiral Koszul duality of Volume I (Theorems A and B) descends to CY Koszul duality.

Conjecture 20.8 *(CY Koszul duality dictionary).* [CJ] For a CY_3 category $\mathcal{C}$ with chiral algebra $A = A_{\mathcal{C}}$:

(i) The chiral Koszul dual $A^!$ corresponds to the *mirror* CY category:

$$A^!_{\mathcal{C}} \;\simeq\; A_{\mathcal{C}^!}$$

where $\mathcal{C}^!$ is the Koszul dual CY category (for $\mathcal{C} = D^b(\mathrm{Coh}(X))$ of a CY manifold X, this is $\mathcal{C}^! \simeq \mathrm{Fuk}(X^\vee)$ under homological mirror symmetry).

(ii) The complementarity theorem (Theorem C) gives:

$$Q_g(A_{\mathcal{C}}) + Q_g(A_{\mathcal{C}^!}) = H^*(\overline{\mathcal{M}}_g, \mathcal{Z}(A_{\mathcal{C}}))$$

where $\mathcal{Z}(A_{\mathcal{C}}) = \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(A_{\mathcal{C}})$ is the chiral derived center.

(iii) The modular characteristics satisfy:

$$\kappa(A_{\mathcal{C}}) + \kappa(A_{\mathcal{C}^!}) = 0$$

when $\mathcal{C}$ arises from a free-field or lattice construction (anti-symmetry), and more generally $\kappa + \kappa' = \rho \cdot K$ for algebras with nontrivial contact terms (Theorem D, AP24).

20.6 The five theorems in the CY setting

We summarize the status of the five main theorems of Volume I when specialized to chiral algebras arising from CY categories.

Theorem *(The five theorems for CY chiral algebras).* Let $A = A_C$ be the chiral algebra of a CY$_3$ category C. Then:

Theorem A (adjunction). The bar-cobar adjunction $B \dashv \Omega$ restricts to CY chiral algebras: $B(A_C)$ is a factorization coalgebra on $\mathrm{Ran}(X)$, and $D_{\mathrm{Ran}}(B(A_C)) \simeq B(A_{C^!})$. The CY identification CY-A(ii) gives $\mathrm{CC}_\bullet(C) \simeq B(A_C)$.

Theorem B (inversion). Bar-cobar inversion $\Omega(B(A_C)) \xrightarrow{\sim} A_C$ holds on the Koszul locus. For CY categories, chirally Koszul is equivalent to the formality of $\mathrm{CC}_\bullet(C)$ as a dg coalgebra.

Theorem C (complementarity). The CY Euler characteristic $\chi(C)$ splits into complementary halves: $Q_g(C) + Q_g(C^!)$ recovers the full Hochschild cohomology.

Theorem D (modular characteristic). The genus-g obstruction is $\mathrm{obs}_g(A_C) = \kappa(A_C) \cdot \lambda_g^{\mathrm{FP}}$ on the uniform-weight lane, where $\kappa(A_C) = \chi^{\mathrm{CY}}(C)$ is the CY modular characteristic. For compact CY$_3$, the BCOV prediction gives $\kappa = \chi_{\mathrm{top}}/24$.

Theorem H (Hochschild). $\mathrm{ChirHoch}^*(A_C)$ is polynomial in degrees $\{0, 1, 2\}$, and the CY Hochschild calculus of C (HKR decomposition, Connes operator, categorical Hodge structure) is faithfully reflected in $\mathrm{ChirHoch}^*(A_C)$.

20.7 Compact CY$_3$ examples

The following compact CY$_3$ families illustrate the range of shadow tower behaviour. In each case, the predicted modular characteristic is $\kappa = \chi_{\mathrm{top}}/24$ (BCOV prediction); the shadow depth class, Hochschild data, and BKM structure vary.

The banana manifold: $\kappa = 0$ with nontrivial tower

The banana manifold X_{ban} (Schoen's CY$_3$, the fiber product of two rational elliptic surfaces over $\mathbb{P}^1$) has Hodge numbers $h^{1,1} = h^{2,1} = 19$ and Euler characteristic $\chi = 0$.

Proposition 20.9 *(Shadow tower of the banana manifold).* The shadow obstruction tower of X_{ban} has:

(i) $\kappa = \chi/24 = 0$. The arity-2 scalar shadow *vanishes*.

(ii) $S_4 = -44$ (quartic shadow from the genus-0 GV invariants $n^0_{d_1,d_2} = -2$ of the banana curve classes).

(iii) The shadow tower starts at arity 4, not arity 2: the cubic shadow is invisible when $\kappa = 0$ (it enters as $\kappa \cdot \alpha$, which vanishes).

(iv) The critical discriminant $\Delta = 8\kappa S_4 = 0$ (since $\kappa = 0$). The standard single-line classification (G/L/C/M) does not directly apply.

(v) Shadow depth class: M (infinite tower) shifted to arity 4. The BKY partition function involves infinitely many banana-curve BPS states, generating an infinite shadow tower whose leading term is quartic.

[PH]

Remark 20.10 (AP31 in action). The banana manifold is the prototypical example of AP31: $\kappa = 0$ does *not* imply $\Theta_A = 0$. The vanishing of κ means the bar complex is uncurved ($d^2 = 0$ at the leading scalar level), but the higher-arity components of Θ_A (the quartic shadow Q and beyond) are nonvanishing, sourced by the instanton contributions (genus-0 GV invariants of the banana curves).

Three CY_3 examples with $\kappa = 0$ show qualitatively different behaviour:

- Heisenberg at $k = 0$: $\kappa = 0$, trivially uncurved, class G.
- Virasoro at $c = 0$: $\kappa = 0$, but higher shadows from contact terms (class M).
- Banana manifold: $\kappa = 0$, but higher shadows from *instantons* (class M, shifted to arity 4).

Verification: `compute/lib/banana_shadow.py` (63 tests).

Enriques $\times E$: the $\mathbb{Z}_2$ quotient of K3 $\times E$

Let $S_{\mathrm{Enr}} = \mathrm{K3}/\iota$ be an Enriques surface (ι a fixed-point-free involution), and E an elliptic curve. The product $S_{\mathrm{Enr}} \times E$ is a generalized CY_3 (the canonical bundle $K_{S_{\mathrm{Enr}}\times E}$ is 2-torsion, so $h^{3,0} = 0$). Nevertheless, the shadow tower machinery applies to the associated vertex algebra, and the Borcherds lift on $O(2, 10)$ provides the automorphic structure.

THEOREM 20.11 *(Shadow tower of Enriques $\times E$).* [PH] The shadow data of Enriques $\times E$ is:

(i) $\kappa(\mathrm{Enr} \times E) = 4$, the weight of the Enriques Borcherds product Ψ on $O(2, 10)$ (Allcock 2000).

(ii) Root multiplicities come from the Enriques elliptic genus $\phi_{\mathrm{Enr}} = \frac{1}{2}\phi_{0,1}$: the Fourier coefficients are $f_{\mathrm{Enr}}(n, l) = f_{\mathrm{K3}}(n, l)/2$ (all integers since f_{K3} has even coefficients).

(iii) The ratio $\kappa(\mathrm{K3} \times E)/\kappa(\mathrm{Enr} \times E) = 5/4$.

(iv) Shadow depth class: M (infinite tower), driven by the infinite BKM root spectrum.

[PH]

Remark 20.12 (The $\mathbb{Z}_2$ quotient on κ). The passage from K3 $\times E$ ($\kappa = 5$) to Enriques $\times E$ ($\kappa = 4$) under the $\mathbb{Z}_2$ quotient does *not* halve κ. Rather, the weight of the Borcherds product drops by 1: the $O(2, 10)$ lattice for the Enriques surface has a different constant term in the Borcherds input form than the $O(3, 19)$ lattice for K3, and the weight formula gives $8/2 = 4$ instead of $10/2 = 5$. The ratio $5/4$ is not a simple topological quotient; it reflects the lattice-theoretic structure of the Borcherds lift.

Verification: `compute/lib/enriques_shadow.py` (72 tests).

The quintic threefold: non-integer κ and BKM obstruction

The quintic hypersurface $Q \subset \mathbb{P}^4$ has $h^{1,1} = 1$, $h^{2,1} = 101$, $\chi = -200$.

PROPOSITION 20.13 *(Quintic shadow tower).* The shadow data of the quintic is:

(i) $\kappa = \chi/24 = -25/3$ (non-integer).

(ii) The non-integrality of κ obstructs a naive BKM superalgebra: BKM root multiplicities must be integers, and the standard denominator-identity machinery requires integral Borcherds lift weight. The quintic has no natural Borcherds–Kac–Moody structure.

(iii) The shadow metric $Q_L(t) = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4)t^2$ is well-defined for fractional κ. With $\kappa = -25/3$: $Q_L(0) = 2500/9 > 0$.

(iv) The GW invariants are known to high degree:

$$n_1^0 = 2875, \quad n_2^0 = 609250, \quad n_3^0 = 317206375, \quad n_4^0 = 242467530000.$$

These are BPS state counts via the Gopakumar–Vafa formula.

(v) Shadow depth class: M (infinite tower). The quintic has an infinite BPS spectrum with curves of all degrees and genera; the shadow tower is controlled by the BCOV holomorphic anomaly equation.

[PH]

Remark 20.14 ($\chi/24$ integrality survey). The quantity $\chi/24$ is an integer if and only if $12 \mid (h^{1,1} - h^{2,1})$. Among standard complete intersections:

- $\mathbb{P}^5[3,3]$: $\chi = -144$, $\kappa = -6$ (integer).
- $\mathbb{P}^5[2,4]$: $\chi = -176$, $\kappa = -22/3$ (non-integer).
- $\mathbb{P}^7[2,2,2,2]$: $\chi = -128$, $\kappa = -16/3$ (non-integer).
- Quintic: $\chi = -200$, $\kappa = -25/3$ (non-integer).

Non-integrality of κ is the generic situation for compact CY$_3$ manifolds. A BKM structure requires additional input beyond the BCOV prediction.

Verification: `compute/lib/quintic_shadow_obstruction.py` (86 tests).

Hochschild homology of compact CY$_3$ examples

The Hochschild homology $\mathrm{HH}_*(D^b(\mathrm{Coh}(X)))$ is computed from the HKR decomposition $\mathrm{HH}_p \simeq \bigoplus_q H^q(X, \Omega_X^{3-p})$ using the CY$_3$ isomorphism $\bigwedge^p T_X \simeq \Omega_X^{3-p}$.

PROPOSITION 20.15 *(Hochschild data of compact CY$_3$ examples).* (i) $X = \mathrm{K3} \times E$: $\dim \mathrm{HH}_* = 96$, with Hodge decomposition

$$(\dim \mathrm{HH}_0, \dim \mathrm{HH}_1, \dim \mathrm{HH}_2, \dim \mathrm{HH}_3) = (4, 44, 44, 4).$$

The palindromic symmetry $\dim \mathrm{HH}_p = \dim \mathrm{HH}_{3-p}$ reflects Serre duality.

(ii) $X = Q$ (quintic): $\dim \mathrm{HH}_* = 208$, with

$$(\dim \mathrm{HH}_0, \dim \mathrm{HH}_1, \dim \mathrm{HH}_2, \dim \mathrm{HH}_3) = (2, 102, 102, 2).$$

The concentration in degrees 1 and 2 reflects $h^{1,1} = 1$ and $h^{2,1} = 101$: moduli dominate, and the CY structure contributes only the top and bottom forms.

[PH]

Verification: `compute/lib/cy3_hochschild.py` (71 tests).

20.8 Integrable systems and lattice models

The shadow obstruction tower admits a dictionary with classical integrable systems, where the conserved quantities I_r correspond to shadow invariants at each arity.

The Calogero–Moser/Bethe dictionary

The Calogero–Moser system with N particles on a line and inverse-square interaction $V(x) = \sum_{i<j} g^2/(x_i - x_j)^2$ has N independent conserved quantities $I_1, I_2, \ldots, I_N$ (the Sekiguchi operators), with $I_2 = H$ (the Hamiltonian) and I_k of degree k in the momenta.

Conjecture 20.16 *(Shadow-integrable dictionary).* [CJ] There is a dictionary between the shadow obstruction tower invariants and the Calogero–Moser conserved quantities:

$$\begin{aligned} I_2 \text{ (quadratic Hamiltonian)} &\longleftrightarrow \kappa \text{ (modular characteristic)}, \\ I_3 \text{ (cubic integral)} &\longleftrightarrow C \text{ (cubic shadow)}, \\ I_4 \text{ (quartic integral)} &\longleftrightarrow Q \text{ (quartic shadow / contact invariant)}. \end{aligned}$$

The Bethe ansatz equations for the Calogero–Moser system correspond to the Maurer–Cartan equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ of the modular convolution algebra, projected to each arity: the r-th conserved quantity I_r is the genus-0 arity-r projection of the MC element Θ_A.

Remark 20.17. The dictionary is *structural*, not a theorem. It captures the parallel between two filtration structures: the arity filtration on Θ_A (controlling the shadow depth classification G/L/C/M) and the degree filtration on the conserved quantities I_r (controlling the integrability class of the Calogero–Moser system). The coupling constant g^2 maps to κ, the number of particles N maps to the number of strong generators, and the Bethe roots map to the shadow metric zeros.

Verification: `compute/lib/cross_volume_shadow_bridge.py` (60 tests).

Lattice finite-size corrections

For a critical lattice model (e.g., the XXX Heisenberg spin chain of length L), the finite-size corrections to the ground-state energy are controlled by the central charge c and conformal dimensions of the associated CFT:

$$E_0(L) = L\varepsilon_\infty - \frac{\pi c}{6L} + O(L^{-3}).$$

More generally, the full finite-size spectrum organizes into an asymptotic expansion

$$E_n(L) - L\varepsilon_\infty = \frac{2\pi}{L}\left(-\frac{c}{12} + h_n + \bar{h}_n\right) + \sum_{k\geq 2} \frac{a_k(n)}{L^{2k-1}}.$$

PROPOSITION 20.18 *(Finite-size corrections as shadow tower).* The finite-size correction coefficients $a_k(n)$ correspond to shadow tower data: the leading coefficient $a_1 = -c/12 = -\kappa/6$ recovers κ, and the higher coefficients a_k for $k \geq 2$ encode the arity-$(2k)$ shadow projections. The decay rate of the corrections is $q = e^{-2\pi/L}$, matching the shadow tower variable. [HE]

Verification: `compute/lib/cross_volume_shadow_bridge.py` (63 tests).

Crystal melting: combinatorial vs bar-algebraic arity

The MacMahon function $M(q)$ counts plane partitions (3D Young diagrams), and the crystal melting model of Okounkov–Reshetikhin–Vafa interprets the DT partition function of $\mathbb{C}^3$ as a statistical-mechanical partition function on the crystal lattice $\mathbb{Z}^3_{\geq 0}$.

Warning 20.19 (Combinatorial arity ≠ bar arity). The crystal melting "arity" (the number of boxes removed from the corner of the crystal) does *not* match the bar complex arity (the tensor length in $B^{(n)}(A)$).

The crystal partition function organizes by the number of boxes:

$$M(q) = \sum_{\pi \in \mathcal{PP}} q^{|\pi|}, \qquad |\pi| = \text{number of boxes in plane partition } \pi.$$

The bar complex organizes by tensor length:

$$B(A) = \bigoplus_{n \geq 1} A^{\otimes n}[n], \qquad \text{arity } n = \text{number of tensor factors.}$$

These two gradings are *not* compatible: a plane partition with k boxes does not correspond to a bar element of arity k. Rather, the crystal melting expansion is an exponential reorganization of the bar complex: $\log M(q) = \sum_k \sigma_2(k) q^k/k$ is the free energy, while the bar complex detects the individual OPE modes.

The fundamental mismatch: crystal melting is a *tensor-algebraic* construction (counting 3D diagrams in $\mathbb{Z}^3$), while the bar complex is a *factorization-algebraic* construction (organized by collision patterns on $\mathrm{Ran}(X)$). The two agree in the graded *character* (both reproduce $M(q)$ or its inverse) but not in the *arity structure*.

Verification: `compute/lib/crystal_bar_identification.py` (70 tests).

20.9 Grand atlas of CY$_3$ shadow invariants

Table 20.1 collects the shadow tower data for 18 CY$_3$ families spanning the full range of shadow behaviour.

Key patterns visible in the atlas:

1. **Toric depth bound**: for toric CY$_3$ with N vertices in the web diagram, the shadow depth satisfies $r_{\max} \leq N$. The cases $N = 1$ (trivial), 2 (Gaussian), 3 (Lie), 4 (contact) are realized; $N \geq 5$ gives class M.

2. **Compact CY$_3$ are generically class M**: all compact CY$_3$ with $\chi \neq 0$ have infinite shadow depth, driven by the infinite BPS spectrum.

3. **κ integrality**: among the atlas families, $\mathbb{P}^5[3,3]$ ($\kappa = -6$), K3 $\times E$ ($\kappa = 5$), Enriques $\times E$ ($\kappa = 4$), and BV$(20,2,0)$ ($\kappa = 5$) have integral κ. For compact CICYs, $\kappa = \chi/24$ is integral only when $24 \mid \chi$; for K3-fibered geometries, κ comes from the automorphic weight and is always integral.

4. **BKM structure requires integral automorphic weight**: K3-fibered CY$_3$s with $\chi = 0$ carry BKM superalgebras via Borcherds lifts; compact CICYs with $\kappa \notin \mathbb{Z}$ do not.

Table 20.1: Grand atlas of CY3 shadow invariants. Columns: CY3 family, topological Euler characteristic χ, Hodge numbers $h^{1,1}$ and $h^{2,1}$, modular characteristic κ, shadow depth class, and data source. For non-compact toric CY3: $\kappa = \chi(S)/2$ where S is the compact base (Proposition 15.17). For compact CICYs: $\kappa = \chi_{\text{top}}/24$. For K3-fibered geometries: κ is the automorphic weight. Families above the first rule are non-compact toric; between the rules are K3-fibered; below are compact complete intersections and special families. BV = Borcea–Voisin.

CY3	χ	$h^{1,1}$	$h^{2,1}$	κ	Class	Source
$\mathbb{C}^3$	—	—	—	1	G	MacMahon
Resolved conifold	—	1	0	1	G	KS wall-crossing
Local $\mathbb{P}^2$	—	1	0	$3/2$	M	McKay($\mathbb{Z}_3$)
Local $\mathbb{P}^1 \times \mathbb{P}^1$	—	2	0	2	M	McKay($\mathbb{Z}_2^2$)
Local F_1	—	2	0	-2	M	Hirzebruch
K3 $\times E$	0	21	21	5	M	Δ_5 / BKM
Enriques $\times E$	0	11	11	4	M	Allcock / $O(2,10)$
Quintic	-200	1	101	$-25/3$	M	BCOV
$\mathbb{P}^5[3,3]$	-144	1	73	-6	M	BCOV
$\mathbb{P}^5[2,4]$	-176	1	89	$-22/3$	M	BCOV
$\mathbb{P}^6[2,2,3]$	-144	1	73	-6	M	BCOV
$\mathbb{P}^7[2,2,2,2]$	-128	1	65	$-16/3$	M	BCOV
Bicubic $\mathbb{P}^2 \times \mathbb{P}^2$	-162	2	83	$-27/4$	M	BCOV
Banana (Schoen)	0	19	19	0	$\mathrm{M}^{\geq 4}$	BKY
$\mathrm{BV}(1,1,1)$	-108	0	54	$-9/2$	M	Voisin–Borcea
$\mathrm{BV}(10,0,0)$	0	35	35	0	?	Voisin–Borcea
$\mathrm{BV}(10,8,0)$	0	19	19	0	?	Voisin–Borcea
$\mathrm{BV}(20,2,0)$	120	61	1	5	M	Voisin–Borcea

5. $\kappa = 0$ **does not imply trivial tower** (AP31): the banana manifold and the balanced Borcea–Voisin families ($r = 10$) have $\kappa = 0$ but potentially nontrivial higher-arity shadows sourced by instantons.

6. **Borcea–Voisin as κ-interpolation**: the BV family parametrized by (r, a, δ) gives $\kappa = (r - 10)/2$, interpolating between $\kappa = -9/2$ (minimal, $r = 1$) and $\kappa = 5$ (maximal, $r = 20$, recovering K3 $\times E$).

Verification: `compute/lib/cy3_grand_atlas.py`, `compute/tests/test_cy3_grand_atlas.py` (119 tests).

Remark 20.20 (Enriques $\times E$ kappa anomaly). The grand atlas uses $\kappa(\mathrm{Enr} \times E) = 4$, the weight of the Allcock Borcherds product on $O(2,10)$, verified by `enriques_shadow.py` (72 tests). This is the authoritative value. The naive BCOV formula $\kappa = \chi/24$ does not apply because Enriques $\times E$ is a *generalized* CY3 ($h^{3,0} = 0$, torsion canonical). The ratio $\kappa(\mathrm{K3} \times E)/\kappa(\mathrm{Enr} \times E) = 5/4$.

Chapter 21

The seven faces of $r_{CY}(z)$ for Calabi–Yau chiral algebras

For a Calabi–Yau chiral algebra arising from the CY-to-chiral functor, the binary collision residue $r_{CY}(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{CY})$ has seven equivalent realizations specific to the CY setting: as the CY bar-cobar twisting morphism, as a CoHA / perverse-coherent-sheaves shadow, as the classical CY Poisson coisson, as the Maulik–Okounkov R-matrix (for $K3 \times E$), as the affine super Yangian $Y(\widehat{\mathfrak{gl}}_1)$ r-matrix (for toric CY$_3$), as the elliptic Sklyanin bracket (for toroidal CY), and as a Gaudin model arising from CY$_3$ collision residues. This chapter establishes the seven-face identification in the CY setting and shows how Vol I's general theorems specialize to the CY$_3$ cases.

21.1 The CY$_3$ collision residue

The collision residue is a single algebraic operation extracted from the universal Maurer–Cartan element Θ_A of a chiral algebra. In the CY setting, this operation acquires geometric meaning: the residue is computed against the holomorphic CY volume form, and the resulting binary operation is the classical limit of the quantum vertex chiral group braiding.

Definition 21.1 *(Binary collision residue, CY version).* Let $\mathcal{C}$ be a smooth proper Calabi–Yau category of dimension d with holomorphic volume form $\Omega_{\mathcal{C}}$. Let $A_{\mathcal{C}}$ denote the chiral algebra produced by the CY-to-chiral functor (Theorem CY-A$_d$, proved for $d = 2$, conditional on chain-level $\mathbb{S}^3$-framing for $d = 3$). Let $\Theta_{A_{\mathcal{C}}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{A_{\mathcal{C}}})$ denote its universal Maurer–Cartan element (cf. Vol I, Theorem III). The *binary CY collision residue* is

$$r_{CY}(z) \;:=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{A_{\mathcal{C}}}) \;\in\; A_{\mathcal{C}} \otimes A_{\mathcal{C}}\,[[z^{-1}]],$$

the genus-0, arity-2 residue along the collision divisor of $\overline{C}_2(\mathrm{Spec}\,\mathbb{C})$, paired against $\Omega_{\mathcal{C}}$ in the CY direction.

Remark 21.2 (The CY direction is a separate slot). The collision residue extracts the z^{-1} part of Θ along the chiral curve direction. In the CY setting, Θ has *two* gradings: arity in the chiral direction (controlled by the curve X) and Hodge weight in the CY direction (controlled by $\mathcal{O}_{\mathcal{C}}$). The CY volume form $\Omega_{\mathcal{C}}$ kills the Hodge weight by integration; what remains is a chiral binary operation valued in $A_{\mathcal{C}} \otimes A_{\mathcal{C}}$. This is the precise sense in which r_{CY} couples the two directions.

Remark 21.3 (Three invariants must not be conflated). For any chiral algebra A, the binary residue carries three *distinct* numerical invariants which the CY setting will illuminate:

(i) $p_{\max}(A)$, the maximal pole order of generator-on-generator OPEs (cf. Vol I, Definition 20.9);

(ii) $k_{\max}(A) = p_{\max}(A) - 1$, the collision depth of r_{CY} itself, after d log-absorption (cf. Vol I, Definition 20.9);

(iii) $r_{\max}(A)$, the shadow depth, the arity at which the obstruction tower $\Theta_A^{\leq r}$ terminates (Vol I, Definition 20.9).

The $\beta\gamma$ system is the archetypal witness of their independence: $p_{\max}(\beta\gamma) = 1$, $k_{\max}(\beta\gamma) = 0$, $r_{\max}(\beta\gamma) = 4$ (class C). Conflating these produces incorrect classifications, and the seven-face programme will distinguish them at every face.

21.2 Face 1: the CY bar-cobar twisting morphism

The first face is the home framework of this volume. The CY-to-chiral functor sends a Calabi–Yau category C to a chiral algebra A_C, and the bar construction sends A_C to a factorization coalgebra $\overline{B}(A_C)$ on $\mathrm{Ran}(X)$. The collision residue is the binary component of the universal twisting morphism between $\overline{B}(A_C)$ and A_C.

Theorem 21.4 *(Face 1: CY bar-cobar realization, $d = 2$).* [PH]Let A_C be the chiral algebra of a CY_2 category C produced by the CY-to-chiral functor Φ (Theorem CY-A_2). Then:

(i) The convolution dg Lie algebra $\mathfrak{g}^{\mathrm{mod}}_{A_C}$ identifies with the homotopy Lie algebra of twisting morphisms $\hom_\alpha(\overline{B}(A_C), A_C)$ between bar coalgebra and chiral algebra (cf. Vol I, Theorem III);

(ii) The universal MC element Θ_{A_C} is the universal twisting morphism;

(iii) The binary collision residue $r_{CY}(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{A_C})$ is the leading arity-2 component of this twisting morphism, evaluated in the CY direction against Ω_C.

[PH]

Proof sketch. For (i), the cyclic bar complex $\mathrm{CC}_\bullet(C)$ identifies with $\overline{B}(A_C)$ as a factorization coalgebra (Theorem 20.1). The convolution identification is the application of Vol I's master twisting-morphism theorem to this coalgebra-algebra pair. For (ii), Θ_{A_C} satisfies the MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ at all levels, with the bar-intrinsic construction $\Theta = D - d_0$. For (iii), the collision residue is computed from Θ by restricting to genus 0, arity 2, and pairing against the holomorphic volume form. □

Conjecture 21.5 *(Face 1: CY bar-cobar realization, $d = 3$).* The statement of Theorem 21.4 extends to $d = 3$, conditional on the chain-level $\mathbb{S}^3$-framing construction of Conjecture CY-A_3. Concretely: if the CY_3-to-chiral functor Φ_3 can be defined on a full subcategory $\mathcal{D} \subset \mathrm{CY}_3$-$\mathbf{Cat}$, then for every $C \in \mathcal{D}$ the convolution identification, the universal MC element, and the binary collision residue extraction of Theorem 21.4(i)–(iii) continue to hold for $A_C := \Phi_3(C)$. [CJ]

Remark 21.6 (Bar-cobar inversion is not the seven-face master move). The cobar functor Ω inverts the bar complex: $\Omega(\overline{B}(A)) \simeq A$ (Vol I, Theorem B). This is a round-trip that recovers A itself. The seven-face programme is not about recovering A from $\overline{B}(A)$; it is about realizing a single algebraic object, the binary residue $r_{CY}(z)$, in seven distinct geometric languages. The closed-string or bulk side is the chiral derived center $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(A_C)$ (Vol I, Theorem H), not the cobar; the seven faces all live on the boundary A_C side.

21.3 Face 2: CoHA and perverse coherent sheaves

The second face is enumerative geometry: the binary residue is the leading non-trivial structure constant of the cohomological Hall algebra of C, or equivalently, the Yoneda product on perverse coherent sheaves. This face is the natural home of Donaldson–Thomas invariants.

CONJECTURE 21.7 *(Face 2: CoHA realization of r_{CY}).* Let C be a smooth proper CY$_3$ category with quiver-with-potential presentation (Q, W) at a stable point of the moduli of stability conditions. Let $\mathcal{H}(Q, W)$ denote the critical cohomological Hall algebra (Definition 15.3). Then the binary collision residue of A_C admits a CoHA realization

$$r_{CY}^{\mathrm{CoHA}}(z) \;=\; \mu^{(2)}_{\mathcal{H}(Q,W)} \otimes z^{-1} \;\in\; \mathcal{H}(Q, W) \otimes \mathcal{H}(Q, W)\,[[z^{-1}]],$$

where $\mu^{(2)}$ is the binary CoHA product, paired with the d log-propagator on $\overline{C}_2$. [CJ]

Remark 21.8 (Status of Conjecture 21.7). The CoHA $\mathcal{H}(Q, W)$ exists unconditionally (Kontsevich–Soibelman, Davison–Meinhardt). The chiral algebra A_C exists for $d = 2$ and is conjectural for $d = 3$. The identification with r_{CY} requires the CY-to-chiral functor to be defined and to commute with critical cohomology, which is the content of Conjecture CY-C and is currently verified for the toric CY$_3$ examples (Theorems 15.4, 15.5). In those cases the identification reduces to the Schiffmann–Vasserot and Rapcak–Soibelman–Yang–Zhao theorems and is ProvedElsewhere with attribution to those two papers.

Remark 21.9 (Perverse coherent sheaves). For a CY$_3$ category arising as $D^b(\mathrm{Coh}(X))$ for X a smooth CY$_3$ variety, the CoHA description above is equivalent to a description in terms of perverse coherent sheaves on X (Bridgeland, Bayer–Macri). The Yoneda product on $\mathrm{Ext}^\bullet$ of perverse coherent sheaves is the binary CoHA product, and the binary collision residue extracts its leading Ext^1 component. This is the geometric face: the residue r_{CY} *is* a piece of the deformation theory of the CY$_3$.

Remark 21.10 (CoHA is not the full quantum vertex chiral group). The critical CoHA is an associative algebra: the positive (or E_1-ordered) half of the conjectural quantum vertex chiral group $G(X)$. The full $G(X)$ would be braided E_2, with the braiding supplied by the R-matrix of the affine super Yangian (Section 21.6). The CoHA face captures the *shadow* of r_{CY} on the ordered side; the braided side is faces 4 and 5.

21.4 Face 3: classical CY Poisson coisson

The third face is the classical limit. In the CY setting, the binary residue r_{CY} has a classical avatar: a coisson bracket on the sheafification of A_C, which in turn is a chiral version of the holomorphic Poisson bracket arising from the CY (2,0)-form (in $d = 2$) or the holomorphic top form via contraction (in $d = 3$).

THEOREM 21.11 *(Face 3: classical Poisson coisson, $d = 2$).* [PH]Let C be a CY$_2$ category and let A_C be its chiral algebraization (Theorem CY-A$_2$). Let $\hbar$ denote a quantization parameter. Then:

(i) The classical limit $A_C^{\mathrm{cl}} := \lim_{\hbar\to 0} A_C$ is a Poisson vertex algebra (PVA) with bracket $\{\cdot,\cdot\}_\lambda$ valued in $A_C^{\mathrm{cl}} \otimes \mathbb{C}[\lambda]$;

(ii) The leading coefficient $\{\cdot,\cdot\}_\lambda^{(0)}$ of the PVA bracket is a chiral Poisson bracket whose classical r-matrix is exactly $r_{CY}(z)$ at $\hbar = 0$;

(iii) This identifies r_{CY}^{cl} as the coisson bracket constructed from the CY volume form Ω_C via the Poisson structure on $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(A_C^{\mathrm{cl}})$ (cf. Vol I, Theorem 20.9).

[PH]

Proof sketch. For a CY_2 category, Φ is the proved CY-to-chiral functor (Chapter 8). The PVA structure on A_C^{cl} is the classical limit of the chiral OPE. The leading λ coefficient of the λ-bracket is the contraction of two factors along the binary collision divisor; this is exactly $\mathrm{Res}_{0,2}^{\mathrm{coll}}$ computed in the classical limit, hence r_{CY}^{cl}. The identification with the coisson bracket arising from Ω_C is the coisson–PVA dictionary (cf. Vol I, Theorem 20.9, applied to the specialization $C \mapsto A_C$). □

Conjecture 21.12 *(Face 3: classical Poisson coisson, $d = 3$).* The statement of Theorem 21.11 extends to $d = 3$, conditional on Conjecture CY-A3: if A_C is a CY_3-chiral algebraization of C, then items (i)–(iii) of Theorem 21.11 hold for A_C^{cl} with the CY_3 volume form $\Omega_C \in H^0(C, \omega_C)$ supplying the coisson pairing. [CJ]

Remark 21.13 (Convention: λ-brackets vs OPE modes (AP49)). Vol I uses OPE modes $a_{(n)}b$. Vol II uses λ-brackets $\{a_\lambda b\} = \sum_n \lambda^{(n)} a_{(n)} b$ with the divided power $\lambda^{(n)} = \lambda^n/n!$. Vol III uses motivic and categorical language. When transcribing r_{CY} between volumes, the $1/n!$ factor must be tracked: the λ^3 coefficient of a λ-bracket is $a_{(3)}b/3! = a_{(3)}b/6$, not $a_{(3)}b$. This convention bridge is the source of several historical errors and is the subject of AP49.

21.5 Face 4: the Maulik–Okounkov R-matrix (for $K3 \times E$)

The fourth face is geometric representation theory: the binary residue is the classical limit of the Maulik–Okounkov stable-envelope R-matrix acting on the equivariant cohomology of a Nakajima quiver variety. For the CY_3 specialization $K3 \times E$, this is the most concrete realization, and it produces r_{CY} explicitly in terms of theta functions on the elliptic factor.

The Maulik–Okounkov stable envelope and its R-matrix $R_{\mathrm{MO}}^{K3\times E}(z)$ exist unconditionally (Maulik–Okounkov 2019); this is the ProvedElsewhere component. The genuinely new content of face 4 is the identification of the classical limit with the binary CY collision residue.

Conjecture 21.14 *(Face 4: MO realization for $K3 \times E$).* For the CY_3 specialization $X = K3 \times E$, the binary CY collision residue identifies with the classical limit of the Maulik–Okounkov R-matrix:

$$r_{CY}^{K3\times E}(z) \;=\; \lim_{\hbar\to 0} \hbar^{-1}\big(R_{\mathrm{MO}}^{K3\times E}(z) - \mathrm{id}\big).$$

The right-hand side is the rational/elliptic stable envelope of Maulik–Okounkov, evaluated on the Hilbert scheme of points on $K3$ along the elliptic direction E. The conjecture is equivalent to Conjecture CY-C for $K3 \times E$. [CJ]

Remark 21.15 (The CY involution $g(z)g(-z) = 1$). For $K3 \times E$, the MO R-matrix is governed by a single Yangian generator $g(z)$ satisfying $g(z)g(-z) = 1$ (Theorem 14.138). This CY involution is the $\mathbb{S}^2$-framing of the CY-to-chiral functor seen at the level of the binary residue: the $z \mapsto -z$ involution is the chiral analogue of the CY Serre functor $\mathbb{S} \simeq [d]$. In particular the CY involution forces the residue to be of *odd* pole order, in agreement with AP19 (the d log-absorption sends even-order poles to odd-order poles in the residue).

Remark 21.16 (Self-dual Gepner trivialization). At the Gepner point of $K3$, where the chiral algebra has $\mathcal{N} = (4,4)$ enhancement, the MO R-matrix trivializes (Proposition 14.139), so r_{CY} vanishes. This is consistent with the seven-face master statement: at the self-dual point of any face, all seven realizations vanish simultaneously. For $K3 \times E$ this is the statement that $\kappa_{\mathrm{eff}} = 0$ in the $\mathcal{N} = (4,4)$ enhancement.

21.6 Face 5: the affine super Yangian $Y(\widehat{\mathfrak{gl}}_1)$ for toric CY_3

The fifth face is the explicit toric face: when the CY_3 is toric and admits a quiver-with-potential presentation, the binary residue is the classical r-matrix of the affine super Yangian $Y(\widehat{\mathfrak{gl}}_1)$ (for $\mathbb{C}^3$, the Jordan quiver) or more generally $Y(\widehat{\mathfrak{g}}_{Q_X})$ (for the toric quiver of X).

THEOREM 21.17 *(Face 5 for toric CY_3 via RSYZ).* [PE] Let X be a toric CY_3 without compact 4-cycles, with toric quiver (Q_X, W_X) and affine super Yangian $Y(\widehat{\mathfrak{g}}_{Q_X})$ (Theorem 15.5). Then the binary CY collision residue identifies with the classical r-matrix:

$$r^X_{CY}(z) \;=\; r^{\mathrm{cl}}_{Y(\widehat{\mathfrak{g}}_{Q_X})}(z) \;=\; \frac{\Omega_{Y(\widehat{\mathfrak{g}}_{Q_X})}}{z} \;+\; (\text{higher poles dictated by } p_{\max}(W_X)),$$

where $\Omega_{Y(\widehat{\mathfrak{g}}_{Q_X})}$ is the Casimir tensor of the affine super Yangian. At the Jordan-quiver specialization $X = \mathbb{C}^3$, the identification reduces to the r-matrix of the affine Yangian $Y(\widehat{\mathfrak{gl}}_1) \cong \mathcal{W}_{1+\infty}$ at the self-dual level. [PE]

Remark 21.18 (Status partition, AP60 watch). Theorem 21.17 is a restatement of the Rapcak–Soibelman–Yang–Zhao theorem (Theorem 15.5) in the collision-residue language of the seven-face programme. AP60 forbids tagging this theorem ProvedHere: the classical content is due to Schiffmann–Vasserot (for $\mathbb{C}^3$, Theorem 15.4) and Rapcak–Soibelman–Yang–Zhao (for general toric without compact 4-cycles, Theorem 15.5). The genuinely new content of face 5 is the translation of their r-matrix into the language of $\mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{A_X})$, which is recorded as the scholium below.

PROPOSITION 21.19 *(Collision-residue translation of face 5, $\mathbb{C}^3$ case).* For $X = \mathbb{C}^3$, the Schiffmann–Vasserot Casimir tensor $\Omega_{Y(\widehat{\mathfrak{gl}}_1)}$ coincides with $\mathrm{Res}_{z=0}\big[r^{\mathbb{C}^3}_{CY}(z)\big]$, where $r^{\mathbb{C}^3}_{CY}$ is the binary CY collision residue of the $\mathbb{C}^3$ chiral algebra. [PH]

Proof sketch. The Jordan quiver superpotential $W = (XYZ - XZY)$ has $p_{\max}(W) = 1$ (each cubic monomial contributes a single contraction in the binary OPE), so $k_{\max} = 0$ and $r^{\mathbb{C}^3}_{CY}(z)$ has a *single* pole at $z = 0$. The residue at this pole is the Schiffmann–Vasserot Casimir $\Omega_{Y(\widehat{\mathfrak{gl}}_1)} \in Y^+(\widehat{\mathfrak{gl}}_1)^{\otimes 2}$ (Theorem 15.4). By the universality of Θ in MC moduli, this is exactly $r^{\mathbb{C}^3}_{CY}$. The matching of higher poles with the toric quiver superpotential follows from the RSYZ theorem for toric quivers without compact 4-cycles. □

Remark 21.20 (Three invariants for toric CY_3 (AP59 watch)). For the standard toric CY_3 family the three invariants are:

Geometry	Quiver	$p_{\max}$	$k_{\max}$	$r_{\max}$
$\mathbb{C}^3$	Jordan	1	0	2 (class G)
Resolved conifold	Klebanov–Witten	1	0	2 (class G)
Local $\mathbb{P}^2$	McKay $\mathbb{Z}_3$	2	1	∞ (class M)
Local $\mathbb{P}^1 \times \mathbb{P}^1$	Beilinson	2	1	$\infty/2$

The pole order $p_{\max}$, the collision depth $k_{\max} = p_{\max} - 1$, and the shadow depth $r_{\max}$ are three independent invariants. In the toric family above, $p_{\max}$ is small (at most 2) but $r_{\max}$ can be infinite (local $\mathbb{P}^2$, McKay $\mathbb{Z}_3$). This separates the pole-order face (face 5, controlled by $p_{\max}$) from the shadow tower face (face 1, controlled by $r_{\max}$). AP59 forbids identifying these.

Gaudin Hamiltonians for toric CY_3 from collision residues

The collision residue at the Jordan quiver ($\mathbb{C}^3$) is the $Y(\widehat{\mathfrak{gl}}_1)$ classical r-matrix; this allows one to write down a chiral Gaudin model whose Hamiltonians are extracted from CY_3 collision data, providing the first instance of face 7 inside face 5. The full development of face 7 occupies Section 21.8.

Construction 21.21 *(Gaudin Hamiltonians from $r_{CY}^{\mathbb{C}^3}$).* Fix marked points $z_1, \ldots, z_n \in \mathbb{C}$ (the chiral curve, with n copies of the Fock representation of $Y(\widehat{\mathfrak{gl}}_1)$). Define

$$H_i^{\mathbb{C}^3} \;:=\; \sum_{j \neq i} \frac{\Omega_{ij}^{Y(\widehat{\mathfrak{gl}}_1)}}{z_i - z_j} \;\in\; Y(\widehat{\mathfrak{gl}}_1)^{\otimes n},$$

where $\Omega_{ij}^{Y(\widehat{\mathfrak{gl}}_1)}$ is the $Y(\widehat{\mathfrak{gl}}_1)$ Casimir $\mathrm{Res}_{z=0}\, r_{CY}^{\mathbb{C}^3}(z)$, acting on the i-th and j-th tensor factors. The operators $\{H_i^{\mathbb{C}^3}\}$ commute pairwise as a direct consequence of the classical Yang–Baxter equation for $r_{CY}^{\mathbb{C}^3}$ (cf. Vol I, Theorem 20.9).

Remark 21.22 (Wall-crossing as gauge equivalence). Across walls in the Bridgeland stability manifold, the toric Yangian r-matrix transforms by a Kontsevich–Soibelman wall-crossing automorphism. In the present language, this is exactly an MC gauge equivalence on Θ_{A_X} (Theorem 15.16): $r_{CY}^{X,\zeta'}(z) = e^{\mathrm{ad}_\alpha} r_{CY}^{X,\zeta}(z)$ for the unique gauge parameter $\alpha \in \mathfrak{n}_W$ supported on the wall divisor. The Gaudin Hamiltonians of Construction 21.21 are gauge-equivalent across walls; their joint spectrum is chamber-independent.

21.7 Face 6: elliptic Sklyanin bracket (for toroidal CY)

The sixth face is the elliptic deformation: when the chiral curve is itself an elliptic curve E_τ and the CY data are toroidal, the binary residue acquires an elliptic structure governed by theta functions. The classical limit of this r-matrix is the elliptic Sklyanin bracket, and the quantization is Felder's elliptic quantum group.

The elliptic r-matrix and its classical Sklyanin bracket are due to Sklyanin (1983) and Belavin–Drinfeld (1984), with the quantization to Felder's elliptic quantum group (Definition 14.14) classical. These components are ProvedElsewhere with explicit attribution. The genuinely new content is the identification of the elliptic r-matrix with the binary CY collision residue $\mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{A_C})$ via the chiral propagator $d\log\theta_1(z\,|\,\tau)$.

Theorem 21.23 *(Face 6 for $\mathfrak{sl}_2$ Heisenberg on E_τ).* [PH] Let E_τ be an elliptic curve and let $A = H_{E_\tau}^{\mathfrak{sl}_2}$ be the $\mathfrak{sl}_2$ Heisenberg chiral algebra realized on E_τ. Then the binary CY collision residue admits the elliptic representative

$$r_{CY}^{\tau}(z) \;=\; \frac{\theta_1'(0\,|\,\tau)}{\theta_1(z\,|\,\tau)}\,\Omega_{\mathfrak{sl}_2},$$

where θ_1 is the odd Jacobi theta function and $\Omega_{\mathfrak{sl}_2}$ is the Casimir tensor of $\mathfrak{sl}_2$ (cf. Section 14.3). The classical limit of r_{CY}^τ defines the Sklyanin bracket $\{f, g\}_\tau = \langle r_{CY}^\tau, [\nabla f, \nabla g]\rangle$ on $E_{q,p}(\mathfrak{sl}_2)^*$, and the quantization $\hbar \to q$ recovers Felder's elliptic quantum group $E_{q,p}(\mathfrak{sl}_2)$ with dynamical parameter λ identified as the B-cycle monodromy of the elliptic propagator. [PH]

Conjecture 21.24 *(Face 6 for general toroidal CY).* The statement of Theorem 21.23 extends to general toroidal CY realizations X on E_τ: the binary residue $r_{CY}^{\tau,X}(z)$ is the elliptic r-matrix of $Y(\widehat{\mathfrak{g}}_X)$ acquiring dynamical theta-function corrections in λ, and its classical limit is the corresponding Sklyanin bracket (cf. Vol I, Theorem 20.9). [CJ]

Remark 21.25 (AP60 status partition). The Sklyanin bracket and Belavin–Drinfeld classification are classical results (1983). The genuinely new content of Theorem 21.23 is the identification of the elliptic r-matrix with the binary CY collision residue $\mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{A_C})$ via the chiral propagator $d\log\theta_1(z\,|\,\tau)$. AP60 forbids tagging the entire theorem ProvedHere; the classical components are ProvedElsewhere with explicit attribution to Sklyanin (1983) and Belavin–Drinfeld (1984).

The three-parameter $\hbar$ identification

A subtle but central observation is that three different $\hbar$ parameters appearing in the literature on elliptic quantum groups are the same parameter under the seven-face dictionary.

Construction 21.26 *(Three-parameter $\hbar$ identification).* The following three quantization parameters coincide:

(i) the boundary coupling $\hbar^{\mathrm{KZ}}$ in the Knizhnik–Zamolodchikov equation (Knizhnik–Zamolodchikov 1984; cf. Vol I, Theorem 20.9);

(ii) the loop parameter $\hbar^{\mathrm{DNP}}$ in the Dimofte–Niu–Py realization of the dg-shifted Yangian (cf. Vol I, Theorem 20.9);

(iii) the prefactor $\hbar^{\mathrm{coll}}$ of the binary CY collision residue $r_{CY}(z)$.

The identification $\hbar^{\mathrm{KZ}} = \hbar^{\mathrm{DNP}} = \hbar^{\mathrm{coll}}$ holds canonically once the CY direction is fixed, and follows from the universality of the binary residue in $\mathfrak{g}^{\mathrm{mod}}_{A_C}$.

Remark 21.27 (Convention sanity check). The identification of Construction 21.26 fails if any of the three parameters carries a hidden $2\pi i$ or $1/n!$. In particular, the KZ parameter is conventionally normalized so that the connection $\nabla = d - \hbar^{-1}\sum_i r_{ij}\, d\log(z_i - z_j)$ has integer monodromy on the configuration space; the DNP parameter is normalized so that the loop algebra has bracket $[a_m, b_n] = [a,b]_{m+n} + m\hbar\delta_{m+n,0}\langle a,b\rangle$; the collision residue is normalized so that $\Omega = r_{CY}(z)\cdot z$ at $z = 0$. These normalizations are mutually compatible; the identification is parameter-by-parameter, not just up to scale.

21.8 Face 7: Gaudin model from CY_3 collision residues

The seventh face is new in this volume. In Vol I, face 7 of the standard seven-face master is the Gaudin model arising from a chiral algebra with primitive generator (Vol I, Theorem 20.9). In the CY setting, the Gaudin model arises from CY_3 collision residues, with Gaudin Hamiltonians constructed directly from the toric or elliptic r-matrix of face 5 or face 6, and with the Bethe ansatz interpreted in terms of DT/PT counts.

THEOREM 21.28 *(Face 7 for $\mathbb{C}^3$: Gaudin from the Jordan-quiver residue).* [PH] Let $X = \mathbb{C}^3$. Fix n marked points $z_1, \dots, z_n$ on the chiral curve and Fock representations $V_1, \dots, V_n$ of $Y(\widehat{\mathfrak{gl}}_1)$. Define

$$H_i^{\mathbb{C}^3} := \sum_{j\neq i} \frac{\Omega_{ij}^{Y(\widehat{\mathfrak{gl}}_1)}}{z_i - z_j},$$

acting on $V_1\otimes\cdots\otimes V_n$, where $\Omega_{ij}^{Y(\widehat{\mathfrak{gl}}_1)}$ is the Casimir tensor $\mathrm{Res}_{z=0}\big[r_{CY}^{\mathbb{C}^3}(z)\big]$ acting on the i-th and j-th factors. Then:

(i) The operators $\{H_i^{\mathbb{C}^3}\}_{i=1}^n$ commute pairwise: $[H_i^{\mathbb{C}^3}, H_j^{\mathbb{C}^3}] = 0$;

(ii) The joint spectrum is governed by the $\mathcal{W}_{1+\infty}$ Bethe ansatz of Litvinov and Maulik–Okounkov;

(iii) The DT counts of $\mathbb{C}^3$ (MacMahon plane-partition generating function) provide the integer multiplicities of the Bethe roots: a Bethe state at the configuration $\{w_a\}$ corresponds to a plane partition of the appropriate shape.

[PH]

Conjecture 21.29 *(Face 7 for general toric/toroidal CY_3).* The statement of Theorem 21.28 extends to any CY3 X admitting a toric (face 5) or toroidal (face 6) chiral realization, with Hamiltonians $H_i^X := \sum_{j\neq i} r_{CY}^X(z_i - z_j)_{ij}$, commutativity following from the classical Yang–Baxter equation for r_{CY}^X, and DT/PT counts of X supplying the Bethe root multiplicities. [CJ]

Remark 21.30 (Proof sketch for $\mathbb{C}^3$). For $X = \mathbb{C}^3$, $r_{CY}^{\mathbb{C}^3}(z) = \Omega_{Y(\widehat{\mathfrak{gl}}_1)}/z$ has a single pole. The Hamiltonians $H_i^{\mathbb{C}^3} = \sum_{j\neq i} \Omega_{ij}/(z_i - z_j)$ are the standard Gaudin Hamiltonians for $Y(\widehat{\mathfrak{gl}}_1)$, and their commutativity follows from the classical Yang–Baxter equation for $r_{CY}^{\mathbb{C}^3}$ (Construction 21.21; cf. Vol I, Theorem 20.9). The Bethe ansatz identification is the Litvinov $\mathcal{W}_{1+\infty}$ Bethe ansatz, which identifies Bethe roots with crystal melting configurations of $\mathbb{C}^3$ (equivalently, plane partitions).

Remark 21.31 (Why CY3 supplies the Bethe roots). The novelty of face 7 in the CY setting is that the Bethe roots are not abstract numbers; they are CY3 DT/PT counts. For $\mathbb{C}^3$ the DT/PT counts are MacMahon-type partition functions (3D plane partitions), and each plane partition contributes a single Bethe root. For more general toric CY3, the topological vertex provides the partition counts and hence the Bethe-root populations. In this sense the CY3 Gaudin model "geometrizes the Bethe ansatz" in the same way that the toric quiver geometrizes the affine Yangian.

Remark 21.32 (Face 7 is genuinely new for CY3). Vol I face 7 (Gaudin from chiral algebra with primitive generator) and Vol II face 7 (Gaudin from twisted holography boundary algebra) both require an algebra with explicit generators. For CY3, the analog must extract Gaudin data from the conjectural quantum vertex chiral group $G(X)$ via its CoHA presentation; the construction passes through the r-matrix of face 5 (or face 6 in the toroidal case), not directly through generators of $G(X)$ itself. This is the genuinely new content of T4 in the CY setting and is the main object of this section.

21.9 The seven-face master theorem in the CY setting

The preceding seven sections each realize the binary CY collision residue in one geometric language. The master theorem asserts that all seven realizations are canonically equivalent, with the equivalences given by explicit specialization or limit, and with the limit/specialization data recorded by the seven-face commuting diagram of Vol I.

Conjecture 21.33 *(Seven faces of $r_{CY}(z)$).* Let C be a Calabi–Yau category of dimension $d \in \{2, 3\}$, and let A_C be its chiral algebraization (proved for $d = 2$; conditional on Conjecture CY-A3 for $d = 3$). Then the seven realizations

$$r_{CY}^{(\mathrm{F1})},\ r_{CY}^{(\mathrm{F2})},\ r_{CY}^{(\mathrm{F3})},\ r_{CY}^{(\mathrm{F4})},\ r_{CY}^{(\mathrm{F5})},\ r_{CY}^{(\mathrm{F6})},\ r_{CY}^{(\mathrm{F7})}$$

constructed in Sections 21.2–21.8 all coincide, in the sense that

$$r_{CY}^{(\mathrm{F}_i)} = \Phi_{i,j}\big(r_{CY}^{(\mathrm{F}_j)}\big) \qquad \forall\, i, j \in \{1, \dots, 7\},$$

for canonical change-of-language morphisms $\Phi_{i,j}$ given explicitly by Theorem CY-A (face 1 to face 3), the Schiffmann–Vasserot/RSYZ identification (face 2 to face 5), the MO stable envelope (face 4), the elliptic specialization $\hbar \to 0$ (face 6 to face 3), and the Litvinov Bethe ansatz (face 5 to face 7). The limits between faces are recorded in the seven-face commuting diagram (cf. Vol I, Theorem 20.9; Vol II, Theorem 20.9).

Remark 21.34 (Status by face). **Face 1** (CY bar-cobar twisting): [PH] for $d = 2$; [CJ] for $d = 3$ (conditional on CY-A3).

Face 2 (CoHA): [PE] for the existence of the CoHA (Kontsevich–Soibelman, Davison–Meinhardt); identification with r_{CY} is [CJ] in general, [PH] for $\mathbb{C}^3$ (Schiffmann–Vasserot, Theorem 15.4) and toric without compact 4-cycles (RSYZ, Theorem 15.5).

Face 3 (classical coisson): [PH] for $d = 2$ via the proved coisson–PVA dictionary; [CJ] for $d = 3$.

Face 4 (MO R-matrix): [PE] for the existence of R_{MO} (Maulik–Okounkov 2019); identification with r_{CY} for $K3 \times E$ is [CJ] in general, with explicit verification at the Gepner self-dual limit (Proposition 14.139).

Face 5 (toric Yangian): [PH] for $\mathbb{C}^3$; [PE] for toric without compact 4-cycles via RSYZ.

Face 6 (elliptic Sklyanin): [PE] for the classical Sklyanin bracket and its quantization to Felder's elliptic quantum group; identification with r_{CY} is [PH] in the $\mathfrak{sl}_2$ Heisenberg case via the Fay identity (Proposition 14.19).

Face 7 (CY_3 Gaudin): [PH] for $\mathbb{C}^3$ (Construction 21.21); [CJ] for general toric/toroidal CY_3.

The status partition above is consistent with AP60: each face's classical components are tagged ProvedElsewhere with explicit attribution, and only the genuinely new identifications carry the ProvedHere tag.

Remark 21.35 (Three invariants across the seven faces). By Remark 21.3, every realization of r_{CY} carries the three invariants $(p_{\max}, k_{\max}, r_{\max})$. AP59 demands that each face state which invariant is being measured, and forbids identifying them.

- Face 5 (toric Yangian) is sensitive to $p_{\max}$: the pole structure of $r_{Y(\hat{\mathfrak{g}})}$ is the OPE pole structure of the toric quiver superpotential.
- Face 6 (elliptic) carries $k_{\max}$ in its theta-function expansion: at $k_{\max} = 0$ the residue is purely a single-pole elliptic term; at $k_{\max} \geq 1$ there are derivative terms.
- Face 1 (bar-cobar) is sensitive to $r_{\max}$: the shadow tower $\Theta_A^{\leq r}$ terminates at arity $r_{\max}$, and the binary residue is the projection to $r = 2$.

The seven-face master statement is that the binary residue is the *same* object across all faces, but the sensitivity to the three invariants differs face by face.

21.10 Cross-volume bridge

This part of Vol III mirrors Vol I Part III and Vol II Part III. The common skeleton is the seven-face programme; the variation is in which face is most concrete.

Remark 21.36 (The three seven-face masters). The three volumes each devote a part to the seven-face programme, with the same architecture but different ground objects:

(1) *Vol I, Part III:* the binary collision residue of a chiral algebra on a curve, in seven languages: bar-cobar twisting, primitive generator, classical r-matrix, KZ connection, Gaudin model, Bethe ansatz, dg-shifted Yangian (cf. Vol I, Theorem 20.9).

(2) *Vol II, Part III:* the binary collision residue of a holomorphic-topological quantum group, in seven languages: open-string brace algebra, derived center, twisted holography boundary, line defect, Wilson line R-matrix, soft graviton coupling, Yangian double (cf. Vol II, Theorem 20.9).

(3) *Vol III, this chapter:* the binary CY collision residue of a Calabi–Yau chiral algebra, in seven CY-specific languages: CY bar-cobar, CoHA / perverse coherent sheaves, classical CY Poisson coisson, MO stable envelope, affine super Yangian for toric CY_3, elliptic Sklyanin for toroidal CY, Gaudin from CY_3 (Theorem **??** above).

The three master theorems are mutually compatible: under the CY-to-chiral functor Φ, face i of Vol III maps to a specialization of face i of Vol I, and similarly for Vol II. The CY setting is the most constrained: each face acquires geometric content that the abstract chiral algebra setting does not see (DT counts, plane partitions, crystal melting, MO stable envelopes).

Remark 21.37 (The CY direction is the new slot). Compared with Vol I and Vol II, the new feature of the CY seven-face master is the presence of the CY direction Ω_C. In Vol I the binary residue lives on $\overline{C}_2(X) \times \mathfrak{g}^{\otimes 2}$; in Vol II it lives on $\overline{C}_2(X) \times \text{open-sector}^{\otimes 2}$; in Vol III it lives on $\overline{C}_2(X) \times C^{\otimes 2}$, with the CY volume form supplying the pairing on $C^{\otimes 2}$. The CY direction is what makes face 4 (MO) and face 7 (CY_3 Gaudin) genuinely geometric: they live on quiver varieties whose definition requires the CY structure.

Remark 21.38 (κ is a face-by-face invariant). The modular characteristic $\kappa(A_C)$ depends on the algebraization, not on the geometry. For $K3 \times E$, four distinct algebraizations give four distinct values: $\kappa_{\mathrm{ch}} = 3$, $\kappa_{\mathrm{cat}} = 2$, $\kappa_{\mathrm{fiber}} = 24$, $\kappa_{\mathrm{BPS}} = 5$ (preface). Each value sits on a different face: κ_{ch} on face 1 (bar-cobar of the chiral de Rham complex), κ_{cat} on face 2 (CoHA), κ_{fiber} on face 3 (classical lattice VOA limit), κ_{BPS} on face 4 (MO/Igusa). The κ-spectrum is the imprint of the geometry on the seven-face master. The rule (from AP48) is that κ depends on the full algebra, not on its Virasoro subalgebra; in the CY setting, κ depends on the full *algebraization*, not on the underlying CY manifold.

Remark 21.39 (Outlook: an eighth face?). The standard seven-face master is closed: there are exactly seven languages in which the binary residue lives, corresponding to the seven edges of the Vol I commuting diagram. In the CY setting one might ask whether the deformation-quantization face (Khan–Zeng 3d PVA sigma model) constitutes an eighth. At present this face reduces to a combination of face 3 (classical coisson) and face 7 (Gaudin from CY_3), with the deformation-quantization parameter playing the role of the loop parameter in face 7. An independent eighth face would require a CY-specific structure not visible in the seven-face commuting diagram; the working notes record the open question of whether the 3d $\mathcal{N} = 2$ holomorphic-twisted theory provides such a structure.

The genus-1 extension, identifying the KZB connection with the elliptic r-matrix on the torus E_τ, is proved for affine KM in Vol I, Chapter `ch:genus1-seven-faces`.

This chapter closes the seven-face programme for Vol III. Subsequent chapters in Part V record the geometric Langlands implications and the bridges to Vols I–II at the level of theorem statements; the present chapter is the algebraic engine that makes the bridges possible.

Chapter 22

Modular Koszul Duality and CY Geometry

The bar-cobar bridge of Chapter 20 translates between the CY cyclic bar complex $\mathrm{CC}_\bullet(\mathcal{C})$ and the chiral bar complex $B(A_\mathcal{C})$. Volume I's modular Koszul duality operates on $B(A)$: Verdier intertwining $D_{\mathrm{Ran}}(B(A)) \simeq B(A^!)$, the complementarity theorem (Theorem C), and the genus expansion controlled by $\kappa(A)$ (Theorem D). This chapter constructs their CY-geometric incarnations: the modular convolution algebra, the CY complementarity pairing, and the shadow CohFT derived from the CY trace.

22.1 The modular convolution algebra for CY categories

22.2 CY complementarity

22.3 The CY shadow CohFT

Part V

The CY Frontier

Chapter 23

Geometric Langlands and CY Quantum Groups

The Feigin–Frenkel center $\mathfrak{z}(\hat{\mathfrak{g}})$ at the critical level $k = -h^\vee$ is $\mathrm{Fun}(\mathrm{Op}_{G^L}(D))$, the spectral side of the geometric Langlands correspondence. Applied to CY categories of Lie-theoretic origin, the functor Φ (Part II) produces chiral algebras whose bar complexes (Chapter 20) and modular Koszul data (Chapter 22) encode the automorphic side. Langlands duality $G \leftrightarrow G^L$ should manifest as a symmetry of the quantum vertex chiral group: the Koszul dual $A^!$ on one side corresponds to the Langlands dual on the other.

23.1 The geometric Langlands programme

23.2 CY categories in the Langlands correspondence

23.3 Quantum geometric Langlands and E_2-chiral algebras

23.4 The Feigin–Frenkel center and the Drinfeld center

Appendix A

Conventions

A.1 Grading conventions

All grading is **cohomological**: differentials have degree $+1$. The bar construction uses desuspension (matching Volumes I–II).

A.2 Sign conventions

Signs follow the Koszul rule throughout. For the curved A_∞-structure: $\mu_1^2(a) = [\mu_0, a]$ (commutator, minus sign).

A.3 Categorical conventions

- **dgCat** denotes the ∞-category of small dg categories over k.
- CY_d-**Cat** denotes the ∞-category of smooth d-dimensional CY categories.
- Tensor products of dg categories are derived unless stated otherwise.
- "Equivalence" means quasi-equivalence of dg categories (equivalence of associated ∞-categories).

A.4 Operadic conventions

- E_n denotes the little n-disks operad (Boardman–Vogt).
- $\overline{\mathrm{FM}}_k(\mathbb{C})$ denotes the Fulton–MacPherson compactification of $\mathrm{Conf}_k(\mathbb{C})$. Note: $\overline{\mathrm{FM}}$ is a blowup along diagonals, NOT the complement of the diagonal (AP, Vol I).
- $\mathrm{SC}^{\mathrm{ch,top}}$ denotes the Swiss-cheese operad from Volume II.

A.5 Bar complex and Koszul duality conventions

- $B(A)$ is the bar complex, a factorization *coalgebra* on $\mathrm{Ran}(X)$.

- $\Omega(B(A)) \simeq A$ is bar-cobar *inversion*: it recovers the original algebra (Vol I, Theorem B). This is NOT Koszul duality.
- $A^! = (H^*(B(A)))^\vee$ is the Koszul dual *algebra* (linear dual of bar cohomology).
- $D_{\mathrm{Ran}}(B(A)) \simeq B(A^!)$ is the Verdier dual (Vol I, Theorem A, Convention conv:bar-coalgebra-identity).
- The collision r-matrix $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_A)$ has pole orders *one less* than the OPE, because the bar construction extracts residues along $d\log(z_i - z_j)$, which absorbs one power of $(z - w)$ (Vol I, AP19).